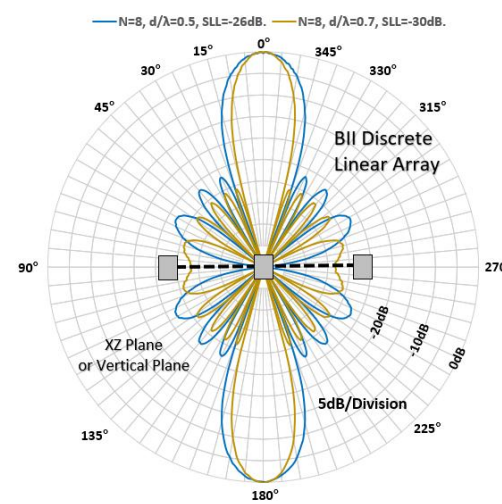
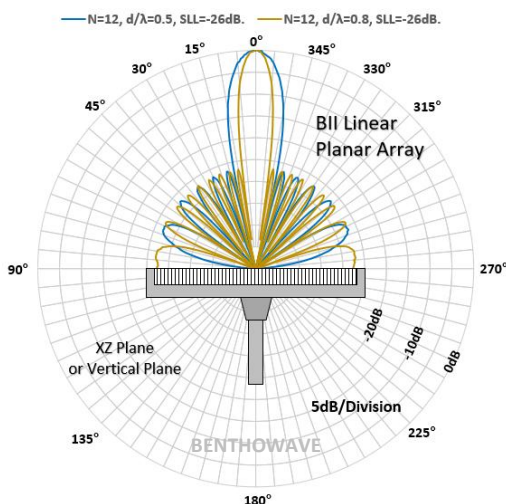
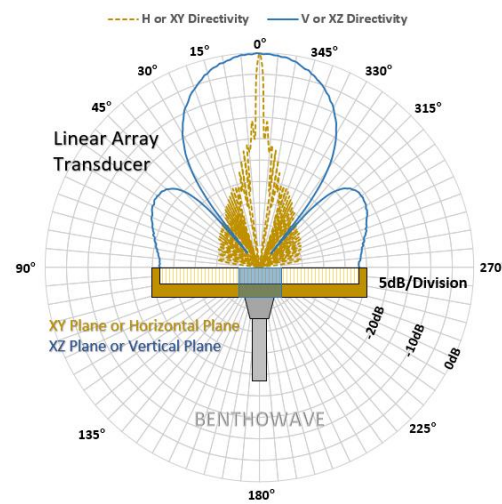
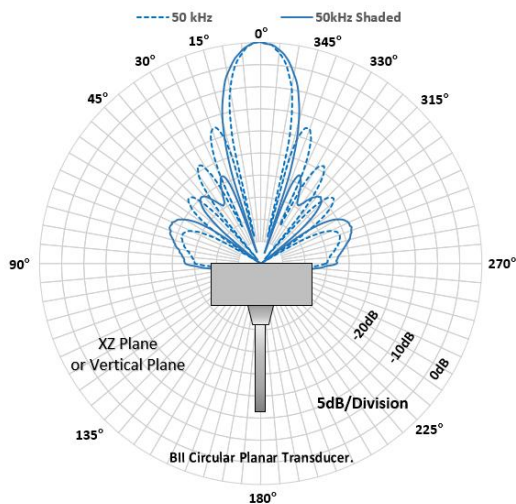
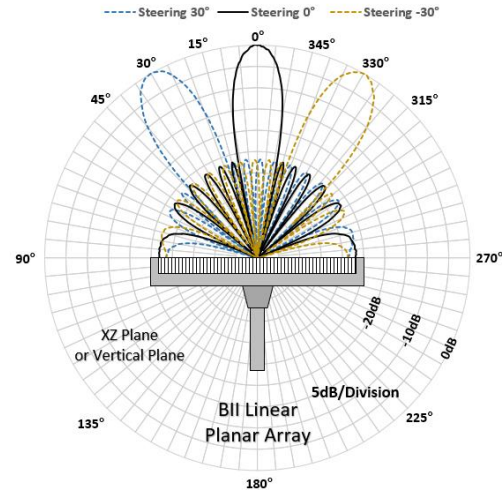
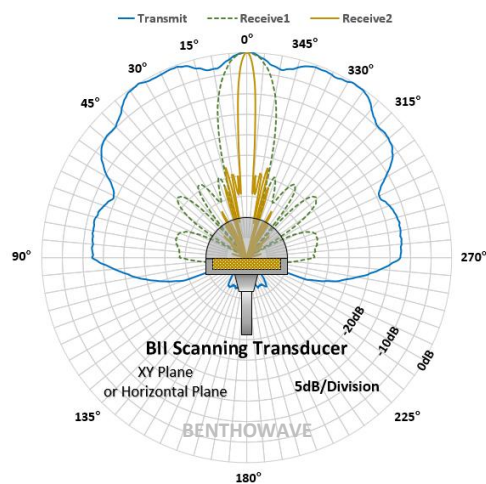


BENTHOWAVE

Acoustic Transducers and Measurement Handbook

A Guide to Measurement Technology and Techniques on SONAR, NDT, AE, HIFU.



Acoustic Transducers and Measurement

BII manufactures acoustic and electronic components for underwater sound (oceanography and hydrography), sonic and ultrasonic engineering, electroacoustics, communication, bioacoustics, exploration seismology and seismic wave, physical acoustics, acoustical imaging, material study, Measurement, Control, and Processing.

Introduction to BII Electroacoustic Components

Page 3

[Acoustic Transducers](#)
[Combination of BII Acoustic Transducers to a New Transducer](#)
[Typical Signals for Active SONAR, Ultrasound, HIFU, and Communication](#)
[SONAR \(Marine and Air\), Power Ultrasonic, and HIFU System](#)
[Receiving Sounds and Waves, Underwater Sound Listening, Recording, and Communication](#)
[Underwater Sound Scanning: Positioning, Tracking, Navigation and Imaging](#)
[Line \(Linear\) Array Hydrophone and Transducer](#)
[Tuned Transducer as a Receiver](#)
[Underwater Beacons: Pingers, Repeater, and Transponders](#)
[Acoustical Measurement: Pulsing and Receiving for Physical Acoustics, NDT, and AE](#)
[Acoustical Imaging: Microscopy, Holography, and Tomography](#)
[Ultrasonic Air Transducer](#)
[Signal Processing Components \(Preamp, Filter, AGC Amplifier, T/R Switch Module, Power Amplifier, Impedance Matching\)](#)
[Piezoelectric Materials and Shapes](#)
[Magnetic Material and Cores](#)

Performance and Designs

Page 31

[Low-Power Acoustical Measurements](#)
[Low-Noise Acoustical Measurements](#)
[Programmable Sensitivity and Automatic Sensitivity Hydrophone](#)
[Overall Conversion Efficiency of Electroacoustics](#)
[Directivity Patterns](#)
[Array Design](#)
[How to Drive Transducers](#)
[How to Choose Components for Long Cable Deployment](#)
[How to Determine Pulse Width, Duty Cycle and Off-time with Input Pulse Power \(Peak Power\)](#)
[Useful Acoustic Formulae of Transducers \(Projectors and Hydrophones\)](#)

Testing and Calibration

Page 42

[Underwater Electroacoustic Measurements](#)
[Measurements of Ferroelectric and Piezoelectric Materials](#)
[Electronic Performance Test](#)

Product Information for Ordering

Page 46

[How to Order Hydrophones, AE Sensors, Ultrasonic Receivers, and Wave Height Sensors](#)
[How to Order Underwater Transducer \(Projector\), HIFU Transducer, and Ultrasonic Power Transducer](#)
[Cable and Connector Information](#) and [Abbreviation List for Electroacoustic Parameters, Cable and Connector Options](#)
[Service Temperature Range, Temperature Sensor](#)
[Operating Static Pressure Range](#)
[Transducer \(Projector and Hydrophone\) Daily Maintenance](#)
[Operating Tips avoiding Electrical Damage](#)
[Choose DC Power Supplies for BII Devices and Electrical Safety](#)
[Preamplifier](#) and [Accessories \(Connector Adaptors\)](#)
[Power Amplifier, Impedance Matching, T/R \(Transmit & Receive\) Switching Modules: Sonar & Ultrasound](#) and [Accessories \(Connector Adaptors\)](#)

Wiring Information

Page 63

[Wirings of BII Hydrophones \(Sound Receiver\)](#)
[Wirings of BII Transducers \(Projectors\)](#)
[Do-It-Yourself or BII Repair and Maintenance](#)

Installation/Mounting

Page 67

[Bolt-Fastening Mount \(BFM and BFM-FH\)](#)
[Bolt-Fastening Front Mount \(BFFM\)](#)
[End Face Mount \(EFM\) \(O-ring Sealing\)](#)
[Flange Mount \(FGM\) \(Marine Sealant or Gasket\)](#)
[Flush Mount \(FSM\) \(Marine Sealant or Gasket\)](#)
[Free-hanging \(FH\)](#)
[Free Hanging with Underwater Connector \(FHUWC\) and Accessories](#)
[Thru-hole Mounting \(Metric Thread\) \(Single O-ring Sealing\)](#)
[Thru-hole Mounting \(Inch Thread\) \(Single O-ring Sealing\)](#)
[Thru-hole Mounting \(Inch Thread\) \(Double O-ring Sealing\)](#)
[Thru-hole Mounting \(O-ring Sealing\) + Free Hanging \(THFH\) for Moorings, Buoys, Drifters and Ocean Observatories](#)
[Thread Mounting, Single O-Ring \(TMSO\)](#)
[NPT Thread Mounting \(NPT\)](#)
[Underwater Connector Assembly](#)

Export, Import, HS Code, Export Compliance, End-Use Statement, Shipping & Delivery

Page 84

[Ordering, Shipping, Customs Clearance, Delivery, and Bespoke Products](#)

IMPORTANT NOTICE AND DISCLAIMER

[Careers at BII](#)
[MANUFACTURER'S AUTHORIZATION](#)

Introduction to BII Electroacoustic Components

Features of BII Electroacoustic Components:

Available Apertures of BII Transducers:

Rectangular, Line	Square	Sphere, Hemisphere	Round Planar, Cylinder, Cylindrical Sector, Spherical Sector (Concave, Convex)
Length ≤ 1.8 m Width ≤ 0.15 m Height ≤ 0.11 m	Length ≤ 0.6 m Width ≤ 0.6 m Height ≤ 0.4 m	Diameter ≤ 105 mm (Resonance $f_s \geq 12$ kHz)	Diameter ≤ 168 mm Height ≤ 400 mm

Broadband: [Broadband transducers](#) with $Q_m = 1.56$ and efficiency $\eta = 0.24$. -3dB bandwidth = f_s/Q_m .

High Power Projector: Greater than 5000 W pulse power.

High Intensity HIFU: [High Intensity Focused Ultrasound](#), greater than 5000 W/cm² at focal zone.

Miniature Aperture: [Acoustic aperture size of \$\Phi 1.0\$ mm](#) is available.

Directivity Pattern: Omnidirectional, Toroidal, Hemispherical, Conical, Cylindrical Sector (Fan-shaped), Spherical Sector, Focusing.

Acoustic Aperture: Spherical, Hemispherical, Rectangular, Circular, Annular, Cylindrical, Linear, Cylindrical (Curved) Segment, Spherical Segment.

Low Noise Detection: [Low noise hydrophones](#) with self-noise levels much below sea-state zero.

Dynamic Range: The [Dynamic Range of signal \(sound\) detection](#) is greater than 100 dB.

Signal Conditioning with Low Power Consumption: [+2.7 to +24 VDC supply with 0.2 to 1.3 mA quiescent current](#). 40 μA quiescent current with +5.2 to +15 VDC supply.

Low Frequency Detection: Down to [0.1 Hz infrasonic sounds](#) can be detected.

Frequency Range: 0.1 Hz to 10 MHz, usable up to 30 MHz at 3rd harmonics.

Calibration: Reciprocal and comparison indoor calibration system in air and water measures TVR, FFVS, and Directivity Pattern from 20 Hz to 10 MHz.

Ocean Depth: up to 1000 m.

Service Temperature: -10 °C to +60 °C or 14 °F to 140 °F by default. Customized on request, available from -15 to 200°C, or 5 to 392°F.

Typical Applications:

Marine (Underwater) Acoustics: SONAR, Communication, Control, Marking, Navigation, Fishery, Oceanography, Seismology, Seafloor-mapping, Sub-bottom Profiling, Marine Animals, Artificial Sound Sources, Underwater Imaging (Microscopy such as surface inspection of underwater structures; Holography such as investigation of noise and vibration of submersibles; ocean acoustic tomography such as remote sensing of the ocean, lakes, rivers, and estuaries; and thermoacoustic tomography for biomedical applications).

Acoustical Testing and Analysis (NDT, AE, Physical Acoustics, Diagnostic Ultrasound, and Acoustic Imaging): Flaw Detection, Thickness Gaging, Process Control, Ultrasonic Systems of A-scan, B-scan (B-mode Imaging, 2D Imaging), C-scan (3D Imaging), Materials Study, Ultrasound Spectroscopy, Shear Wave Impedometry, Examination of Structures, Control and Monitoring of Manufacturing Process, Health Monitoring of Structure, Acoustic Microscopy, Holography, and Tomography.

Ultrasonic Processing (High Intensity Focused Ultrasound or HIFU): Physical, Chemical, Biological, Thermal and High-stress Effects of Ultrasonic Energy, Sonic Radiation in Sonochemistry, Material Processing, and Sonoluminescence, Cavitation, Acoustic Streaming, Acoustic Wave Interaction, Dispersion, Emulsification, Coagulation, Anti-algae and Anti-bacteria, Fluid Dynamics, Nonlinear Acoustics.

Air Acoustics: Air-coupled Ultrasonic Testing and Analysis; Material Study, Thermo-acoustics, Measurands influence reflection, refraction, scattering and transmission; Robotics, Proximity Detection, Sound Ranging, Counting, Monitoring, Remote Control, Alarming, Motion Detection, Level Measurement, Speed Measurement, Automatic Sizing, Sorting & Positioning of Parts, Edge Detection, Web Guiding System, Surface/Profile Characterization and Quick Scanning for Quality Control.

Acoustical Technologies:

Vibration Compensation: When suspended from a ship or boat, buoy, or used in towed array, the hydrophone experiences a large movement and induced vibration resulting from surface waves, currents, hydrodynamic flow turbulence, cable movement, etc... The translational acceleration in axial direction can be cancelled with special design and construction, and acceleration sensitivity in other directions are also lower (partially cancelled). Spurious signals caused by induced vibration can be reduced. Acceleration Sensitivity with Compensation: 1. ≤ 40 to 100 dB in axial direction of the hydrophone. 2. ≤ 80 to 110 dB in other directions of the hydrophone.

High-Resolution Acoustic System: BII's broadband transducers provide high axial resolution. BII's focusing transducers provide high Lateral Resolution.

Beamforming: Sidelobe suppression (weighting, amplitude shading) and beam steering (phase shift, or Time Delay). BII manufactures amplitude-shaded transducers whose side lobes is of -21dB (Linear or Line Transducers), -28dB (Disk Planar Transducer), -20dB to non-sidelobe (Multi-element Linear Array Transducer).

Array Focusing: By phase-shifting or time-delaying the elements to compensate the phase differences of a spherical wave, an acoustic focal zone in near field will be formed to receive or transmit sounds.

Wide Beams: Insonify and listen a wide field of interest (up to 120°) underwater which is valuable in such fields as signaling, communication, positioning, etc....

Interferometric SONAR (Phase-measuring, Interferometry): The phase differences of received signals are detected by multiple linear receive arrays paralleling to the linear transmit array. After ambiguity is removed with proper techniques, the direction of arrival (DOA) and location of the scatterer can be accurately determined.

Goniometric and High-Resolution Methods for Direction of Arrival (DOA): The amplitudes and phases of outputs of a sensor array are used to estimate the spatial location of a sound source and the number of sound sources.

Parametric Transducers: When two underwater sound waves of different primary frequencies f_{p1} and f_{p2} ($f_{p1} > f_{p2}$) propagate in the same direction, they interact with each other to create low frequency sound wave of secondary frequency f_{sec} . ($f_{sec} = f_{p1} - f_{p2}$). The directivity of secondary frequency is close to the ones of primary frequencies. This difference frequency sound is useful for practical applications in sediment profiling, depth sounding and communication. Parametric array gain or efficiency (generally $\leq 1\%$) is better as primary sound powers are higher, secondary frequency f_{sec} is higher, down shift ratio $(f_{p1} + f_{p2}) / (2f_{sec})$ is lower, and $(\alpha p * R_r)$ is lower (αp : mean primary sound attenuation coefficient; R_r : rayleigh distance).

Split-aperture Correlation: Target Angle Estimation. Beams of a quadrant array (apertures) are separately formed. The signals from the direction (azimuth: Ψ , Φ) will result in the phase contributions to the single-frequency directivity patterns functions of the array. With signal processing technologies, the spatial angle (Ψ , Φ), which can be computed quantitatively, indicates the target direction.

Doppler Effect: The Doppler Shift will be detected by BII broadband transducers to measure ship speeds, characterize marine currents, and track targets.

Mills Cross and Multibeam: Two lines array are perpendicular to each other, the outputs are multiplied, or correlated together for scanning and positioning.

Side Scan: The fan-shaped beam covers wide across-track swath and provides high along-track spatial resolution in tens or hundreds meter range. Acoustic image of underwater bottom is achieved at grazing angles of incidence.

Synthetic Aperture SONAR (SRS) and Synthetic Aperture Ultrasound (SAU): A long artificial array aperture is created from a single transducer in post-processing of the recorded signals to achieve higher spatial resolution.

Synthetic Aperture Sequential Imaging: a high-resolution image is formed by coherently compounding low-resolution images created by an array transducer.

3D Imaging: Mechanically scanned 3D transducers or 2D Matrix-array Transducers.

Acoustic Imaging: Microscopy, Holography, and Tomography.

Thermoacoustics: Conversion between acoustic (mechanical) energy and thermal energy which may be induced from electricity or microwave such as thermoacoustic tomography.



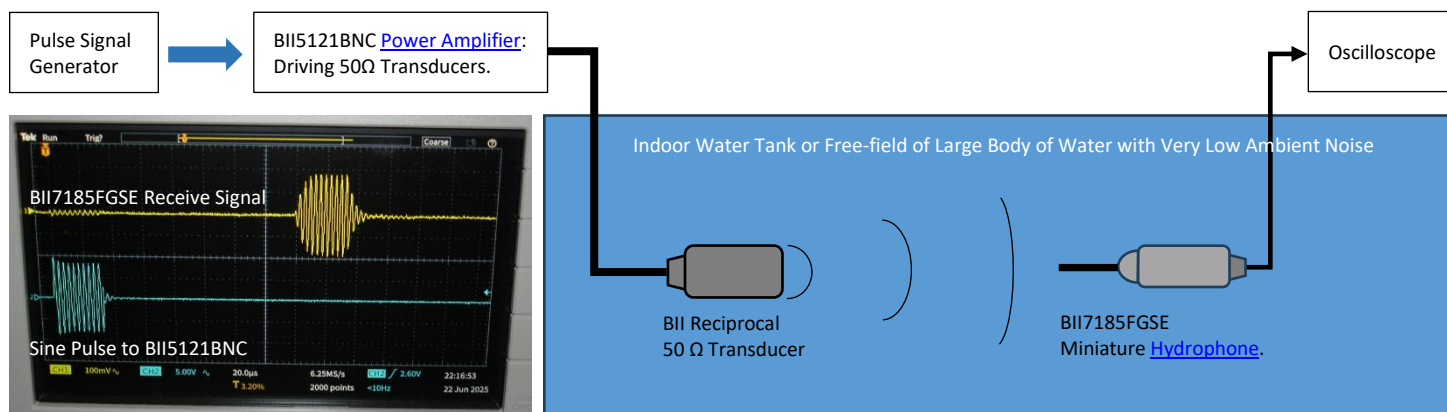
Ψ Acoustic Transducers (0.1 Hz to 10 MHz): Underwater & Air SONAR, NDT, AE & HIFU Ψ

TYPICAL APPLICATIONS

Echo Sounder (Navigation/Object Avoidance, Depth/Distance Sounder, Wave-height Sensor), Target Strength Measurement, Seafloor-mapping, Sub-bottom Profilers, Side-scan SONAR, Fishery SONAR, Transponders, Artificial Acoustic Target, Positioning, Beacon, Communication and Telemetry, Seismology, Oceanography, Ocean Current Profiling, Acoustic Speedometers (Doppler SONAR), Sound Velocity Profiler, Marine Bioacoustics, Marine Animals, Acoustic Deterrent Devices, Flow Meter, NDT (Non-destructive Test), Acoustic Emission Detection, Diagnostic Ultrasounds, Ultrasonic Test and Analysis, High Intensity Focused Ultrasound, Material Study.

Hydrophone & Acoustic Sensor: Low Noise, Low Power, Miniature, Omni & Highly Directional, Beamforming, Deep Water, High Temperature.			
» BII7000	Omnidirectional Spherical Hydrophone	» BII7010	Low Noise Broadband Hydrophone: 0.1Hz to 500kHz
» BII7020	Line Array Hydrophone: 1Hz to 160kHz	» BII7040	Hemispherical Hydrophone
» BII7550	Split Beam Transducer: Target Angle Estimation	» BII7060	Vector Hydrophone Array: Measurement of Bearing
» BII7070	Directional Hydrophone (Conical Beam)	» BII7110	Flush-Mounting Hydrophone
» BII7120	Ultra-Low Noise Hydrophone: Below Sea-state Zero	» BII7140	Acoustic Element for Array/Streamer/Beacon
» BII7180	Miniature Hydrophone: Φ1.0 to Φ3.5mm Aperture	» BII7230	AE Sensor: Detect Acoustic Emission
» BII7630	Phased Array Transducer: Beamforming & Imaging	» BII7680	Wide Beam Directional Transducer: 120°x30° View
Marine Transducer (Transmit & Receive): Broadband, High Power, Omni & Highly Directional, Beamforming, High Temperature.			
» BII7500	High Power Piston Transducer	» BII7510	Communication and Miniature Transducer
» BII7520	Omnidirectional Spherical Transducer	» BII7530	Low Frequency Transducer
» BII7540	Parametric Array Transducer: Sediment Profiling	» BII7550	Split Beam Transducer: Target Angle Estimation
» BII7560	Echo Sounder Transducer up to 2MHz	» BII7570	Side Scan Transducer: Interferometric and Parametric
» BII7580	Ultrasonic Power Transducer: High Q	» BII7590	Free Flooded Ring Transducer: Broadband, Deep Water
» BII7600	Communication Transducer: Conical Beam	» BII7610	Doppler Transducer: Speed Measurement
» BII7620	Directional Low Frequency Transducer	» BII7630	Phased Array Transducer: Beamforming & Imaging
» BII7640	Scanning Transducer & Omnidirectional Transceiver	» BII7650	High Intensity (HIFU) Transducer
» BII7660	Multibeam Transducer: Imaging Sonar	» BII7670	Underwater Annular Transducer in Pipes
» BII7680	Wide Beam Directional Transducer: 120°x30° View	» BII7690	NDT & Imaging Transducers: High Resolutions
» BII7700	Hemispherical Transducer	» BII7710	Scanning SONAR Transducer: 360° Search
» BII7720	Flush-Mounting Transducer	» BII7730	Broadband Single or Dual-Element Transducer
» BII7740	Annular Array Transducer: Sidelobe Suppression & Focusing	» BII7760	Spherical Sector Directivity Transducer: up to 60°
		» BII7770	Mechanical Scanning SONAR Transducer
Air Transducer (Transmit & Receive): Air-coupled NDT, Material Study, Navigation, Ranging, Measurement and Control.			
» BII7900	Ultrasonic Air Transducer: up to 140°C (284°F)	» BII7910	Air Array Transducer: Electronic Beam Focusing
High Temperature Hydrophones and Transducers: Service Temperature up to 200°C/392°F.			
» BIIHTTX	High Temperature Hydrophones and Transducers (Up to 198°C/392°F)		
Acoustic Instrumentation: Preamplifier, Filter, Power Amplifier, Impedance Matching, Transmit and Receive System, Pulsing-Echo System,...			
» BII1000	Hydrophone & Ultrasonic Preamplifier	» BII4000	Signal Generator for SONAR, NDT, NDT Ultrasound
» BII2000	AGC (Automatic Gain Control) Amplifier	» BII5000	Power Amplifier: Sonar, HIFU, NDT, ...
» BII2100	Transmit & Receive System: Sonar & Ultrasound	» BII6000	Impedance Matching for Transducers
		» BII8010	Ultrasound Pulsar & Receiver

System Performance of an Underwater Acoustic System in a Small Body of Water (Indoor Water Tank).



Basics on Choosing Acoustic Transducers.

1. **Frequency Range:** Filters, -3dB Bandwidth/Qm, fs; 2. **Noise:** Ambient Noise, Hydrophone Self-noise, Filters, Acceleration Response; 3. **Directional Response Pattern:** Single, Dipole, Linear/Planar/Conformal Arrays, Elements in Series or Parallel; 4. **Sounds Level/Dynamical Range/Cavitation/Focusing:** Sensitivity/FFVS, Preamps; TVR, HIFU, Sinusoidal and Pulsing Power; 5. **Output/Connectors:** Differential, Single-ended; Coax, XLR, TRS, DIN, Wire Leads, Underwater Connector, MIL-5015, etc.; 6. **Cable:** Wire Bundle, Coax, PUR/PVC/Rubber Jacket, Twisted Pair, Shielded, Unshielded. 7. **DC Power and Consumption:** Working/Service Time; 8. **Environmental Factors:** Hydrostatic Pressure/Deployment Depth, Service Temperature, Cable, Mounting/Installation, Connectors; 9. **Impedance Matching:** 100Ω/50Ω/8Ω/5Ω; 10. **Transmit & Receive:** TR Switch, Separated High Power Projector and Low Noise Receiver, Pulse-Echo/Echo-sounding/NDT; 11. **Subject Properties:** Liquids, Ocean/Lake/River Water, Air/Gas, Solids/Plastics/Metal/Ceramic/Composites, etc.; 12. **Signal Conditioning/Array Processing:** Preamps, Power Amplifiers, Filters, T/R Switching, Beamforming/Amplitude Shading/Beam Steering/Side-lobe Suppressing, etc.. 13. **Signal Type:** Tone Burst (Sinusoidal) Excitation, Chirp Pulse, Square Wave Pulse, Spike (Shock) Pulse, Coded Excitation (Barker/Golay), Continuous Signals, Arbitrary Signals, Aquatic/Marine Animal Sounds, Ambient and Ship/Vehicle Noises, etc.

Low Frequency Sounds in the Ocean at 1 Hz to 1 kHz.

Low Frequency Sounds between 1Hz to 1kHz undergo little attenuation in ocean and lakes: 1.2×10^{-7} at 1Hz, 0.0002 at 10Hz, 0.0015 at 100Hz, 0.07 at 1kHz, in dB/km, proximately. The major propagation losses are Spreading Loss, Absorption and Scattering at boundaries such as seafloor sediments. The sounds being trapped in deep sound channel or SOFAR channel avoiding losses from both top and bottom boundaries can propagate very big area with very low loss. These low frequency sounds contain valuable information on their sources and propagation areas such as Weathers (Winds, Rains, Surface Waves, Ice, Storms, ...), Geological Process (Earthquakes, Tsunamis, Moving of Tectonic Plates and Crusts,...), Marine Animals (Whales, Dolphins, Fish, Invertebrates, ...), Anthropogenic Activities (Shipping, Harbor, Oil and Wind Energy Extraction, Air Guns, ...).

Besides sounds detection or monitoring with deployment at 1000m depth, the distributed hydrophone network in shallow water (at a specific depth) can also be used to detect and monitor upward sounds being refracted from SOFAR channel.

Signal Processing

Sonar Signal Envelope Detector According to Nyquist–Shannon sampling theorem: If the highest frequency in the original signal is f , the lower bound on the sampling frequency for perfect signal reconstruction is $2f$, called Nyquist sampling rate. To get better resolution such as underwater acoustic image, high carrier frequency is used in several hundreds kHz to several MHz. This imposes cost on electronic processing circuit such as A/D converter and data storage. A Envelope Detector removes high frequency carrier signal and extract the low frequency envelope. Low cost electronic system can be used to sample and analyze the envelope.

Typical Applications

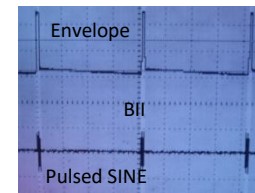
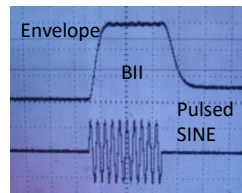
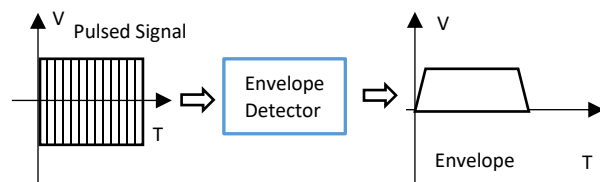
High Frequency Scanning, Side Scan and Imaging Sonar

Fishery/Plankton Sonar, Sediment Penetration/Parametric Sonar

Echosounder, Navigation Sonar, Sea-floor Mapping Sonar

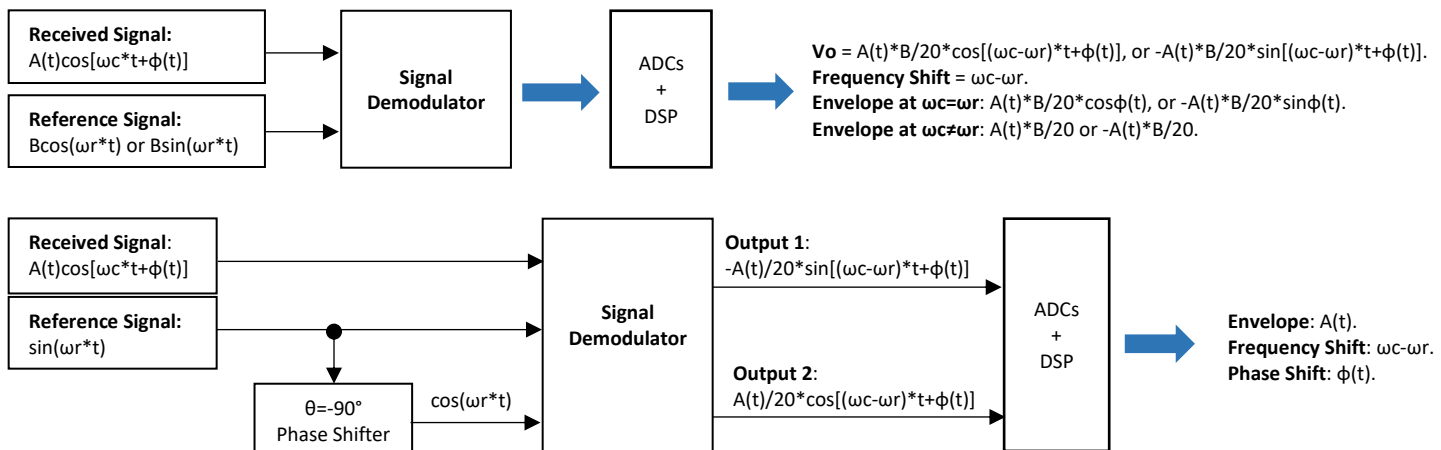
Target Strength Measurement, Abundance Estimate

Envelope Detection



Signal Demodulator The useful information of a known narrowband signal is not contained in the carrier frequency, but in its modulation. A Signal Demodulator demodulates a known narrowband signal such as pulsed SINE, FSK and CHIRP around its carrier frequency f_c in time domain and keeps only the meaningful low frequency content of the signal, and thus lowers the sampling rate of A/D converter and reduces the amount of sampled digital data. The demodulated signal contains in-phase and quadrature components which describes the received signal truthfully. The envelope, frequency shift and phase shift can be extracted with signal processing in software for estimation of target range, bearing, speed and size. Besides, A Signal Demodulator can also be used to convert ultrasound to audible sounds for real-time listening and searching of marine animals, SONAR sources, acoustic tags, pingers, etc.

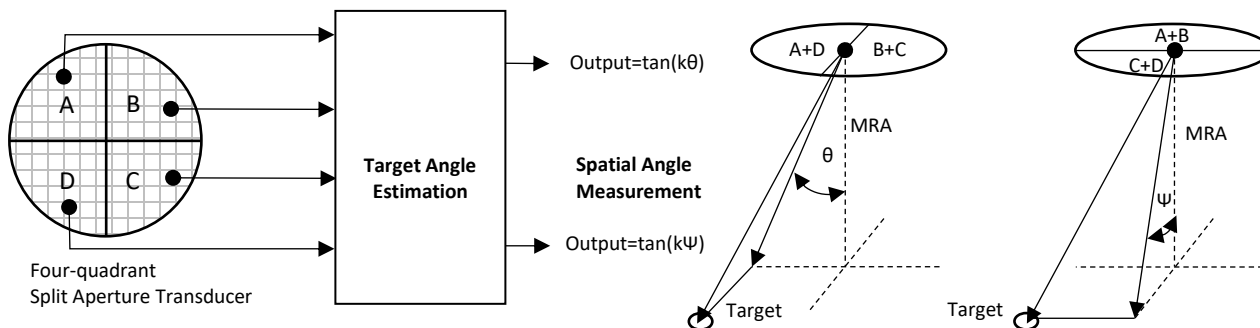
Typical Applications: Communication, Echo Sounding; Correlation Receiver, Doppler SONAR, Speed Measurement; Target Angle Measurement, Split-Aperture SONAR.



t : Time; ω_c : Frequency of Received Signal; ω_r : Frequency of Reference Signal; $A(t)$: Amplitude of Received Signal; B : Amplitude of Reference Signal; $\phi(t)$: Phase Shift between Received Signal and Reference Signal.

Target Angle Estimation is a four-quadrant split aperture correlator which works with a four-quadrant split beam transducer to estimate the direction of arrival of a plane wave sound (or, measure the spatial angle of the sound) and determine the angular position of the target relative to the acoustic axis of the transducer.

Typical Applications: Target Tracking and Searching, Homing Torpedoes; Navigation, Positioning, Obstacle Avoidance; Fishery Sonar; Seafloor Slope Mapping.



K: Angle Sensitivity = Phase Shift/Spatial Angle, transducer's parameter. **θ and ψ:** Spatial Angles of a Sound Source Relative to MRA of the Transducer. **MRA:** Maximum Response Axis of Transducers, or **Acoustic Axis**.

Array Beamformer

Array Beamforming: Beam Steering and Array Shading/Weighting (Sidelobe Suppression) for Narrowband Signals

Appropriate phase shifts can be imposed on the array signal paths to steer the main lobe to the desirable direction. Aperture response is tuned with amplitude shading/weighting to provide the most desirable directional pattern for a particular application: a desirable beam width and response at main lobe, and suppressed responses at sidelobes, or vice versa. Array elements could be discrete elements of one transducer, or discrete hydrophones and projectors.

Array beamformers for receiving and transmitting narrowband sounds can be implemented with digital beamforming in software.

Wide View of Field $\pm 90^\circ$ (frequency dependant); Weighting Laws: Triangular, Dolph-Chebyshev, Cos, Cos^2 , Hamming ...; Linear, Planar and Conformal (Spherical and Cylindrical) Array.

Typical Applications

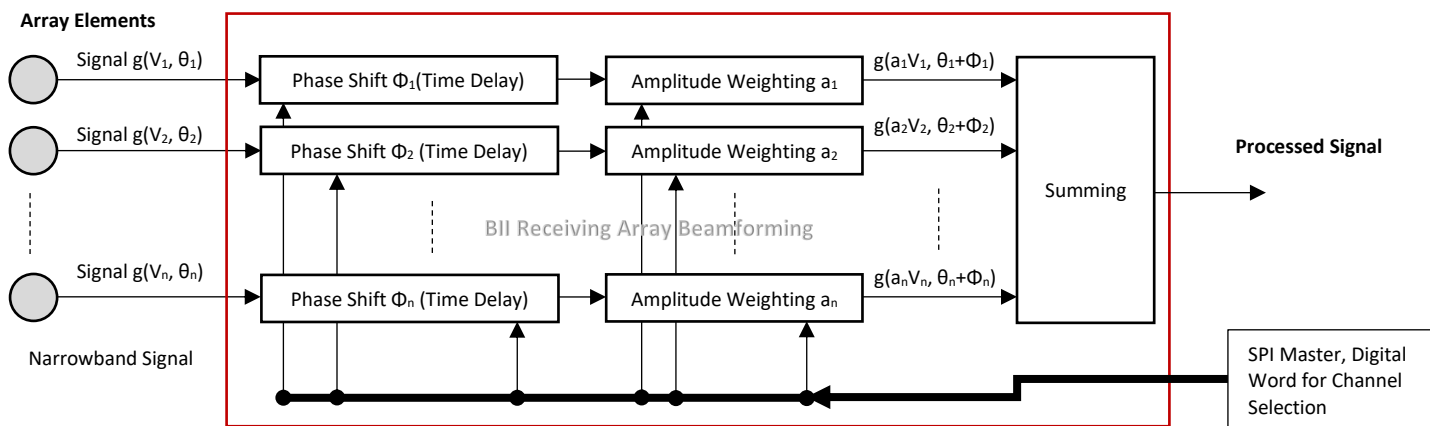
Array Focusing, Mills Cross
Navigation, Obstacle Avoidance

Delay Compensation Caused by "Moving"
Scanning and Imaging Sonar, Fishery SONAR Scanning View of Field

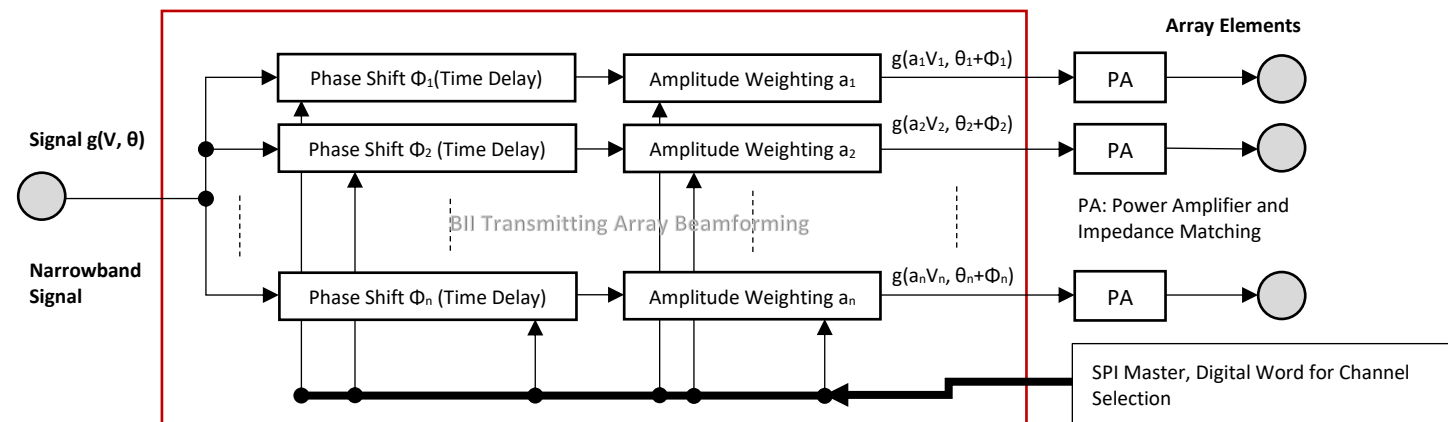
Bearing, Location, Acoustic Positioning
Communication, Beacons

Structure of Receiving Array Beamforming

It is recommended that the signals are bandpass-filtered before they pass to digital beamformer.

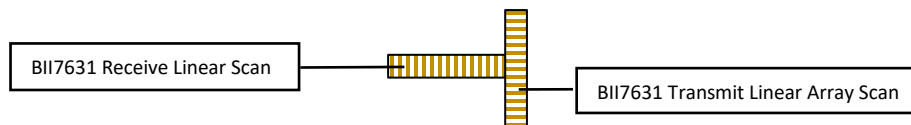


Structure of Transmitting Array Beamforming

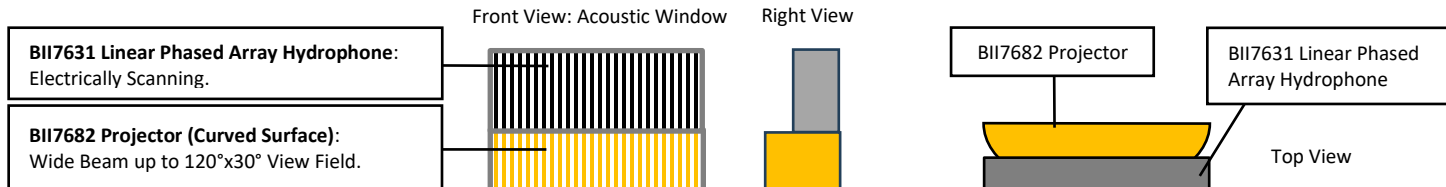


Combination of BII Acoustic Transducers to a New Transducer (illustration ONLY)

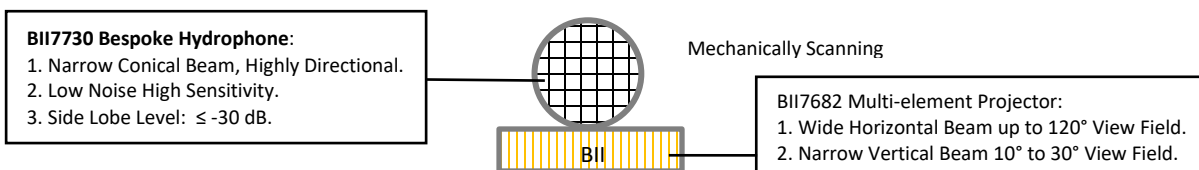
3D "T" Type Imaging Multibeam Transducer: Two [BII7631](#) Linear Phased Array (Rectangular Aperture).



2D Imaging Multibeam Transducer: one [BII7631](#) Linear Phased Array (Rectangular Aperture) and one [BII7682](#) Wide Beam Directional Transducer (Curvilinear or Cylindrical Sector Aperture).



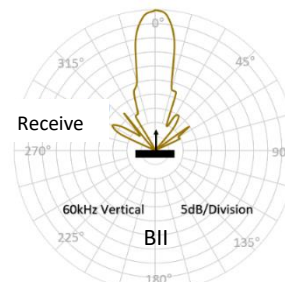
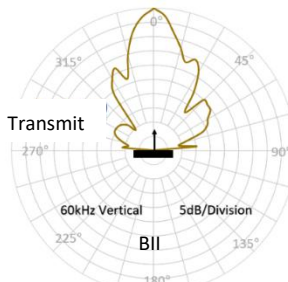
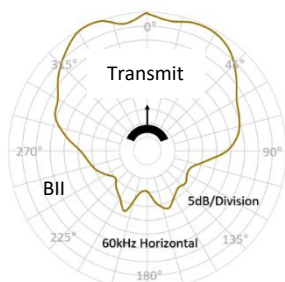
Echo-ranging or Scanning Transducer: one [BII7682](#) Wide Beam Directional Projector (Cylindrical Sector Aperture) and one bespoke [BII7730](#) Hydrophone.



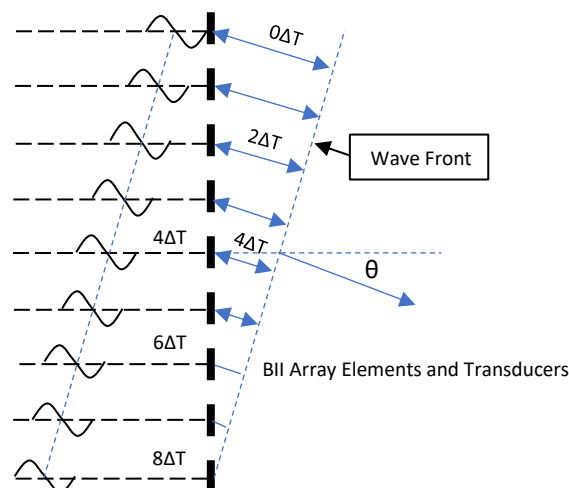
Echo Ranging Transducer, typical directivity Pattern. Illustration ONLY at 60 kHz.

Transmit Beam Pattern

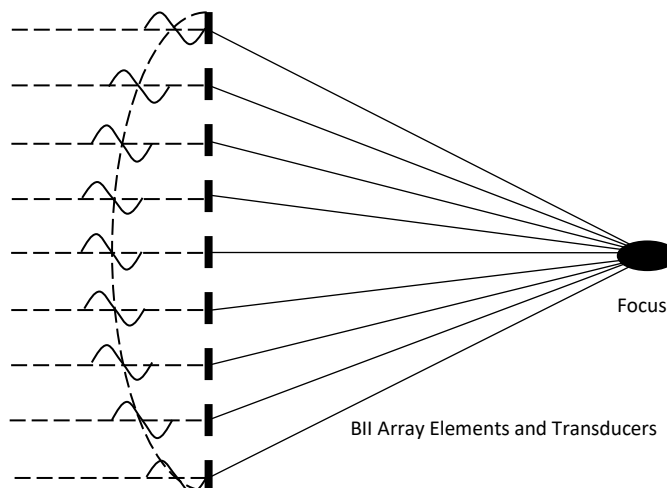
Receive Conical Beam Pattern



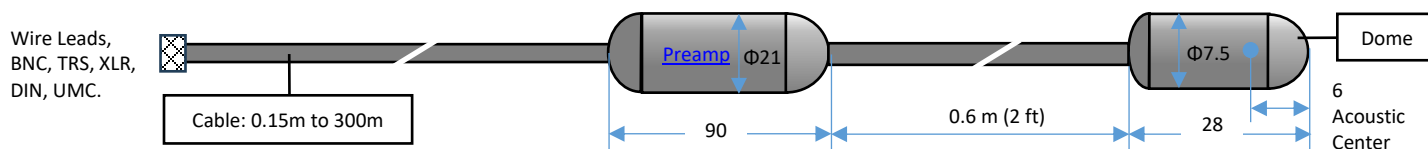
Linear and Planar Array Beam Steering



Linear and Planar Array Beam Focusing



Hydrophone as Array Element: Free Hanging with Smooth Domes for an Linear Discrete Array. Illustration Only, Size Scale is NOT 1:1.

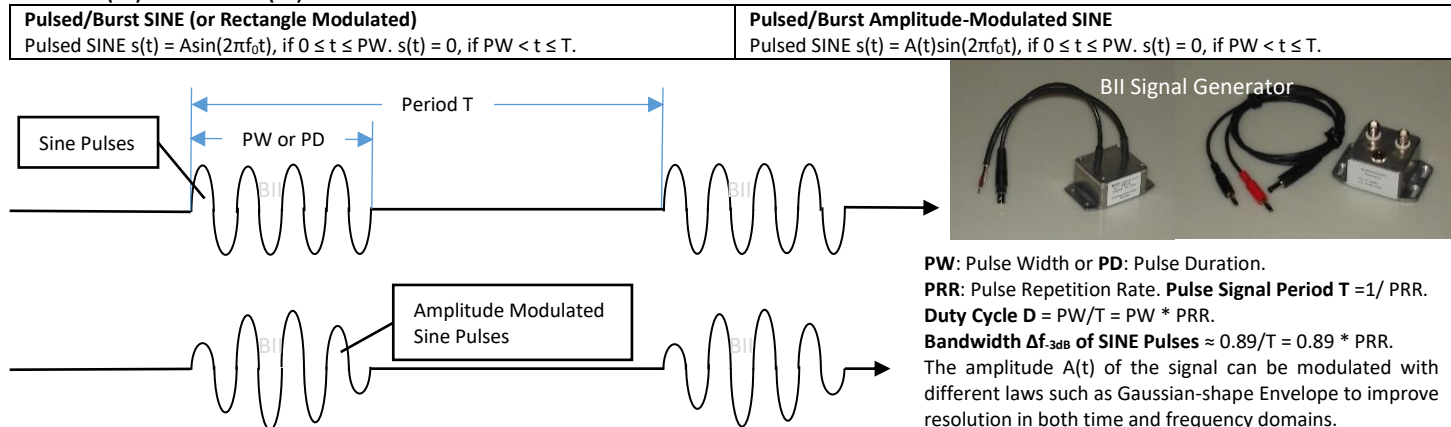


The streamlined hemispherical domes minimize drag forces and hydrodynamic noises caused by the hydrophone in motion or the flow past the hydrophone.

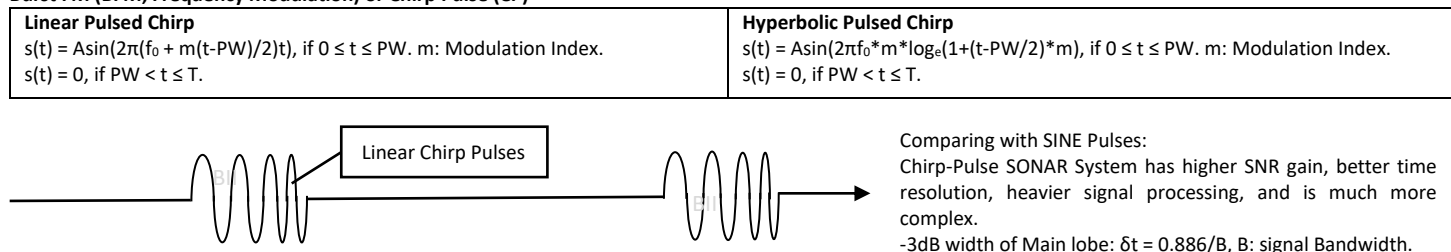
Typical Signals for Active SONAR, Ultrasound, HIFU (High Intensity Focused Ultrasound) and Communication

BII4000 series signal generators are embedded and standalone components/modules which generate SINE pulses, Pulse Trains, or Spike Pulses for Pulsing-Echo acoustic systems (SONAR, NDT, Diagnostic ultrasound...) and underwater communication system, and feature low power, high accuracy, and long-term stability.

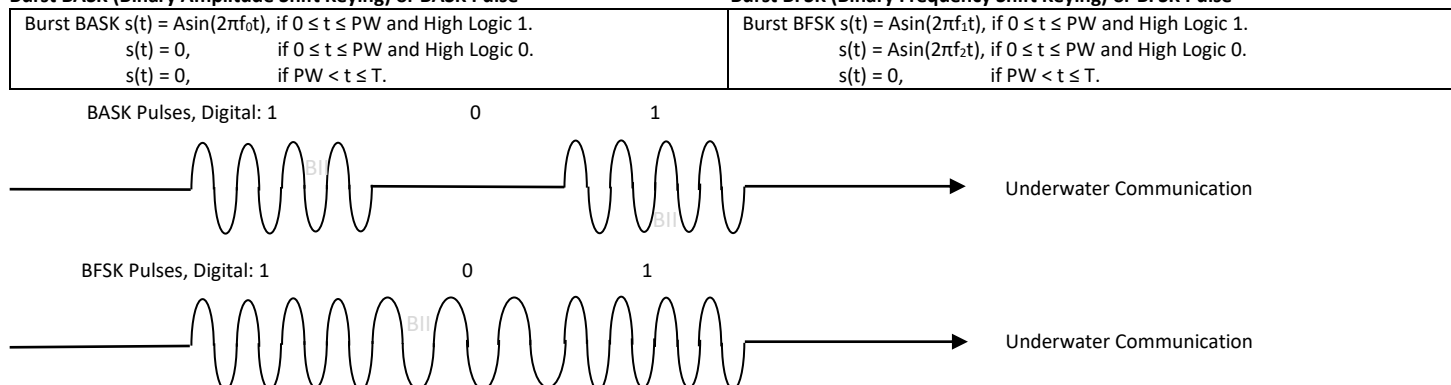
Burst SINE (BS) or SINE Pulse (BS)



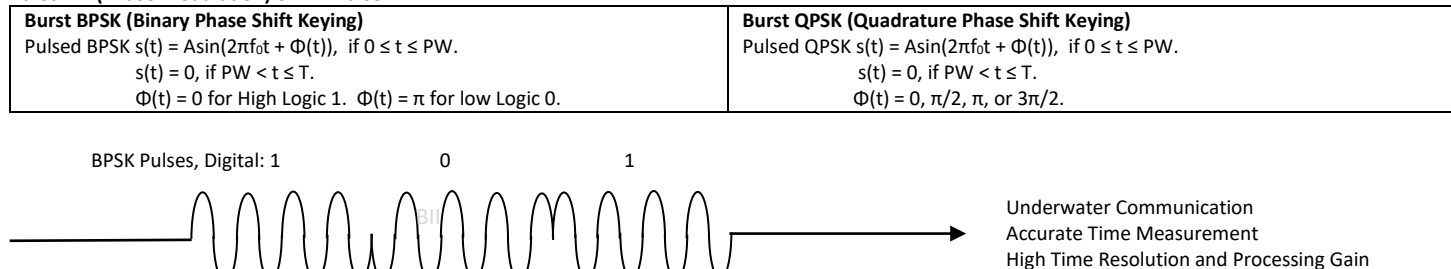
Burst FM (BFM, Frequency Modulation) or Chirp Pulse (CP)



Burst BASK (Binary Amplitude Shift Keying) or BASK Pulse



Burst PM (Phase Modulation) or PM Pulse



Burst Rectangular Pulse (BRP), Burst Square Signal (BSS), and Positive or Negative Voltage Spike (PVS or NVS).

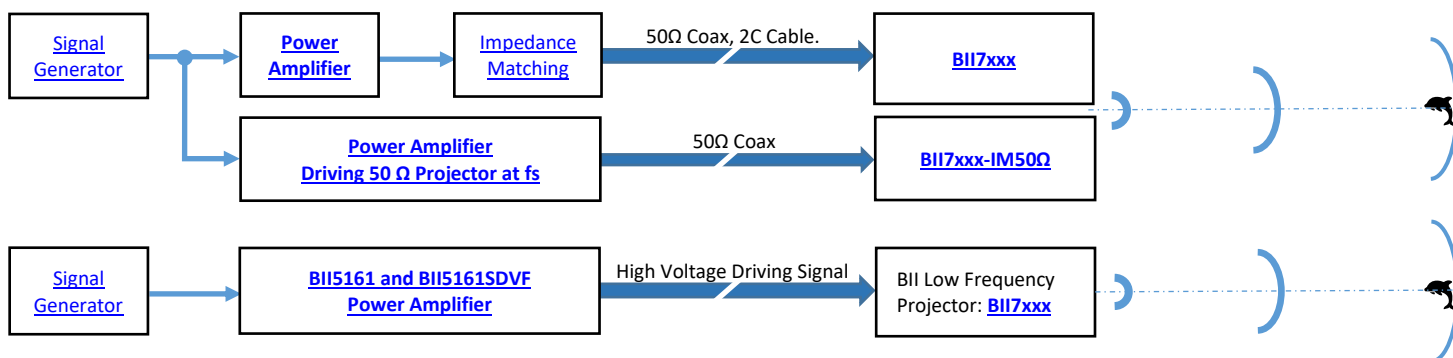


Square Signal can deliver 1.414 times electrical energy to transducer comparing with SINE Pulses.
Pulse and Square Signal system has much higher electrical efficiency than SINE pulsing system.
Voltage Spike has highest resolution in time domain for ultrasonic measurement.

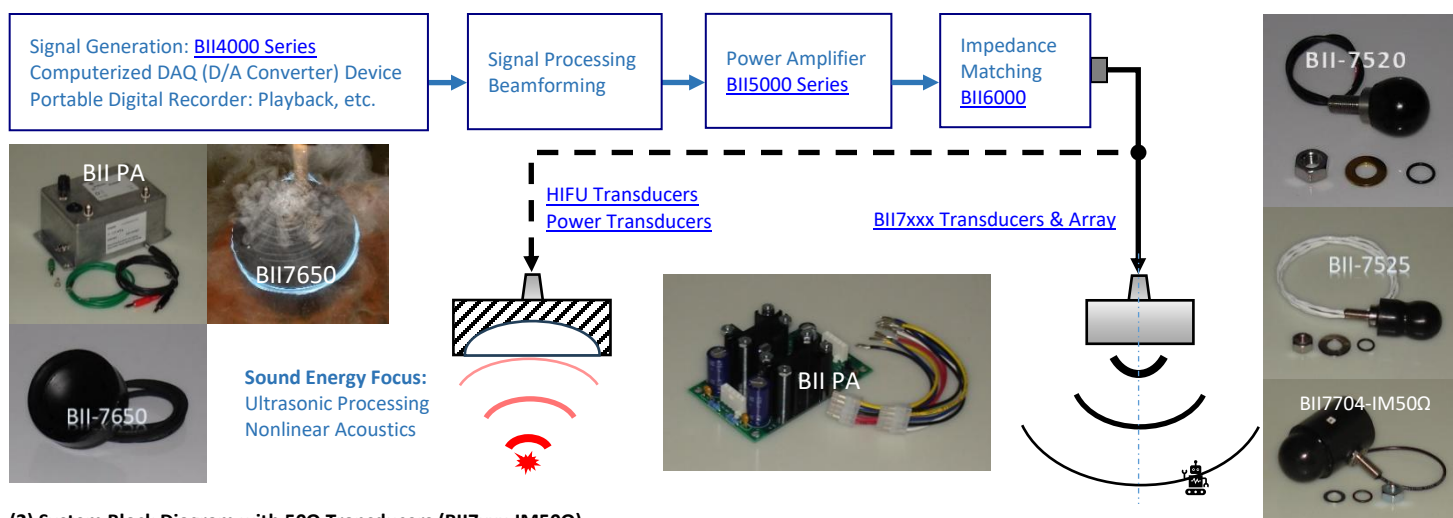
BII Components in SONAR (Marine and Air), Power Ultrasonic, and HIFU (High Intensity Focused Ultrasound) System

Linear & Nonlinear Acoustics: SONAR/NDT/AE/HIFU, Bioacoustics, Navigation, Communication, Oceanography, Seafloor-Mapping, Sub-bottom Profiling, Material Study.

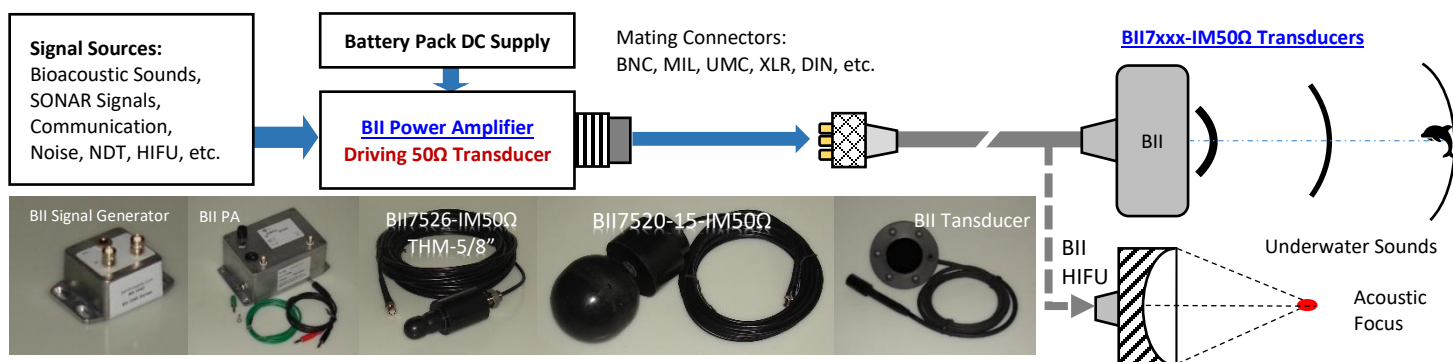
Generation of Underwater or Ultrasonic Sounds (IM50Q: Impedance Matching to 50Ω)



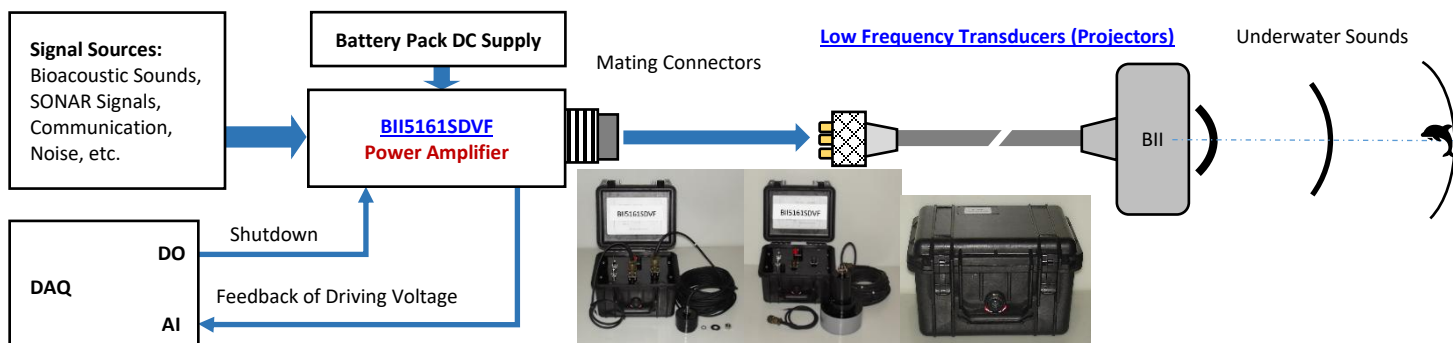
(1) System Block Diagram with Individual Components.



(2) System Block Diagram with 50Ω Transducers (BII7xxx-IM50Q).



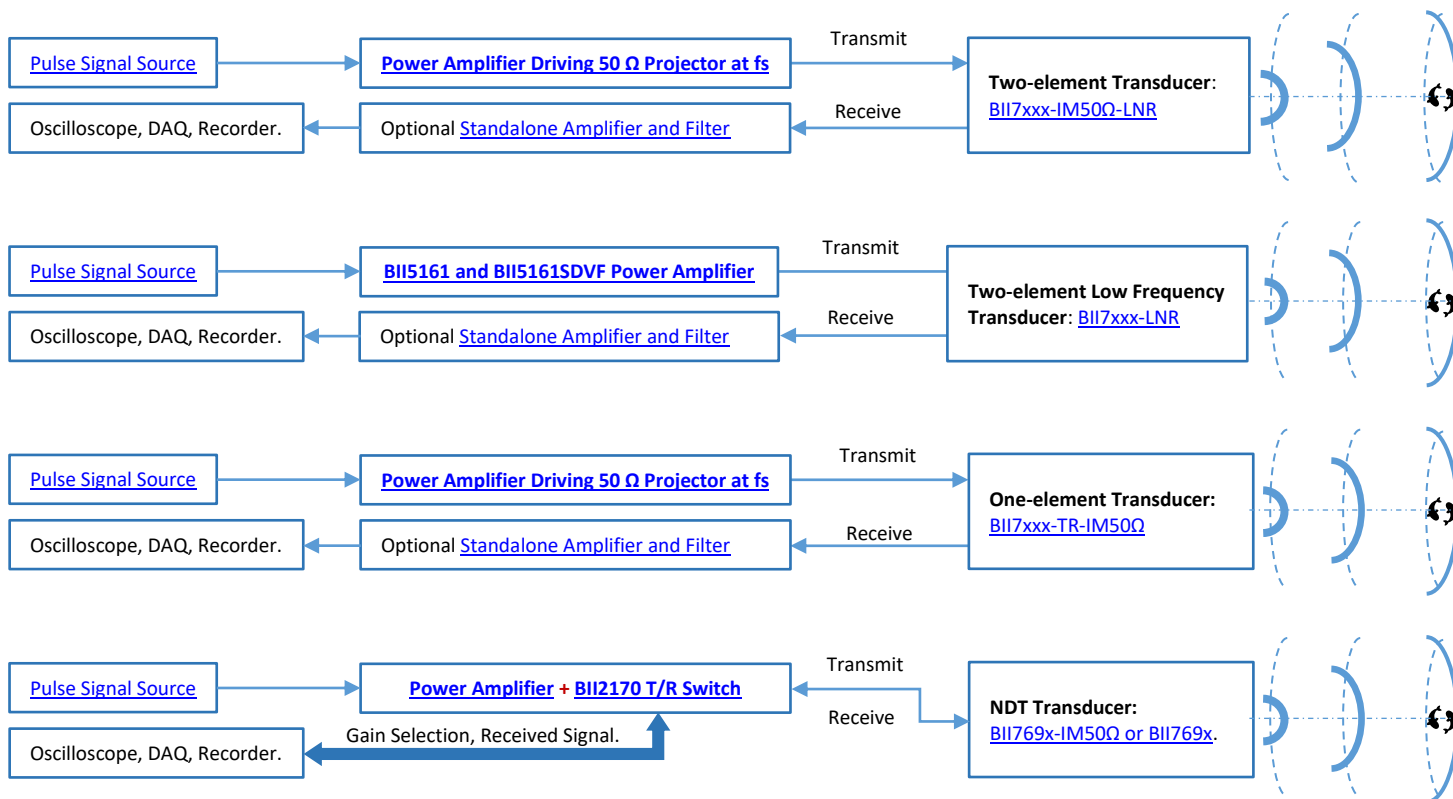
(3) Generate Low Frequency Sounds as low as 100Hz.



Transmitting and Receiving Sounds (Pulsing Acoustics): Echo-sounding, NDT, Communication,...

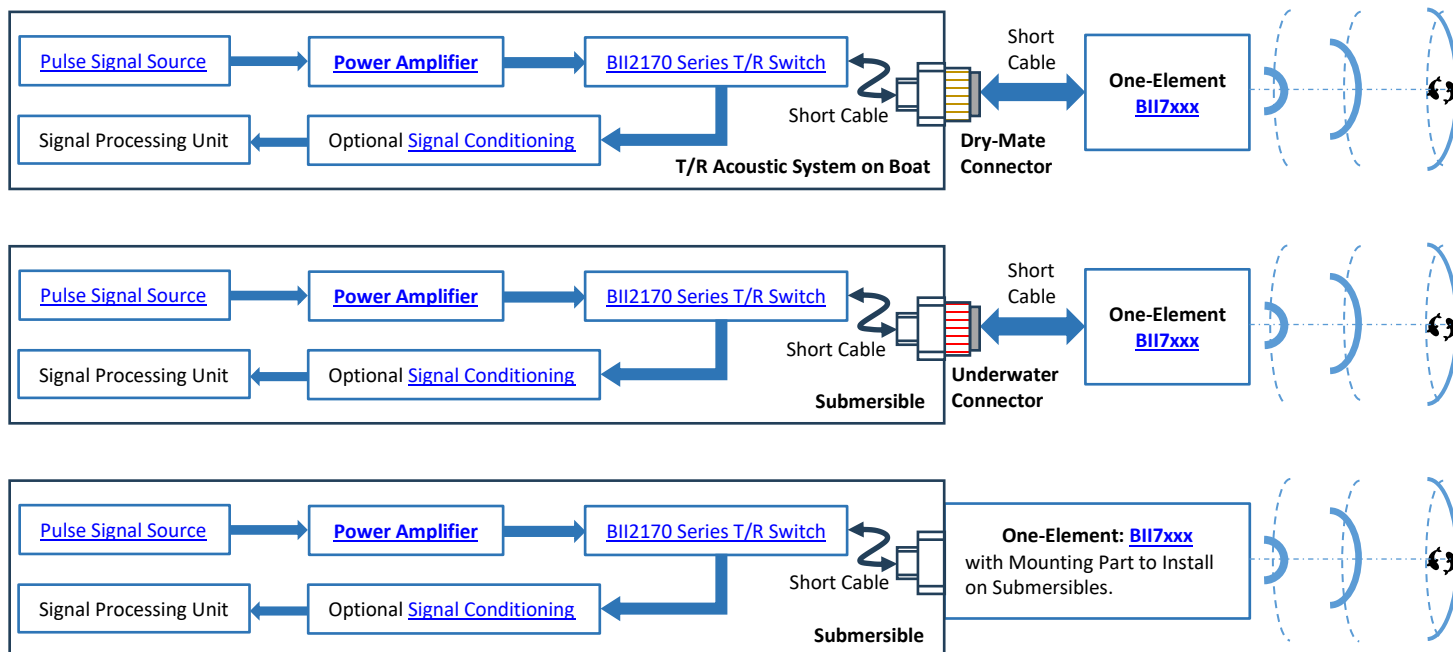
IM50Ω: Impedance Matching to 50Ω; **TR**: Transmit and Receive Switching with built-in low noise preamp and bandpass filter; **LNR**: Built-in separated Low Noise Receiver.

Transmitting and Receiving Underwater and Ultrasonic Sounds:

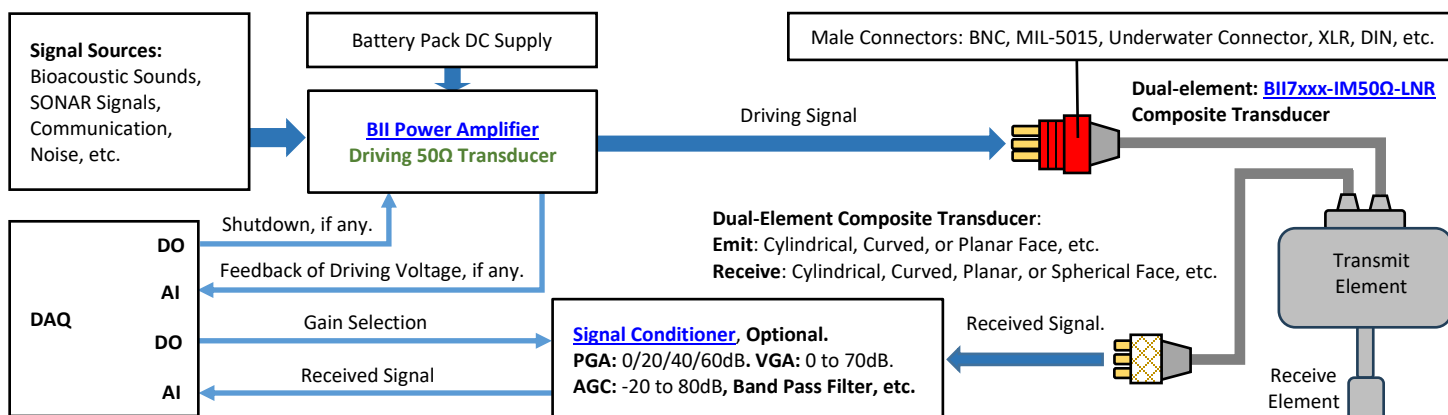
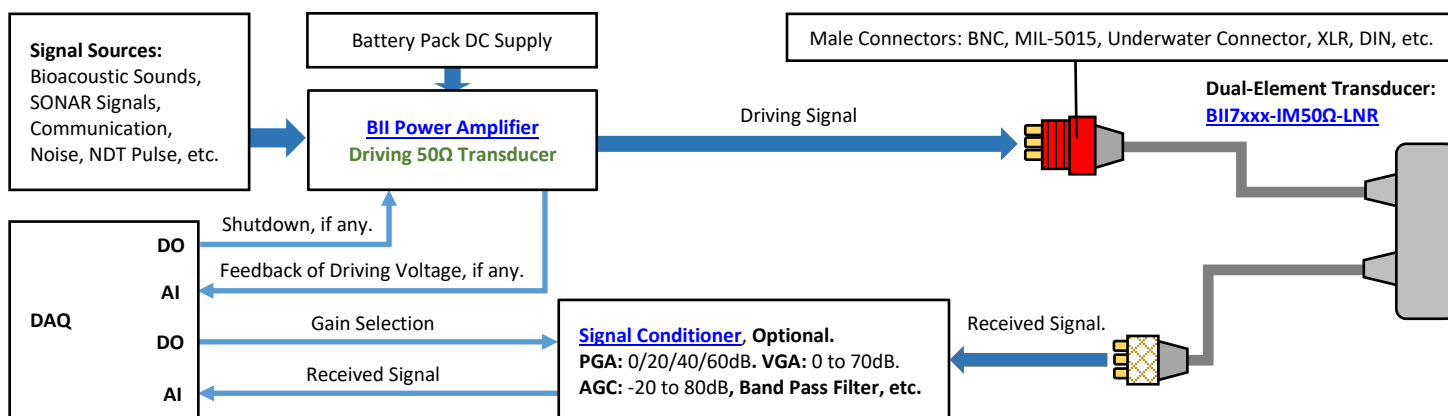


Alternatives of Transmitting and Receiving Underwater or Ultrasonic Sounds for Short Cables between Transducers and Power Amplifiers.

Short Cable here means its capacitance has little effect on the transducer sensitivity, and its adverse effects on the transducer can be neglected.

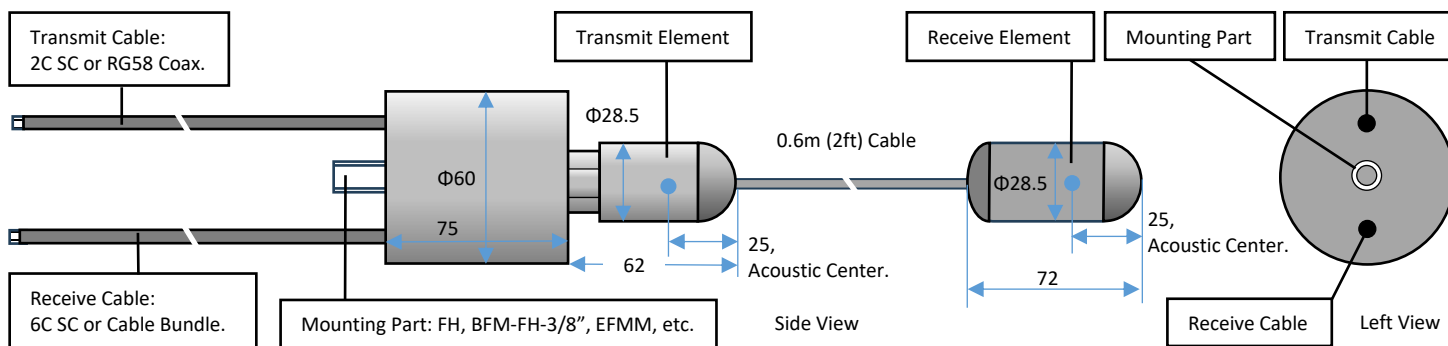


System Block Diagram of Transmitting and Receiving Sounds of Dual-Element Transducers: a Simple Configuration. DO: Digital Outputs, AI: Analog Inputs.

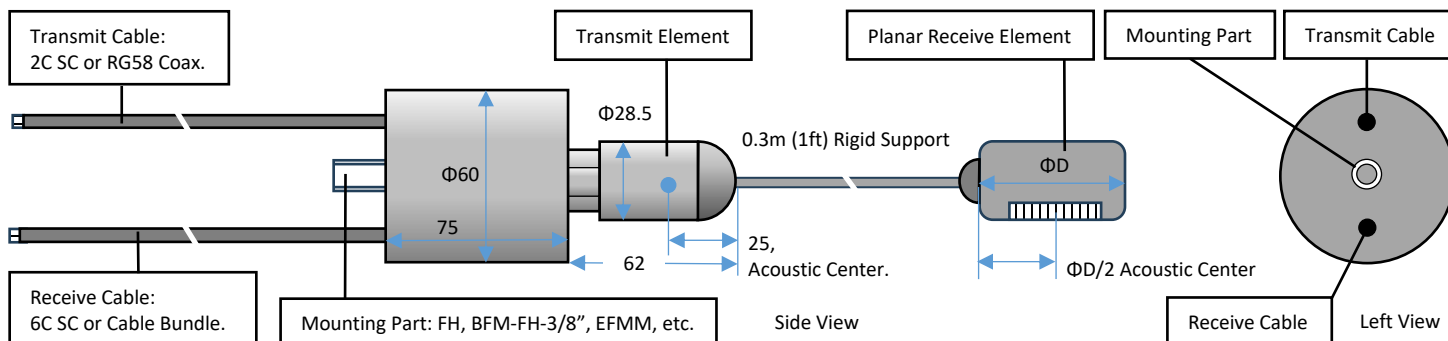


Some Configurations of Dual-Element Transducers [BII75xx-IM50Q-LNR](#) (Sizes are for illustration ONLY and vary with different transducers.)

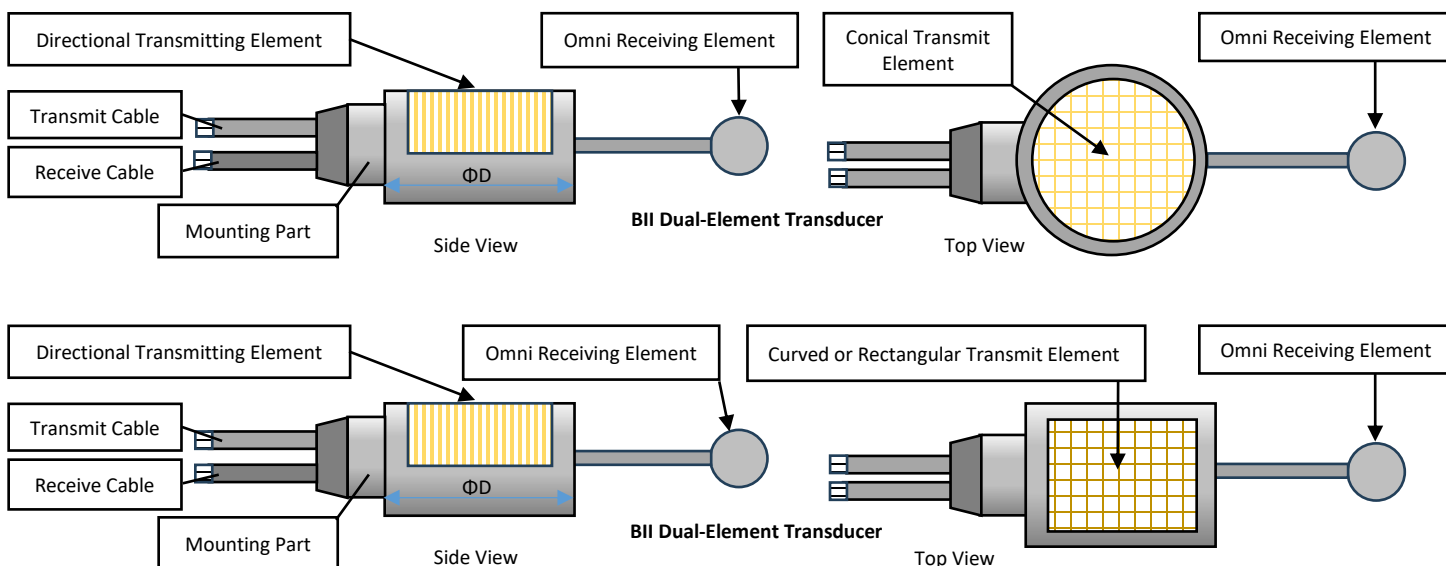
1. Omnidirectional Transceiver: Transmit Omnidirectional Sounds in Horizontal Plane, Receive Omnidirectional Sounds in Horizontal and Vertical Planes.



2. Omnidirectional Projector and Directional Receiver: Transmit Omnidirectional Sounds in Horizontal Plane, Receive Sounds in Vertical or Horizontal Plane.

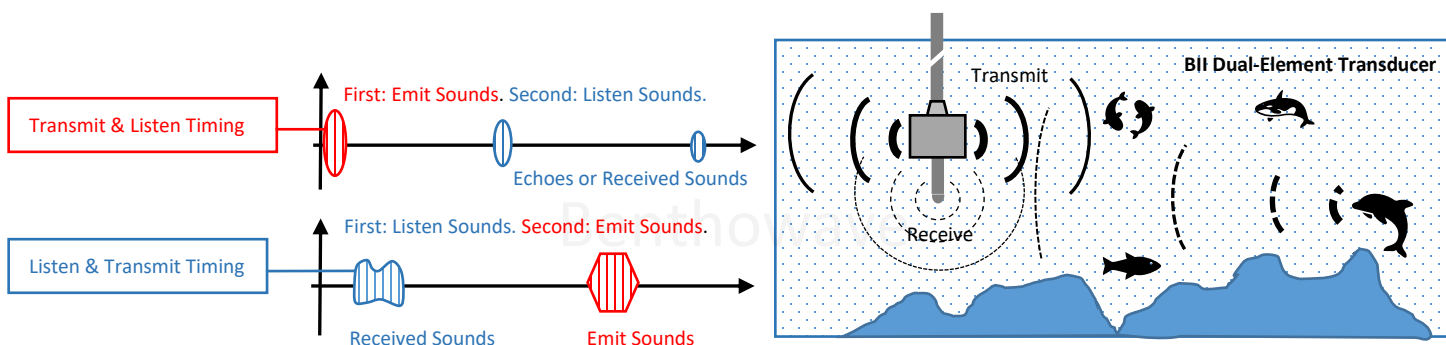


3. Directional Projector and Omni Receiver: Transmit Directional Sounds in Horizontal Plane, Receive Omnidirectional Sounds in Vertical and Horizontal Plane.

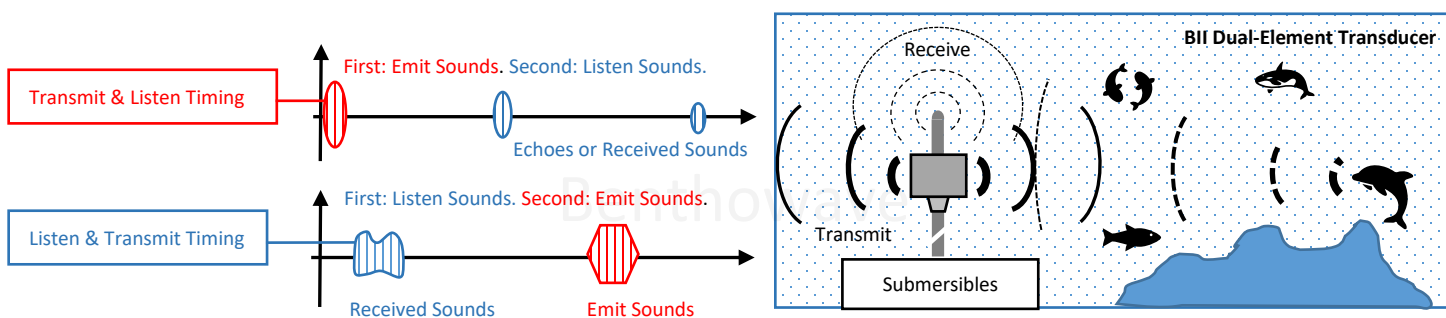


Some Applications of Dual-Element Transducers on Underwater Communications

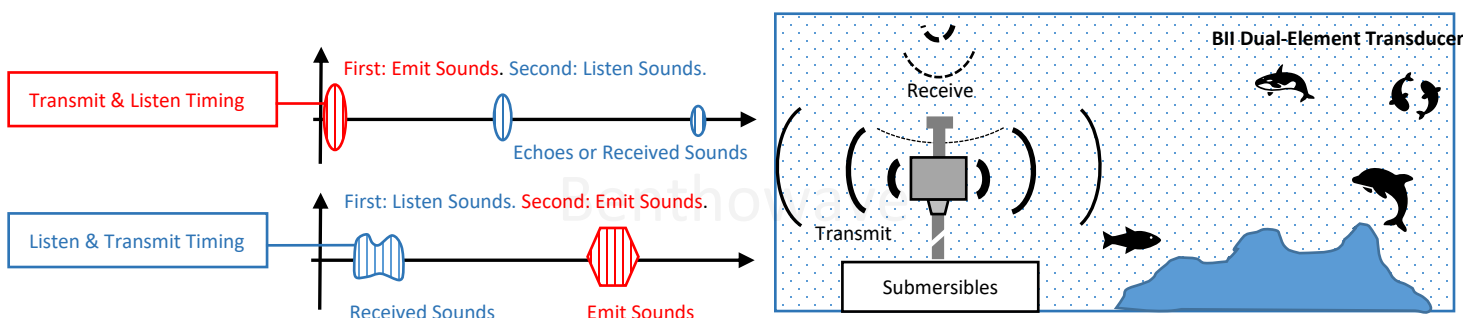
1. Communication System: Transmitting Omnidirectional Sounds Horizontally and Receiving Omnidirectional Sounds in Horizontal and/or Vertical Planes.



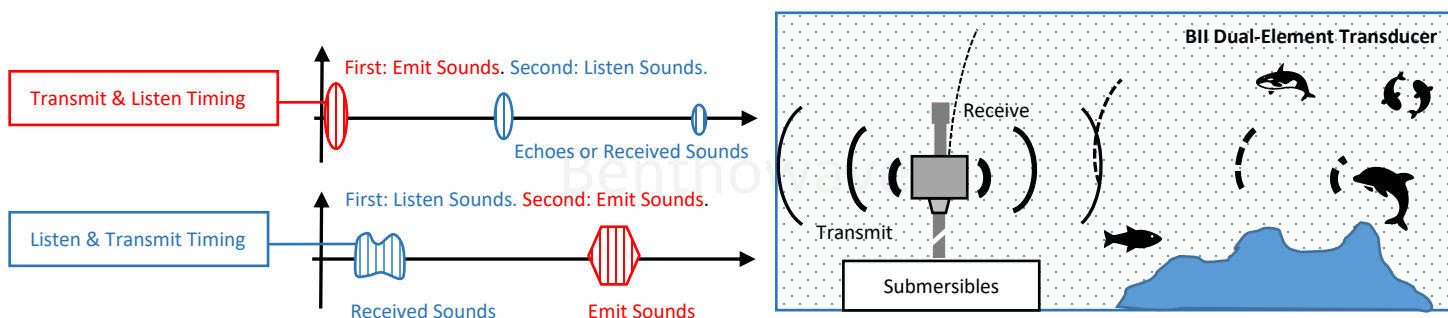
2. Communication System: Transmitting Omnidirectional Sounds Horizontally and Receiving Omnidirectional Sounds in Horizontal and/or Vertical Planes.



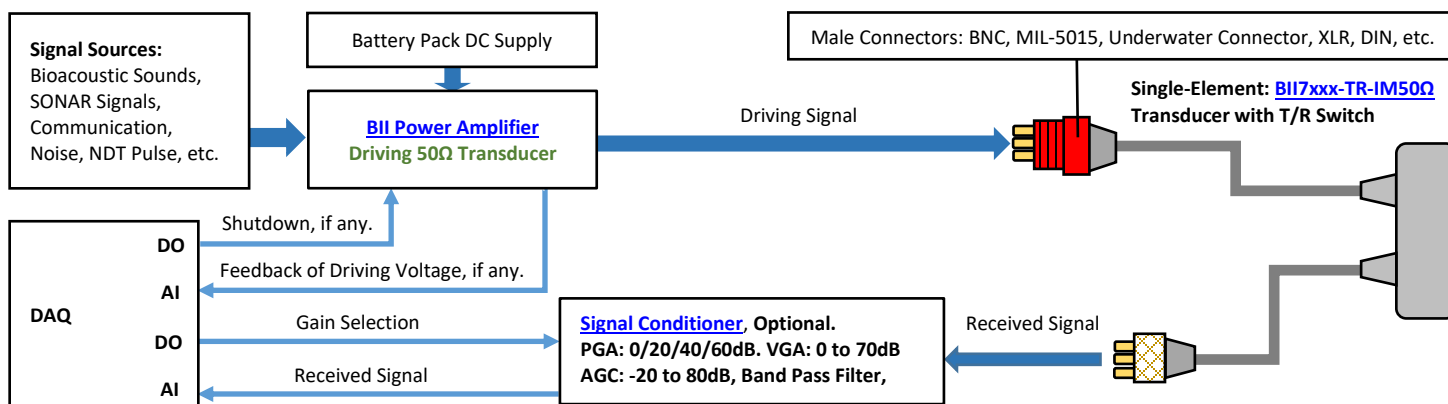
3. Communication System: Transmitting Omnidirectional Sounds Horizontally and Receiving Directional Sounds in Vertical Plane.



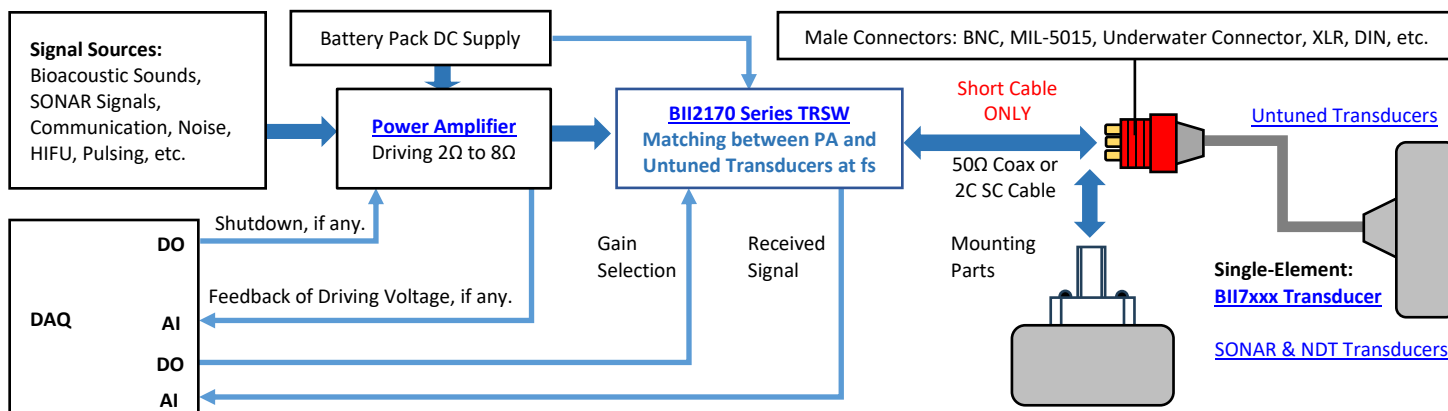
4. Communication System: Transmitting Omnidirectional Sounds Horizontally and Receiving Directional Sounds in Horizontal Plane.



System Block Diagram of Transmitting and Receiving Sounds for **Single-Element Transducers**: a Simple Configuration. DO: Digital Outputs, AI: Analog Inputs.



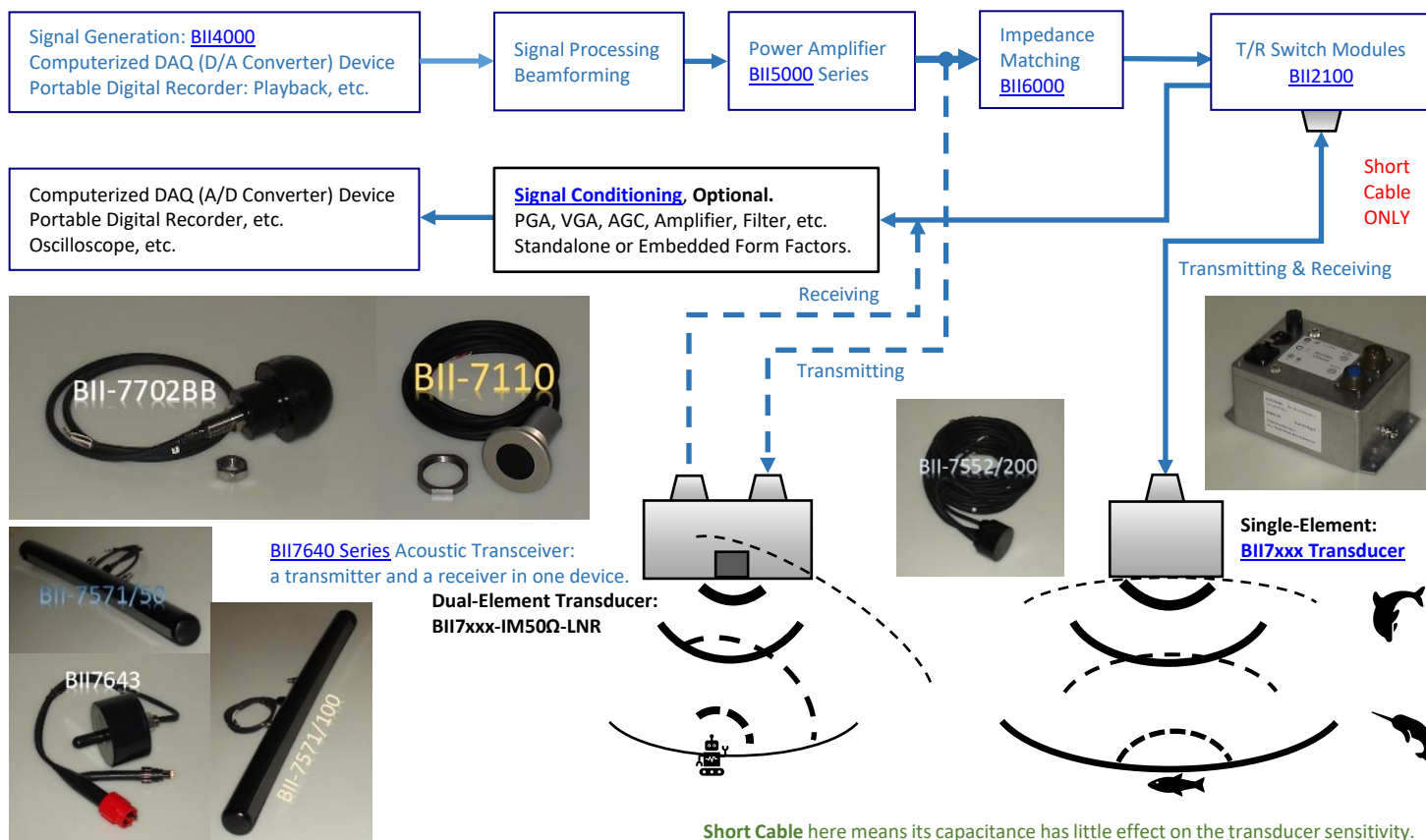
Alternatives of System Block Diagram of Transmitting and Receiving Sounds of **Single-Element Transducers**: a Simple Configuration for SONAR and NDT Transducers



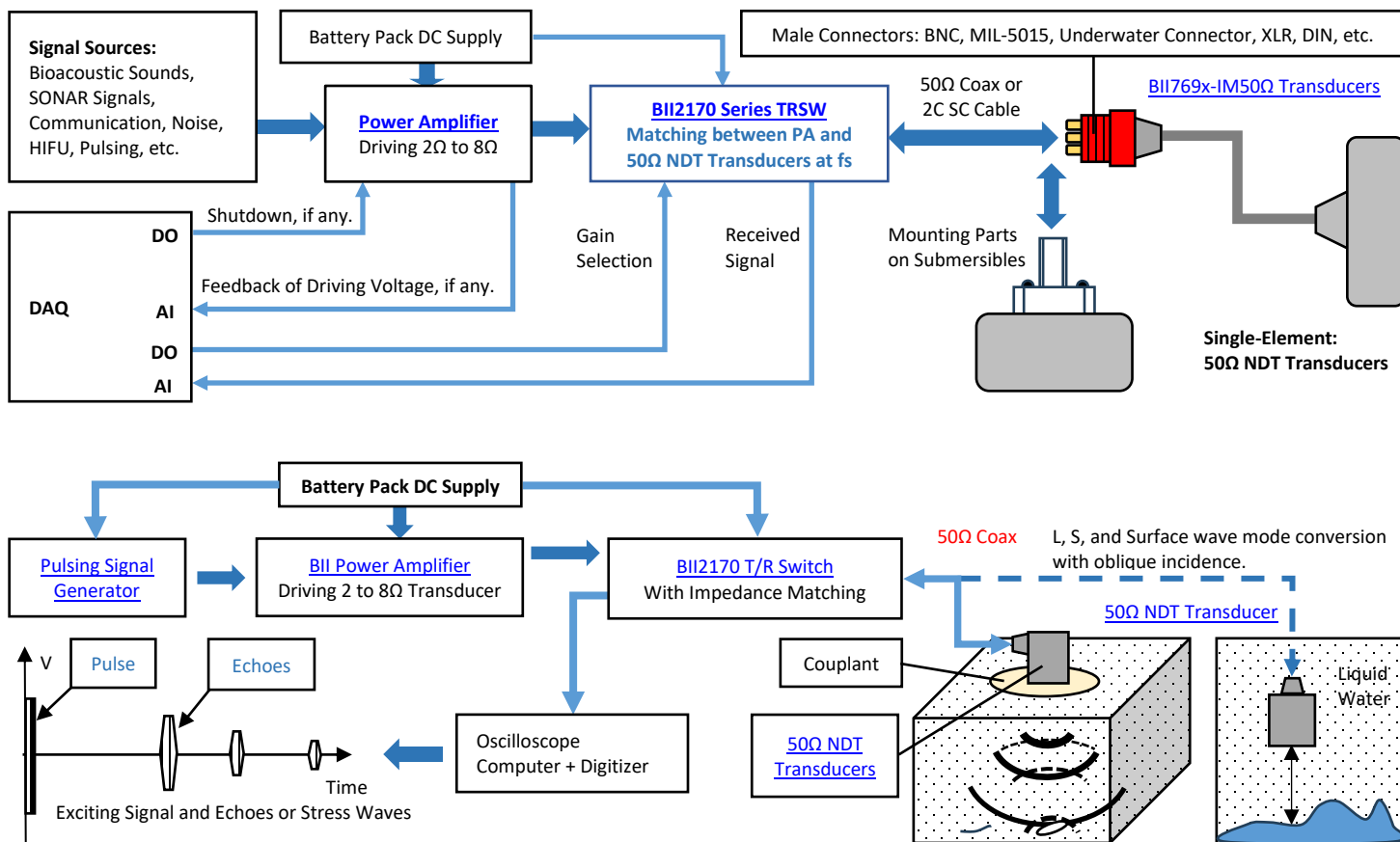
Single-Element NDT Transducers BII7690 Series.



System Block Diagram with Individual Components (Single-Element Untuned Transducers BII7xxx and Dual-Element Transducer: BII7xxx-IM50Q-LNR).



System Block Diagram with 50Ω NDT Transducers (BII769x-IM50Q).



Receiving Sounds and Waves

BII manufactures wide range of hydrophones (Underwater Microphones) and AE/NDT sensors for underwater and ultrasonic applications from 0.1 Hz to 10 MHz, and they are completely waterproof, corrosion-resistant and withstand high levels of salinity. **Two packages are available:** **Hydrophones (Sensors) without Signal Conditioner** and **Hydrophones (Sensors) with [built-in Signal Conditioner](#).**

Low Noise: Lower than sea-state 0

Miniature: Acoustic Aperture size $\Phi 0.8\text{mm}$

Sidelobe Suppression: $\leq -30\text{dB}$, Bespoke

Phased Array: Beamforming/Steering/Focusing

Ocean Depth: up to 1000m

Low Power: 40 to 350 μA quiescent current for battery powered instruments

Broadband: Detect signal up to 3MHz

Directivity: Omnidirectional and Directional, Custom-fit

High Temperature: up to 120°C or 248°F and 198°C/392°F

Projector: Hydrophones without preamp can be used as low power sound sources

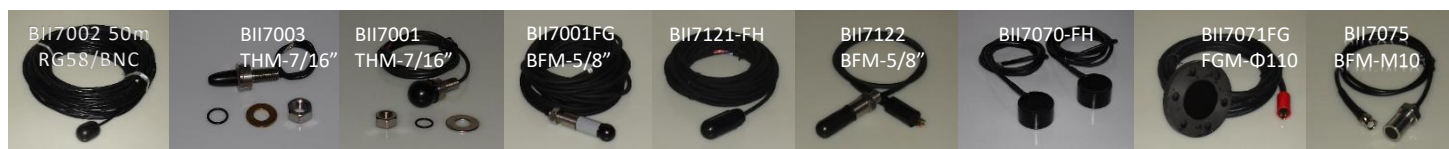
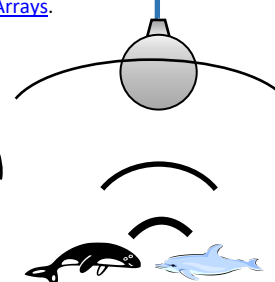
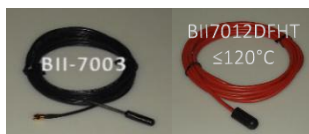
Computerized DAQ (A/D Converter) Device
Portable Digital Recorder, etc.
Oscilloscope, etc.

[Signal Conditioning, Optional.](#)
PGA, VGA, AGC, Filtering, etc.

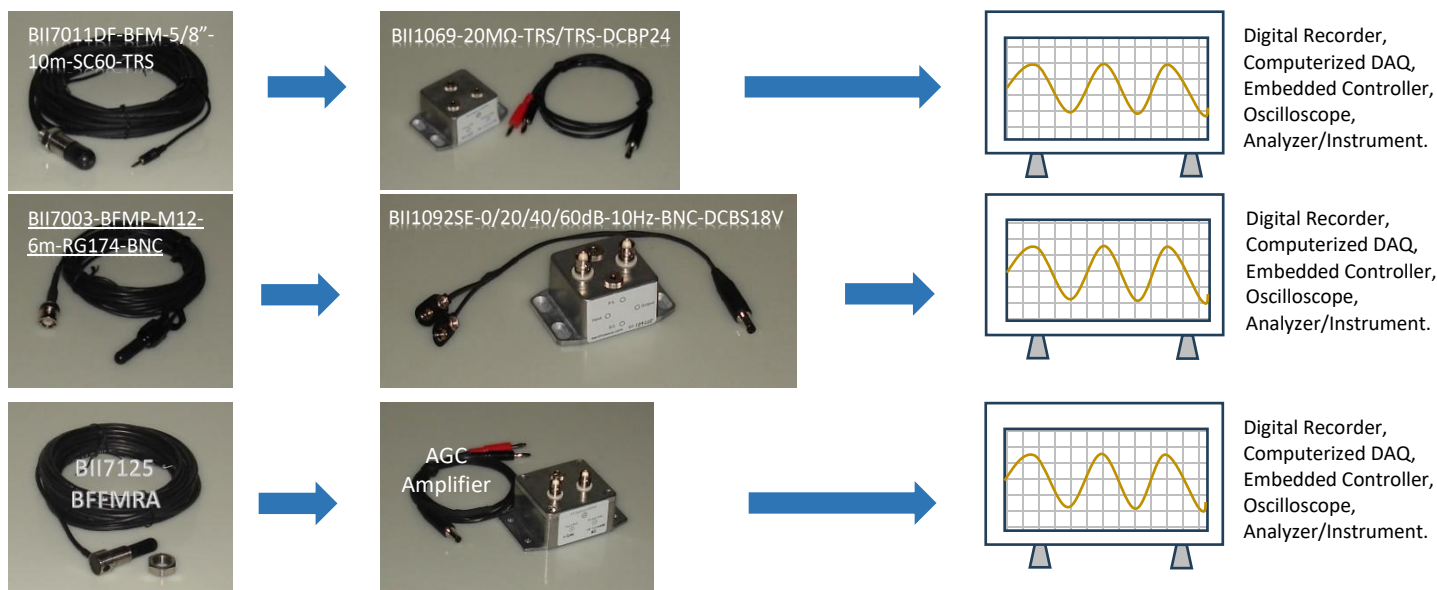
[Hydrophones,](#)
[Transducers,](#)
[Arrays.](#)

Low Noise Hydrophone: Detect unknown sounds and narrowband known sounds.

1. 0.01 Hz to 300 kHz.
2. Self-Noise level below Sea State Zero.
3. Programmable Sensitivity FFVS.
4. Custom-fit to specific known sound sources such as pingers, locators, acoustic tags, etc.



Typical Components of an Acoustic Receiving System. Depending on the requirements of the system, the [signal conditioner](#) is optional.



Thru-Hole Mounting, End-face Mounting, Flush Mount, or Underwater Connector, ...

Autonomous underwater vehicle (AUV), Unmanned underwater vehicles (UUV), Diving bell, Submersible container, etc.

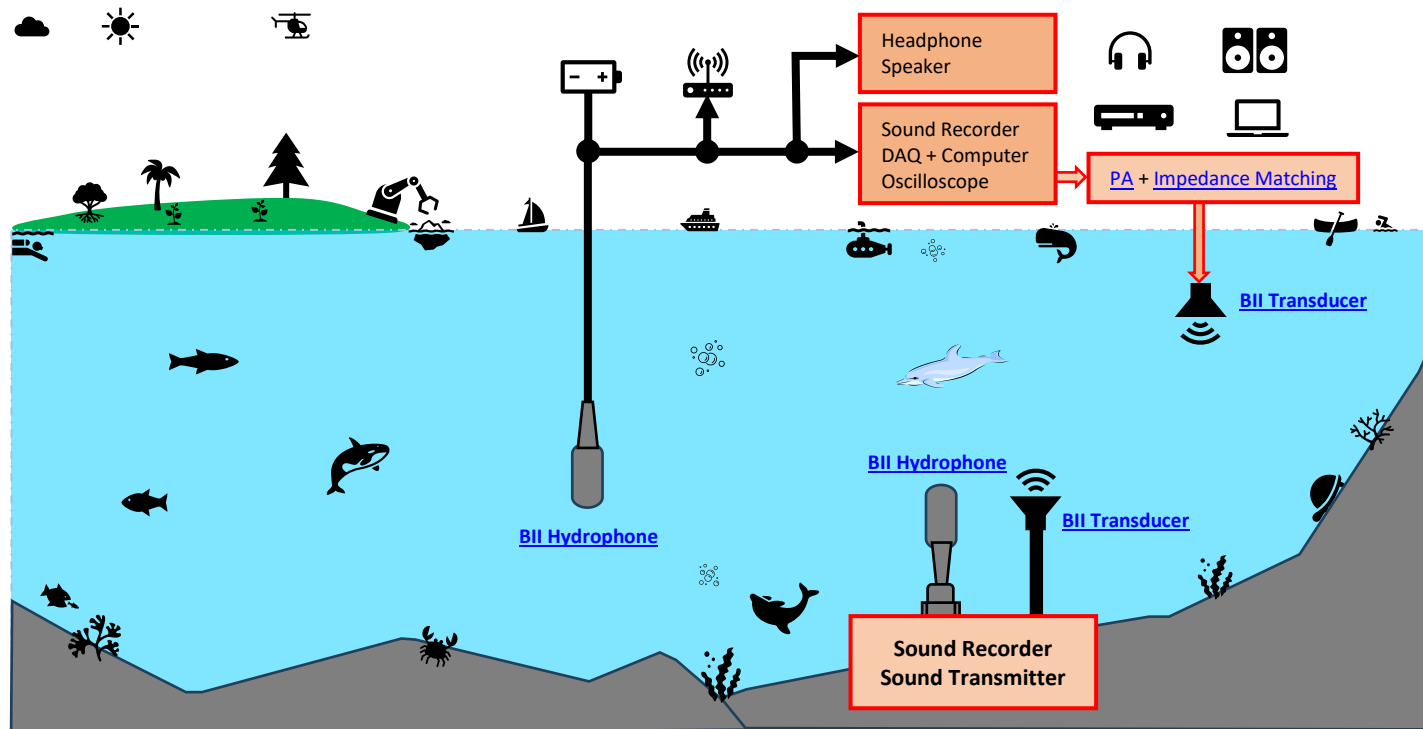
[Signal Conditioning:](#)
Amplification, PGA, VGA, AGC, Filtering, ...

Computerized Instruments:
Data Recording, Analysis, Control, Communication, etc.

Underwater Sound Listening, Recording, and Communication

BII7120 Series Low Noise and Low Frequency Hydrophone: Noise Level Below Sea State Zero

BII's low noise hydrophones are optimized to possess self noise levels below sea-state zero with omnidirectional response in low frequency range and toroidal response in high frequency range. Its streamlined hemispherical dome minimizes drag force and hydrodynamic noise. The power consumption can be customized to be 600 μ A quiescent current at 5VDC for battery powered underwater instrumentation. A spatial array of multiple hydrophones can be set up for directional measurement system. The hydrophones can measure underwater sounds and pressure fluctuations down to 0.1Hz: Surface Waves (Wave-height Sensor), Turbulences, seismic, ocean traffics, industrial noises, precipitations, biologics, ... With these low power hydrophones, battery and system lifetimes are extended, and lighter portable systems with lower-capacity batteries can be achieved. Its compact small size avoids interferences to acoustic field under test. The preamplifier integrated in the hydrophone can drive cable up to 200m without signal loss. Available cable terminals include audio connectors (TRS, XLR), BNC, and underwater mateable connectors. The housing and mounting part are corrosion resistant plastics and/or stainless steels.

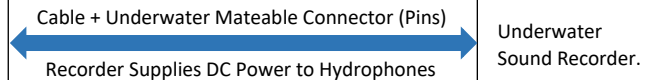


Typical Applications

Underwater Sounds Recording, Listening, and Communication, Noise Measurement, Marine Bioacoustics, Passive Acoustic Monitoring (PAM System).
Coastal/Offshore Processes, Engineering & Management, Wave-Structure Interaction, Wave-height Sensor, Wave and Tide Recorder/Logger.
Surface Waves, Ocean Turbulences, Hydrodynamics, Marine Geophysics, Battery-Powered Instruments: Sonobuoy, Recorder, Transponder, Acoustic Release...

Typical Configuration for Long Term Deployments of Underwater Sound Recorder.

[BII7000 Spherical Series](#), [BII7010 Cylindrical Series](#), [BII7040 Hemispherical Series](#), [BII7070 Planar Series](#), [BII7120 Cylindrical Low Noise Series](#) + Low Power Preamp [BII1067](#) (1.1mA, +2.7 to +24VDC, 20 to 40dB Gain, 1Hz to 350kHz Bandwidth.) or [BII1011](#).



Questions

How do I set up my professional sound recorders to work with BII Hydrophones?

- BII hydrophones have their own DC power supply to support **Line Input** of recorders, and **Do NOT** use phantom power 48V which may destroy the hydrophones.
- Maximum Input Level (Line Input)** of recorders should be large enough to avoid saturation or clipping during recording.
Equivalent Input Noise of recorders should be low enough for the recorders to be sensitive to weak signal of the interest.
- Sampling Rate** of the recorder should be fast enough to avoid missing high frequency sound of the interest. Generally, the **Sampling Rate** should be at least two times greater than the maximum frequency of sound.
- Calculate the **memory size of data storage** according to sampling rate, resolution, sampling channels, and recording time, and use suitable recording media.
- Calculate **battery service life** according to battery power and consuming current.
- When the cable is greater than 5m, **twisted pair cable and balanced signal or differential signal** are recommended to reduce EMI noise.

How do I playback the recorded sounds in water?

System Setup: Recorder (Recorded Sounds) with **Line or Phone Output** -> [Audio Power Amplifiers](#) -> [Impedance Matching Device](#) -> [Transducers \(Projectors\)](#).

What if the mating connector of my DAQ module or recording device is NOT available from BII?

- Buyer may order BII products with wire leads, and buyer assembles the mating connector to the cable end.
- A connector adaptor might be assembled by BII by customization, and BII ships the adaptor to buyer as accessory of the device. Please contact BII for customizations.
- Many adaptors for standard connectors are available in worldwide electronic suppliers such as BNC to SMA, BNC to SMC, XLR to TRS, etc. Check out your local suppliers.

How do I use Gain Selection wires in field?

1. Manual Gain Selection.

When a **Gain Selection wire** is floating or open, its digital logic is High or "1". When a **Gain Selection wire** is short to **Digital Common**, its digital logic is Low or "0".

Sensitivity of a Hydrophone is fixed when its Gain Selection wires are fixed to **Digital Common** or open (floating) during operation.

- Gain Selection with Digital Outputs.** Digital Outputs of a DAQ (data acquisition device) select gains with TTL/CMOS logic levels.

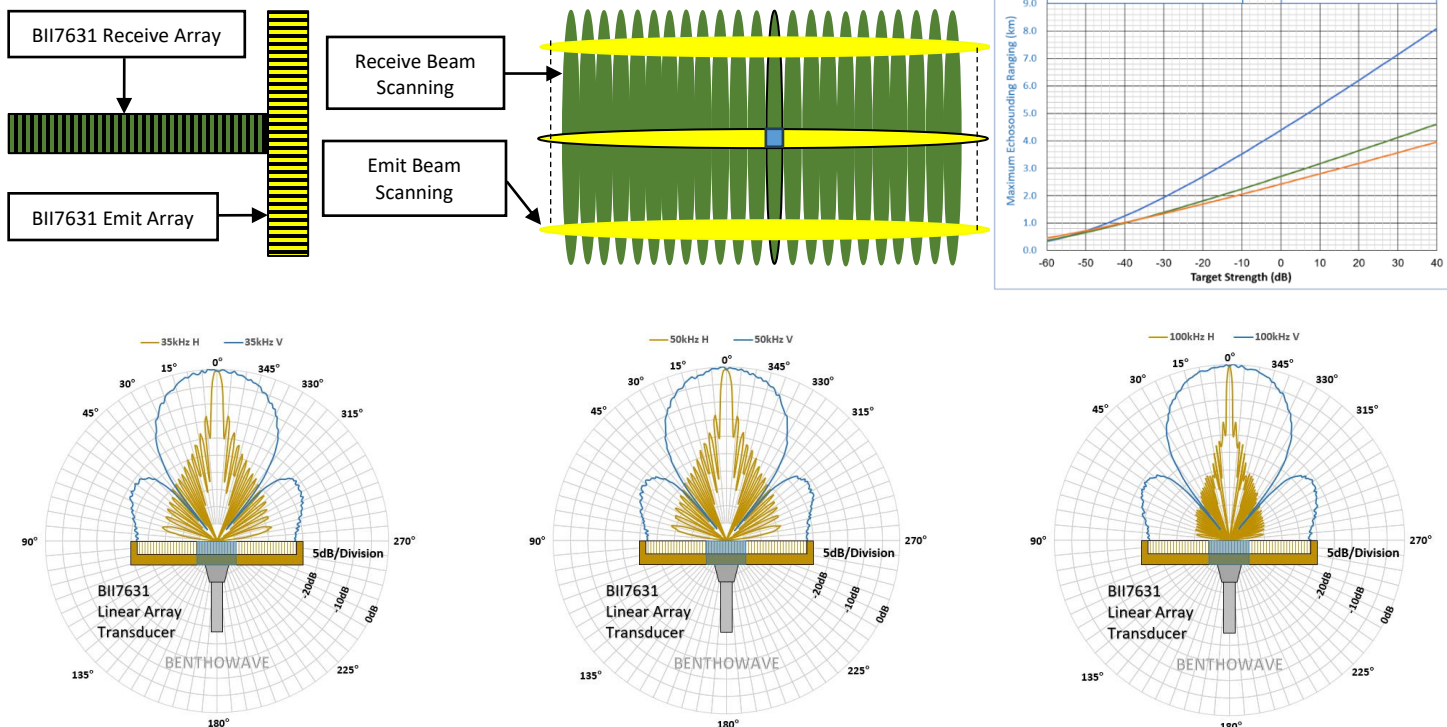
Underwater Sounds Scanning: Positioning, Tracking, Navigation and Imaging

Typical Applications: 2D Imaging Sonar, Object/Target Positioning and Tracking, Obstacle Avoidance, Fishery Sonar, Navigation, Underwater Robotic and Vehicle.

BII7630 Series Phased Array Transducer, Underwater Scanning: Positioning, Tracking, Navigation and Imaging

Beamforming and Total Focusing Method (TFM): Phased array transducers are rectangular (linear) array with custom-fit along-length beamwidth and cross-length beamwidth for uses in oceanography such as location, tracking and search of underwater sound sources in hundreds or thousands meter range, and in acoustical imaging and diagnosis in millimeter range such as biomedical study, NDT & AE, and material study. With dynamic focusing (Total Focusing Method TFM), both resolution and sensitivity are improved in detecting flaws or defects and acoustic imaging. Along-length beam can be steered and focused in $\pm 75^\circ$ range with array beamforming technology. Multiple beams at different directions can also be formed simultaneously with digital beamforming technology. The side lobes along length can be suppressed with amplitude shading or weighting. Two or four array hydrophones can be used to set up "T" type cross array functioning as **Target Angle Estimation System** with Mills Cross technique. High resolution image can be formed with the technology of **Synthetic Aperture Sequential Imaging**. A phased line array transducer (projector) and a phased array hydrophone can work as a multibeam SONAR with Mills Cross technique. Multiple transducers can be wired in parallel electrically to set up a longer line array for reducing along-length beam width in low frequency range.

2D "T" Type Imaging Multibeam Transducer: Two BII7631 Linear Phased Array (Rectangular Aperture).



Echo intensity of blue intersection area of a emit beam (yellow colour) and a receive beam (green colour) provides the average depth and backscattering information of the intersection area. Hundreds of tightly spaced echoes can be implemented from a single acoustic ping.

BII7660 Series Multibeam Transducer: 2D Imaging Sonar, Positioning, Tracking, and Navigation.

BII7660 series high frequency high resolution multibeam transducers are designed for uses in 2D image Sonar, underwater floor/bottom mapping, sector scanning, navigation, object detection, target tracking, obstacle avoidance in tens or hundreds meter range. Emitting aperture (one element) and receiving aperture (32 or 64 elements) operate independently. The curved emitting element emits a broad fan-shape beam to insonify a large sector of interest (field of view), the linear receiving array has fan-shape beam perpendicular to the emit-beam and can be steered electrically to scan the area of interest. Both emit and receive array are broadband to support broadband SONAR signals (such as SINE Pulses, FM/Chirp Pulses, Ricker Pulses, etc.). Acoustic sweeping (2D scanning) of the horizon or vertical can be achieved with digital beamforming technology.

Transducer

Receiving (Linear Aperture to Scan)

Steering Fan-Shaped Beam

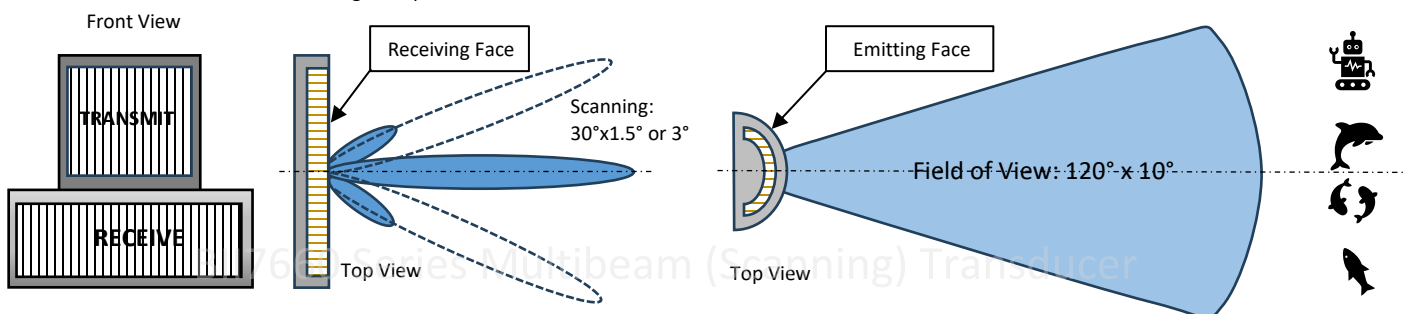
Rectangular Aperture, 32 or 64 Elements.

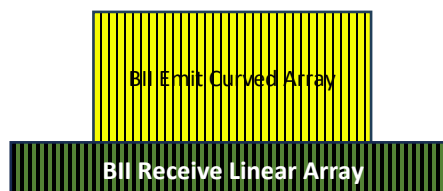
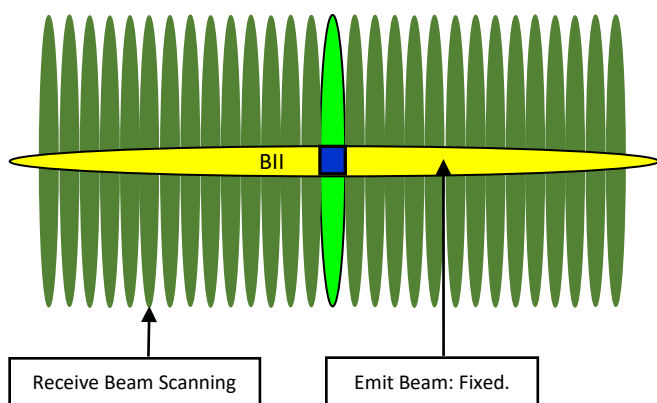
Transmit: Curved Array (Wide Beam, Insonifying Wide Area of the interest)

Curved Aperture

1 Element

Subjects





BII MultiBeam Transducer = Curved Array + Linear Array

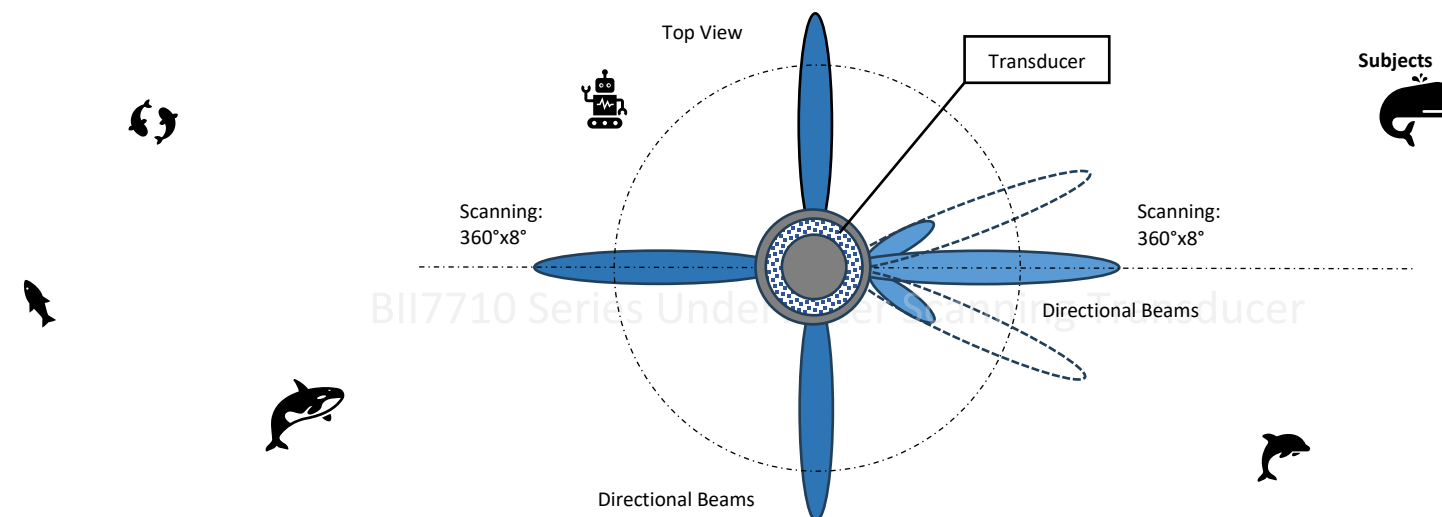
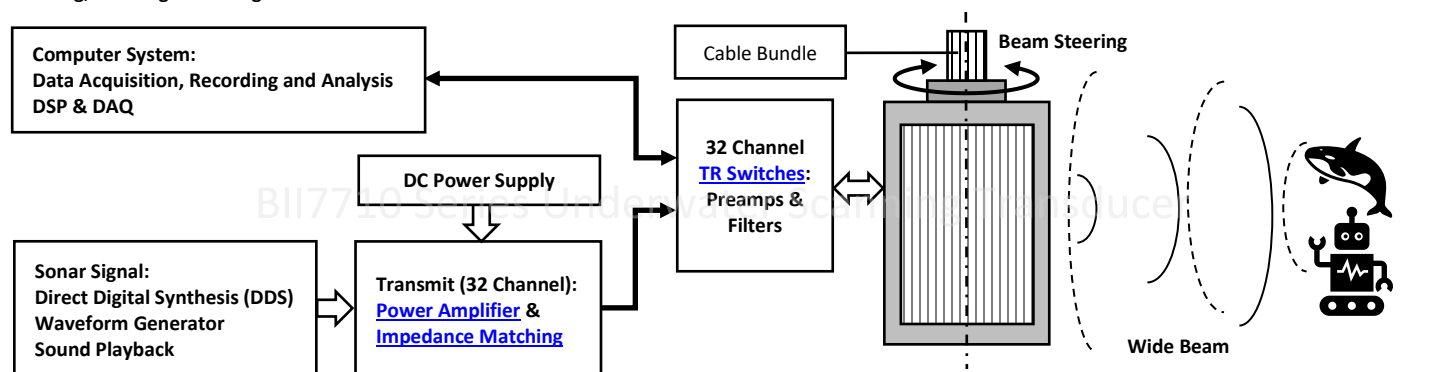
Echo intensity of the **blue intersection area** of a emit beam (**yellow colour**) and a receive beam (**green colour**) provides the average depth and backscattering information of the intersection area. Hundreds of tightly spaced echoes can be implemented from a single acoustic ping.

BII7716 Series Scanning SONAR Transducer: 360° View Field in Horizontal Plane, 360° Search: Tracking, Locating and Navigation.

Scanning SONAR Transducer for 360° Search

Underwater Scanning Transducers are underwater 32-element cylindrical array transducers (Transmitting Sounds and Listening sounds), and directional beams are implemented with subarrays for scanning. Typical applications are acoustic positioning, tracking, echo locating, and navigation in horizontal plane in the ocean, rivers, and lakes. The Scanning SONAR Transducers provide efficient solution to detect and locate fish schools and targets in the horizontal plane about hundreds and thousands meters with 360° searching. The transducers have superior performances for use in shallow water and near surface fishing. The subarray that operator determines sends out a pulse of narrow beam sound in specific directions and receive the echo reflected from targets; or, in passive listening mode, it searches the sounds emitted from sound sources with highly directional beam.

Tracking, Locating and Navigation:



BII7770 Series Underwater Mechanical Scanning Transducer: Tracking, Locating, Navigation, Communication, and Artificial Acoustic Target.

The Underwater Scanning Transducer integrates a wide beam projector and a narrow beam low noise directional hydrophone for mechanical scanning SONAR. Typical applications are long range acoustic positioning, tracking, echo locating, navigation, communication, and artificial acoustic target in horizontal or vertical plane in the ocean, rivers, and lakes.

Sound Ranging of Full-duplex and Half-duplex Communication: around 4.0 km depending on the transducer types, ambient noise, receiver's noise and sensitivity, etc.

Target strength of a 10"-length (0.25m) fish is about -35 dB. Echo Ranging of BII7771 and BII7772 to this fish is around 1.2 km.

Target strength of a dolphin is about -20 dB.

Target strength of a whale is about 0 dB.

Target strength of seamounts is from +30 to +60 dB, say 30dB.

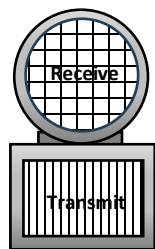
Echo Ranging of BII7771 and BII7772 to this dolphin is around 1.7 km.

Echo Ranging of BII7771 and BII7772 to this whale is around 2.6 km.

Echo Ranging of BII7771 and BII7772 to seamounts is around 3.6 to 4.1 km.

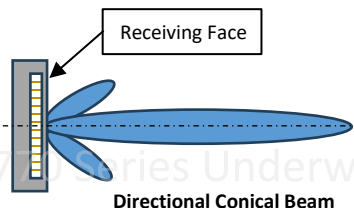
Transducer

Front View



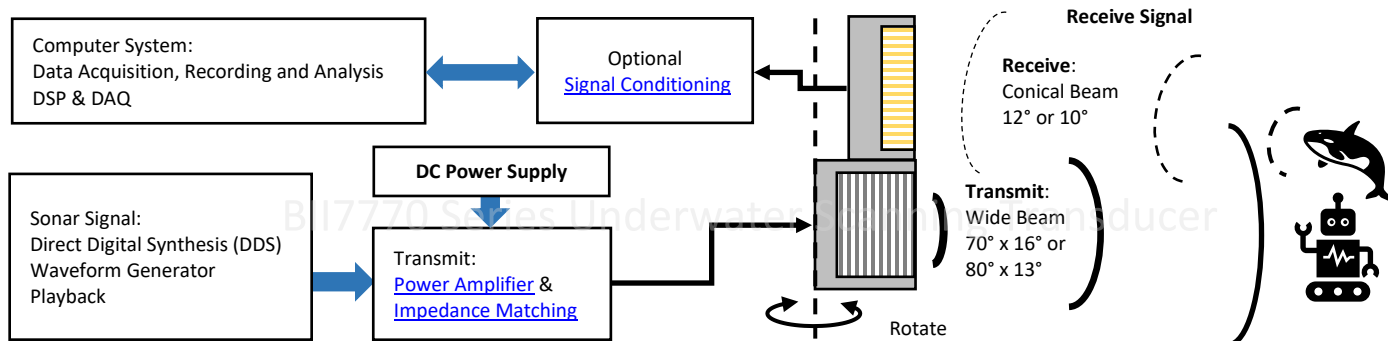
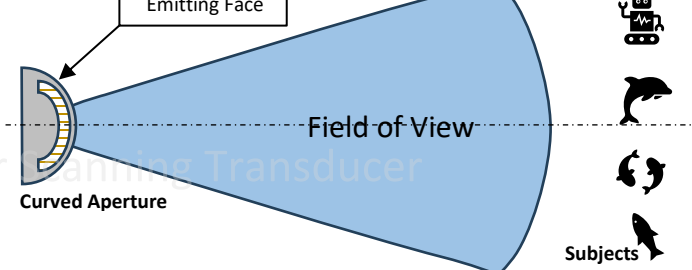
Receiving (Circular Aperture)

Top View

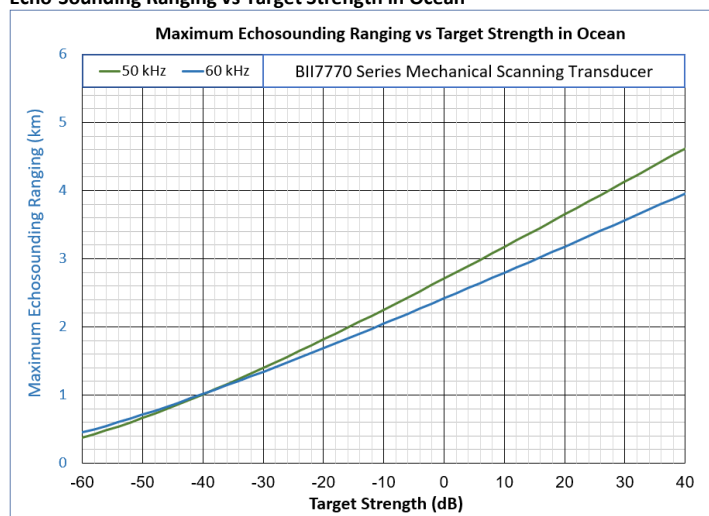


Transmit: Curved Array (Wide Beam, Insonifying Wide Area of the interest)

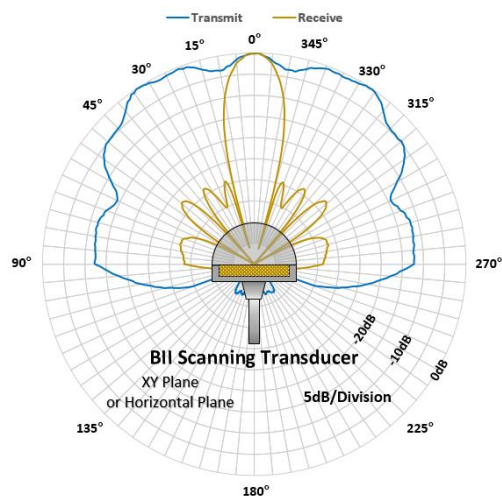
Top View



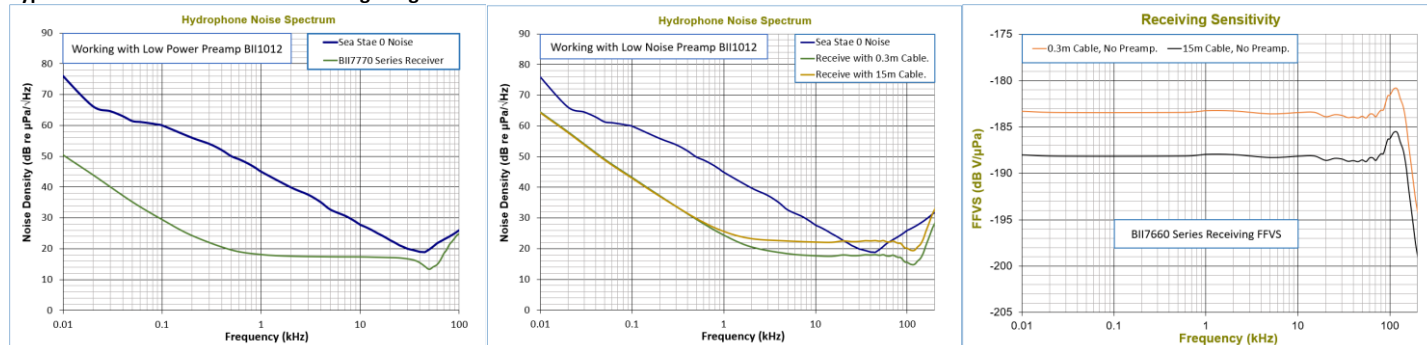
Echo-Sounding Ranging vs Target Strength in Ocean

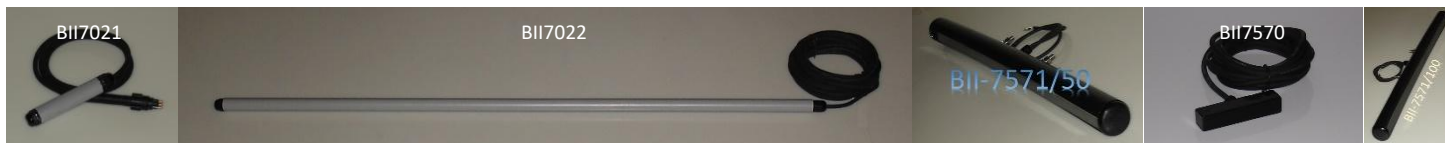


Directional Response in Horizontal Plane: BII7772 60kHz.



Typical Receiver Performance of BII Long Range SONAR Transducers.





Line (Linear) Array Hydrophone and Transducer

BII7020 Series Line Array Hydrophones: Toroidal Directivity Response.

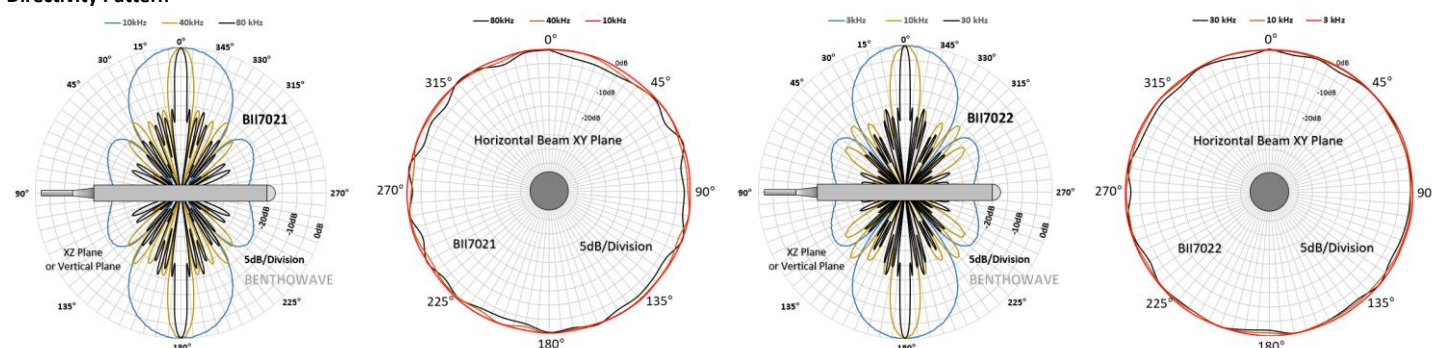
A BII's line (linear) array hydrophone averages sound pressures along its length as a spatial filter, and utilize built-in amplitude shading/weighting to reduce side-lobes. Its toroidal directivity response makes it a great gear to detect and search underwater sound sources omnidirectionally with its main lobe and reject the noises in other directions. Smooth domes and housing reduce interferences to acoustic field under test.

Split-aperture correlators based on line arrays implement target angle estimation. The narrow beamwidth along the length provide direction information of the sound sources and reject the unwanted noises coming from other directions, the omnidirectional response cross the length covers large area to search. Two or four line arrays can be used to set up "T" or "+" type cross array functioning as a Target Angle Estimation System with Mills Cross techniques which provides accurate bearing information of sound sources.

The hydrophone body has streamlined hemispherical domes which minimize drag forces and hydrodynamic noises caused by the hydrophone in motion or the flow past the hydrophone.

For applications of beam steering (Digital or FFT Beamforming) , please deploy multiple [BII7021](#) arrays, [BII7010](#) hydrophones, or [BII7140](#) hydrophones to configure discrete linear or planar arrays.

Directivity Pattern



BII7570 Series Side-scan Transducer and Linear Transducer

BII7570 Series Linear, Side Scan, Interferometric (Phase-measuring) and Parametric Transducers: Sea-floor Mapping and Sub-bottom Profiling.

Linear Transducer: Bespoke fan-shaped beam provides wide view area in across-track plane and high spatial resolution in along-track plane. Sidelobe ≤ -21 dB.

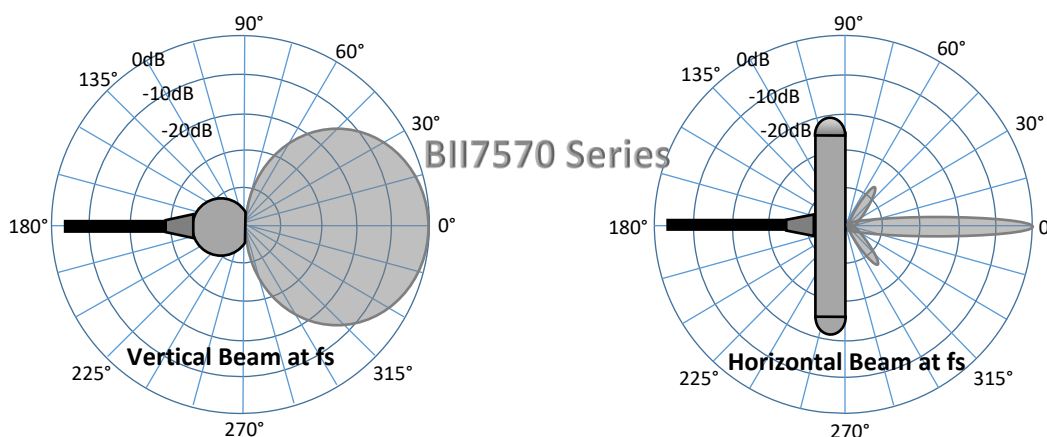
Conventional Side Scan: BII's side scan transducers with fan-shaped beam are designed for use in underwater imaging, sea/river/lake-floor mapping, target location, mine hunting, fisheries... The beam covers wide across-track swath and provides high along-track spatial resolution in tens or hundreds meter range. Acoustic image of underwater bottom is achieved at grazing angles of incidence. High resolution image can be formed with the technique of **Synthetic Aperture Imaging**. Multiple frequencies are available in one transducer.

Interferometric (Phase-measuring) Side Scan: The phase differences of received signals are detected by multiple linear receive arrays paralleling to the linear transmit array. After ambiguity is removed with proper techniques, the direction of arrival (DOA) and location of the scatterer can be accurately determined.

Parametric Side Scan: When two underwater sound waves of different primary frequencies f_{p1} and f_{p2} ($f_{p1} > f_{p2}$) propagate in the same direction, they interact with each other to create low frequency sound wave of secondary frequency f_{sec} ($f_{sec} = f_{p1} - f_{p2}$). The directivity of secondary frequency is close to the ones of primary frequencies. This difference frequency sound is useful for practical applications in sediment profiling, depth sounding and communication. Parametric array gain or efficiency (generally $\leq 1\%$) is better as primary sound powers are higher, secondary frequency f_{sec} is higher, down shift ratio $(f_{p1} + f_{p2}) / (2f_{sec})$ is lower, and $(\alpha p \cdot R_r)$ is lower (αp : mean primary sound attenuation coefficient; R_r : rayleigh distance). Attenuation/absorption coefficient of sediments is frequency dependent around $0.06f$ to $0.6f$ (dB/(m $\cdot$ kHz)).

Typical Applications: Sea/River/Lake-floor Mapping, Target Location Underwater, Fisheries, Direction of Arrival, Mine Hunting.

Directivity Pattern: illustration ONLY. Please refer to -3 dB beam width of a specific transducer.

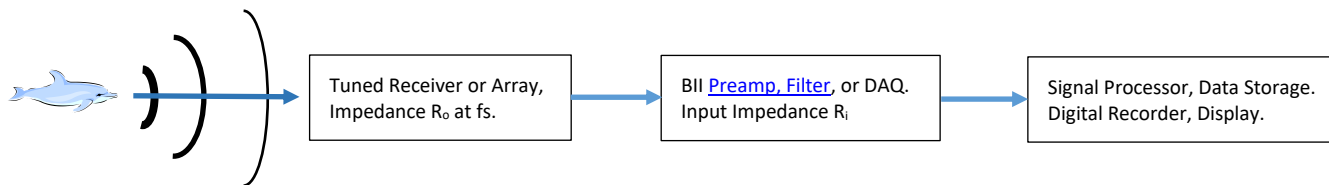


Tuned Transducer: Optimum Receiver with Maximum SNR

Tuned Transducer as a Receiver

Tuned transducers (hydrophones, AE sensor, NDT receivers, etc...) are band pass sound receivers and projector operating around resonance f_s , which feature broadband, possible maximum power transfer from the transducer to preamplifier, and offer flexible solutions to special transducer demanding in underwater acoustics and NDT (Non-destructive Test).

Impedance of a tuned transducer is resistive at f_s .



Three Major Operations of tuned receivers: Impedance matching $R_o = R_i$, Open Circuit $R_i \gg R_o$, and $R_o < R_i \ll \infty$.

a. Impedance matching between the transducer and its signal conditioning circuit:

1. Sound energy being reflected from the transducer is minimized.
2. Maximum power transfer from transducer to preamplifier or signal conditioner.

Advantages:

Broaden the bandwidth, reduce impulse rings, increase spatial resolution in tracking, positioning, and NDT applications. Reduce mutual interaction among array elements in an array. Reduce interference to incident acoustic waves.

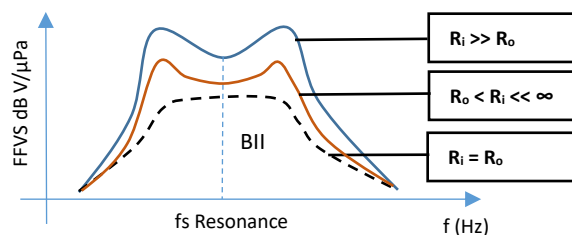
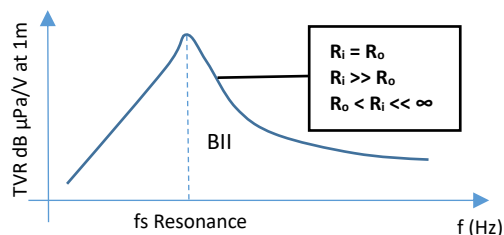
Disadvantage:

The sensitivity drops 6 dB around f_s comparing to OCV (open circuit voltage).

b. Input impedance of signal conditioner $R_i \gg$ transducer impedance R_o :

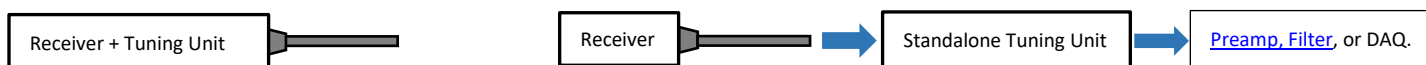
Output voltage of receiver is OCV (open circuit voltage).

Typical TVR and FFVS of Tuned Transducer:



Tuned Receiver Structure:

1. Tuning Unit integrated inside receiver housing
2. Receiver + Standalone Tuning Unit



How to order:

a. Customize BII Standard Transducers:

1. First, check online datasheet or contact BII to get f_s information of the transducer (hydrophones, AE sensors, NDT Transducers, etc...).

2. Append **TT** to part number of a transducer. for example:

BII7011TT is tuned BII7011 hydrophone.

BII7004TT is tuned BII7004 hydrophone.

BII7074TT is tuned BII7074 hydrophone.

Note: 1. The tuned transducer (or receiver) is NOT recommended to operate far from its f_s resonance.

2. The size of tuned transducer might be bigger or longer than its original ones. Contact BII if the size is important to your applications such as array spacing.

b. Customize a New Transducer.

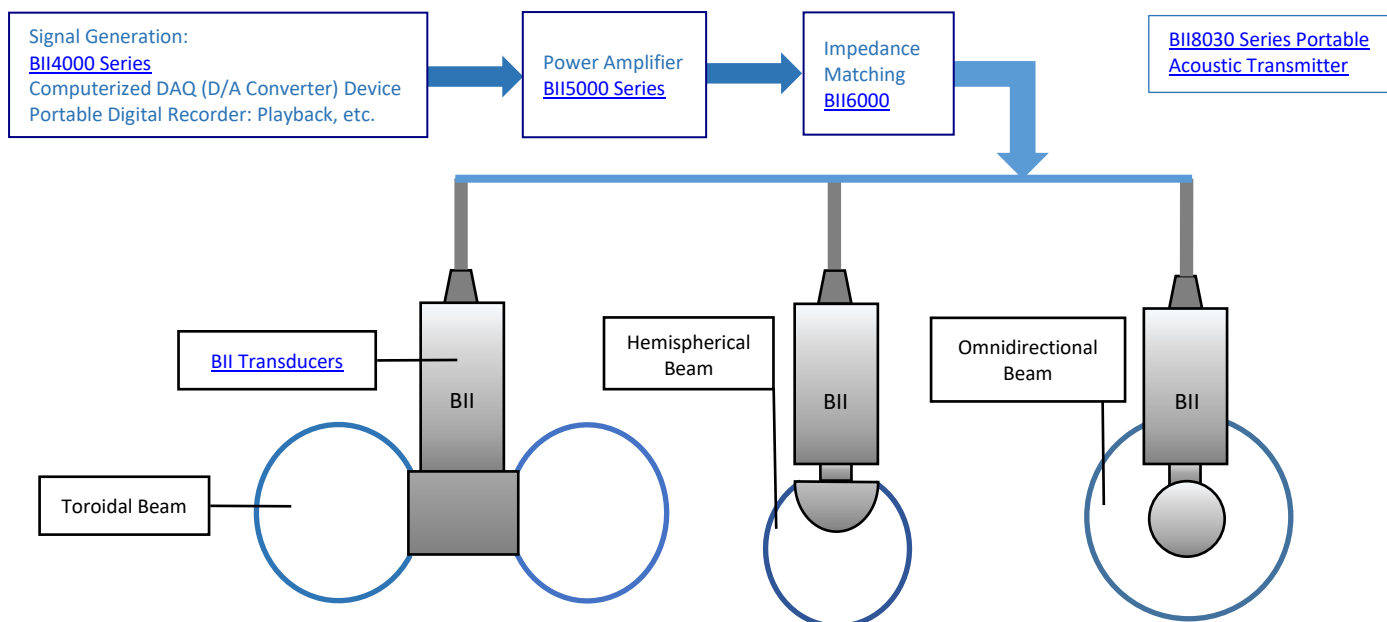
1. Determine frequency of the interest or f_s .
2. Determine directivity response of the receiver.
3. Is a preamplifier necessary? Yes, if the signal is weak. No, if the sound level is strong.
4. Determine or compromise bandwidth, reflection, and overshoot with operating condition: $R_o = R_i$, $R_i \gg R_o$, or $R_o < R_i \ll \infty$.
5. Integration or standalone tuning unit?
6. Contact BII with your specs for new customized receivers.

Underwater Beacons: Pingers, Repeaters, and Transponders

DESCRIPTION

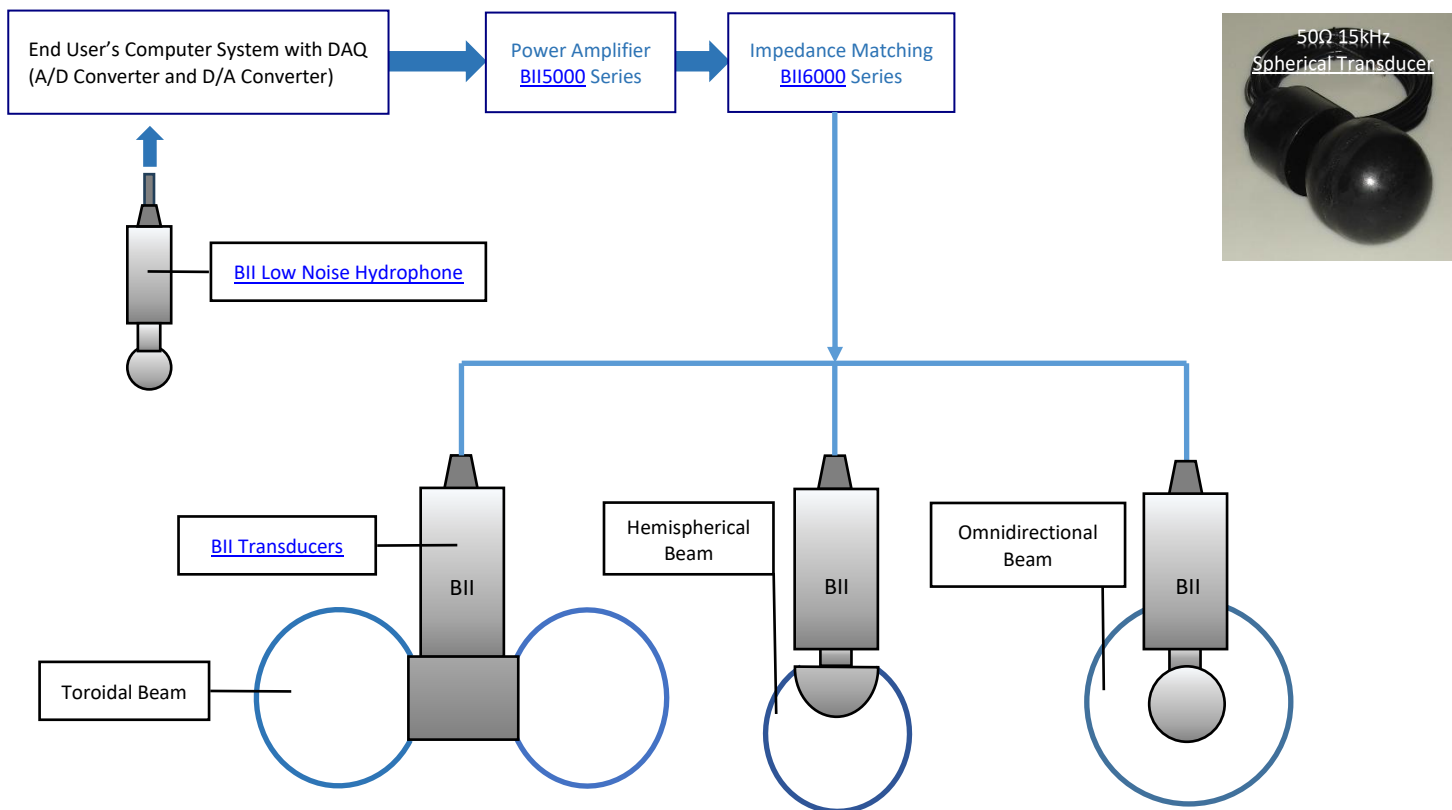
Underwater Beacons are broadband acoustic transmitters and/or receivers. BII components are ready for you to implement and develop unique and proprietary pingers repeaters, and transponders: Underwater Communication and Positioning, Acoustic Locator and Release, Pilot Sound Sources, Artificial Acoustic Target, Long and Short-Range Navigation.

Pingers Setup:



Repeater and Transponder Setup:

Repeaters and Transponders includes a low noise hydrophone with suitable directivity pattern for receiving an incoming acoustic sound from a remote station or a sound source such as a pinger and a submarine. Buyer's computer samples and processes incoming signals to synthesize a new signal as an echo signal, an "answer" signal, or a "command" signal. the computer outputs the new signal to sound transmitting system which consists of a power amplifier, an impedance matching device, and a transducer (projector).



Acoustical Measurements: Pulsing and Receiving for Physical Acoustics, NDT, and AE.

Acoustical measurements include:

1. Time Measurements such as Frequency $f = 1/T_0$, Bandwidth $B = \Delta f$ at -3dB Points, Time Delay Δt .
2. Spatial-Size Measurements such as Sound Path Length L , Wavelength λ , Schlieren Image Distribution, etc.
3. Measurement of Energy Dissipation such as Attenuation of Sound Amplitude during Spatial Propagation, Impedance Change, Damping Decrement, Bandwidth, etc.

Sound velocity C and attenuation coefficient α are two fundamental quantities. Propagation of a plane wave in a medium:

$$A_x = A_0 e^{j\omega(t-x/C)} e^{-\alpha x}$$

A_0 : Amplitude of Sound Pressure, Particle Velocity, or Density. $\omega = 2\pi f$. t : Propagation Time. x : Displacement from Origin to Measuring Position.

C and α of sound waves propagating in a medium are the macroscopic manifestation arising from the interaction between sound waves and one or more properties of the material. With accurate C and/or α , some of mechanical, thermal, electrical, magnetic, and optical properties of the material can be determined qualitatively or quantitatively.

Typical ultrasonic (elastic or stress) waves used in measurements:

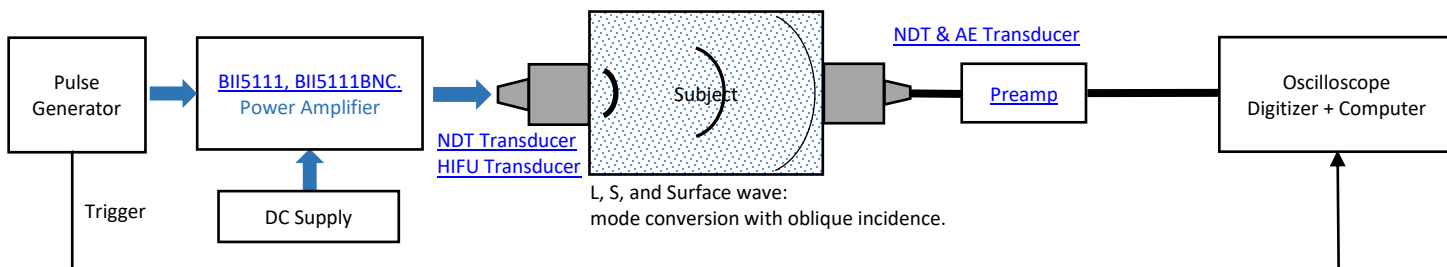
1. Compressional (Longitudinal, or L) Wave.
2. Shear (Transverse or S) Wave.
3. Surface (Rayleigh) Wave.
4. Lamb, Stonely, and Scholte Waves, etc.
5. Wave Mode Conversion with Wedge at Critical Incident Angles.



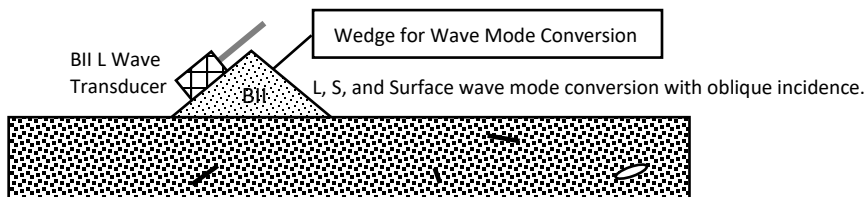
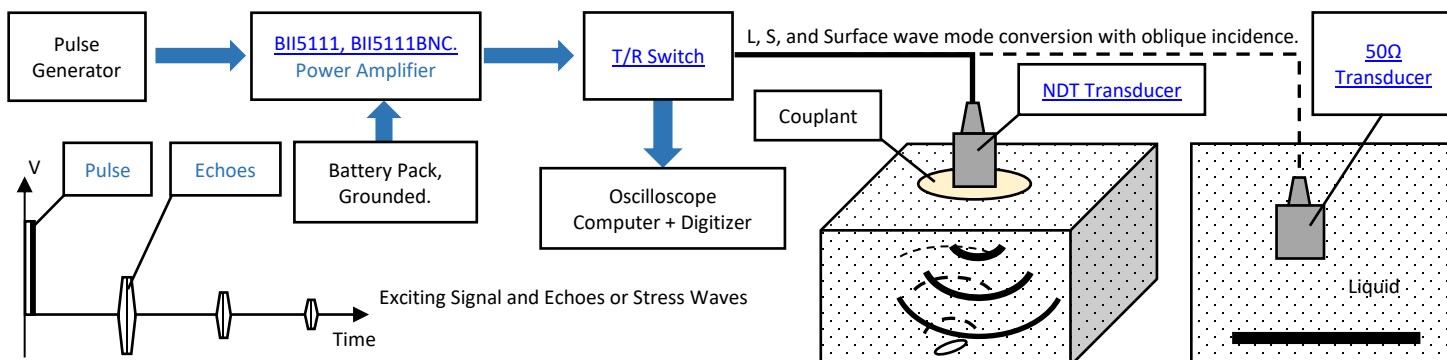
BII manufactures broadband (low Q_m) longitudinal transducers (planar wave and focused wave) with fundamental f_s up to 10 MHz (usable at 3rd harmonics or up to 30 MHz), and bespoke Acrylic or Plexiglas (or other engineering plastics such as Polystyrene, Nylon, PTFE, etc...) Wedge for wave mode conversion.

Useful Formula and Parameters for Ultrasonic Transducers		
Q : Lumped System Quality Factor; λ : Wavelength; D : Aperture Diameter; FWHM : Full Width at Half Maximum.		
Best Axial Resolution $\approx 0.95Q \cdot \lambda$.	Best Lateral Resolution = Minimum Beam Width $\approx \text{FWHM} = 1.4\lambda \cdot F/D$.	Near Field Length $N = D^2/(4\lambda)$.
Rayleigh Distance $= \pi D^2/(4\lambda)$.	Fraunhofer Zone Distance $= 2.3D^2/(4\lambda)$.	Angle of Divergence $= \sin^{-1}(1.22\lambda/D)$.
Wave Mode Conversion at Oblique Incidence : L, S, Rayleigh (Surface), Lamb, Stonely, and Scholte Waves, Snell's Law: $\sin\theta_L/C_L = \sin\theta_S/C_S = \sin\theta_{RL}/C_{RL}$.		
Immersion Testing from Water to Steel: First Critical Angle: 15°, compressive wave to shear wave mode conversion. Second Critical Angle: 27°, shear wave to surface wave mode conversion.		Contact Testing from Plexiglas to Steel: First Critical Angle: 28°, compressive wave to shear wave mode conversion. Second Critical Angle: 58°, shear wave to surface wave mode conversion.

(1) NDT Pulser.



(2) NDT Pulser and Receiver.



AE Detection System

BII7230 Series: AE Sensor without built-in preamp. Acoustic aperture: $\Phi 2$ to $\Phi 12.7$ mm, 1 kHz to 1.2 MHz. FFVS sensitivity is calibrated in water.

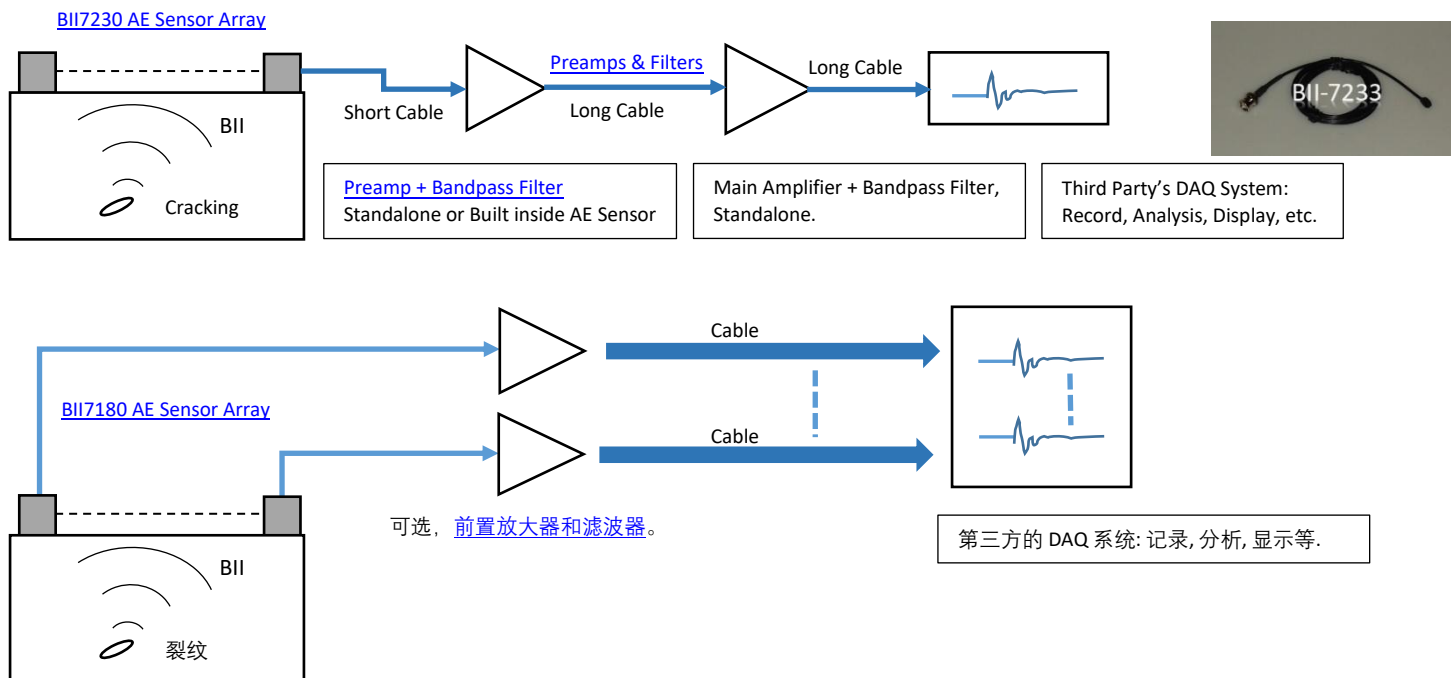
BII7180 Series: AE Sensor with built-in preamp. Acoustic aperture: $\Phi 1$ to $\Phi 3$ mm, 90 Hz to 3.5 MHz. FFVS sensitivity is calibrated in water.

BII AE acoustic sensors detect wideband stress wave released by the structure (Macro or Micro) deformation, cracking, leaking, chemical reaction etc... They can be used as contact sensors and immersion sensors to detect AE sources up to 200°C (392°F) in air and liquids (or water), or as embedded AE sensors. Phase arrays can be configured with these small-aperture AE sensors to position the AE sources. When AE sensors are used in air, the couplant (water, gel, grease, oils, adhesives, and commercial couplant.) is a necessary material to provide efficient acoustic coupling between the sensor face and the piece under test (DUT).

Typical Applications

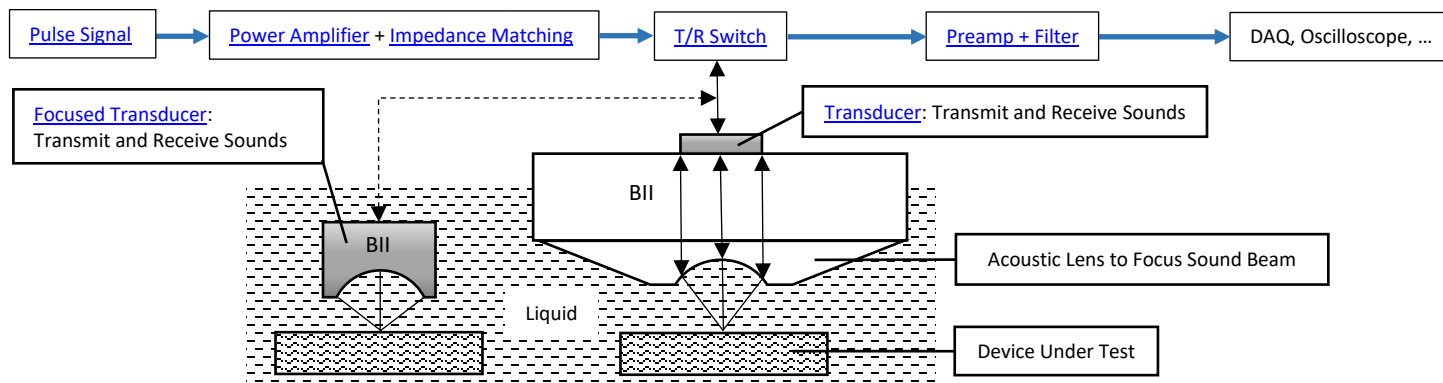
Examination of Structure	Control and Monitoring of Manufacturing Process	Material Study	Health Monitoring of Structure
--------------------------	---	----------------	--------------------------------

AE Detection System

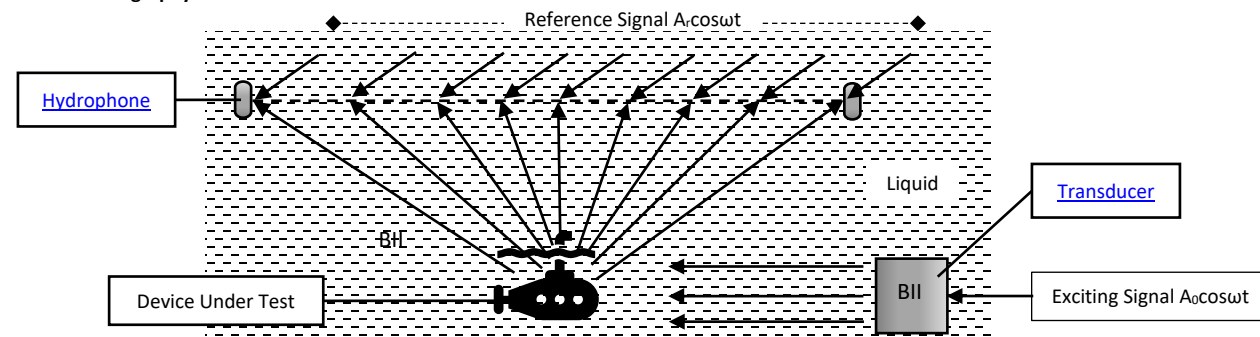


Acoustical Imaging: Microscopy, Holography, and Tomography

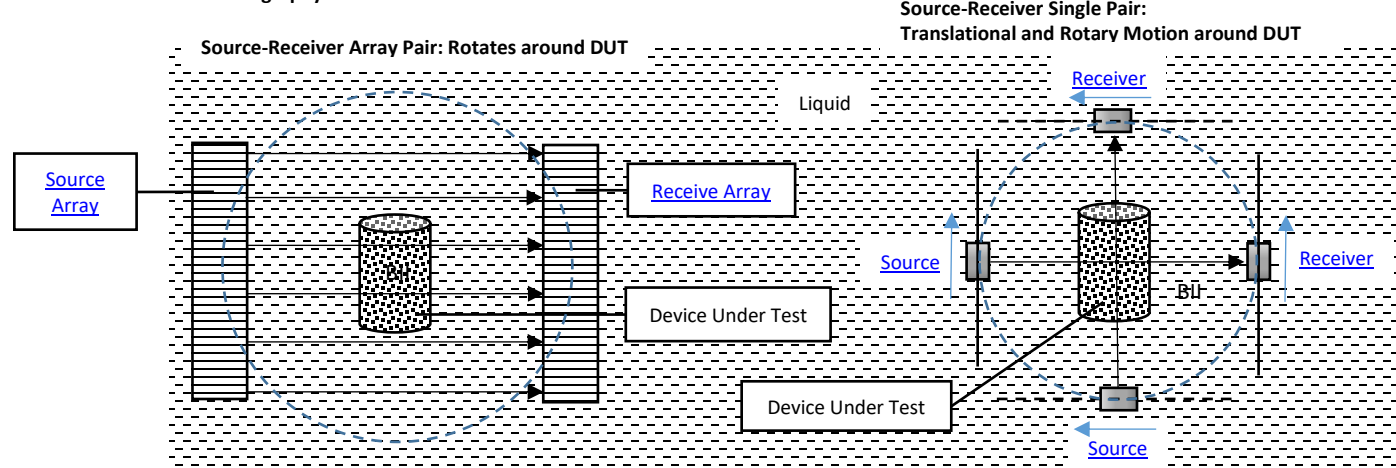
Acoustic Reflection Microscope



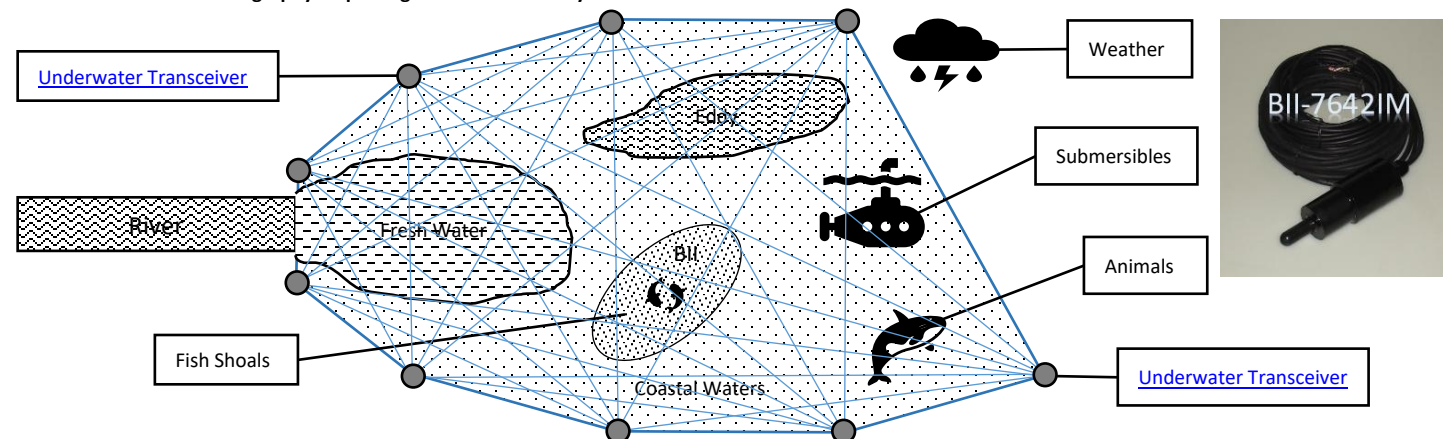
Acoustic Holography



Acoustic Transmission Tomography



Underwater Acoustic Tomography: Capturing Acoustic Variability of an Area.

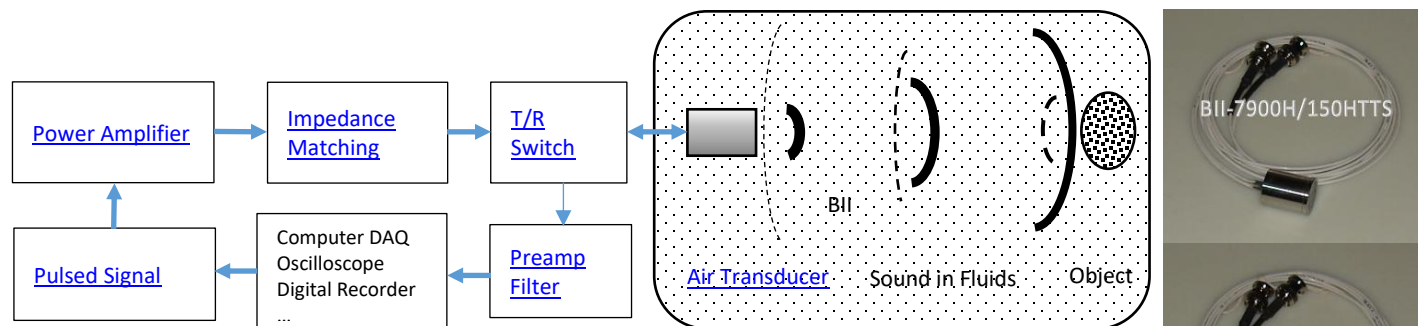


Ultrasonic Air Transducer: Planar and Electronic Focusing, 30 to 300 kHz, -40°C to 140°C (-40°F to 284°F) is available.

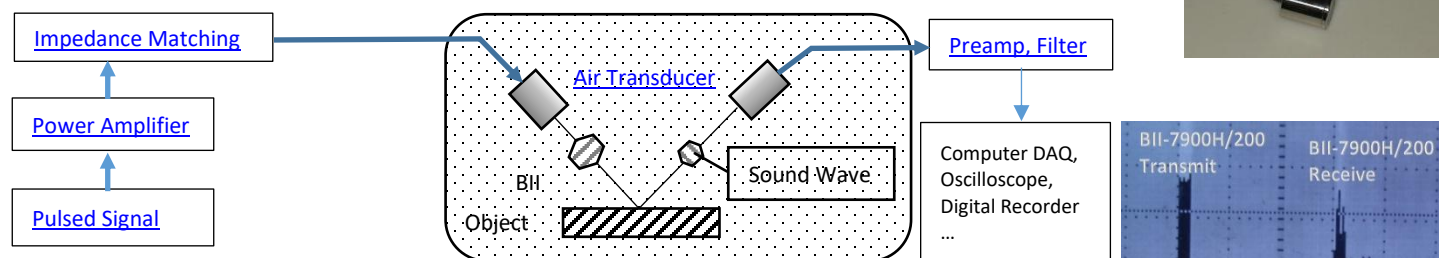
BII's air transducers featuring low to medium Qm in air are for air-coupled NDT (Non-destructive Testing), navigation, ranging, measurement and control, and characterization of airlike fluids (gases) and materials such as woods, plastics, rubber, foam, and composites. Transducers are water-proofed to withstand 50m water depth at least. Sounds generated with Ultrasonic Air Transducer propagate in air or airlike fluids (gases) and materials.

Typical Applications	
Measurands influence propagation time, phase, and attenuation.	Measurands influence reflection, refraction, scattering and transmission.
Robotics, Proximity Detection, Sound Ranging, Material Study	Counting, Monitoring, Remote Control, Alarming, Motion Detection
Level Measurement, Speed Measurement, Leak Detection	Automatic Sizing, Sorting & Positioning of Parts, Ultrasonic Testing and Analysis
Edge Detection, Web Guiding System, Air-Coupled NDT	Surface/Profile Characterization and Quick Scanning for Quality Control

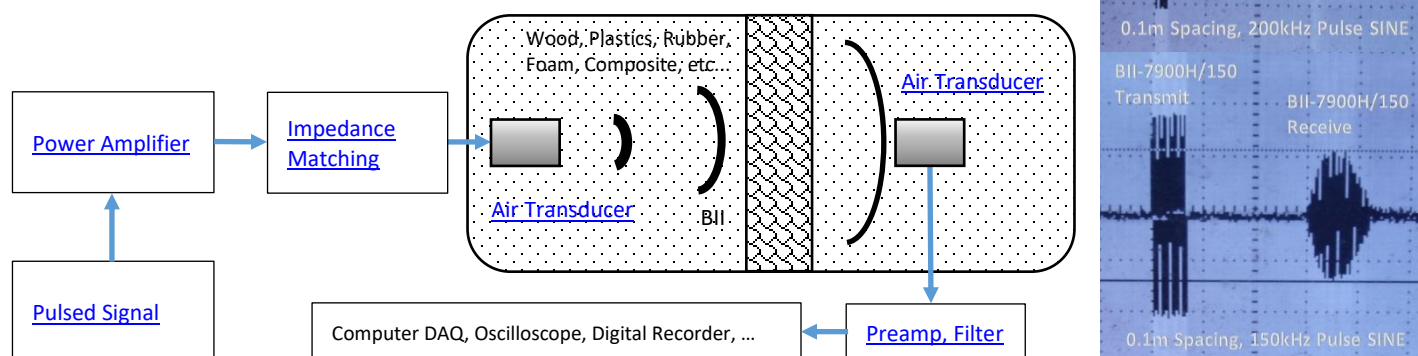
Pulse-Echo System: SONAR in Air and Airlike Fluids



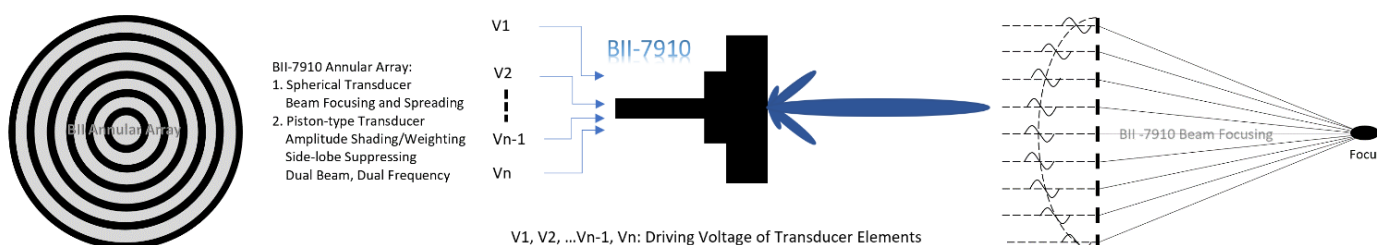
Pulse-Reflection System in Air and Airlike Fluids



Pulse Transmission System: Information Transmission through Air, Airlike Fluids and Material.



Air Array Transducer: Electronic Beam Focusing



Absorption of Sound in Air at 20°C (68°F), Relative Humidity: 10%, 1 atm.												
Frequency (kHz)	30	40	50	70	100	120	150	200	250	300	500	1000
Absorption (dB/m)	0.3	0.4	0.5	0.7	1.8	2.5	4.0	6.5	10	16	43	200

The radiation impedance of air or airlike fluids is quite low to cause energy efficiency of an air transducer is quite low. More than 96% of electrical energy of the driving signal is transformed to heat inside the transducer. When the temperature inside a transducer approximates curie temperature of the piezoelectric material at high driving voltage, the piezo material is destroyed beyond repair.

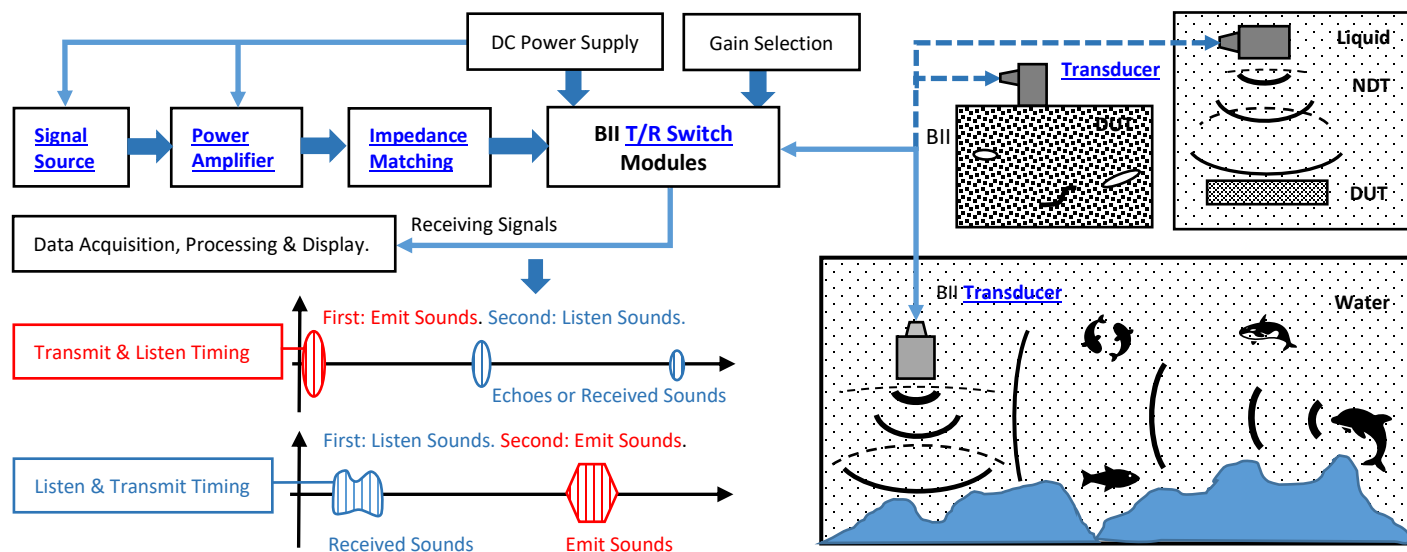
Pulsed driving signal allows the transducer to cool down by itself in ambience. But generally, the materials used in an air transducer are non-metal which have low thermal conductivity comparing to metals such as aluminum. Therefore, **the pulse duration (width) and duty cycle of a driving signal must be chosen carefully to allow the transducer cooling down by itself in surroundings. Otherwise, the transducer will be damaged beyond repair.**



BII Signal Processing Components

BII Components	Main Functions
BII1000 Series	Preamplifiers with built-in Filters: amplify or attenuate electrical signals generated by piezoelectric sensing elements. VGA (Variable Gain Amplifier) for TVG (Time Variable Gain) Applications: compensate transmission loss in active SONAR and ultrasound system. Preamp Types: Fixed Gain, Digitally Programmable Gain, Variable Gain.
BII2000 Series	AGC (Automatic Gain Control) Amplifier: Compensate the propagation losses of water, air, and solids automatically.
BII2100 Series	Transmit and Receive Switching Module with built-in impedance matching, preamp, and filter.
BII4000 Series	Signal Generator for Acoustic Systems of SONAR, NDT, HIFU...
BII5000 Series	Power Amplifiers: Drive SONAR, HIFU, and Ultrasonic NDT Transducers.
BII6000 Series	Impedance matching and tuning between power amplifiers and piezoelectric transducers.

Block Diagram of a Typical Pulse-Echo Acoustic System and a Transponder/Communication Acoustic System.



Piezoelectric Materials and Shapes

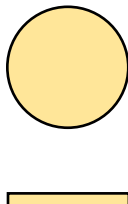
High d constant (strain or charge constant) is desirable for piezoelectric materials to emit large sound pressures or stress waves, high g constant (voltage constant) is desirable in materials to have high voltage or charge sensitivity in response to a sound pressure or stress, high k (electromechanical coupling factor) and low Q_m (mechanical quality factor) are desirable in materials to have a broadband response of emitting and/or receiving sounds (or stress waves). BII manufactures acoustic transducers (Hydrophones and Projectors) with following piezoelectric ceramics.

Material:	Features (exclusive to the piezoelectric materials used in BII products.)
Composites: PZT, PT, or PN + Passive Material Matrix.	Highly anisotropic. High k_{33} , g_{33} and g_h . Low Q_m , broadband. High power as sound projectors; High sensitivity as sound receivers. -40 to 140 °C (-40 to 284 °F). 
Hard PZT, Modified $Pb(Ti,Zr)O_3$:	High k , high d , low loss, high power transducers. -100 to 250 °C (-148 to 482 °F) applications.
Soft PZT, Modified $Pb(Ti,Zr)O_3$:	High k , high g , broadband, high sensitivity hydrophones and sensors. -100 to 250 °C (-148 to 482 °F).
Lead Titanate PT, Modified $PbTiO_3$:	Highly anisotropic, $k_t/k_p \geq 10$, high g_h , high power. Great resistance to depoling by electric field. -40 to 300 °C (-40 to 572 °F).
Lead Metaniobate PN, Modified $PbNb_2O_6$:	Low Q_m , high frequency broadband and pulsing transducers. -100 to 400 °C (-148 to 752 °F).
Barium Titanate BT, Modified $BaTiO_3$:	High power SONAR (duty cycle $\leq 1\%$). Medium d_{33} and g_{33} , low g_{31} , high N . Great resistance to depoling by compressive stress. -20 to 80 °C (-4 to 176 °F). Lead free.
Bismuth Layer Structure BiT, Modified $Bi_4Ti_3O_{12}$:	High frequency transducers. Anisotropic, $k_t/k_p \geq 4$. Low k_t and k_{33} , low d constant, medium g constant. -100 to 500 °C (-148 to 932 °F). Lead free.
The features mentioned above are for piezoelectric materials only. The features of BII transducers (Hydrophones and Projectors) depend on piezoelectric materials, passive materials, mechanical and electrical structure, loading medium, etc..	

Besides standard product lines, BII manufactures transducers with bespoke piezoelectric materials listed above to meet your requirements. Shapes of piezoelectric materials: Bar/Rod, Round Disk, Rectangular Plate, Curved Shell, Tube (Cylindrical Shell), Sphere, Hemisphere, Convex Shell, Concave Shell, Composite, and custom-fit piezo shapes. Following piezo elements are listed with Top View and Front View.



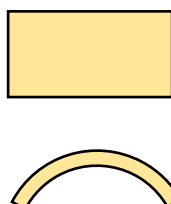
Bar



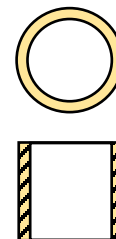
Disk (Round Plate)



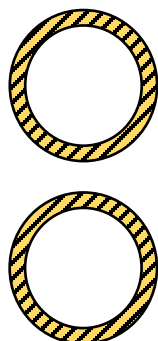
Rectangular Plate



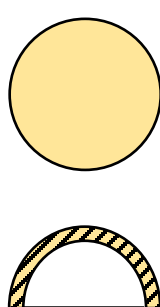
Cylindrical Shell (Curved Shell)



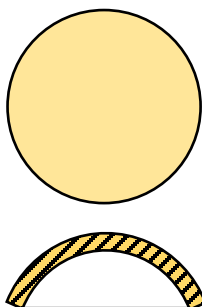
Tube (Cylindrical Shell)



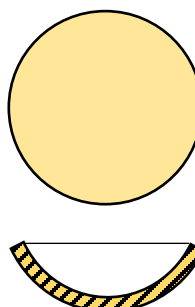
Spherical Shell



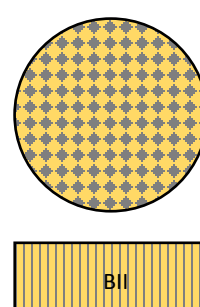
Hemispherical Shell



Concave Spherical Shell



Convex Spherical Shell



Piezoelectric Composite

Magnetic Materials and Cores

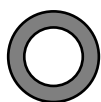
The impedance (or admittance) of a piezoelectric transducer (a projector or a hydrophone) is frequency-dependent ranging from hundreds k Ω to several Ω , and is capacitive, resistive, or inductive at different frequency ranges. An [impedance matching and tuning unit](#) is a necessary device to change the impedance of the transducer in a specific narrow frequency range to meet the load requirements of a power amplifier for maximum and efficient power transfer (high power factor) from the electric to the mechanical, or to match the input impedance of a preamplifier for maximum and efficient power transfer from the mechanical to the electric, or achieve optimum source resistance for minimum noise figure NF. Besides, BII also manufactures custom-fit voltage step-up transformers for measurement of ferroelectric and piezoelectric materials.

Magnetic and air cores manufactured at BII are designed for impedance matching between piezoelectric transducers (SONAR, NDT, HIFU, AE) and power amplifiers. The core geometries include Toroid, Cylinder and Plane (PCB Based), and are customized to fit for BII transducers.

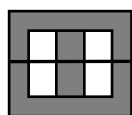
Features of BII's magnetic cores:

- Frequency.** 30 Hz to 50 MHz with ferromagnetic cores; up to 100 MHz with air core.
- Cooling.** Thermal conductive performances are improved with special thermal compounds to avoid over-heating.
- Bs and Air Gap.** Magnetic B/H curve are customized to increase the capacity of power conversion between electric field and magnetic field.
- Core Loss.** The hysteresis and eddy-current losses are minimized with suitable materials and sizes.
- Insulation.** Insulation voltage among windings ≥ 1200 Vrms. Insulation voltage between windings and core surfaces ≥ 2000 Vrms.
- EMI Radiation.** Toroid cores minimize the RF radiation.
- Miniaturization.** The cylinder cores provide the smallest size options for small transducers.
- Service Temperature.** -40 to 140 °C (-40 to 284 °F). Custom-made cores working from -40 to 287 °C (-40 to 550 °F) are available.
- Planar Cores.** Inductive planar devices (based on PCB) are available to fit for some special transducer constraints in high frequency range.

Toroid



EE



Cylinder



Plane (PCB Based)



Manufacturing of Cores:

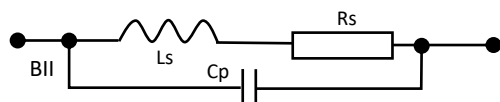
1. **Stacked Laminations**
2. **Tape-wound Cores**
3. **Powder Cores**
4. **Printed Circuit Boards (PCB)**

Besides BII's cores, soft magnetic cores from world-wide manufacturers are used to make inductive components for impedance matching from 30 Hz to 10 MHz.

Testing of the Inductive Components:

1. Inductance, Q, Turn Ratio of Transformer, Leakage Inductance, Frequency Response, Impedance vs. Frequency, Lumped Equivalent Circuit Model.
2. Insulation.
3. Service Temperature.

Equivalent Circuit of Magnetic Cores with Windings:



Typical B-H Curve (Hysteresis)

Eddy Current Loss: $W_{ec} \propto B^2 f^2 A_g^2$

Hysteresis Loss $W_H \propto B_{max}^n f$, $n = 1.6 \sim 2.0$.

Anomalous Loss W_a is determined by core geometry, H, and dB/dt.

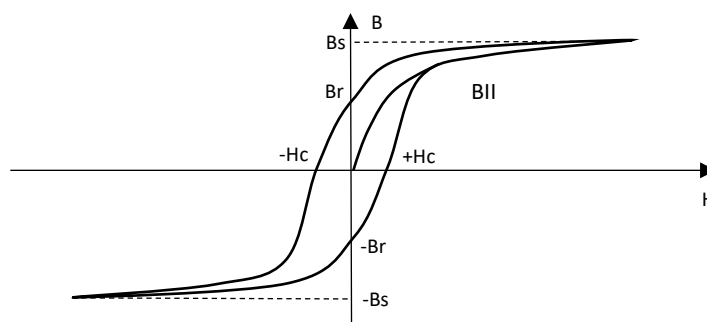
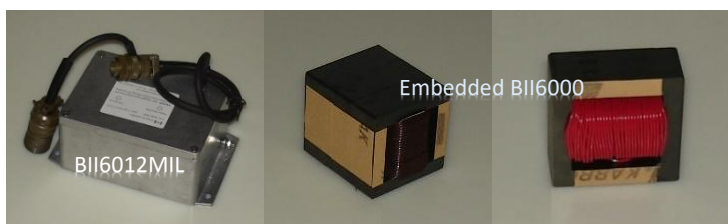
High-power magnetic material for uses from 30 Hz to 20 kHz:

Amplitude $\mu_a = 1300$, maximum $\mu_{max} = 7500$, $B_s = 1.4$ T at $H = 800$ A/m.

Air - gapped Core: Effective Permeability $\mu_e \approx \frac{\text{Effective Length}}{\text{Air Gap Length}} = \frac{L_e}{G}$

Energy stored in magnetic field $w_m = \frac{B^2 V_c}{2\mu_r \mu_0} = \frac{1}{2} L i^2$

Energy stored in shunt capacitance C_0 of piezoelectric transducer $w_c = \frac{1}{2} C_0 V^2$



Magnetic Properties of Typical Ferromagnetic Materials Used at BII, at 25°C unless Specified Otherwise.

Material	μ_{max}	μ_i	Saturation B_s (T)	Hysteresis/Core Loss W_H	Coercivity H_c (A/m)	Curie Temperature
Purified Iron	180,000	10,000	2.15	30 J/m ³ at d.c.	4	770 °C
Iron	5,000	150	2.15	500 J/m ³ at d.c.	80	770 °C
Electrical Steel	50,000	350	1.5	32 W/lbs at 1kHz, 1.5T, 100°C	48	740 °C
Powder Alloy	20~300	14~150	1.0~1.5	1.7 W/cm ³ at 0.2MHz, 0.1T, 100°C	20 ~ 180	500 °C
Soft Ferrite MnZn	4800	2200	0.49	0.215 W/cm ³ at 0.5MHz, 50mT, 100°C	21	210 ~ 300 °C
Soft Ferrite NiZn	2800	750	0.52	0.500 W/cm ³ at 3.0MHz, 30mT, 100°C	40	260 ~ 350 °C

Acoustic System Performance and Designs

Low-Power Acoustical Measurements

Generally, field-deployed, hand-held, and portable acoustical measuring instruments (Underwater Sound Recording, Sonobuoy, Bottom-moored Hydrophones, Autonomous Underwater Vehicles, NDT Tester, AE Material Health Monitors, etc.) use batteries (non-rechargeable or rechargeable) as DC power sources which requires low-power components to extend service time.

BII manufactures low current and low power preamplifier with built-in electronic filters.

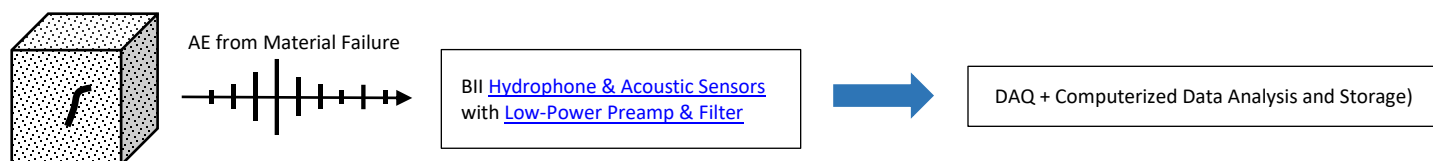
1. Hydrophones or acoustic sensors without signal conditioning components detect loud sounds.

In this case, the hydrophones or acoustic sensors does not need any electronic energy, or its power consumption is 0 watt.



2. Hydrophones or acoustic sensors with signal conditioning components detect weak sounds.

In this case, the hydrophones or acoustic sensors consume electronic energy to amplify the signals and reject noises.



3. Low-Power Preamplifier and Filters.

[BII1060 Series](#) Low Power Low Noise Preamplifiers for acoustical instruments: Hydrophones, Air Transducers, Acoustic Emission Sensors, and Ultrasonic Transducers (Non-destructive Testing). Uses in underwater sound (oceanography and hydrography), sonic and ultrasonic engineering, electroacoustics, communication, bioacoustics, exploration seismology and seismic wave, physical acoustics, acoustical Imaging, and measurements of ferroelectric and piezoelectric materials.

Features	
Low Supply Current: 40 μ A.	Single-ended and Differential Output.
Wide Supply Voltage: +2.7 to +30 VDC.	Driving Long Cable: 200 m.
Low Voltage and Current Noise: 7nV/VHz, 1.0fA/VHz, at 1 kHz.	Bespoke Gain 0 to 60 dB and Bespoke Built-in Filters.

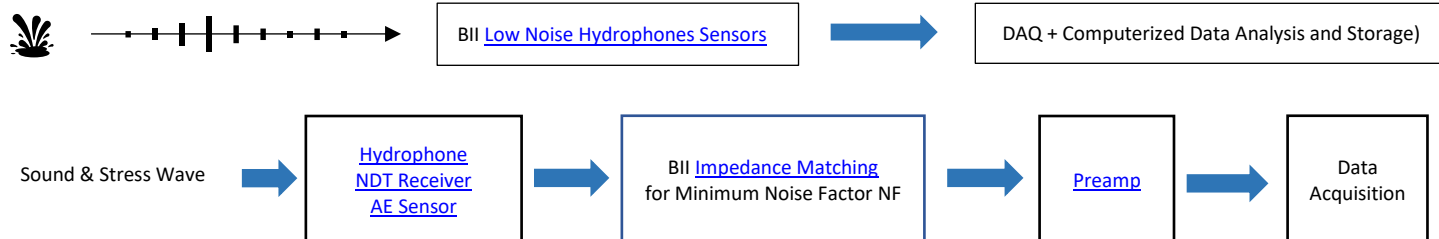
Please refer to [Choose a DC Power Supply for BII Devices](#) for more info on calculation of service time of a battery-operated system.

Low-Noise Acoustical Measurements

In acoustic measurement the ultimate limit of detectability of weak sounds is confined by noises (ambient noises and electrical noises) - unwanted signals that obscure the desired signals.

Roughly, both ocean ambient noises and electrical noises decrease when frequency increases. Besides, when bandwidth of a system is wider, more noises couple into the system to sum up to be higher noise level at its output. Therefore, a bandpass filter is necessary to reject noises outside the frequency range of the interest.

To avoid noise variation resulting from different signal conditioning circuit, hydrophones (acoustic sensors) with built-in preamp should be used in low-noise acoustical measurements.



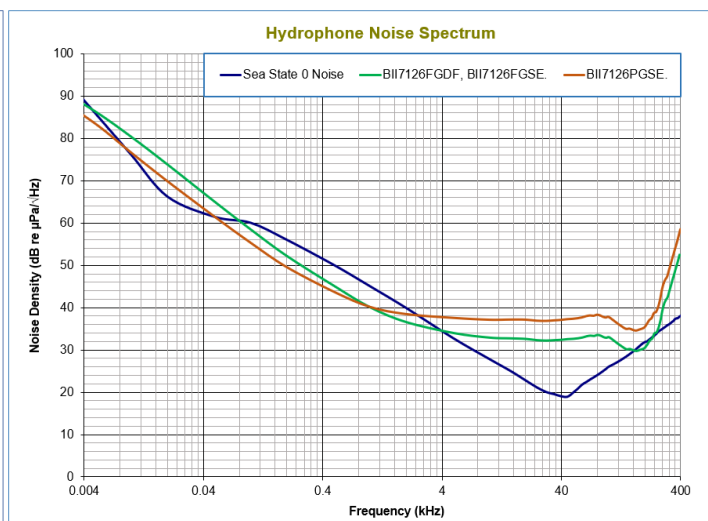
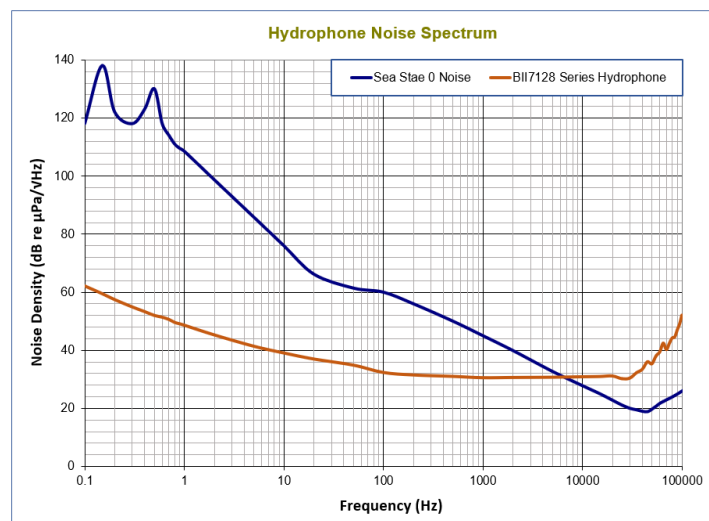
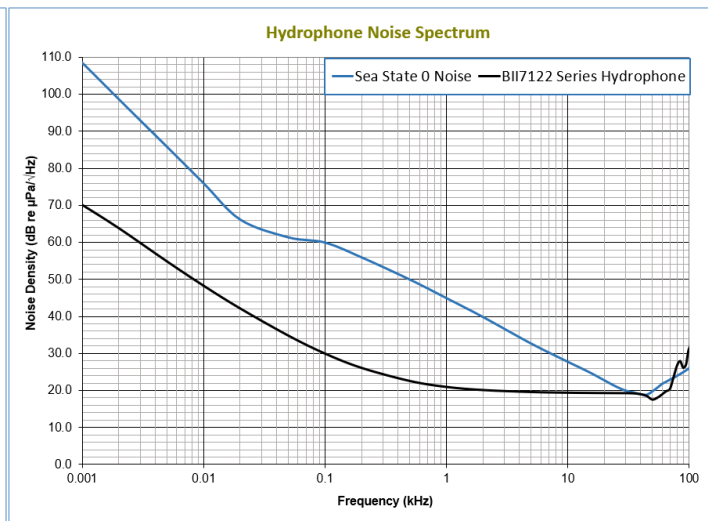
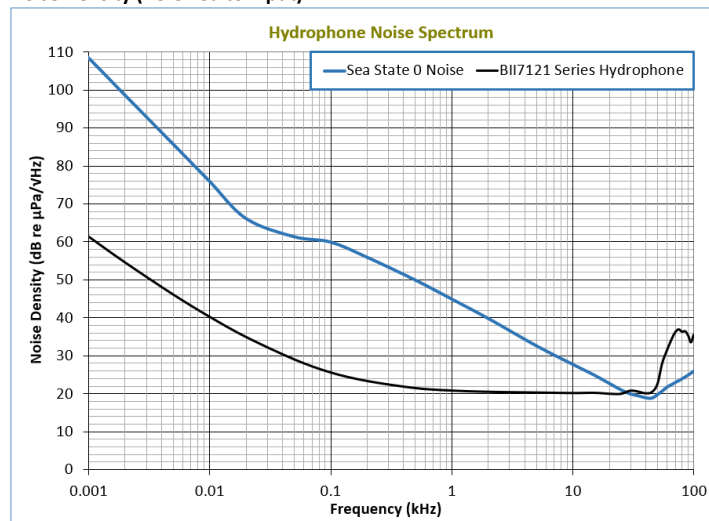
[BII7120 Series](#) Omnidirectional Low Noise Hydrophone and [BII7079](#) Directional Low Noise Hydrophone (Acoustic Sensor): Noise Level below Sea State Zero.

A low-noise acoustic sensor consists of an [acoustic transducer](#) and a [low-noise preamp and filter](#). these low noise sensors are widely applied in sonic and ultrasonic engineering, electroacoustics, communication, bioacoustics, exploration seismology and seismic wave, physical acoustics, acoustical Imaging, and material study.

Approximately, electronic noise density of a preamp hooking up with a transducer: refer to input, RTI, $V_n^2 = e_n^2 + [i_n * \text{impedance of the transducer (or hydrophone)}]^2$. e_n : voltage noise density of the preamp, i_n : current noise density of the preamp.

In high frequency range such as greater than 100 kHz, the pressure noise density of a acoustic sensor with [low-noise preamp](#) BII1051 ≈ 180 dB – FFVS of transducer, in dB $\mu\text{Pa}/\text{VHz}$. For example, a 1 MHz [NDT transducer](#) has FFVS of -197.6 dB V/ μPa . Working with BII1051, the pressure noise density referring to input ≈ 17.6 dB $\mu\text{Pa}/\text{VHz}$.

Noise Density (Referred to Input):



Programmable Sensitivity Hydrophone: 60dB Selection Range

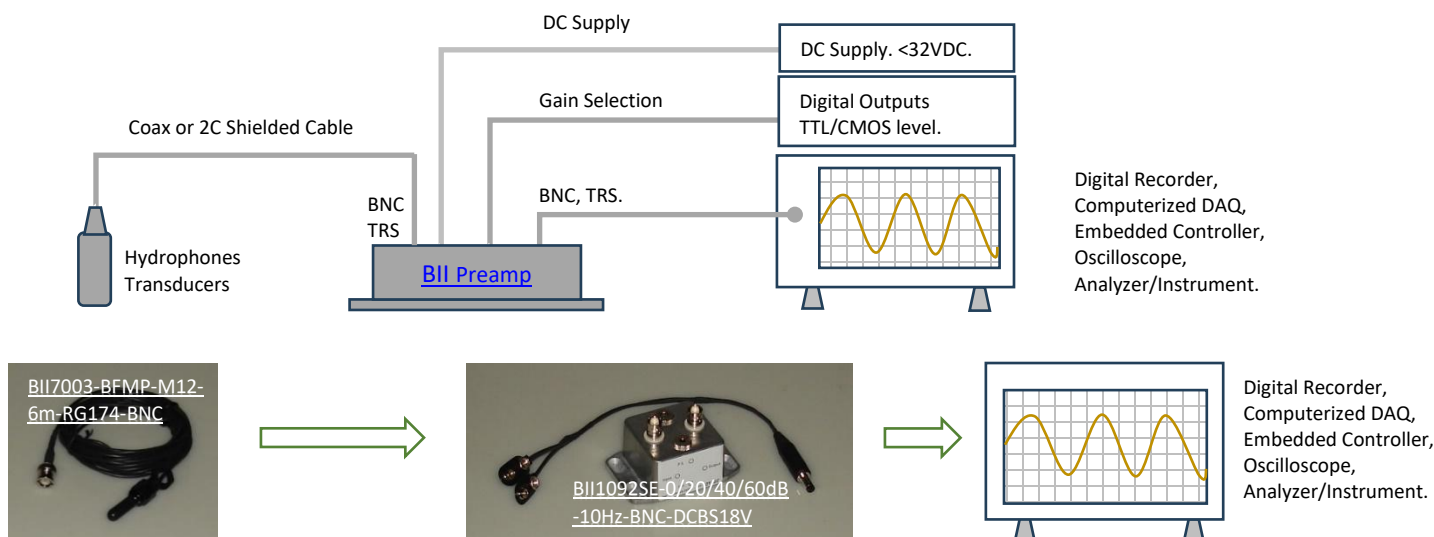
Sensitivity (FFVS) of a Programmable-sensitivity hydrophones (**PSH**) is digitally or manually selected with three lines (Two-bit Word, A0, A1 and Digital COMMON) or two lines (One-bit Word, A0 and Digital COMMON), or is continuously controlled by an analog voltage. The logic level of digital signal is CMOS/TTL-compatible and power supply rail-compatible. In applications of the active Sonar and sound-source tracking (marine animals, acoustic beacon, pingers, locator, acoustic tags...), the sound level at detection spot varies greater than 40dB (60 dB or higher) because of the variation of propagation loss, target strength, source level, etc. and the sound level could be too high or too weak for a sensitivity-fixed hydrophone. BII's programmable sensitivity hydrophone (**PSH**) is one of the solutions to solve the issue. BII provide various PSH with different frequency ranges (0.1Hz to 10MHz) and different directivity patterns (Omnidirectional, Toroidal, Hemispherical, Conical, Fan-shaped, etc.)

Nominal Sensitivities (FFVS) Variation: 0 ~ 60/70 dB, 0/30 dB, 0/40 dB, 10/40 dB, 10/50, 20/50 dB, 20/60 dB, 30/70 dB, 0/20/40/60 dB, 20/40/60/80 dB.

Related Preamps: [BII1090](#) PGA Amplifier and [BII1030](#) VGA Amplifier.

Applications: Compensate Propagation Loss & Target Strength; SONAR; NDT; Detection of Acoustic Tag/Pinger/Locator/Marine Mammals; Passive Acoustic Monitoring.

1. PSH: Standalone Hydrophones (AE Sensors, NDT Transducers) + Standalone PGA or VGA Preamp.



2. PSH: Piezoelectric Sensing Element + Built-in PGA and VGA amplifier.

- (1). Drive long cables up to 1000 m.
- (2). Refer to [BII7000](#), [BII7010](#), [BII7070](#), [BII7120](#), and [BII7180](#).

Automatic Sensitivity Hydrophone (ASH) for SONAR and NDT: 80dB Range

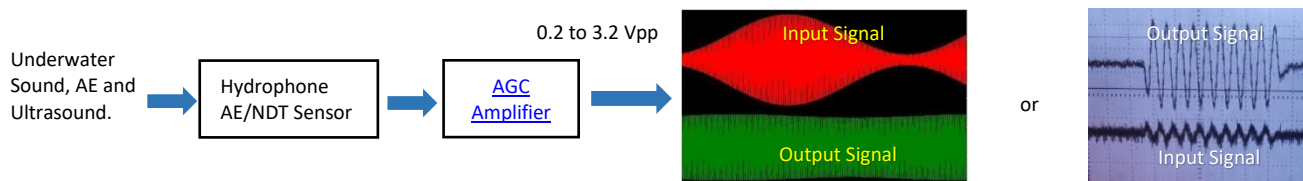
An automatic Sensitivity Hydrophones (AE Sensors, NDT Transducers) (**ASH**) maintains suitable (or constant) voltage amplitude at output despite the variations of the detected sound pressures by automatically reducing its sensitivity when the sound is strong and raising it when sound is weak. It has 80dB dynamic range to equalize the differences in received sound intensities emitted by different sound sources, amplitude variations in a single sound source due to fading or reduced power, as well as varying echo levels caused by sound propagation loss in SONAR and ultrasonic testing. Besides, it is useful to detect the information being carried on by signal frequencies such as Doppler sonar for speed measurement, ASK and FSK digital communication, etc. The built-in bandpass filter allows the receiver to detect the sound of the interest and reject the sounds out of its bandwidth.

Related Amplifiers: [BII2000](#) AGC Amplifier.

Typical Applications

Compensate Propagation Loss & Target Strength	Search of Acoustic Beacons: Tag, Pinger and Locator	Underwater Digital and Voice Communication
Passive Acoustic Monitoring (PAM System)	Doppler SONAR, Echo Sounding	Pulsing Ultrasonic Testing, Material Study.

1. ASH: Standalone Hydrophones (AE Sensors, NDT Transducers) + Standalone AGC Preamp.



Usable Frequency: 100 Hz to 1 MHz, **Input Dynamic Range:** 80 dB, **Output Voltage Level:** 0.2 to 3.2 Vpp, **Sound Level Range ΔP :** 80 dB μ Pa or $P_i = 110$ to 190 dB μ Pa.

Note: Minimum detection threshold is limited by noise levels in passband, which include electronic self-noise of the hydrophone and ambient sound noises if the ambient sound noises are not the measurand. The wider passband is, the higher the noise level is. **Pressure Noise (RTI) P_n :** $P_n \approx (-162 - \text{Sensitivity of Sensing Element})$ at $f \geq 1$ kHz, in dB μ Pa/VHz. RTI: Refer to Input. (Most sensitive element is used in [BII-7120 series](#) low noise hydrophone, and is around -182 dB V/ μ Pa). **Output Type:** Single Ended or Differential, **Output Level Error:** $< \pm 2.0$ dB, **Current (Quiescent):** 17 mA @+24 VDC, DC Supply Voltage: +16 to +32 VDC.

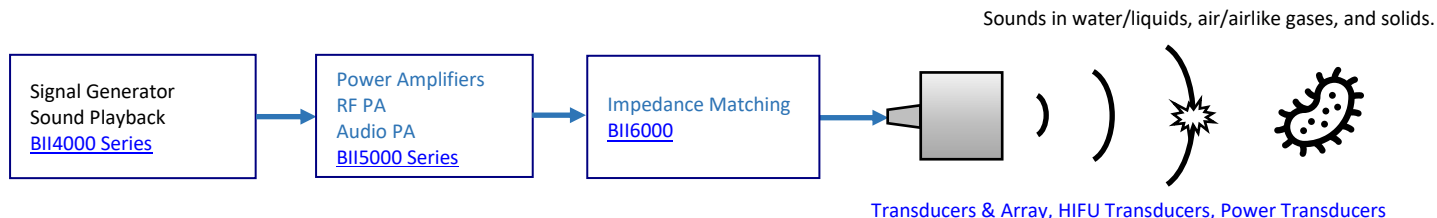
2. ASH: Piezoelectric Sensing Element + Built-in AGC Amplifier.

- (1). Drive long cables up to 1000 m.
- (2). Refer to [BII7000](#), [BII7010](#), [BII7070](#), [BII7120](#), and [BII7180](#).

Overall Conversion Efficiency from Electric Power to Sound Power

Generally, field-deployed, hand-held, and portable **acoustical analytical instruments** (Hand-held Diver SONAR, Bottom-Moored Beacons/Transponders, Autonomous Underwater Vehicles, Echosounder, Sound Telemetry, NDT Flaw Tester, Material Study, etc...) use batteries (non-rechargeable or rechargeable) or a gasoline/propane portable generator as power sources which requires low-power components to extend service time.

Sonic processing equipment produce high sound intensity to modify the load medium, such as heating, surface cleaning, cavitation, degassing, emulsification, coagulation, destruction of bacteria, production of chemical reaction, etc... Most of these equipment operate in laboratories and factories.



Overall Conversion Efficiency from Electric Power to Sound Power

Efficiency of a System except Signal Generator = PA Efficiency * Impedance-Matching Efficiency * Transducer Efficiency.

PA Efficiency

BII manufacture two types of power amplifiers to drive transducers to produce sounds: Class AB (Linear Mode) and Class D (Switching Mode).

Efficiency of Class AB Linear PA $\eta = 0.6$ to 0.7 .

Efficiency of Class D Switching Mode PA $\eta = 0.91$ to 0.985 .

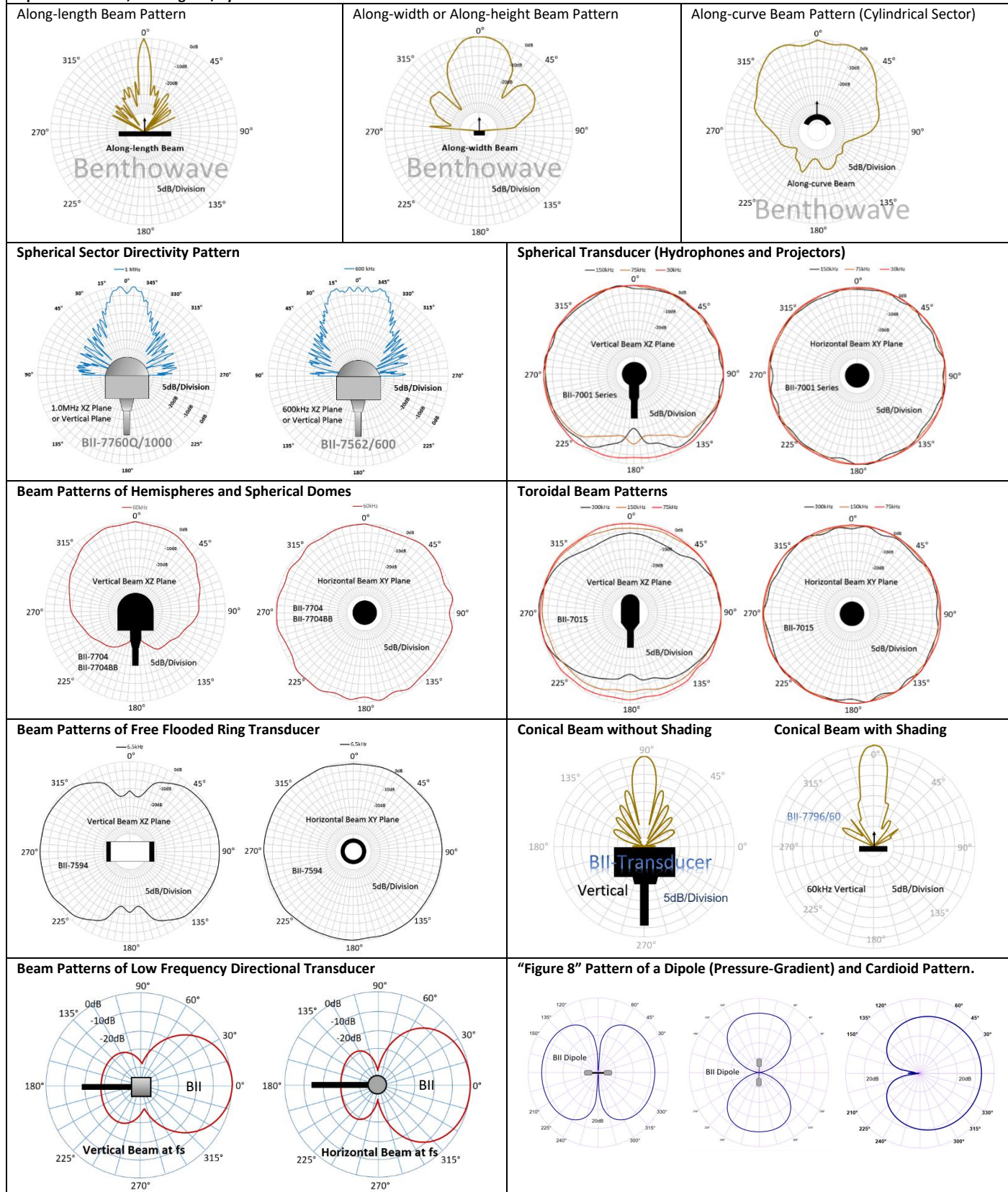
Impedance-Matching Efficiency $\eta = 0.91$ to 0.99 at $25\text{ }^{\circ}\text{C}$ ($77\text{ }^{\circ}\text{F}$). The efficiency increases with temperature increases till $100\text{ }^{\circ}\text{C}$ ($212\text{ }^{\circ}\text{F}$).

Transducer Efficiency $\eta = 0.1$ to 0.93 . Generally, cylindrical, and spherical transducers possess highest efficiency. Broadband NDT transducers possess lowest efficiency. Other kind of disk-shape planar transducers possess median efficiency around 0.3 to 0.6 depending on their bandwidth.

Directivity Patterns:

Following beam patterns are for illustration only. BII manufactures bespoke beam patterns for transducers (hydrophones and projectors) and arrays.

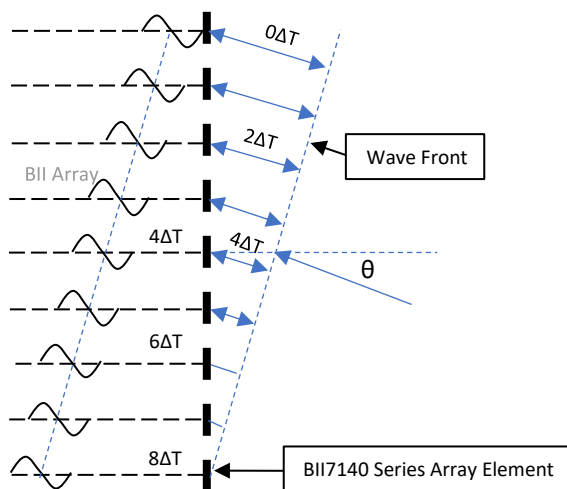
Apertures of Line, Rectangular, Cylindrical Sector and Curved Surface.



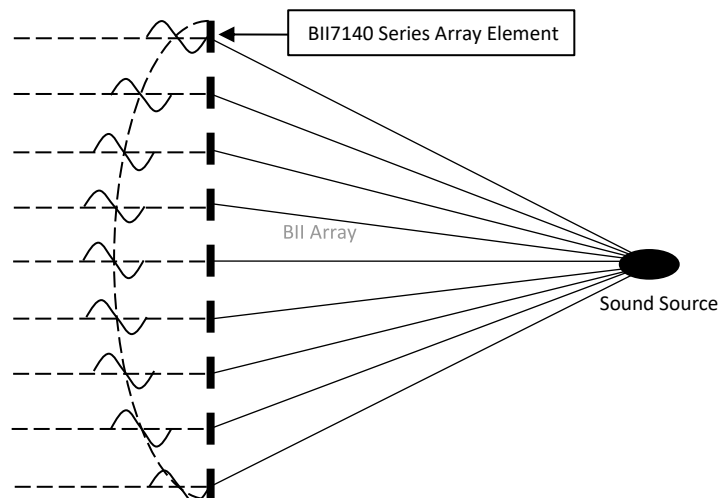


Transducer and Hydrophone Array & Aperture Design - Spatial Filtering

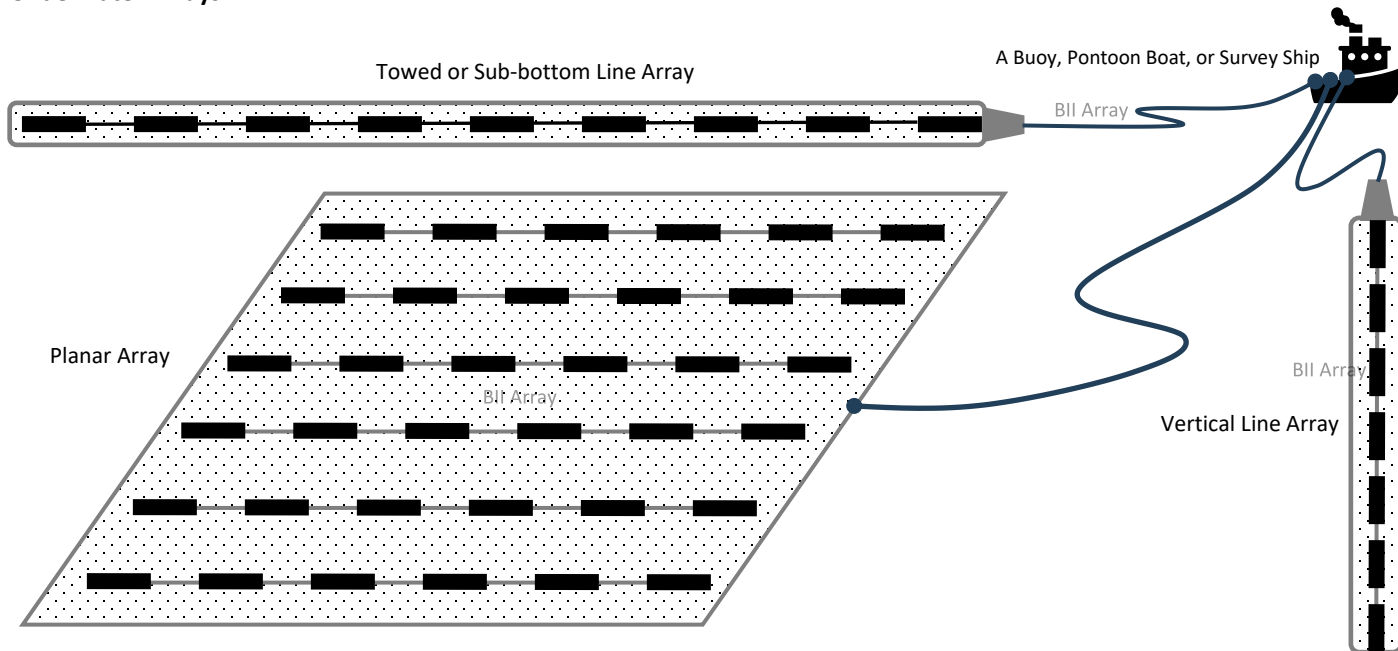
Linear (Rectangular) Array Beam Steering



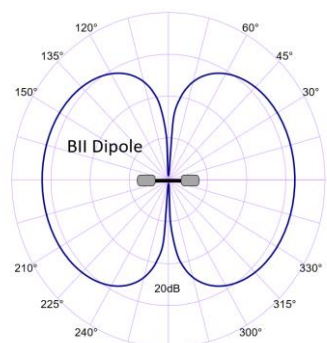
Linear, Annular, and Planar Array Beam Focusing



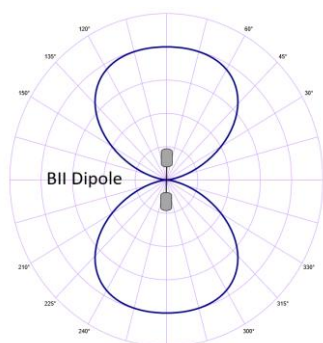
Underwater Arrays



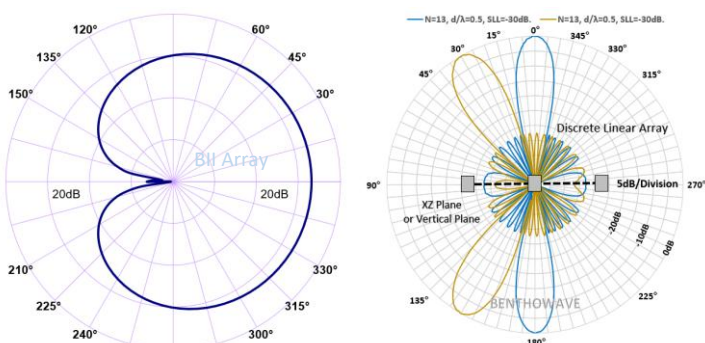
"Figure 8" Pattern of a Dipole (Pressure-Gradient).



Cardioid Pattern



Linear Array with BII Discrete Hydrophones



Benthowave's [transducers](#) and [hydrophones](#) can be used as array elements, which, depending on the operating frequency, can be simplified as Point, Line or Rectangle Aperture elements in Linear, Planar, Curved, and Conformal Array. Different mounting options are available from BII: Thru-hole/Hull, Bolt-fastening, End-face Mounting, Free Hanging, Flush Mounting and Flange Mounting. With the signals of array elements, tracking of sound source, beam-steering and sidelobe suppression can be done in signal processing. Standard dry and underwater mateable connectors are supported for quick interfacing among data acquisition devices. The performance of an array is determined by the performance of the elements, quantity of elements, spacing among elements and array pattern. Using an array of elements instead of single element underwater, the high array gain can be achieved or the signal-to-noise ratio (SNR) can be improved.

With space-time signal processing, the -3dB beam width of the main lobe of the array can be controlled with weighting/shading techniques, and the beam of the main lobe can be steered to a specific direction with time-delay techniques. 2D or 3D images underwater can be captured with the arrays and the echo signal processing.

BII Bolt-fastening Mount is designed to be installed on portable-mounting apparatus for easy deployment, and also make configuration of the array pattern be easy, flexible and simple. Different array pattern can be set up quickly and easily in the field. Thru-hole Mount is designed to be installed on the wall/hull of underwater submersibles, pipes, tanks and vessels to detect or produce acoustic signals outside a submersible or inside a pipe, tank or vessel. Hull mounted array (such as linear, planar and cylindrical array) can be configured.

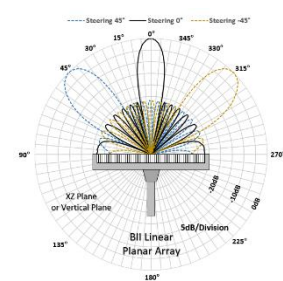
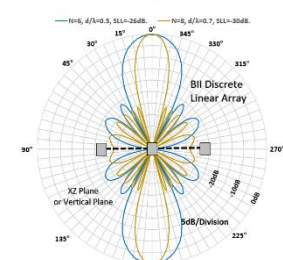
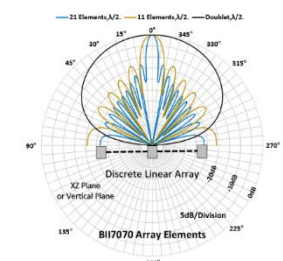
How to use BII Transducer/Hydrophone as discrete array elements? To achieve narrow beam width at a specific frequency and steer the beam to the direction of interest, multiple BII transducers or hydrophones shall be used to set up line array or planar array. For example, to detect 20kHz underwater sound, seven [BII7070](#), [BII7010](#) or [BII7140](#) series hydrophones are used in half-wavelength spacing linear array to achieve beamwidth of 18° and suppressed sidelobe level of -30dB comparing to main lobe. Preamplifiers can also be integrated to drive long cable. Similarly, using the same array pattern, seven BII low-frequency acoustic transducers (projectors) can generate underwater sounds with a beamwidth of 18° and suppressed sidelobe level of -30dB.

BII provide design service of Line (Linear) Array, Rectangular Planar Array, and Circular Planar Array: Side Lobe Level SLL, -3dB Beam Angle θ_{-3dB} ; Array Element Spacing d ; Array Element Number N , Beam Steering, FFVS or TVR loss, and mutual effects among these parameters.

Please contact us for more info about Array Design Services.

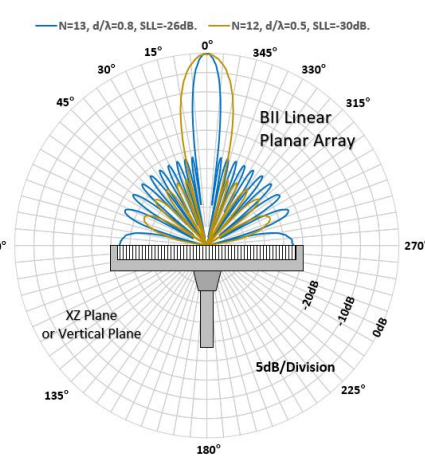
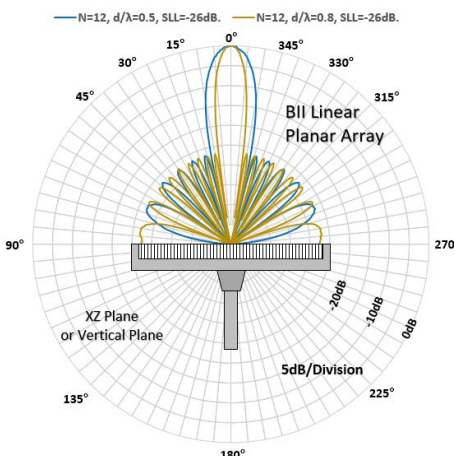
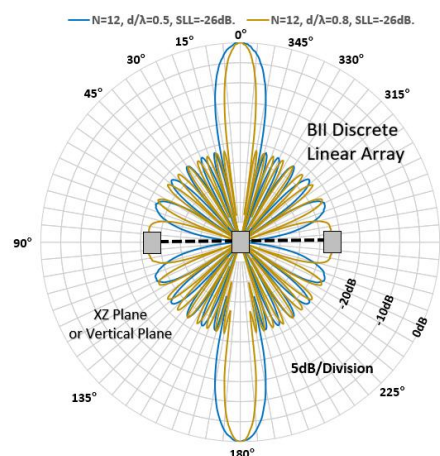
Related Products with Array Technology for Scanning SONAR: Imaging and long range acoustic positioning, tracking, echo locating, navigation, communication, and artificial acoustic target in horizontal or vertical plane in the ocean, rivers, and lakes.

- [BII7630](#) Series Phased Array Transducer: Beamforming, Dynamic Focusing.
- [BII7660](#) Series Multibeam SONAR Transducers: 2D Imaging Sonar
- [BII7710](#) Series Scanning SONAR Transducer: 360° View Field in Horizontal Plane
- [BII7770](#) Series Underwater Mechanical Scanning Transducer
- [BII7740](#) Series Annular Array: Amplitude Shading, Array Focusing & Spreading



Some Designs of Discrete Line Array and Linear Planar Array.

N : Quantity of Array Elements, d : Element Spacing, λ : Wavelength, θ_{-3dB} : Half Power Beamwidth, SLL: Sidelobe Level referring to Main Lobe.

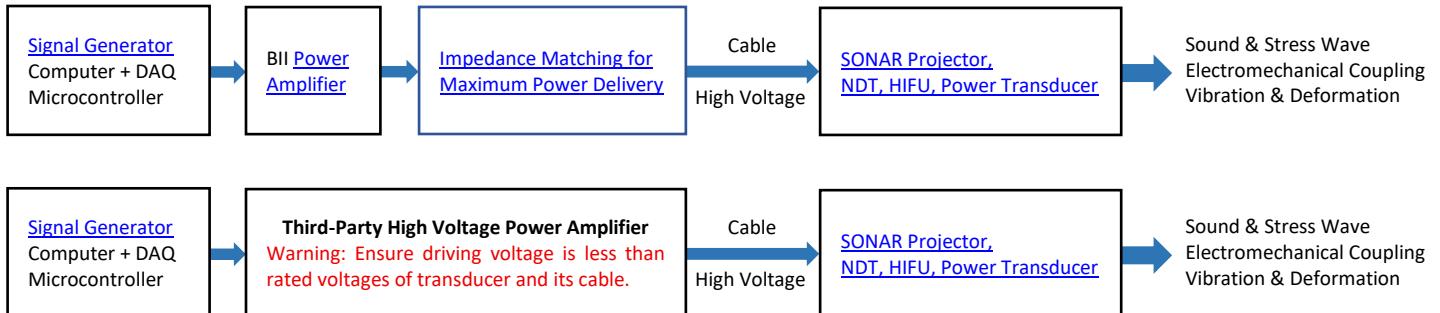


How to Drive Transducers (Projector)

Impedance Matching between Power Amplifiers and Transducers.

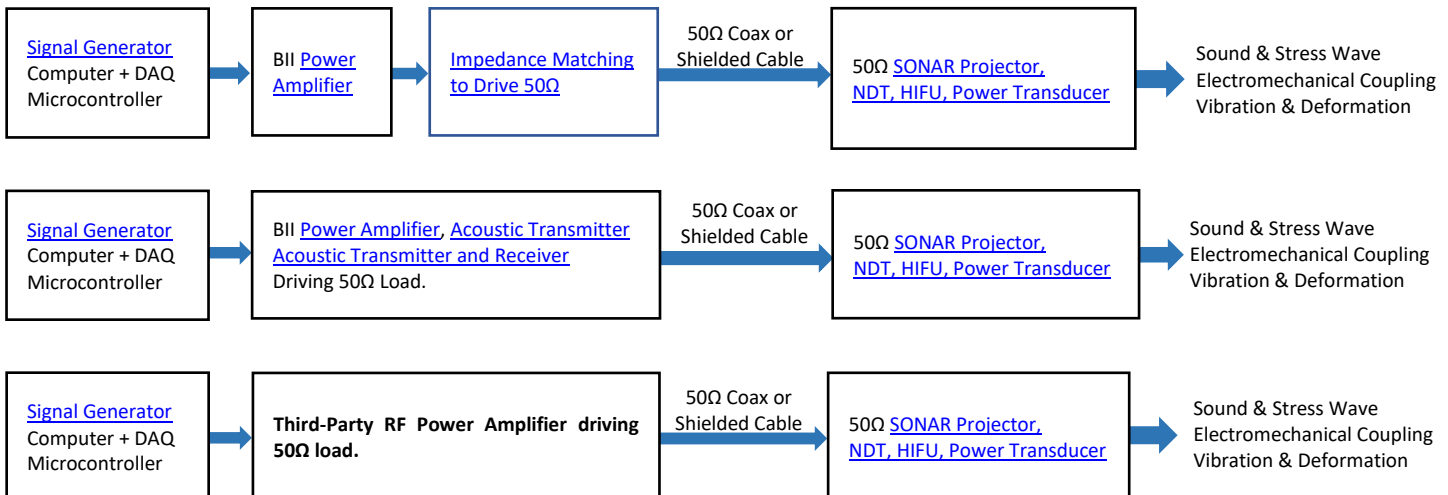
The cable has adverse effect on power transmission between the power amplifier and the transducer caused by cable's resistive, capacitive, and inductive performance in different frequency range. Transducer itself is resistive, capacitive, and inductive load at different frequency range. Therefore, an impedance matching unit is a necessary part between power amplifier and transducer for maximum power transmission and the best power factor.

(1). Transducers without Built-in Impedance Matching



- Driving voltage of transducer is limited by voltage ratings of both cables and transducers.
- Grounding are necessary for operating safety.
- Impedance mismatch exists between cable and transducer. The mismatch can be ignored in low frequency range when cable length is much less than electromagnetic wavelength.
- This setup is favorable to hull-mounted or wall-mounted transducers. The cables and/or wires are short among transducers and electronic driving circuits. In low frequency range, "Short Cable" means the adverse effects of its capacitance, resistance, and inductance on power transmission are little and can be ignored.
- Exchangeability among parts is poor. Each transducer needs a bespoke impedance matching device to match its impedance to the one of a power amplifier.

(2). 50Ω Impedance Matching



Advantage: The exchangeability among parts is excellent for easy maintenance, fast system recovery, lower cost, etc.

BII transducers with built-in impedance matching are available in the product numbers IM (impedance equipped with the default 50Ω) or -IM50Ω (50Ω impedance matching). 50Ω is the most commonly used impedance matching and is convenient for use with 50Ω coaxial cable.

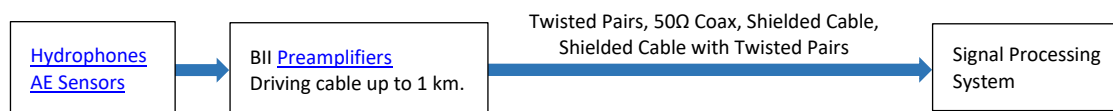
According to the customer's specific power amplifier load requirements, customers can also customize transducers with other resistance values, such as IM2Ω, IM3Ω, IM5Ω, IM8Ω, IM16Ω, IM75Ω, IM100Ω, etc. When the cable characteristic impedance value, transducer impedance value and power amplifier load requirements are different, short cables must be used between the power amplifier and transducer to prevent the adverse effects of the cable on power transmission.



How to Choose Components for Long Cable Deployment

Much information carried by underwater sounds, ultrasounds, vibrations, stress and shock waves is far from the sites where operators or recording and signal processing systems stand, or, the subjects under investigation or test are dangerous to human being or far from the central operating systems. These field applications are characterized by remote signal (or energy) detection, monitoring, transmitting, and control. As the distance increases between a transducer (hydrophone and projector) and its signal process system, a special set of instrumentation problems is encountered such as EMI noise pickup, signal attenuation over cable, cable cost, voltage potential differences among "grounds", etc.

Sound Detection



BII preamps provide flexible and bespoke cable driving solutions with current, single-ended voltage, or differential voltage to drive Twisted Pairs, 50Ω Coax, Shielded Cable, Shielded Cable with Twisted Pairs.

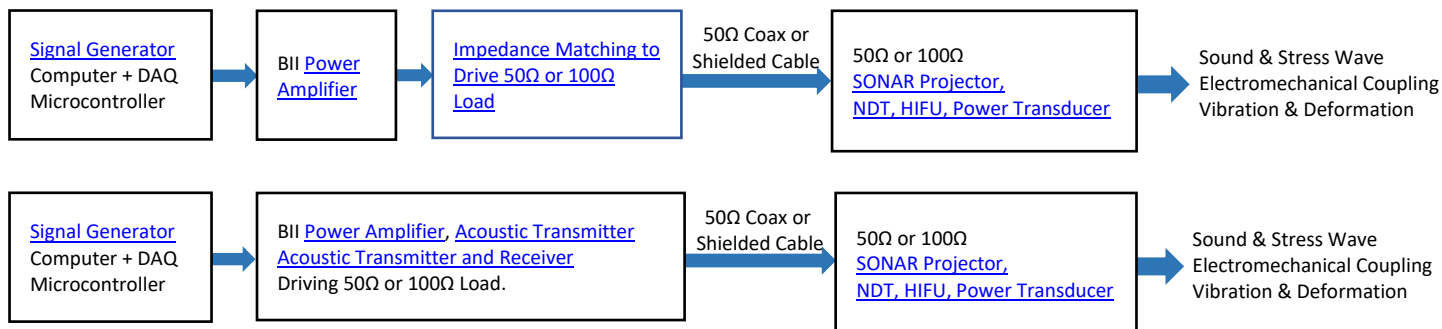
A Twisted Pair is best for lowest cost wiring solutions in low frequency range. Signals in current form is immune to voltage noise pickup, so shielding is not needed. The conductor resistance does not reduce signal current, so small gauge wire can be used. Buyer may make twisted pairs with single wire for special applications such as towed array wirings inside a streamer housing or conduit. [BII differential preamp](#) drives a twisted pair.

A Shielded Cable with Twisted Pairs is best for EMI rejection for low frequency voltage or current signal transmission over long cable, but generally it is much more expensive than other cables. BII stocks shielded cable with a twisted pair and PUR or PVC jacket.

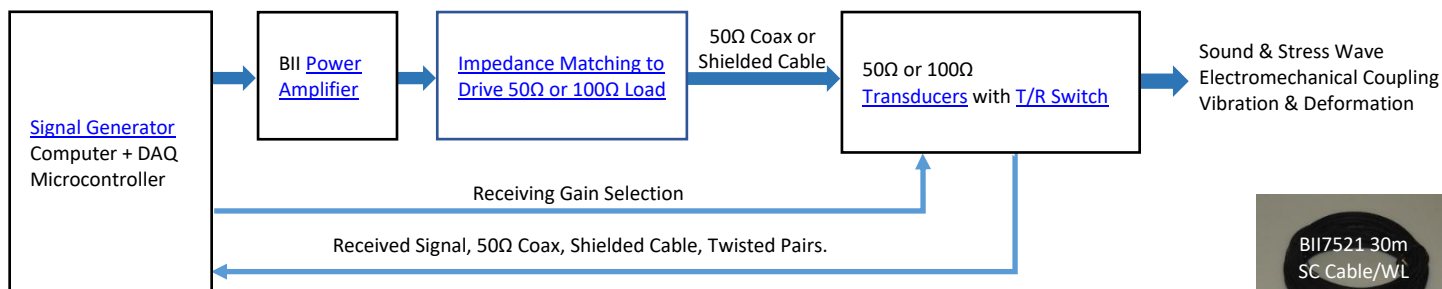
A Shielded Cable with Multiple Conductors is for single-ended and differential voltage signal transmission. Differential signal is superior to single ended signal with cancellation of common-mode noise voltage from magnetic induction, capacitive coupling, and ground potential differences or loops, etc. [BII preamps](#) drive shielded cable up to 1 km.

A Coax Cable such as RG58, RG174, RG178, or RG316 is for high frequency signal transmission. The coax shield provides adequate rejection to high frequency EMI. [BII1040](#) and [BII1090](#) series preamps achieve 20 MHz bandwidth.

Sound Radiation



Sound Radiation and Detection: SONAR and NDT



1. The shielded cable or coax provide means for grounding which is critical to operating safety.
2. EMI generated by conductors is reduced by grounded shielding.
3. Magnetic flux generated by a twisted pair will cancel each other.
4. Maximum transmitting power is limited by voltage and current ratings of the cables besides the power rating of the power amplifier.
5. In low frequency range such as 0.1 to 10 kHz, the high-power impedance matching unit is bulky and may not be suitable to be integrated into transducer housing.

How to determine pulse width, duty cycle and off-time with input pulse power (peak power)

This document is exclusively for choosing BII transducers and related BII products.

Refer to [Typical Signals for Active SONAR, Ultrasound, HIFU \(High Intensity Focused Ultrasound\) and Communication](#) for more information.

f_s: Series (Motional) Resonance Frequency. Generally, TVR and efficiency η of a transducer reach maximum values at f_s and the transducer (projector) operates at f_s to produce sound. **SL**: Source Level, in dB $\mu\text{Pa}\cdot\text{m}$; **T**: Temperature of the loading medium such as sea water, in °C, $[(^{\circ}\text{C} \times 9/5) + 32 = ^{\circ}\text{F}]$; **TVR**: Transmitting Voltage Response, in dB $\mu\text{Pa}\cdot\text{m}/\text{V}$; **G_p**: Equivalent Parallel Conductance of the Transducer, in Siemens S (or Ω^{-1}). $G_p = G_{\text{max}}$ at f_s . **R_{TL}**: normalized Load Resistance of the transducer such as a 50 Ω transducer which, generally, has a built-in impedance matching network.

IPP: Input pulse power (peak power) driving a transducer, in RMS Watt; **MIPP**: Maximum Input Pulse Power of a transducer at f_s , in RMS Watt; **MCIP**: Maximum Continuous Input Power at f_s , in RMS Watt; **MPW**: Maximum Pulse Width at MIPP and at f_s , in Second; **PW**: Pulse Width (or Pulse Duration), in Second; **D**: Duty Cycle.

Note: **MIPP**, **MCIP** and **MPW** are maximum values allowable to drive a transducer without over-heating and over-voltage.

1. Determine SL or RMS drive voltage V_{rms} required by the application and obtain the temperature T of the loading medium in field.

2. Calculate IPP (input pulse power):

$$10 * \log \text{IPP} = \text{SL} + 10 * \log G_p - \text{TVR}, \quad \text{IPP} = V_{\text{rms}}^2 * G_p, \quad \text{or}, \quad \text{IPP} = \frac{V_{\text{rms}}^2}{R_{\text{TL}}}$$

If G_p and TVR of a transducer are not published online, please contact BII for the information.

If Impedance of the transducer (Z , θ , or R , X) are published, G_p can be deduced from impedance.

$$G_p = \frac{\cos \theta}{Z} = \frac{R}{R^2 + X^2}$$

Generally, the voltage and current rating of a transducer cable is 600 V_{rms} and 4 A or 10 A.

If the driving voltage required to achieve the SL is greater than 600 V_{rms} or the voltage rating of the transducer cable, a built-in impedance matching network should be used to boost driving voltage for achieving greater output acoustic power (or SL).

3. Choose suitable transducers with IPP and MIPP at f_s .

Transducers with MIPP greater than IPP are suitable for the application. **Warning: if IPP > MIPP at f_s , the transducer would be damaged beyond repair with over-heat or over-voltage.**

For example, an application needs IPP of 500 W to achieve required sound level, and we choose a BII transducer with MIPP of 1000 Watt, MPW of 1 Second, and MCIP of 10 Watt, at f_s . Loading medium is sea water, and its temperature is 17°C.

$$4. \text{PW} \leq \frac{\text{MIPP} * \text{MPW} * \left(\frac{120^{\circ}\text{C} - T}{103^{\circ}\text{C}} \right)}{\text{IPP}}$$

$\text{PW} \leq 1000 * 1 * 1 / 500 = 2$ (Seconds). That is, pulse width (duration) of 500W driving signal at 17°C sea water must be less than 2 seconds.

$$5. D \leq \frac{\text{MCIP} * \left(\frac{120^{\circ}\text{C} - T}{103^{\circ}\text{C}} \right)}{\text{IPP}}$$

$D \leq 10 * 1 / 500 = 2\%$. That is, duty cycle of 500W driving signal at 17°C sea water must be less than 2%.

6. Off-time $\geq$ Pulse Width * (1-D)/D.

Off-time $\geq$ 98 (Seconds) when PW=2 seconds.

Off-time $\geq$ 49 (Seconds) when PW=1 second.

Off-time $\geq$ 4.9 (Seconds) when PW=100 ms.

Off-time $\geq$ 490 (ms) when PW=10 ms.

Off-time $\geq$ 49 (ms) when PW=1 ms.

etc ...

After driving the transducer for 2s at 500W, the transducer must be off and allow it to cool down in the 17°C water for 98 seconds at least before driving it again.

...

After driving the transducer for 1ms at 500W, the transducer must be off and allow it to cool down in the 17°C water for 49 milliseconds at least before driving it again.

7. If calculated PW and D do not meet the requirements of the application, please choose another transducer and re-evaluate it by repeating step 2 to step 6 listed above; or, consider using an array. **For a planar array of N element parallel-mounted transducers, the sound level of the array will increase by $20 * \log N$ and the narrower beam width will be achieved by comparing the sound level and beam width of the single element.**

8. After a suitable transducer is chosen, and IPP, PW and D are checked out completely, it is time to choose [power amplifier](#) and [impedance matching network](#).

a. Choose the power amplifier which can be used with the IPP, PW and D.

Note: power amplifiers with maximum power rating less than transducers' MIPP are recommended to use. This can reduce the possibility of damaging transducers with over-power (or over-voltage) by accident.

b. Find out the load requirements from the specification of the power amplifier.

Most RF power amplifiers drive 50 Ω load, and most audio power amplifiers drive 2 Ω to 32 Ω .

BII power amplifiers drive 2 Ω to 32 Ω from 20Hz to 3MHz, and BII's bespoke power amplifier drive 50 Ω up to 10MHz.

c. BII's BII6000 series provide impedance matching between transducers and power amplifiers at specified operating frequency (generally, at f_s). There are two packages available: an impedance matching unit is built inside transducer housing or is manufactured inside a separate housing as a standalone unit.

9. Contact BII about your request on transducers and impedance matching units.

Useful Acoustic Formulae of Transducers (Projectors and Hydrophones)**Unit Conversion**

Free-field Voltage Sensitivity FFVS: $-180 \text{ dB V}/\mu\text{Pa} = 10^{-180/20} \text{ V}/\mu\text{Pa} = 10^{-9} \text{ V}/\mu\text{Pa} = 10^{-3} \text{ V}/\text{Pa} = 1 \text{ mV}/\text{Pa} = 10^{-3.6} \text{ V}/\mu\text{Pa} = 20 \cdot \log_{10}(10^{-9}) \text{ dB V}/\mu\text{Pa} = -180 \text{ dB V}/\mu\text{Pa}$.

Transmitting Voltage Response TVR: $180 \text{ dB } \mu\text{Pa}/\text{V at } 1\text{m} = 10^{180/20} \mu\text{Pa}/\text{V@1m} = 10^9 \mu\text{Pa}/\text{V@1m} = 10^3 \text{ Pa}/\text{V@1m} = 20 \cdot \log_{10} 10^{3+6} \text{ dB } \mu\text{Pa}/\text{V@1m} = 180 \text{ dB } \mu\text{Pa}/\text{V at } 1\text{m}$.

Impedance Z and Admittance Y.

$$Z = R + jX = \frac{G}{G^2 + B^2} - \frac{jB}{G^2 + B^2}; \quad Y = G + jB = \frac{R}{R^2 + X^2} + \frac{jX}{R^2 + X^2}; \quad |Z| = \frac{1}{|Y|} = \sqrt{R^2 + X^2} = \frac{1}{\sqrt{G^2 + B^2}}; \quad \text{Impedance Phase } \theta = \tan^{-1} \frac{X}{R} = -\tan^{-1} \frac{B}{G}.$$

R: Resistance, X: Reactance, G: Conductance, B: Susceptance.

Bandwidth or Quality Factor Q_e and Q_m .

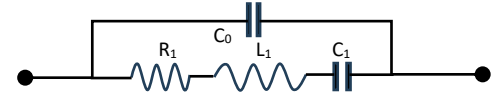
Untuned Transducer: Electrical Q_e at $f_s = \omega_s C_0 R_1 = B_0/G_{\max}$; Mechanical Q_m at $f_s = 1/(\omega_s C_1 R_1) = G_{\max}/B_1$.

Tuned Transducer (C_0 is cancelled out by Inductance) -3dB Bandwidth at $f_s = f_s/Q_m = f_s \cdot B_1/G_{\max}$.

Parallel Resonance Frequency $f_p = f_s \sqrt{1 + C_1/C_0}$, which **maximum FFVS** is achieved at.

f_s : **Mechanical Resonance Frequency or Series resonance frequency which maximum TVR is achieved at.**

The info above-mentioned is useful to choose transducers (hydrophone, projector, and T/R Transducers.).



Equivalent Circuit of BII Transducer around Resonance

Parameters of a Broadband (Low Q) Transducer (300kHz, $\Phi 18\text{mm}$ Planar Aperture): $C_0 = 0.568 \text{ nF}$, $C_1 = 0.517 \text{ nF}$, $L_1 = 0.611 \text{ mH}$, R_1 (Material Loss and Radiation Resistance) = 260Ω .

Useful Acoustic Formulae:

Directivity, Main Lobe Width:	Linear Transducer: $\theta_{-3\text{dB}} (^\circ) = 50.8 \cdot \lambda/L$ at $L > \lambda$, λ : Wavelength, L: Length. Disk Transducer: $\theta_{-3\text{dB}} (^\circ) = 58.9 \cdot \lambda/D$ at $D > \lambda$, λ : Wavelength, D: Disk Diameter.		
Transducer SL (Source Level):	SL = TVR + $20 \cdot \log(V_{\text{rms}})$, dB re μPa @ 1m. TVR: Transmitting Voltage Response. V_{rms} : (Root Mean Square) Driving Voltage of Transducers.		
Transducer SL (Source Level):	SL = $10 \cdot \log(P_{\text{in}}) + 10 \cdot \log(\eta) + \text{DI} + 170.8$, dB re μPa @1m. P_{in} : Input Electrical Power. η : Electroacoustic Efficiency. DI: Directivity Index.		
Input Power:	$P_{\text{in}} = V_{\text{rms}}^2 \cdot G$. G: Parallel Conductance of Transducer.		
Distance d of Acoustic Far Field of a Transducer:	Planar Transducer: $d \geq \text{Radiation Area}/\lambda$. Line (linear) or Thin Cylinder: $d \geq (\text{Length} \cdot \text{Length})/\lambda$ and $d \geq \text{Length}$.		
Hydrophone/Acoustic Sensor V_{rms}:	$V_{\text{rms}} = \text{SPL} + \text{FFVS}$, dBV. SPL: Sound Pressure Level (dB re μPa). FFVS: Free Field Voltage Sensitivity (dB re $\text{V}/\mu\text{Pa}$). dBV = $20 \cdot \log(\text{Voltage})$.		
Transducer as a High Pass Filter:	A transducer or hydrophone (no preamp) with a parallel resistor is a high pass filter: $f_{-3\text{dB}} = 1/(2\pi R C_h)$. R : Input Resistance or Impedance of a Preamp or A/D Converter. C_h : Capacitance of a Transducer or Hydrophone at 1 kHz or f_s .		
Signal Loss Over Extension Cable or a Capacitor:	$20 \cdot \log[C_h/(C_h + C_c)]$, dB. C_h : Hydrophone Capacitance. C_c : Capacitance of Extension Cables or Capacitors in Parallel.		
Underwater Sound Transmission Loss:	Spherical Spreading: TL = $20 \cdot \log R$. R – Range, m. Absorption Coefficient ($\leq 50 \text{ kHz}$): $\alpha \text{ (dB/km)} = 1.0936 [0.1 \cdot f^2/(1 + f^2) + 40 \cdot f^2/(4100 + f^2)]$. f: frequency (kHz).		
Attenuation/Absorption Coefficient of Sediments:	around 0.06f to 0.6f (dB/(m*kHz)).		
Electrical Power Loss over Cable:	Power Loss over cable = $2 \cdot I_{\text{rms}}^2 \cdot R \cdot L$. I_{rms} : RMS Cable Current. R: Nominal Conductor Resistance, about 0.035 Ω /meter. L: Cable Length (meter).		
Signal Detection:	$C = B \cdot \log(1 + \text{SNR})/\log 2$. SNR = SL – TL – NL – DI, or SNR = SL – 2TL + TS – NL + AG. C: Channel Capacity. B: Channel Bandwidth. SNR: Signal-to-Noise Ratio.		
Constants:	Free Dielectric $\epsilon_0 = 8.842 \times 10^{-12} \text{ C/mV}$. Sound Speed in Sea Water: $C = 1449.2 + 4.6T - 0.055T^2 + 0.00029T^3 + (1.34 - 0.010T) \cdot (S - 35) + 0.012z$ (m/s). T: Temperature, $^\circ\text{C}$. S: Salinity, parts per thousand of dissolved weight of salts. z: Depth, meter. Sound Speed in Fresh Water: $C = 1481 \text{ m/s}$. Density of Sea Water: 1026 kg/m^3 . Fresh Water: $\rho = 998 \text{ kg/m}^3$.		
Ocean Sound Sources	Frequency Range	Ocean Sound Sources	Frequency Range
Vessel Traffic:	5 to 500 Hz	Drill Ship:	10 to 20,000 Hz
Air-gun (Seismic Reflection Profiling):	5 to 1,000 Hz	Ice:	10 to 1,000 Hz
Earthquake:	Up to 100 Hz	Ocean Surfaces:	1 to 50,000 Hz
Biologic (Sea Animal) Sounds:	Several Hz to 150 kHz	Turbulent Pressure Fluctuation:	1 to 100 Hz

Absorption of Sound in Air at 20°C (68°F), Relative Humidity: 10%, 1 atm.												
Frequency (kHz)	30	40	50	70	100	120	150	200	250	300	500	1000
Absorption (dB/m)	0.3	0.4	0.5	0.7	1.8	2.5	4.0	6.5	10	16	43	200

Planar Transducer:	33 kHz	50 kHz	70 kHz	120 kHz	200 kHz	300 kHz	420 kHz	600 kHz
Cavitation Threshold: at Sea Surface, (dB μPa)	203	206	209	214	214	217.5	220	223

Testing and Calibration

Underwater Electroacoustic Measurements

FFVS (Free-field Voltage Response) or **RVR** (Receiving Voltage Response), **TVR** (Transmitting Voltage Response), **Directivity Pattern** (Beam Pattern).

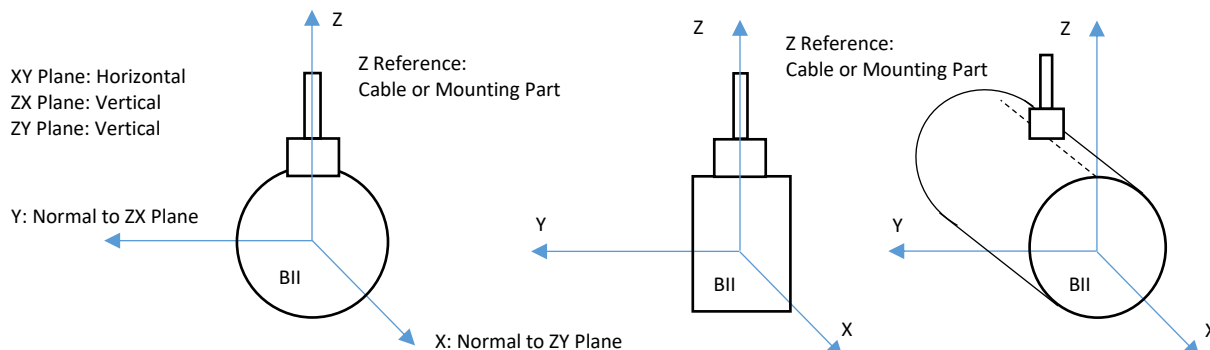
Impedance Measurement: Z- θ , R-X, G-B, Equivalent Circuit Model.

Power: Efficiency, Power Factor, Quality Factor (Bandwidth).

Environmental Testing: Pressure, Thermal (Temperature), Electric Field (or Voltage Rating).

Calibration Methods: BII uses primary methods (or reciprocal calibration) and secondary method (or comparison calibration) from 26 Hz to 10 MHz.

Hydrophone and Transducer (Projector) Orientation



BII's TVR and FFVS Testing:

Spherical and cylindrical hydrophones/transducers: in XY plane

Hemispherical hydrophones/transducers: on acoustic axis normal to the hemispherical face.

Curved/sector hydrophones/transducers: on acoustic axis normal to the curved face.

Circular planar hydrophones/transducers: on acoustic axis normal to the receiving or transmitting face.

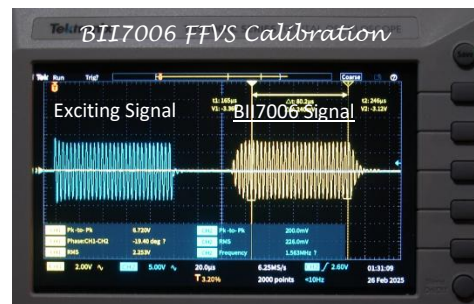
Rectangular planar hydrophones/transducers: on acoustic axis normal to the receiving or transmitting face.

Note: Acoustic axis is perpendicular to the transmitting/receiving face and passes through the geometry center of the face.

Signal Level is too weak:

If the signal level generated from transducers or hydrophones is much less than the expected. The major reasons are followings:

1. Spot of signal detection is NOT on the acoustic axis of maximum response. Make sure the detection spot is on acoustical axis of directional transducers or hydrophones.
2. Standing waves or echo interferences in small body of water. Use larger body of water to reduce reflection from boundaries.



Tolerance and Accuracy of Electroacoustic Measurement

Ambient Temperature:	23 $\pm$ 2 $^{\circ}$ C		
Transducer Parameters	Tolerance ¹	Accuracy	Note
Resonant Frequency in water:	Nominal $\pm$ 10%	7 ppm $\pm$ 1 mHz (at 5 to 40 $^{\circ}$ C, typical) ²	Loaded with Water
Capacitance in water:	Nominal $\pm$ 10%	$\pm$ 0.08% ($\pm$ 0.045% typical) ²	Test with Low Electrical Field at 1 kHz
Impedance/Admittance in water:	Nominal $\pm$ 10%	$\pm$ 0.08% ($\pm$ 0.045% typical) ²	G-B, Z- θ , or R-X
-3 dB Beam width ($^{\circ}$):	Nominal $\pm$ 2 $^{\circ}$	$\pm$ 0.3 $^{\circ}$ ($\pm$ 0.1 $^{\circ}$ to $\pm$ 0.2 $^{\circ}$ typical) ²	-3 dB Angle of Main Lobe
TVR (dB μPa/V@1m):	Nominal $\pm$ 3 dB	$\pm$ 1.0 dB ($\pm$ 0.5 dB typical) ³	Transmitting Voltage Response
FFVS or RVR (dB V/μPa):	Nominal $\pm$ 3 dB	$\pm$ 1.0 dB ($\pm$ 0.5 dB typical) ³	Receiving Sensitivity

Note:

1. The tolerance is the variation from the nominal value stated in datasheet and is caused by repeatability of manufacturing processes and parameter variations of raw materials from batch to batch.
2. These values of accuracies are from third-party's measuring instrument.
3. Theoretical calculation from specifications of third-parties' measuring instruments of Voltage, Frequency, Time, Impedance and Dimension. Handbook values of densities are used. Sound velocities measured at BII are used.

How Often Should the Transducer (Projector and Hydrophone) be Recalibrated?

If a major disturbance (a field parameter exceeds the maximum value stated in datasheet) such as a high temperature, high driving voltage or a high pressure is applied to a transducer, the transducer needs to be recalibrated.

Piezoelectric materials used in BII standard transducers (Projector and Hydrophone) have average ageing time of 1 to 3 years. BII carefully manages the material inventory to allow enough ageing time to stabilize piezoelectric materials. FFVS variation of a BII reference hydrophone used in comparison calibration Δ FFVS $\leq$ $\pm$ 0.2 dB at 1kHz in five years. BII does not have suggestions on recalibration of these standard/reference transducers. The recalibration shall reference to the metrology policies or regulations of end user's country.

Average ageing time of customized new transducers is one month. It is suggested to recalibrate these transducers in field after one-year field use.

Does BII Provide Calibration Service?

Yes, BII provides calibration services to BII transducers with reasonable fees and different calibration packages to suit end users' different requirements.

Certified Underwater Sound Measurements

The Institute for National Measurement Standards is a country's national metrology institute responsible for evaluation of standards and methods of measurement. Its Calibration Laboratory Assessment Service provides quality system, technical assessment services and certification of specific measurement capabilities of calibration laboratories.

BII's Underwater Electroacoustic Measurement Laboratory is NOT certified by National Metrology Institute in Canada. If a buyer needs measurement certified by National Metrology Institute, the buyer should have BII's products calibrated by certified calibration laboratories at buyer's cost in buyer's country. The national acoustic and naval laboratories in most countries provide certified underwater sound calibration services.

Measurements of Ferroelectric and Piezoelectric Materials

Ferroelectric and Piezoelectric Material Test:

D-E or P-E Hysteresis Loop. Dielectric Constant and Loss in low and high filed (Electric Field, Stress and Temperature). Piezoelectric Constants: d and g constants.

The Equations of state of piezoelectric effect: $D = dT + \epsilon^T E$ and $S = s^E T + dE$

Thermodynamics of Ferroelectric/Piezoelectric Medium: Gibbs Functions $G = U - S^*T - E^*D - H^*B - \sigma^*\theta$.

Energy Loss Density per Hysteresis Loop of Ferroelectric/Piezoelectric Materials = $\int DdE$.

Piezoelectric Strain Constant d, Polarization P, Dielectric Constant ϵ and Electrostrictive Constant q: $d \approx 2q^*\epsilon^*P$.

Dielectric Susceptibility χ , Internal Field Constant γ , Polarizability α , Entity Density N: $\chi = N^*\alpha/(\epsilon_0^*(1-N^*\alpha^*\gamma))$.

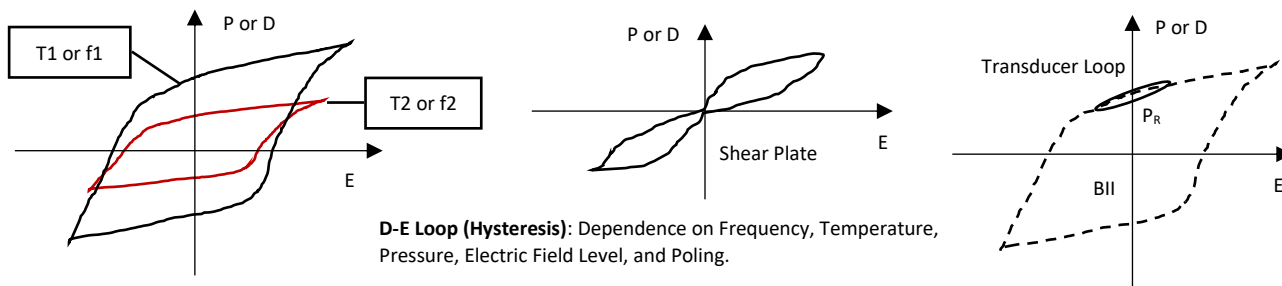
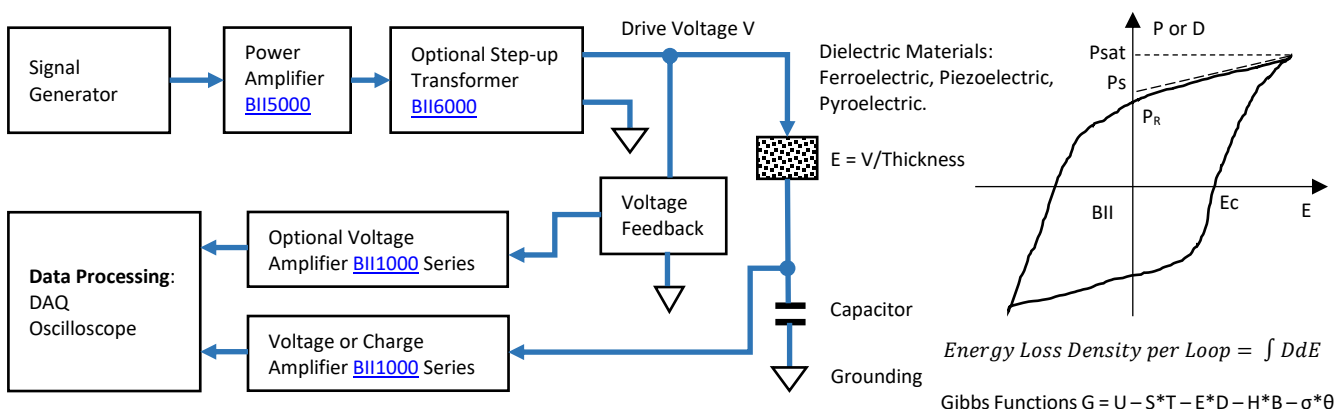
Debye Equation: $\epsilon_r^* = \epsilon_{r\infty}^* + (\epsilon_{rs}^* - \epsilon_{r\infty}^*)/(1 + j\omega/\omega_r)$. ω_r : Relaxation Frequency.

Note: 1. BII electronic components are custom-fit to a specific measurement system. Please contact and discuss with BII about your specs.

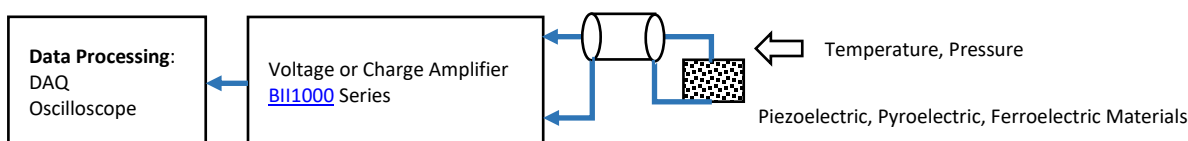
2. The block diagrams are for illustration ONLY, some parts are not shown such as DC power suppliers.

Studies of Ferroelectric and Piezoelectric Materials. BII manufactures electronic components measuring properties of dielectrics (pyroelectric, ferroelectric and piezoelectric materials) from 30 Hz to 10 MHz: Power Amplifiers, Voltage Step-up Transformers, Charge and Voltage Amplifiers. BII power amplifiers can drive the materials directly down to 1 Hz if transformers are not used to step up voltage. Parameters and performances of the material are measured at working conditions. Therefore, the performance prediction of transducers is accurate in the design phase with these material data.

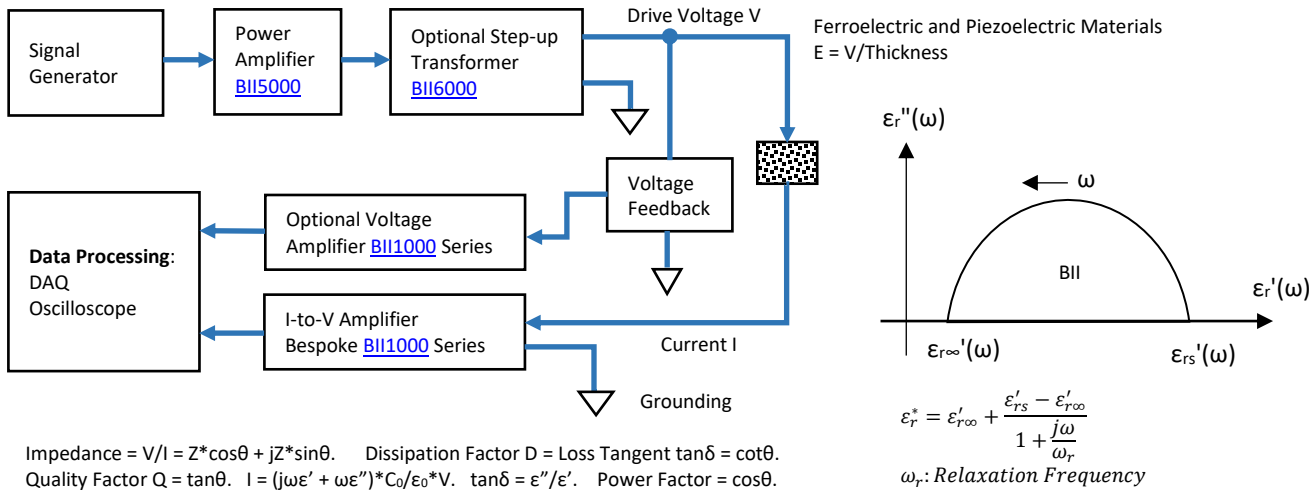
D-E or P-E Hysteresis Loop and S-E. A wideband measurement system of high electric field and/or low electric field can be set up with these components to test P-E (or D-E) hysteresis (Sawyer-Tower Circuit), Strain-Electric Field, dielectric constant, energy loss (dissipation factor or loss tangent $\tan\delta$), and their dependence on driving field, frequency, temperature and pressure (axial or hydrostatic).



Direct Measurement of Piezoelectric Constants. With standard test weights and BII amplifiers, a low-cost simple quasi-static absolute measurement system can be set up to measure piezoelectric strain and/or voltage constants with reasonable accuracy: d_{31} , d_{33} , d_h , g_{31} , g_{33} and g_h . The direct method has no limitation on material sizes and shape comparing to the resonance method and does not require a piezoelectric reference material comparing to the comparison method. To maintain high measurement accuracy, the amplifiers can be calibrated whenever needed with standard laboratory electronic instruments such as signal generators and oscilloscopes, etc...Thanks to the high gain of the operational amplifier, the capacitances of the material under test and the cable have little effect on the charge measurement, or BII's charge amplifiers provide "short circuit" condition or constant E for d constant measurement. BII's high impedance voltage amplifiers provide "open circuit" condition or constant D for g constant measurement. Besides, the amplifiers can also be used to measure the effects of thermal/temperature, pressure, electrical field on Ferroelectric and Piezoelectric materials such as thermal and pressure depoling.

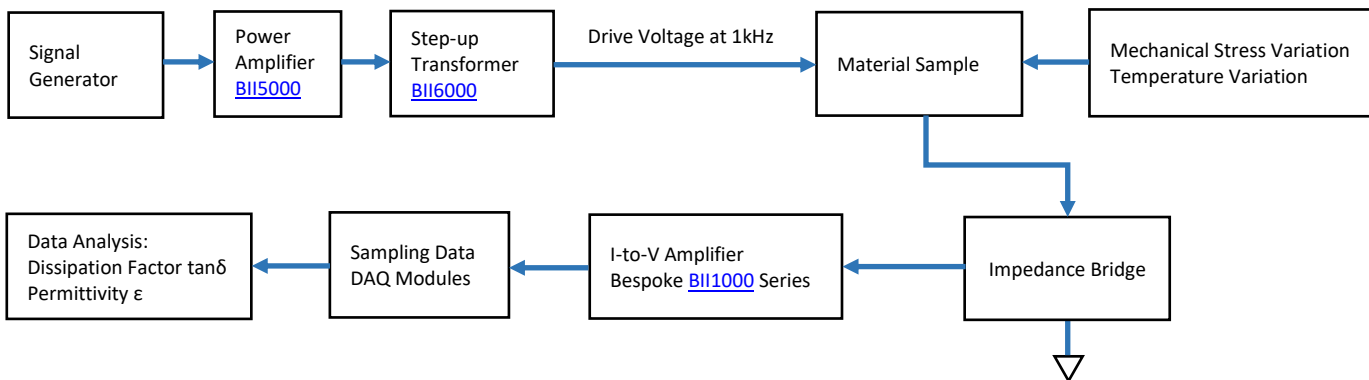


Impedance Spectroscopy. A customized impedance measurement system can be configured with BII's electronic components to study the material response to the applied electric field from 1 Hz to 10 MHz. The impedance of a material is computed over a range of frequencies with measured V and I. Material properties such as energy storage, dissipation, relaxation and polarization are revealed from the impedance.



Large Signal Piezoelectric Properties. High driving electrical field increases dielectric constant and loss factor of the piezoelectric material. The piezoelectric constants d and k are functions of dielectric constants. The increase of $\tan\delta$ dissipation factor results in higher temperature which may destroy the piezoelectric transducer. Dielectric constants and loss factors of the materials for high power operation may be measured at various field levels (up to 600 Vrms/mm) at 1 kHz under isothermal conditions. Effects of mechanical stress (pressure) and/or temperature on piezoelectric properties need to be measured as well.

High Field Measurement:



Low Field and Ageing Measurement:



Electronic Performance Test

Parameter Measurements of Preamplifiers, Power Amplifiers, Impedance Matching Unit and T/R Switches

Beside standard tests such as Gain vs Frequency, -3dB cut-off frequencies, power consumption, etc., following performance tests are available up to 1.5MHz upon request when ordering:

1. THD+N ratio and THD+N level measurements

$$THD + N \text{ ratio} = 20\log_{10}\left(\frac{\text{rms value of noise and distortion}}{\text{rms value of signal, noise, and distortion}}\right)$$

2. SINAD measurement

$$SINAD = 20\log_{10}\left(\frac{\text{rms value of signal, noise, and distortion}}{\text{rms value of noise and distortion}}\right)$$

3. THD ratio and THD level measurements

$$THD \text{ ratio} = 20\log_{10}\left(\frac{\text{rms value of distortion}}{\text{rms value of signal and distortion}}\right)$$

4. SMPTE IMD measurements

The SMPTE IMD function provides a measure of the second and third order intermodulation distortion introduced by the DUT by injecting two pure tones (tone 1 and tone 2, where tone 1 is at a much lower frequency than tone 2, for example, 20 Hz and 3.5 kHz respectively) into the DUT. SMPTE IMD is expressed in dB (default) or as a percentage.

If tone 1 = f1 and tone 2 = f2, the following harmonics are considered.

- f2 - f1
- f2 + f1
- f2 - 2f1
- f2 + 2f1

The SMPTE IMD value is computed as the ratio of the sum of the intermodulation harmonics amplitude to the upper frequency tone amplitude.

5. DFD measurements

The DFD measurement is similar to SMPTE IMD, except that the two tones in the stimulus signal are of equal amplitude and are spaced closer to each other (typically 28 kHz and 30 kHz).

6. SNR measurement

$$SNR = 20\log_{10}\left(\frac{\text{rms value of signal}}{\text{rms value of noise}}\right)$$

7. Phase Measurement

Phase measurements are used to describe the positive or negative time offset in a periodic waveform cycle (such as a sine waveform), measured from a reference waveform. Phase is expressed in degrees (°) and varies with frequency.

8. Crosstalk Measurement

In Sonar systems with more than one channel such as an array with beam steering and/or amplitude shading, it is common for a signal in one channel to appear at the output of another channel at a reduced level. Crosstalk refers to this signal leakage across channels, and it is expressed in dB (default) or as a percentage. Crosstalk is a measurement of the ratio of the signal amplitude in an unused channel relative to that of a channel driven with a signal. Crosstalk is largely due to capacitive coupling between the channel conductors in the device and generally varies with frequency. Crosstalk can be computed as follows.

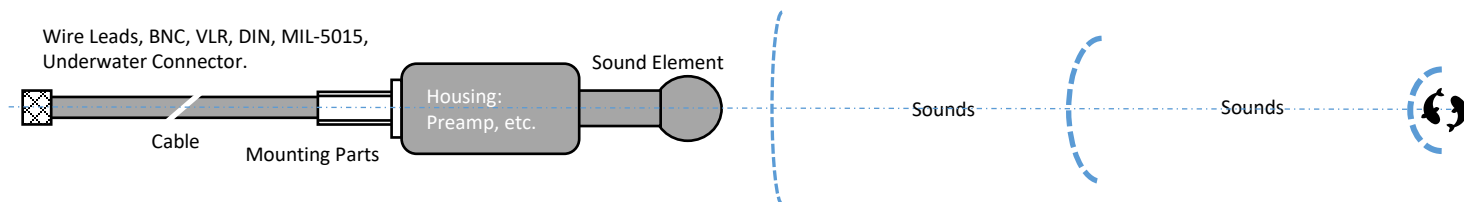
$$Crosstalk = 20\log_{10}\left(\frac{\text{rms value of signal measured}}{\text{rms value of signal driven}}\right)$$

Ordering Information

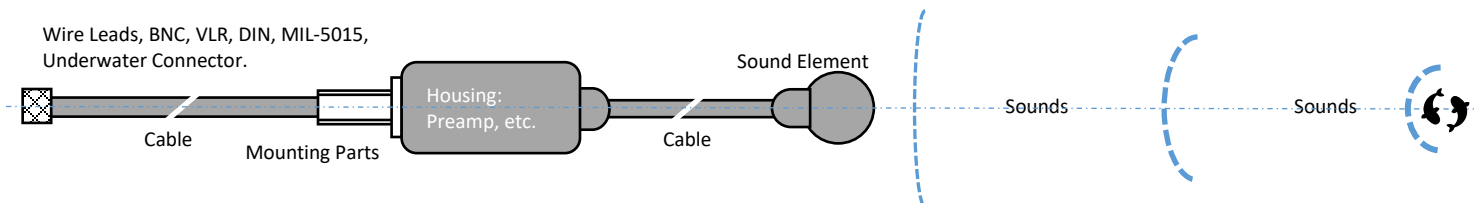
How to Order Hydrophones, AE Sensors, Ultrasonic Receivers, and Wave Height Sensors

Transducer Structure with Electronic Module (Preamp, Filters, AGC Amp, etc.) and Sound Element.

1. **Default Transducer Structure:** Sound Element is potted in Housing of Electronics.



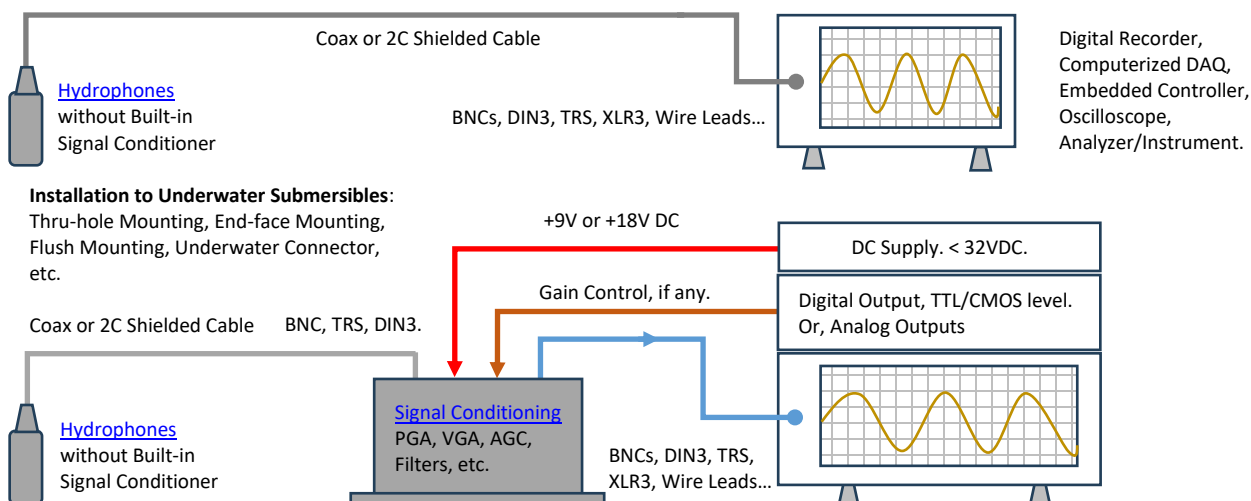
2. **Bespoke Transducer Structure:** Sound Element is separated from Housing of Electronics with cable of bespoke length.



How to Order Sound Receiving Devices Which Do NOT Have Built-in Preamp and Filters

Hydrophone Part Number	-Mounting Part	-Cable Length	-Cable Type	-Connector Type
BII7xxx	Mounting Options.	in meter.	Cable Options.	Connector Options.
Example:	Description			
BII7xxx-FH-6m-RG174-BNC	BII7xxx Hydrophone, Free Hanging, 6m RG174 Coax, BNC Male.			
BII7xxx-BFMP-NPT3/8"-6m-RG174-BNC	BII7xxx Hydrophone, Bolt-fastening Mounting: BFMP-NPT3/8", 6m RG174 Coax, BNC Male.			
BII7xxxDW-THM-7/16"-0.6m-SC36-WL	BII7xxxDW Hydrophone, Thru-hole Mounting THM-7/16", 0.6m Shielded Cable SC36, Wire Leads.			
BII7xxx-HT-FH-6m-RG178-BNC	BII7xxx Hydrophone, Service Temperature: -10 °C to 120 °C (14 °F to 248 °F). Free Hanging, 6m RG178 Coax, BNC Male.			
BII7xxxDF-FH-0.6m-SC65-UMC3P	BII7xxxDF Hydrophone, Free Hanging, 0.6m Shielded Cable SC65, 3-pin Underwater Mateable Connector.			

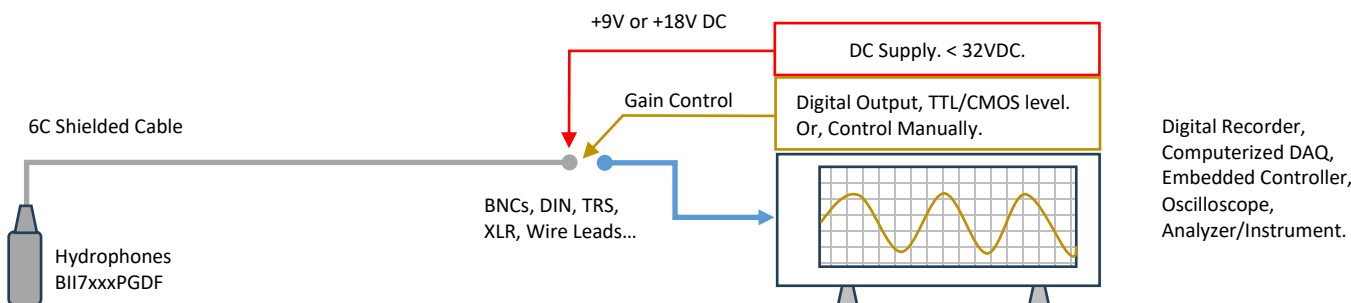
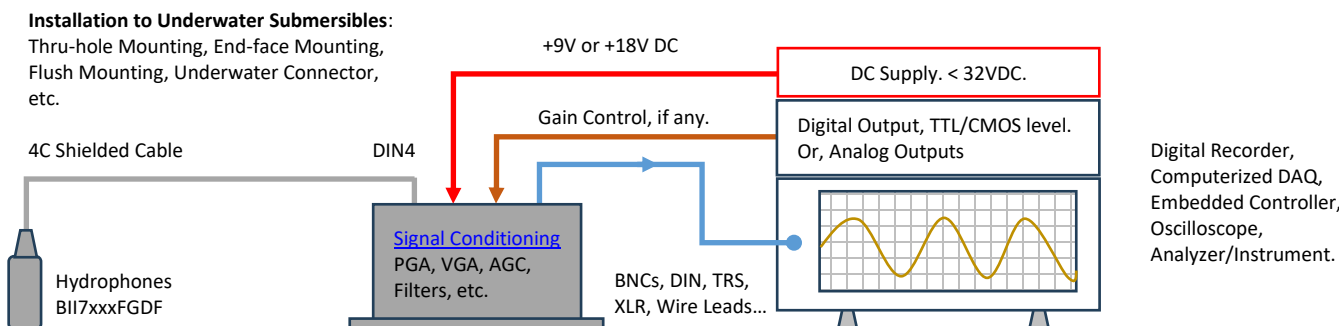
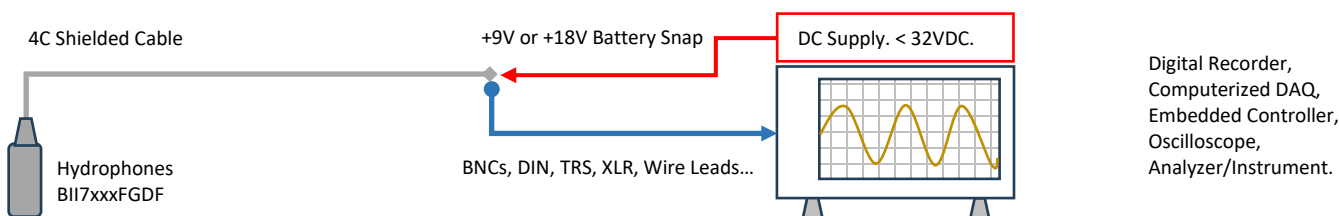
System Configuration of Receiving Sounds and Waves.



How to Order Sound Receiving Devices Which Have Built-in Preamp and Filters

FG: Fixed Gain; PG: Programmable Gain; DF: Differential Output; SE: Single Ended Output; BPF: Band Pass Filter; HPF: High Pass Filter; LPF: Low Pass Filter.					
Part Number	-Preamp Gain	-HPF or HPF/LPF	-Mounting	- Cable Length	-Connectors for Signal/Gain Selection/DC Supply
BII7XXXFGDF	Fixed, in dB.	-3dB Filter frequencies	Mounting Options.	in meter. Up to 305m (1000 ft).	Connector for Signals, Gain Selection (if any), and DC Supply.
BII7XXXPGDF	Gain Options.				
Example of Part Number:			Description		
BII7XXXFGDF-26dB-10Hz-BFM-7/16"-100m-DIN3/BS			BII7XXXFGDF Hydrophone, Gain: 26dB, High Pass Filter: 10Hz, Bolt Fastening Mounting BFM-7/16", 100m Shielded Cable, Connector: 3-pin DIN for Signals and Battery Snap for +9VDC Batteries.		
BII7XXXFGDF-26dB-10Hz-BFM-7/16"-100m-XLR3/BS			BII7XXXFGDF Hydrophone, Gain: 26dB, High Pass Filter: 10Hz, Bolt Fastening Mounting BFM-7/16", 100m Shielded Cable, Connector: 3-pin XLR for Signals and Battery Snap for +9VDC Batteries.		
BII7XXXFGDF-26dB-10Hz-BFM-7/16"-100m-DIN4			BII7XXXFGDF Hydrophone, Gain: 26dB, High Pass Filter: 10Hz, Bolt Fastening Mounting BFM-7/16", 100m Shielded Cable, Connector: 4-pin DIN for Signals and DC Power Supply.		
BII7XXXFGDF-26dB-10Hz-FH-0.6m-UMC4P			BII7XXXFGDF Hydrophone, Gain: 26dB, High Pass Filter: 10Hz, Free Hanging, 0.6m Shielded Cable, Connector: 4-pin Underwater Mateable Connector for Signals and DC Power Supply.		
BII7XXXPGDF-10/50dB-10Hz-BFM-7/16"-100m-DIN3/WL/BS			BII7XXXPGDF Hydrophone, Gain: 10 & 50dB, High Pass Filter: 10Hz, Bolt Fastening Mounting BFM-7/16", 100m Shielded Cable, Connector: 3-pin DIN for Signals, Wire leads for Gain Selection, and Battery Snap for +9VDC Batteries.		
BII7XXXPGDF-20/50dB-10Hz/300kHz-BFM-7/16"-100m-XLR3/WL/BS			BII7XXXPGDF Hydrophone, Gain: 20 & 50dB, Bandpass Filter: 10Hz to 300kHz, Bolt Fastening Mounting BFM-7/16", 100m Shielded Cable, Connector: 3-pin XLR for Signals, Wire leads for Gain Selection, and Battery Snap for +9VDC Batteries.		

System Configuration of Receiving Sounds and Waves.

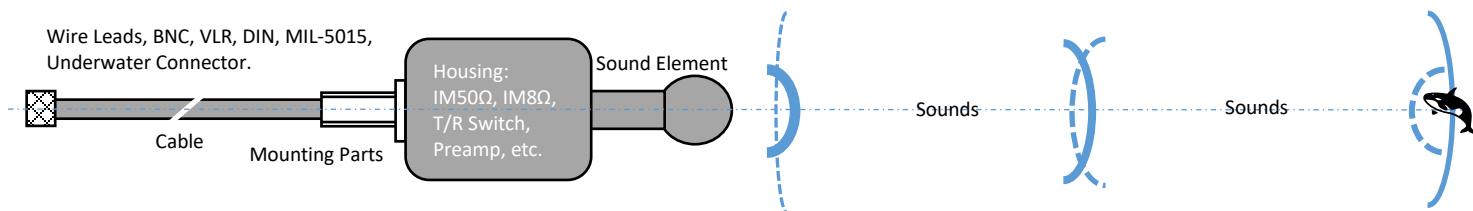


How to Order Underwater Transducer (Projector), HIFU Transducer, and Ultrasonic Power Transducer

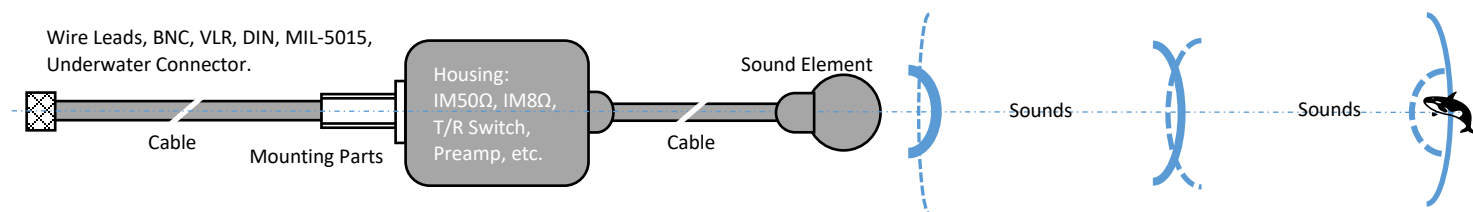
If a hydrophone or transducer does not support a specific parameter, or the default parameter stated in the specification is to be ordered, please ignore that parameter or use blank. If your applications have requirements about physical size, operating depth, service temperature, special regulations, ..., please contact us for custom-fit products.

Transducer Structure with Electronic Module (Impedance Matching Network and Transmit/Receive Switch, Preamp, etc.) and Sound Element.

1. **Default Transducer Structure:** Sound Element is potted in Housing of Electronics.

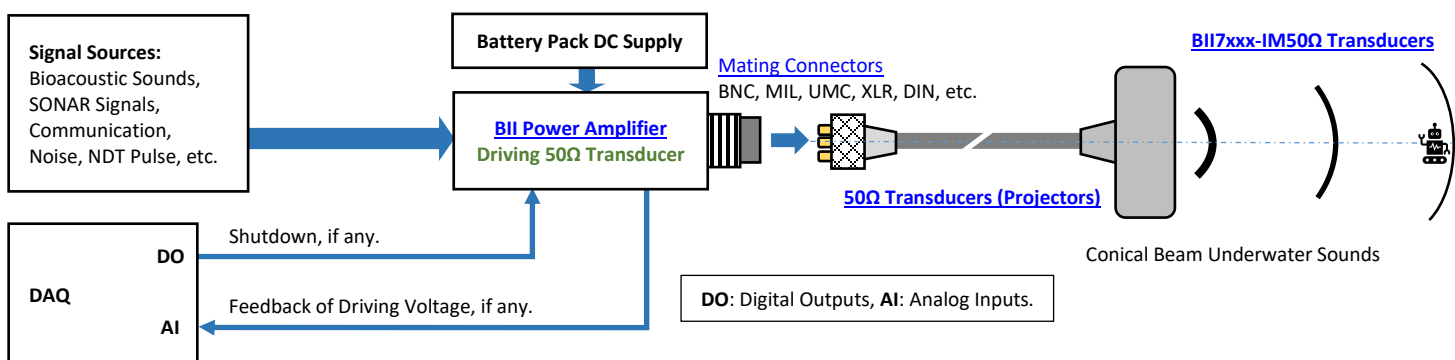
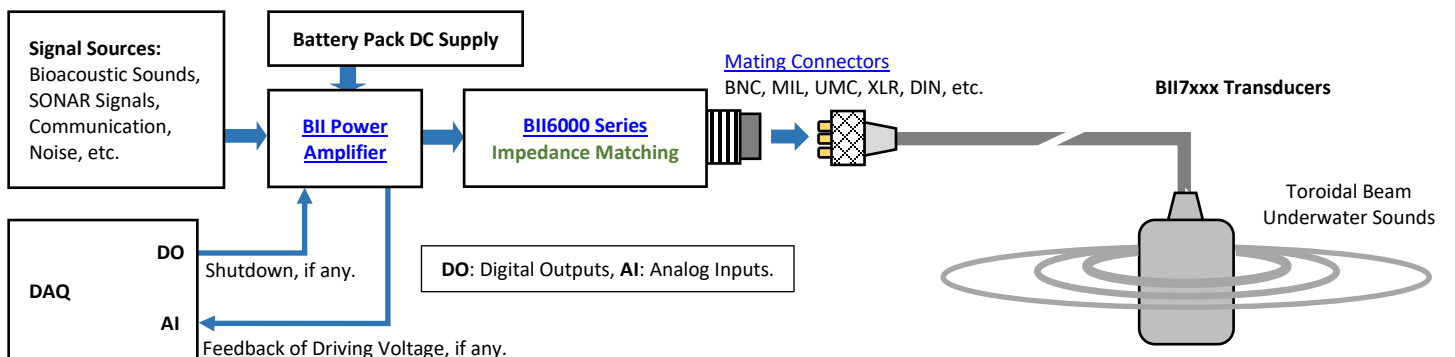


2. **Bespoke Transducer Structure:** Sound Element is separated from Housing of Electronics with cable of bespoke length.



System Block Diagram of Generate Sounds

This setup is a compact and low cost solution for end users' acoustic system by integrating BII's Power amplifier (PCB form factor) and Impedance Matching device (PCB or small-enclosure form factor) into end users' grounded enclosures.

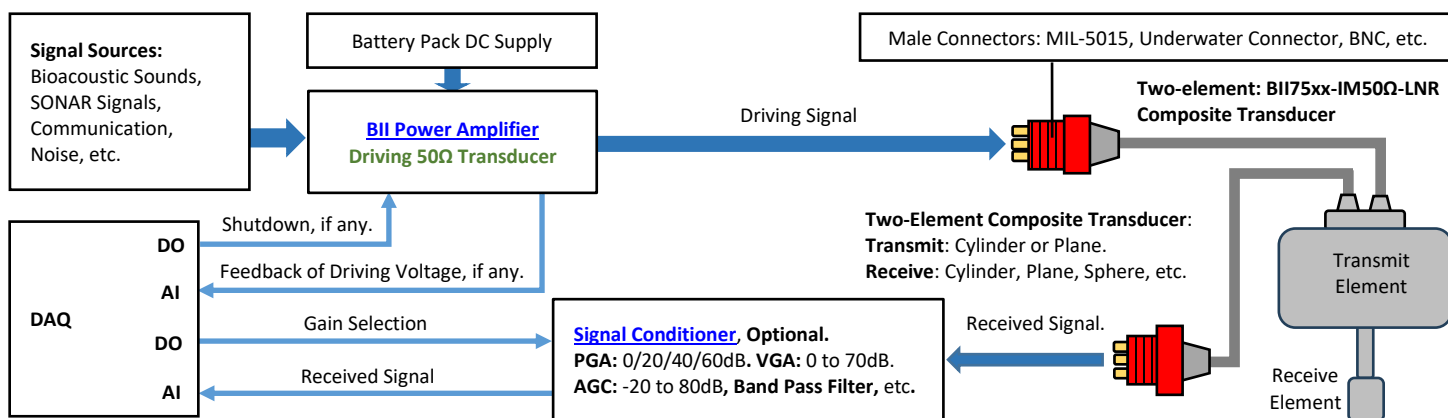
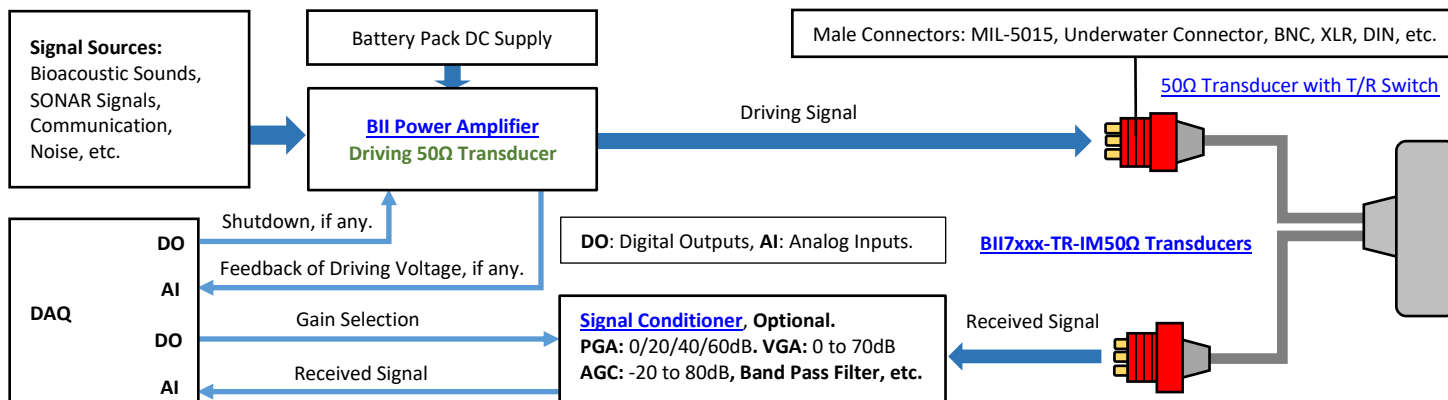


How to Order Transducers Generating Sounds. The default options are for stock items which are regularly available.

FH: Free Hanging. SC for Transmit: Shielded Cable (Rubber Jacket, 600V) with 2 conductors. Coax: 50 Ω Coaxial Cable. WL: Wire Leads.					
Underwater Mateable Connector UMC2P is fixed with 0.6m unshielded cable (USC).					
Part Number	-Appendage	-Mounting	-Cable Length	-Cable Type	-Connector for signals of Transmit and Temperature Sensor
BII75xx BII75XX-IM50Ω	Default: None.	Default: BFM-FH-M8, or BFM-FH-3/8".	Default: 15m or 0.6m.	SC for low frequency signal. USC for UMC2P Connector. Coax for high frequency signal.	Default: WL.
Example:		Description			

BII75xx-BFM-FH-15m-SC-WL	BII75XX Transducer, Bolt-Fastening Mounting with Free Hanging: BFM-FH, 15m Shielded Cable, Wire Leads.
BII75xx-BFM-FH-M8-0.6m-USC-UMC2P	BII75XX Transducer, Bolt Fastening Mounting with Free Hanging: BFM-FH-M8, 0.6m Unshielded Cable, Male Underwater Mateable Connector with Locking Sleeve: DLSA-M.
BII75XX-HT-FH-6m-RG178-BNC	BII75XX Transducer, Service Temperature: -10 °C to 120 °C, or 14 °F to 248 °F. Free Hanging, 6m RG178 Coax, BNC Male.
BII75XX-IM50Ω-FH-20m-RG58-BNC	BII75XX Transducer, Built-in Impedance Matching Network as 50Ω load at fs, Free Hanging, 20m RG58 Coax, Male BNC.
BII75XX-IM8Ω-FH-15m-SC-XLR3P	BII75XX Transducer, Built-in Impedance Matching Network as 8Ω load at fs, Free Hanging, 15m Shielded Cable, XLR Plug.

System Setup of Transmitting and Receiving Sounds.



How to Order Transmitting and Receiving Transducers. The default options are for stock items which are regularly available.

FH: Free Hanging. **SC for Low Frequency Transmit:** Shielded Cable (Rubber Jacket, 600V) with 2 conductors. **Coax for High Frequency Transmit:** 50 Ω Coaxial Cable. **SC for Low Frequency Receive:** Shielded Cable with 4 conductors. **Coax for High Frequency Receive:** 50 Ω Coaxial Cable. **WL:** Wire Leads. **HPF:** -3dB High Pass Filter Frequency. **LPF:** -3dB Low Pass Filter Frequency. **Cable of Temperature sensor** is two-conductor shielded cable. **Cable of DC Supply** is two-conductor shielded cable in case that receive cable is coax.

Receiving Cable is fixed to be four-conductor Shielded cable. Transmitting cable can be customized to be Coax or two-conductor shielded cable.

Length of Transmitting and receiving cables are same in default.

Underwater Mateable Connector UMC2P and UMC4P are fixed with 0.6m unshielded cables.

Part Number	-Preamp Gain	-HPF/LPF	-Mounting	-Cable Length	-Transmit Cable	-Connector for signals of Transmit/Receive/DC Supply/Temperature
BII75xx-TR-IM50Ω BII75xx-IM50Ω-LNR	Default: xxdB	-3dB Receive bandpass Frequencies. Default: 2kHz to xxxkHz	Default: BFM-FH-M8, or, BFM-FH-3/8".	Default: 15m.	SC or Coax. Default: SC.	Default: WL.
Example:		Description				
BII75xx-TR-IM50Ω-xxdB-2kHz/xxxkHz-BFM-FH-M8-15m-SC-WL		BII75xx Transducer, Built-in T/R Switch, Built-in Impedance Matching Network as 50Ω load at fs, Receive Gain: xxdB, Receive Bandpass Filter: 2kHz to xxxkHz. Bolt-Fastening Mounting with Free Hanging: BFM-FH-M8, 2 x 15m cables, Transmitting Cable: Shielded Cable, Wire Leads.				
BII75xx-IM50Ω-LNR-xxdB-2kHz/xxxkHz-BFM-FH-M8-15m-SC-WL		BII75xx Transducer, Built-in Impedance Matching Network as 50Ω load at fs, Built-in Low Noise Receiver, Receive Gain: xxdB, Receive Bandpass Filter: 2kHz to xxxkHz. Bolt-Fastening Mounting with Free Hanging: BFM-FH-M8, 2 x 15m cables, Transmitting Cable: Shielded Cable, Wire Leads.				
BII75xx-TR-IM50Ω-xxdB-2kHz/xxxkHz-BFM-FH-M8-15m-SC-MIL3P/XLR4P/BS		BII75xx Transducer, Built-in T/R Switch, Built-in Impedance Matching Network as 50Ω load at fs, Receive Gain: xxdB, Receive Bandpass Filter: 2kHz to xxxkHz. Bolt-Fastening Mounting with Free Hanging: BFM-FH-M8, 2 x 15m cables, Transmitting Cable: Shielded Cable, 3 Pin MIL-5015 Connector for Transmit Signal, 4 Pin XLR for Receive Signal, 9V Battery Snap for DC Supply.				
BII75xx-TR-IM50Ω-xxdB-2kHz/xxxkHz-FH-15m-RG58-BNC/BNC/BS/TRS		BII75xx Transducer, Built-in T/R Switch, Built-in Impedance Matching Network as 50Ω load at fs, Receive Gain: xxdB, Receive Bandpass Filter: 2kHz to xxxkHz. Free Hanging, 2 x 15m cables, Transmitting Cable: RG58 Coax, BNC Male Connector for Transmit Signal, BNC Male for Receive Signal, 9V Battery Snap for DC Supply, TRS for Temperature Signal.				

Cable and Connector Information for Signals of Hydrophones and Power Transducers (Projectors). Non-UL Uses.

	Wire and Cable Types	Ratings of Voltage, Current or Power, and Temperature.
	AWG18 Wires (WR)	3000 V _{rms} , 10 A _{rms} .
Cables:	Two Conductor Unshielded Cable (USC). AWG18 Conductors.	600 V _{rms} , 10 A _{rms} . -40°C To +90°C, or -40°F to 194°F.
	Two Conductor Shielded Cable (SC). AWG20 Conductors.	600 V _{rms} , 4 A _{rms} . -50°C To +90°C, or -58°F to 194°F.
	Two Two-conductor Shielded Cable Bundle (2SC). AWG20 Conductors.	600 V _{rms} , 10 A _{rms} . -50°C To +90°C, or -58°F to 194°F.
	Two, Four or Six Conductor Shielded Cable (SCxx). AWG20, AWG24, AWG26, AWG28 Conductors.	60 to 300 V _{rms} , 0.2 A _{rms} to 10A, for Hydrophone Use ONLY. -40°C to +80°C or -40°F to 176°F.
	High Temperature Shielded Cable (HTSC200) or (SC40)	600 V _{rms} , 4.0 A _{rms} , up to +200°C or 392°F, Non-waterproof.
	High Temperature Shielded Cable (HTSC200) or (SC32)	600 V _{rms} , 0.8 A _{rms} , up to +200°C or 392°F, Non-waterproof.
	High Temperature Shielded Cable (HTSC150).	600 V _{rms} , 2.8 A _{rms} , -70°C to +150°C or -94°F to 302°F.
	High Temperature Twisted Pair Shielded Cable (HTTSC125).	300 V _{rms} , 0.92 Arms, -50°C to +125°C or -58°F to 257°F.
	Coax RG58 (50Ω) (RG58)	1400 V _{rms} , 4 A _{rms} . -40°C To +80°C or -40°F to 176°F.
	Coax RG174/U (50Ω) (RG174)	1100 V _{rms} , 1.6 A _{rms} . -40°C To +75°C or -40°F to 167°F.
	Coax RG316/U (50Ω) (RG316)	900 V _{rms} , 2.0 A _{rms} , -50°C To +200°C or -58°F to 392°F.
	Coax RG178B/U (50Ω) (RG178)	750 V _{rms} , 0.86 A _{rms} , -70°C To +200°C or -94°F to 392°F.
	Connector Type	Ratings of Voltage, Current or Power, and Temperature.
	1. Wire Leads (WL)	Used for Cables or Wires.
Connectors:	2. 50Ω BNC (BNC), Bayonet Lock. Panel Mount or In-line. In-line BNC: Input uses Pin, output uses Socket. Panel Mount BNC: Both Input and Output use BNC Jacks.	500 V _{rms} , 100 ~ 316 W, 1 A _{rms} Continuously. Pulse Current: 5 ~ 10 A _{peak} at range of μS to mS, D ≤ 1%. 10 ~ 20 A _{peak} at range of nS to μS, D ≤ 1%. Used for Grounded Signal with Metal Enclosures or Coax Cables.
	3. MIL-5015 Type Connector (MIL), Thread Fastening. Panel Mount or In-line. Input uses Pin, output uses Socket.	500 V _{rms} , 13 A; Up to +125°C or 257°F, or, 900 V _{rms} , 13 A; Up to +125°C or 257°F. Used for Metal Enclosures or Shielded Cables.
	4. Circular Connector DIN EN (DIN), Thread Fastening. Panel Mount or In-line. Input uses Pin, Output uses Socket.	250 V _{rms} , 10 A; -40°C to +100°C or -40°F to 212°F. Used for Metal Enclosures or Shielded Cables.
	5. XLR Connector (XLR), Positive Latchlock. Panel Mount or In-line. Input uses Pin, Output uses Socket.	133 V _{rms} , 15 A; -25°C to +75°C or -13°F to +167°F. Used for Metal Enclosures or Shielded Cables.
	6. 3.5mm or 1/8" TRS (TRS35), Panel Mount with Jack, In-line with Plug, for analog audio signals.	30 Vrms, 0.3A; -25°C to +75°C or -13°F to +167°F. Used for Metal Enclosures or Shielded Cables.
	7. Underwater Mateable Connector (UMC), Thread Fastening DLSA-F or DLSA-M. Panel Mount or In-line. Input uses Pin, Output uses Socket.	600 V _{rms} , 10 A. Waterproof, IP68. Max. Diameter Ø35.5mm. Used for Metal Enclosures or Transducers (Projectors).
	8. Underwater Mateable Connector (UMC), Thread Fastening MCDLS-F or MCDLS-M. Panel Mount or In-line. Input uses Pin, Output uses Socket.	300 V _{rms} , 10 A. Waterproof, IP68. Max. Diameter Ø22mm. Used for Metal Enclosures or Hydrophones.
	9. Sheathed Banana Jack. Panel Mount or In-line.	1000V CAT III/600V CAT IV, 25A. IEC61010-031 rated. NOT USED in this device.
	10. SMA (Plug, Male Pin) (SMA). Thread Fastening. In-line.	Voltage Rating: 335 V _{rms} Continuous. (Max. Diameter Ø9.24 mm). Up to 155°C or 311°F.
	11. SMC (Plug, Female Socket) (SMC). Thread Fastening. In-line.	Voltage Rating: 335 V _{rms} Continuous. (Max. Diameter Ø6.4 mm). Up to 155°C or 311°F.
	12. LEMO (Plug Male Pins) (LEMO). Push-Pull Fastening. Panel Mount or In-line.	900 V (AC), 1270 V (DC), 8A, (Max. Diameter Ø9.5 mm with 3 contacts). Temp (min / max) -55°C / +250°C.
How to choose cable and connector for BII devices: Driving Voltage $V_{drive} (V_{rms}) = \sqrt{RMS\ Power * \frac{G}{G^2 + B^2}}$. BII lists G-B data at f_s and/or the graph of G-B vs Frequency in online datasheet. Case 1. Deliver 1000 W_{rms} to 3 kΩ transducer at f_s. Note: $G/(G^2+B^2) = 3\ k\Omega$ is the resistive load of the transducer in load medium at f_s. Driving voltage to transducer $V_{drive} = \sqrt{1000 * 3000} = 1732\ V_{rms}$. The current to 3 kΩ transducer $I_{drive} = V_{drive}/R_L = 1732V_{rms}/3000\Omega = 0.57733\ A_{rms}$. Therefore, AWG18 Wire and Wire leads are suitable. Case 2. Deliver 500 W_{rms} to 300 Ω transducer at f_s. Note: $G/(G^2+B^2) = 300\ \Omega$ is the resistive load of the transducer in load medium at f_s. Driving voltage to transducer $V_{drive} = \sqrt{500 * 300} = 387.3\ V_{rms}$. The current to 300 Ω transducer $I_{drive} = V_{drive}/R_L = 387.3V_{rms}/300\Omega = 1.291\ A_{rms}$. Therefore, Two Conductor Shielded Cable and MIL-5015 Type Connector or Underwater Mateable Connector (UMC) are suitable. Case 3. Deliver 300 W_{rms} to 50 Ω transducer at f_s. Driving voltage to transducer $V_{drive} = \sqrt{300 * 50} = 122.5\ V_{rms}$. The current to 50 Ω transducer $I_{drive} = V_{drive}/R_L = 122.5V_{rms}/50\Omega = 2.45A_{rms}$. Therefore, 50Ω RG58 Coax and BNC are suitable. Please contact us for bespoke wirings of differential transducers such as dipole, quadrupole, multimode rings, and flextensional sources.		

Abbreviation List for Ordering

Principal Parameters										
DRx: Depth Rating x meters			SL: Source Level, in dB re $\mu\text{Pa}\cdot\text{m}$				Q _m : Quality factor, -3dB Bandwidth = f_s/Q_m			
fs: Resonance Frequency in kHz			PW: Pulse Width				D: Duty Cycle			
CW: Continuous Wave (Operating)			f _m : Impedance Matching Frequency in kHz. f _m = fs by default.							
BW: -3dB Beam Width			SLL: Side Lobe Level in dB, Nominal to Main lobe.				TS: Built-in Temperature Sensor (NTC thermistor)			
Built-in Impedance Matching and T/R Switch										
IM: Impedance Matching			TR: Transmit & Receive Switch					TRIM: T/R Switch and Impedance Matching		
Beam Pattern										
SPH:		Beam generated by spherical transducer. Generally, -3dB Beam Width at fr: 360°x250° in XY and ZX plane, frequency dependent.								
HSPH:		Beam generated by hemispherical transducer. Generally, -3dB Beam Width at fr: 360°x70° in XY and ZX plane, frequency dependent.								
CLD:		Beam generated by cylindrical transducer. Generally, -3dB Beam Width at fr: 360°x(40°~80°) in XY and ZX plane, frequency dependent.								
BW or $\Delta\theta$:		-3dB conical Beam Width $\Delta\theta$ in ZX and ZY plane, in °, generated by transducers of circular plate or convex shell.								
$\Delta\theta_H \times \Delta\theta_V$: (Planar)		-3dB fan-shaped Beam Width, generated by transducers with rectangular (or square) apertures. $\Delta\theta_H$ is in ZX plane (Along-track, Along-length or Horizontal Plane) and $\Delta\theta_V$ is in ZY plane (Cross-track, Cross-length or Vertical Plane), such as -3dB Along-Length x Cross-Length Beam Width = 5° x 50°.								
$\Delta\theta_H \times \Delta\theta_V$: (Curved)		-3dB fan-shaped Beam Width, generated by transducers with cylindrical curve apertures. $\Delta\theta_H$ is in ZX plane (Along-curve or Horizontal Plane) and $\Delta\theta_V$ is in ZY plane (Cross-curve or Vertical Plane), such as -3dB Along-curve x Cross-curve Beam Width = 120° x 20°.								
Signal Conditioning of Received Signals										
SE: Single Ended Signal/Unbalanced			DF: Differential Signal/Balanced			HPF: High Pass Filter		LPF: Low Pass Filter		BPF: Band Pass Filter
AGC: Automatic Gain Control Amplifier			VGA: Variable Gain Amplifier			PG: Programmable Gain Preamp		FG: Fixed Gain Preamp		Reserved
Mounting Parts: Detail information at Mounting Parts .										
Cable: -4° to 140° F (-20° to 60° C) Typically										
WR: Wires			MiniCoax: Coax (ΦD=1.4mm)				RG178: RG178B/U 50Ω Coax, -70°C To +200°C			
RG316: RG316/U 50Ω Coax, -50°C To +200°C			RG174: RG174/U 50Ω Coax				RG58: RG58 50Ω Coax (see Note 1)			
SCB: Shielded Cable Bundle			WCB: Wire/Cable Bundle							
SC26: Shielded Cable with Polyurethane Jacket, ΦD = 2.6 mm. Temperature: -40 to 70°C, Voltage: ≤ 150 V.										
SC32: Shielded Cable with Twisted Pair and Teflon (PTFE) Jacket, ΦD=3.2 mm (SC32), up to 200°C. Not water-proof.										
SC36: Shielded Cable with Twisted Pair and PVC Jacket, ΦD = 3.6 mm. ≤ 300Vrms, -20 to 60°C.										
SC47: Shielded Cable with Twisted Pair and Polyurethane Jacket, ΦD = 4.7 mm. ≤ 300 Vrms, -30 to 90°C.										
SC60: Shielded Cable with Twisted Pair and PVC Jacket, ΦD = 6.0 mm. ≤ 300Vrms, -20 to 60°C.										
SC65: Shielded Cable with Rubber Jacket, ΦD = 6.5 mm. -50°C to +90°C, ≤ 600 Vrms.										
MiniSC: Miniature Shielded Cable, ΦD=2.438 mm. -20°C to +105°C, or -4°F to 221°F. Max. Operating Voltage: 200 V RMS, Max. Current: 0.9 Amps.										
SC: Shielded Cable (ΦD = 2.4 to 9.7 mm). Uses: Water Depth ≤ 300 m; Max. Operating Voltage: 300 Vrms or 600 Vrms available, Max. Current: 4 A.										
USC: Unshielded Cable (ΦD = 8.7 to 15.3 mm), AWG18 Conductors. Uses: Water Depth ≤ 3000 m; Max. Operating Voltage: 600 Vrms, Max. 10 A.										
Shielded Cable with Twisted Pair (SCTP): Recommended for hydrophone (receiver) without preamplifier.										
Cable Jacket: PVC, Polyurethane, EPDM rubber, FEP, PTFE, etc...										
Cable Length: Customized, up to 305 m or 1000 ft.										
1. The cable has adverse effects on the transducer (projector) performances in high frequency range. The longer the cable is, the worse the adverse effects on the transducer (projector) performance is. If a long cable is needed for the project (comparing to electromagnetic wavelength), impedance matching is necessary between the transducer (projector) and the signal source (power amplifier).										
2. To transmit received signals (receivers, hydrophones) over a long cable, using differential signal transmission is a good option, and matching impedances between receiver (hydrophone) and the cable.										
High Temperature Cables										
RG178: RG178B/U 50Ω Coax, 750 Vrms, 0.86 Arms, -70°C To +200°C, Diameter: Φ1.8 mm.										
RG316: RG316/U 50Ω Coax, 900 Vrms, 2.0 Arms, -50°C To +200°C or -58°F to 392°F. Diameter: Φ2.5 mm.										
HTWR200: High Temperature Wire, 14 to 392°F (-10° to 200° C), Maximum 600VAC.										
HTWR449: High Temperature Wire, -40 to 840°F (-40° to 449° C), Maximum 600VAC. Dry Use ONLY.										
HTPSC125: High Temperature Twisted Pair Shielded Cable, 300 Vrms, 0.92 Arms, -50°C to +125°C or -58°F to 257°F.										
HTSC150: High Temperature Shielded Cable, -94 to 302°F (-70° to 150° C), Maximum 600VAC.										
HTSC200: High Temperature Shielded Cable, -65 to 392°F(-54° to 200°C), Maximum 600VAC.										
HTUSC: High Temperature Unshielded Cable, -40 to 840°F (-40° to 449° C), Maximum 600VAC. Dry Use ONLY.										
Connectors (In-line or Panel Mount)										
WL: Wire Leads.			BNC: BNC Male (or Plug) 50Ω.				BNCf or BNCJ: BNC Female (or Jack) 50Ω.			
DIN3/4: DIN Receptacle with 3/4 Male Pins.			SBJ or SBP: Sheathed Banana Jack or Plug.				TRS: 3.5mm or 1/8" TRS Plug or Jack.			
LEMO: PLUG 3 Contacts, Socket or Pin.			SMA: Plug, Male Pin, 50Ω.				SMC: Plug, Female Socket, 50Ω.			
XLR3/4/5/6: XLR Receptacle with 3/4/5/6 Male Pins.			5015: MIL-5015 style Circular Connector, Socket or Pin.				SP: Solder Pins.			
UMC: Underwater Mateable Connector (See Note 2).			BS: +9V Battery Snap.				Custom: Customized Connector.			
Underwater Connectors are for uses underwater, other connectors and wire leads listed above are for dry uses ONLY in air.										
Maximum Diameter Size of Connectors: useful to choose mounting/installation parts of the transducers/hydrophones, Unit: mm.										
Connector:	Male BNC	SMA (Plug, Male Pin)	SMC (Plug, Female Socket)	LEMO (PLUG MALE 3 Pins)	1/8" (3.5mm) TRS	XLR	DIN	MIL-5015	9V Battery Snap	Underwater Connector
Biggest Size:	Φ14.3	Φ9.24	Φ6.4	Φ9.5	Φ10.5	Φ20.2	Φ17	Φ30	Φ13	Φ21.5 or Φ35
Note 1: RG58 Coax with BNC connector. Available lengths: 1m, 2m, 3m, 5m, 10m, 15m, 20m, 30m. Please choose cable length which is much less than the Electromagnetic wavelength $\lambda \approx c/f$. c: speed of light in Vacuum.										
Warning: It is buyer's sole responsibility to make sure that BNC/SMA/SMC shield of the signal source is firmly grounded for operating safety before hooking up the transducer (projector) to the signal source. Coax with BNC/SMA/SMC is not intended for hand-held use at voltages above 30Vac/60Vdc.										

Note 2: Underwater Mateable Connectors (UMC). BII stocks wet-mateable connectors from global underwater connector manufacturers whose connectors and accessories are available worldwide. Buyers may purchase mating connectors and accessories from local suppliers or representatives of these global companies in buyers' countries or purchase these mating connectors from BII as integral parts of BII products. Please contact BII for details.	
UMC2P: Underwater Mateable Connectors with 2 pins.	Major Uses: (1) Hydrophones/AEs, no built-in preamp; (2) Transducers, no built-in T/R Switch.
UMC3P: Underwater Mateable Connectors with 3 pins.	Major Uses: (1) Hydrophones/AEs, no built-in preamp; (2) Transducers, no built-in T/R Switch.
UMC4P: Underwater Mateable Connectors with 4 pins.	Major Uses: Hydrophones/AEs with built-in fixed gain preamps.
UMC6P: Underwater Mateable Connectors with 6 pins.	Major Uses: (1) Hydrophones/AEs with built-in Programmable gain preamps. (2) Transducers with built-in T/R Switch.
Note 3: Underwater Mateable Connectors with 2/3/4/6 Sockets are available too.	
Note 4: Necessary accessories such as Locking Sleeve (MCDLS-F, MCDLS-M, DLSA-F, or DLSA-M), Nut, Washer, O-ring, if any, are included for installation.	
Underwater Mateable Connector: Handling Instructions	
BII stocks underwater mateable connectors of Standard Circular and Micro Series for signal transmission and power supply underwater.	
Always apply grease before mating.	Disconnect by pulling straight, not at an angle.
Do not pull on the cable and avoid sharp bends at cable entry.	Do not over-tighten the nuts.
Not be exposed to heat or direct sunlight.	Wet in fresh water before use.

Service Temperature Range:

- Default Operating Temperature: -10°C to +60°C (14°F to 140°F); Storage Temperature: -20°C to +60°C (-4°F to 140°F);
- Bespoke high temperature transducers and hydrophones are available, please refer to respective datasheet for their service and storage temperatures.
 - Service Temperature: -10°C to 120°C (14°F to 248°F).
 - Service Temperature: -10°C to 200°C (14°F to 392°F).

**Characteristics of Temperature Sensor Built inside Transducers (Hydrophones and Projectors):**

-40 °C to 125 °C NTC Temperature Sensor: Resistance Vs. Temperature							
Temp. (°C)	Resistance (kΩ)	Temp. (°C)	Resistance (kΩ)	Temp. (°C)	Resistance (kΩ)	Temp. (°C)	Resistance (kΩ)
-40	197.388	5	22.165	50	4.160	95	1.112
-35	149.395	10	18.010	55	3.539	100	0.976
-30	114.345	15	14.720	60	3.024	105	0.860
-25	88.381	20	12.099	65	2.593	110	0.759
-20	68.915	25	10.000	70	2.233	115	0.673
-15	54.166	30	8.309	75	1.929	120	0.598
-10	42.889	35	6.939	80	1.673	125	0.532
-5	34.196	40	5.824	85	1.455		
0	27.445	45	4.911	90	1.270		

-40 °C to 250 °C NTC Temperature Sensor: Resistance Vs. Temperature							
Temp. (°C)	Resistance (kΩ)	Temp. (°C)	Resistance (kΩ)	Temp. (°C)	Resistance (kΩ)	Temp. (°C)	Resistance (kΩ)
-40	204.7	35	6.944	110	0.7483	185	0.1439
-35	154.4	40	5.830	115	0.6603	190	0.1313
-30	117.6	45	4.918	120	0.5840	195	0.1202
-25	90.44	50	4.166	125	0.5176	200	0.1103
-20	70.15	55	3.545	130	0.4598	205	0.1015
-15	54.87	60	3.028	135	0.4093	210	0.0937
-10	43.27	65	2.596	140	0.3651	215	0.0868
-5	34.39	70	2.234	145	0.3263	220	0.0807
0	27.53	75	1.929	150	0.2923	225	0.0754
5	22.2	80	1.671	155	0.2624	230	0.0706
10	18.02	85	1.451	160	0.2361	235	0.0665
15	14.72	90	1.265	165	0.2128	240	0.0628
20	12.10	95	1.105	170	0.1923	245	0.0597
25	10.00	100	0.9679	175	0.1742	250	0.0570
30	8.311	105	0.8500	180	0.1581		

- By default, the temperature sensor measures the inner temperature of a transducer (sound projectors).

When a transducer (or a projector) transmits sound into water or load medium, fractional electrical energy is being converted to heat by piezoelectric materials. If a transducer (projector) transmits sound continuously underwater for a long time and the transducer can not cool down by ambient water or medium, the materials of the transducer might be overheated and damaged. The maximum inner temperature of BII transducers (projectors) varies from 120 to 200°C, or 248 to 392°F.

There is a formula on [How to determine pulse width, duty cycle and off-time with input pulse power \(peak power\)](#) in the data sheet of each BII transducer (projector).

The temperature inside the transducer (projector) is less than 120 °C or 248 °F during operation by following the formula.

If the transducer (projector) must operate continuously with high power in your application, please contact BII for custom-fit high temperature transducer.

- The temperature sensor can also be built into a hydrophone to measure the ambient temperature.

Operating Static Pressure Range:

The maximum Operating Static Pressure varies with transducer types: ≤ 9.8 MPa, or 1000 m Ocean Depth. 1 m Water Depth = 9.8 kPa.

Underwater Transducer (Projector and Hydrophone) Daily Maintenance (Handling)

Warning: Wrongful handling may damage the transducer and hydrophone beyond repair.

- Do not drive underwater transducers and hydrophones when they are in air. Do not drive projector (transducer) with high power in non-resonance frequency range.
- Do not drive the hydrophones with built-in preamplifiers and/or signal conditioning amplifiers.

3. Keep transducers, hydrophones, and the cable away from any sharp item.
4. Remove saltwater spray and moisture with cloth.
5. Do not move or lift the transducers and hydrophones by the cable in air.
6. Do not squeeze or step on the transducers, hydrophones, and the cable.
7. Handle the transducers and hydrophones gently, avoid impacts and collision.
8. Do not use strong solvents to clean the transducer faces and hydrophone faces.
9. Do not expose the transducers and hydrophones to direct sunlight or excessive heat.
10. After use, rinse the transducers, hydrophones, and the cable in fresh clean water, dry them with cloth.
11. Do not use high pressure water, sand blasting or metal tools to clean the transducer faces and hydrophone faces.

Air Transducer Daily Maintenance (Handling)

Warning: Wrongful handling may damage the air transducer beyond repair.

1. Do not use the air transducers in water.
2. Do not drive the transducers if it has built-in preamplifiers and/or signal conditioning amplifiers.
3. Keep transducers and the cable away from any sharp item.
4. After use, clean and remove moisture, dusts or dirties on the transducers, if any, with cloth.
Do not use strong solvents to clean the transducer faces.
5. Do not move or lift the transducers by the cable in air.
6. Do not squeeze or step on the transducers and the cable.
7. Handle the transducers gently, avoid impacts and collision.
8. Do not expose the transducers to direct sunlight or excessive heat.
9. Do not use high pressure water, sand blasting or metal tools to clean the transducer faces.

Operating Tips to Avoid Damage of BII Devices.

Many electrical breakdowns and damages (overheating, over-voltage, over-current) are caused by accidental wrong wiring and improper power level of a DC supply. Observing following procedures may reduce the chances of the damage.

1. Read the datasheet carefully before wiring and powering the devices. Know well about the maximum and minimum ratings.
2. To avoid physical damage of transducers, please refer to [Underwater Transducer \(Projector & Hydrophone\) Handling](#).
3. Choose suitable DC Supply.
 - a. Use a DC Power Supply with adjustable limits for current and voltage. Make sure its maximum value is less than MAXIMUM RATINGS of BII devices. This can avoid accidental damage.
 - b. Use DC Power Supply with fixed current and voltage which are less than MAXIMUM RATINGS of BII devices.
 - c. Use Batteries whose maximum supply voltages are known and fixed such as 1.5V, 3V, 9V, 12V, 22V, 44V, standard, marine, or automobile Batteries.
Warning: Fully charged and new batteries present higher voltage than the ratings. For example, fully charged 12V Automobile or Marine Battery are from 12.6 to 14.4 VDC. Measure voltage of batteries before assembly and wiring, ensure that voltage of battery pack is less than maximum DC supply voltage of the devices.
4. Calculate the electrical power (current and voltage) suitable for the system setup. Choose accessory components suitable to the power (current and voltage).
5. Operation:
 - a. Set the current limit of DC supply to lowest permitted level, set voltage of DC supply to lowest operating level.
Shut down the power.
 - b. Start wiring among the system components.
Check out the wirings carefully to avoid short circuit, wrong wiring, wrong polarity, etc...
Make sure firm grounding for safety.
Any exposed bare wires must be insulated with insulation material to the electrical safety level.
 - c. Wire the system to DC Power supply.
Turn on power.
 - d. Run the system at the lowest power level. Check out the operation and see if the system runs normally.
Turn off power supply immediately and disconnect power supply if the abnormal occurs such as current limiting from DC supply, etc... Trouble shooting.
6. If operation is normal, and wirings and setup are OK at lowest power level, increase power level gradually by adjusting power supply's voltage and current levels.

Warning: It is buyer's responsibility and liability to make sure all electrical wirings and setup observe local electrical code for safety.

Choose DC Power Supplies for BII Devices and Electrical Safety.

- BII devices are designed to be used with Battery-powered DC Supply for field applications (Ocean, Lake, River, Shore, or NDT).
- Every country has Electrical Safety Code Rule, and in some country such as Canada, each province and territory in Canada has its own legislation and dedicated authority to regulate electrical safety. Please check out details of Electrical Safety Code with electrical safety regulatory authorities in your country and/or province before you purchase products from BII.

We take Electricity Safety Authority (ESA) in Ontario, Canada as an example. **Please carefully read details of clause (b), (c), and (d) which are related to BII devices.**

"You must get approval for electrical products sold or used in Ontario, according to the Ontario Electrical Safety Code Rule 2-022. The following are exceptions:

- (a) Equipment displayed at trade shows
- (b) Certain Equipment connected to the output of a Class 2 power supply or equal
- (c) Portable battery-operated equipment
- (d) Equipment designed for use outside Ontario
- (e) Fluorescent and HID lamps (not self-ballasted)
- (f) Revenue billing devices and electricity meters.

(b) Electrical equipment connected to the output of an approved Class 2 power supply or equal

You don't need approval for electrical equipment connected to the output of:

- an approved Class 2 power supply (certified to CAN/CSA-C22.2 No. 223, or both CAN/CSA C22.2 No. 66.1 and CAN/CSA C22.2 No. 66.3). The output of this power supply may not exceed 100 VA, with the operating voltage not more than 30 V rms, 42.4 V peak, or 60 V dc.
- a limited power supply certified to CAN/CSA-C22.2 No. 60950-1.

Approval is still required for:

- Medical electrical equipment. (Note: Such equipment used in a veterinary practice is not considered medical electrical equipment.)
- Equipment for hazardous locations. The Ontario Electrical Safety Code defines these under Sections 18 and 20.

(c) Portable battery-operated equipment

You don't need approval for battery-operated, portable electrical equipment unless it is:

- located in hazardous locations;
- used to test, measure or show one or more electrical quantities and marked as Category III or IV.

You don't need approval for equipment in hazardous locations powered by at most two button-cell batteries (such as electronic wristwatches, hearing aids, or calculators). This is because they can't cause ignition under normal conditions.

For battery-operated medical devices as defined by the Food and Drugs Act, see Health Canada's approval requirements.

(d) Equipment designed for use outside Ontario

You don't need approval for electrical equipment intended solely for use outside Ontario, because no applicable Canadian standard or certification program applies. An example would be travel adapter plugs for using electrical equipment overseas. "

Please refer to following links for more details on Electricity Safety Authority (ESA) in Ontario, Canada:

<https://esasafe.com/electrical-products/product-approval-exceptions/>

https://esasafe.com/assets/files/esasafe/pdf/Electrical_Safety_Products/Exceptions-Product-Approval-Requirements.pdf

100 VA rated in clause (b), generally, refers to apparent power. VA (volt-amperes) = RMS voltage × RMS current.

Real Power (W) = VA × power factor = RMS voltage × RMS current × cos(φ).

Pulse Power (W) = Energy/Time. Energy is stored in a long period and released quickly as short bursts much higher than continuous rating.

Most BII devices is employed with pulse signal to deliver pulse power such as NDT, SONAR, Echo Sounding with portable batteries.

Batteries, Linear DC Power Supplies, and Switching Mode DC Power Supplies.

1. Batteries such as +1.5V (AA, AAA, C and D), +3V (CR123A and CR2032), +9V, +12V (Marine, Automobile and 23A), 22V and 44V are suitable for low noise and portable applications. The capacity can be increased with parallel or series of multiple batteries (Battery Pack). **Please refer to ESA Clause (c).**

Warning: Fully charged and new batteries present higher voltage than the ratings. For example, fully charged 12V Automobile or Marine Battery are from 12.6 to 14.4 VDC. Measure voltage of batteries before assembly and wiring, ensure that voltage of battery pack is less than maximum DC supply voltage of the devices.

2. A linear DC Power Supply is suitable for indoor applications such as a laboratory acoustic system. **Please refer to ESA Clause (b).**

3. Switching mode Power supply (SMPS) is NOT recommended for low noise applications. SMPS is a noisy source in the system due to its switching frequency (Turning on and off the switches). EMI/RFI is produced due to the current being switched on and off. Large peak "in-rush" surge current may be produced. Some types of SMPS might output very high voltage to destroy other electronic devices in the system if SMPS fails in some extreme cases.

How to Calculate Service Time of a Battery:

1. **Compute the total DC current draw of components.** For example, [BII1067](#) preamp draws 1.55mA of quiescent current. The load current to driving 50m cable is 0.45mA_{rms} with SINE signal of 0.2Vpp and 200kHz. So, the total DC current draw is 2mA.

2. **Check out battery datasheet for battery capacity.** Battery Capacity is measured in milliamp-hours (mAh), or amp-hours (AH). For example, from the datasheet of a 9V battery, it has capacity 700mAh when its voltage drops to 4.8V at constant current draw 10mA. Generally, the datasheet will specify several capacities with different constant current draw. Choose the one close to the current draw of your circuit.

3. **Calculate Service Time with Battery Voltage dropping to 4.8V.**

Service Time of a Battery = Battery Capacity/Current Draw of a Circuit = 700mAh/2mA = 350 hours

Because [BII1067](#) can work at 2.7V, the service time of the battery will be much longer than the one calculated above. If the datasheet of a battery has capacity data with supply voltage dropping to 2.7V, the accurate service time can be computed.

4. Same procedures can be applied to calculation of the service time of a battery or battery pack for high power SONAR system which needs high battery capacity.



Preamplifiers for Hydrophones and Ultrasonics, Low Noise, Low Power, Broadband and Built-in Filter

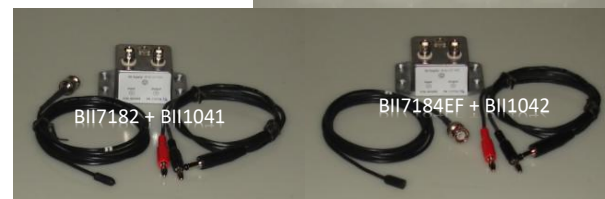
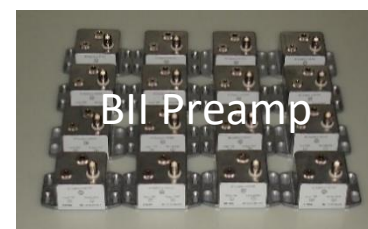
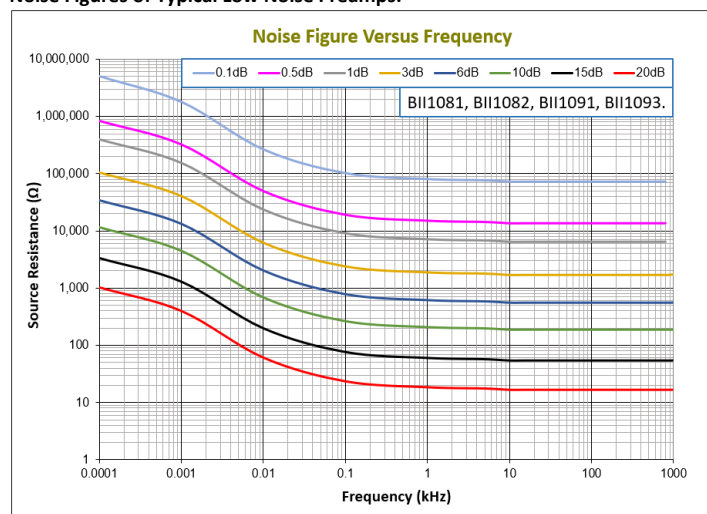
BII's low noise low power preamplifiers (amplifiers) have built-in filters and fixed or digitally programmable gains for uses in broadband (wideband) underwater SONAR, ultrasonic (Ultrasound, NDT, AE.) system and material study.

Typical Applications

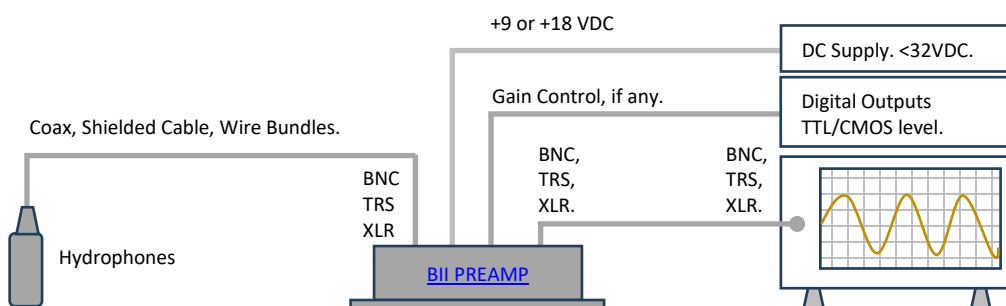
Hydrophone, SONAR, Underwater Communication.	Ultrasonic (Ultrasound, AE, NDT) Testing, Material Characterization.
--	--

Preamplifier	Description	Maximum Cable Length	Features
» BII1030	Variable Gain Voltage Amplifier	1000m or RG58 Coax	1.8 nV/√Hz, 0.8 fA/√Hz, 30 Hz to 10 MHz, 70 dB Variable Gain Range.
» BII1040	Fixed Gain Voltage Amplifier	200m or RG58 Coax	3 nV/√Hz, 4 fA/√Hz, 12 Hz to 17 MHz, 40 dB Gain.
» BII1050	Fixed Gain Voltage Amplifier	305m or RG58 Coax	1 nV/√Hz, 0.8 fA/√Hz, 0.02 Hz to 10 MHz, 40 to 60 dB Gain.
» BII1060 » BII1010	Fixed Gain Voltage Amplifier	1000m	0.1 Hz to 800 kHz, 40 dB Gain, Battery-operated Systems.
» BII1080	Fixed Gain Voltage Amplifier	200m	5.2 nV/√Hz, 3.1 fA/√Hz, 20 to 60 dB Gain, 1 Hz to 2.2 MHz.
» BII1090	Digitally Programmable Gain Voltage Amplifier	1000m or RG58 Coax	1 nV/√Hz, 0.8 fA/√Hz, 0.02 Hz to 8 MHz. Gains: 0, 20, 40, 60, 80 dB.
» BII2000	Automatic Gain Control (AGC) Amplifier	200m	-20 to 80 dB Automatic Gain Control, 100 Hz to 1 MHz Bandwidth.

Noise Figures of Typical Low Noise Preamps.

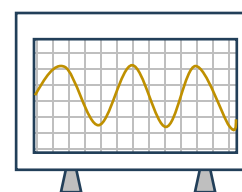


System Wirings of Standalone Preamp.



Digital Recorder,
Computerized DAQ,
Embedded Controller,
Oscilloscope,
Analyzer/Instrument.

Typical Components of an Acoustic Receiving System. Depending on the system requirements, the signal conditioner is optional.



Digital Recorder,
Computerized DAQ,
Embedded Controller,
Oscilloscope,
Analyzer/Instrument.

Accessories of Preamps, Hydrophones, or Acoustical Receiving Instruments:

DC Supply Cable: DCBP24. One 1m DC supply cable with Red and Black Banana Plugs, and DC Power Plug.

To Terminals of DC Supply:

- One Red 4mm Banana Plug.
- One Black 4mm Banana Plug.

Default 1m. Bespoke Length Available.



DC Power Plug.
To DC Power Jack
of the Device.

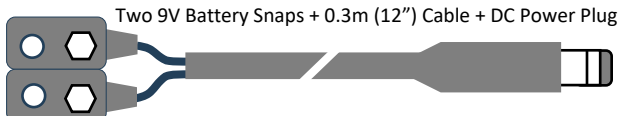


Red Banana Plug: +VDC. **Black Banana Plug:** Common.

One 1m DC supply cable. One end of the cable is with DC Power Plug, another end is Red and Black Banana Plugs. Depending on output terminals of buyer's DC Supply, buyer may assemble other type of connectors to DC supply cable at buyer's cost.

DC Supply Cable: DCBS18V. One 0.3m (12") DC supply cable with two 9V Battery Snaps which supplies +18VDC to amplifiers, and one DC Power Plug.

To Two 9V
Batteries.

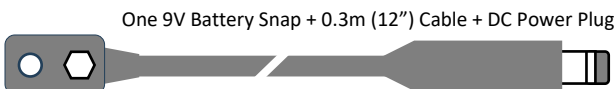


DC Power Plug.
To DC Power Jack
of the Device.



DC Supply Cable: DCBS9V. One 0.3m (12") DC supply cable with one 9V Battery Snaps which supplies +9VDC to amplifiers, and one DC Power Plug.

To One 9V
Battery.



DC Power Plug.
To DC Power Jack
of the Device.

DC Supply Cable: DCWL-24. One 1m DC supply cable with Wire Leads and DC Power Plug.

Default 1m. Bespoke Length Available.

To Clamp Terminals
of DC Supply:



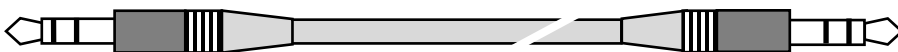
DC Power Plug.
To DC Power Jack
of the Device.

Accessory A1: BNCM-BNCM. Bespoke length RG58, RG174, or RG178 Coax with BNC Male to BNC Male. Default: 0.6m.



→ Recorders and Analyzers with BNC Jack.

Accessory A2: TRS-TRS. Bespoke length cable with 3.5mm TRS Plug to 3.5mm TRS Plug. Default: 1.828m.



→ Recorders and Analyzers with TRS Jack.

Accessory A3: TRS-WL. Bespoke length cable with 3.5mm TRS Plug to Wire Leads. Default: 0.9m.



→ Recorders and Analyzers with Screw-Tighten or Clamp Terminals.

Accessory A4: TRS-XLR3P. Bespoke length cable with 3.5mm TRS Plug to XLR Receptacle with 3 Male Pins. Default: 0.9m.



→ Recorders and Analyzers with XLR Plug with 3 Female Sockets on front panel.

Accessory A5: DIN3P-XLR3P. Bespoke length cable with DIN Receptacle with 3 Male Pins to XLR3 Receptacle with 3 Male Pins. Default: 1m.

DIN, Receptacle,
3 Male Pins.



→ Recorders and Analyzers with XLR Plug with 3 Female Sockets on front panel.

Accessory A6: DIN3P-WL. Bespoke length cable with DIN Receptacle with 3 Male Pins to Wire Leads. Default: 1m.

DIN, Receptacle,
3 Male Pins.



→ Recorders and Analyzers with Screw-Tighten or Clamp Terminals.

Accessory A7: XLR3P-WL. Bespoke length cable with XLR Receptacle Male Pin to Wire Leads. Default: 1m.

XLR Receptacle
with 3 Male



→ Recorders and Analyzers with Screw-Tighten or Clamp Terminals.

Power Amplifiers for SONAR, NDT, HIFU & Dielectric Material Test

BII's power amplifiers are designed to drive underwater and ultrasonic transducers up to 10MHz: SONAR, NDT, HIFU and Ultrasonic Power Transducers. Linear and switch mode (Class D or Switching) power amplifiers are available.

Typical Applications

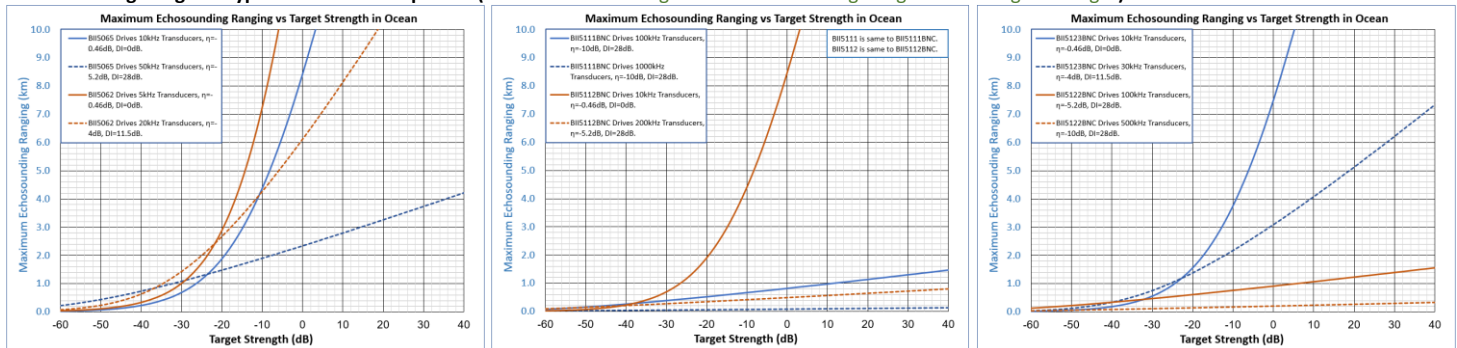
Sub-bottom Profiler, Sea-floor Mapping, Depth Sounder, Noise Generation.	Ultrasonic (NDT) Testing, Dielectric Material Characterization.
Speedometer, Fishing Finding, Navigation, Acoustic Positioning.	Sonic Processing, Process Control, Cavitation, HIFU.
Underwater Communication, Telemetry, Acoustic Beacons, Remote Control.	Marine Animal Acoustics, Bioacoustics, Playback of Marine Sounds.

Power Amplifier	Output Power	Bandwidth	Source Level in Water	Quiescent Current	Type	Package
» BII5000	32 Watt	10 Hz to 120 kHz	184.0 + DI	100mA active, 10μA shutdown.	Linear	PCB
» BII5020	86 Watt	0.1 to 500 kHz	187.0 + DI	36mA active, 16mA shutdown.	Linear	PCB, Metal Enclosure.
» BII5040	31.3 Watt	50 kHz to 10 MHz	182.7 + DI	53mA active, 0.8mA shutdown.	Linear, Pulse.	PCB, Metal Enclosure.
» BII5060	208, 415 Watt	0.1 to 70 kHz	196.0 + DI	59mA Active, 24mA Shutdown.	Linear	PCB, Metal Enclosure.
» BII5100	309 Watt	1 to 500 kHz	194.2 + DI	13mA	Switch Mode (Class D)	PCB, Metal Enclosure.
» BII5110	108/216 Watt	1 kHz to 10 MHz	194.1 + DI	1mA and 2mA	Switch Mode (Class D)	PCB, Metal Enclosure.
» BII5120	138 Watt	20 Hz to 2 MHz	189.2 + DI	65mA active, 6mA shutdown.	Linear	PCB, Metal Enclosure
» BII5160	208 Watt	0.1 to 20 kHz	194.0 + DI	59mA Active, 24mA Shutdown.	Linear	Portable Case

Switch Mode Operation: Input signal of the PA is pulse or square signals with TTL/CMOS level or 2Vpp to 5Vpp.

Source Level (dB μPa*m) in water. **DI** is Directivity Index of a transducer, in dB. **η**: Transducer Efficiency, in dB.

$\eta = -1.55\text{dB}$ ($\eta = 0.7$) is used in the table above. η of broadband planar transducers such as NDT Transducers could be $\leq -5\text{dB}$, η of broadband spherical or cylindrical transducers could be $\geq -0.5\text{dB}$.

Echo Sounding Range of Typical BII Power Amplifiers (Communication Range = 2 * Echo Sounding Range at 0dB Target Strength)**BII6000 Series Matching Network: Impedance Matching between Transducers and Amplifiers**

The complex impedance of a piezoelectric transducer (a projector or a hydrophone) is non-linear and frequency dependent. Its magnitude ranges from hundreds kΩ to several Ω, and its phase varies from -90° to +90°. The transducer is a capacitive, resistive, or inductive load at different frequency ranges. An impedance matching & tuning device is necessary to change the complex impedance of the transducer in a specific frequency range (generally around resonance frequency f_s of the transducer) to meet load requirement of a power amplifier for **maximum and efficient power transfer from the electric to the mechanical**, or to match input impedance of a preamplifier for **maximum and efficient power transfer from the mechanical to the electric**, or achieve the optimum source resistance for **minimum noise factor NF**. Generally, -3dB bandwidth of a transducer with impedance matching and tuning is greater than that of original transducer. Besides, BII also manufactures custom-fit voltage step-up transformers for study of dielectric, ferroelectric, and piezoelectric materials.

Generally there are two kind of acoustic system: Broadband (Low Q) and Narrowband (High Q). **Narrowband Acoustic System** possesses merits of high quality factor, and very low energy loss on the device itself. High sound energy with high efficiency can be delivered to subjects such as HIFU system, Ultrasonic Degassing/Cleaning/Physical & Chemical Processing, etc.. Generally the sounds are continuous sounds, or burst sounds with pulse width greater than 200ms. **Broadband Acoustic System** possesses merits of low quality factor, wide bandwidth, much shorter "ringing" to reach stable state. High sound energy with low or median efficiency can be delivered to subjects such as Ultrasonic NDT system, SONAR Echo-sounding, Communication, AE Detection, etc.. Generally the sounds are pulsing sounds, burst sounds with pulse width of 10ns to 10ms, or continuous/burst communication signals. **BII chooses BII6010 parts (Low Q broadband or High Q Narrowband) suitable for BII transducers. When buyer orders BII6010 for buyer's own transducers, buyer should specify Low Q or High Q BII6010 clearly when ordering.**

Transmit and Receive Switch: SONAR & Ultrasound

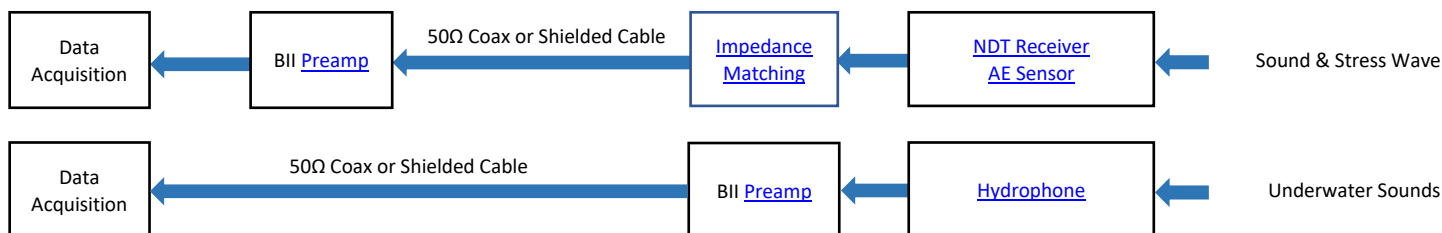


BII T/R switch module provides an integrated solution for a wide range of acoustic applications based on Emitting and Listening Timing Techniques. The device works at active mode (Transmitting Sounds) and passive mode (Listening Sounds) in a half-duplex acoustic system. It integrates an Impedance Matching Network (except BII2100 series), a T/R switch, a bandpass filter, and a low noise DPGA preamplifier (Digitally Programmable Gain Amplifier) or an AGC Amplifier into one compact housing. Gain-selection is accomplished by a two-bit or one-bit digital word (TTL/CMOS level compatible). BII T/R Switches feature low noise level and large dynamic range suitable to detect large signals from short distance and the highly attenuated signals from long distance.

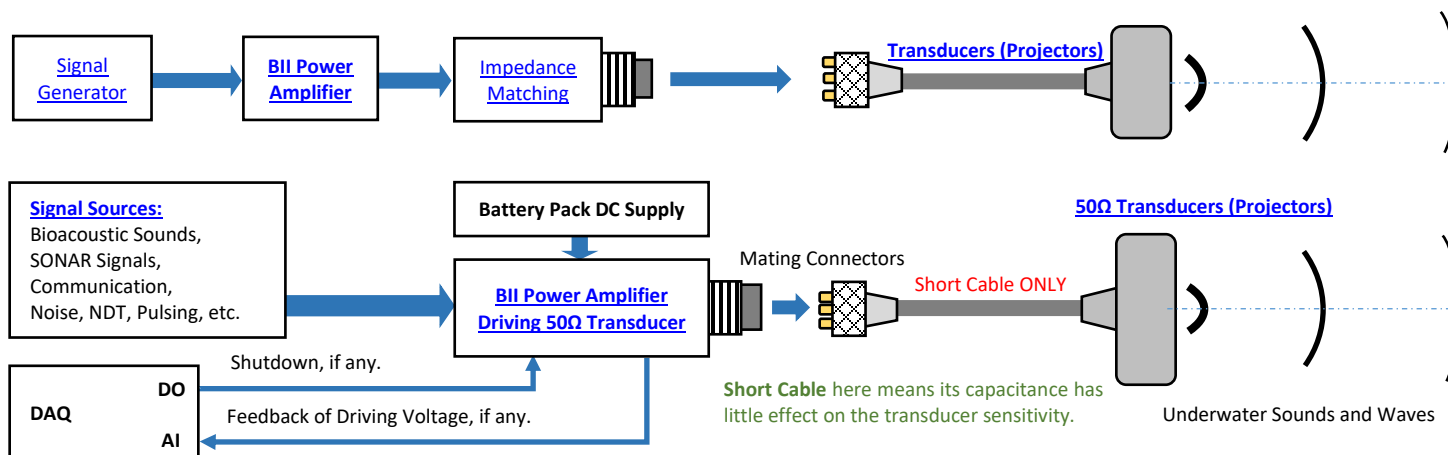
Typical Applications: Echo Sounding SONAR, Positioning/Communication, Artificial Acoustic Target, Marine Bioacoustics, NDT/Diagnostic Ultrasounds.

T/R Switch Modules	Frequency Range	Receiving Gain Range	Gain Control	Built-in Impedance Matching
BII2100 Series	10 Hz to 10 MHz.	20, 40, 60, 80dB.	Digital, TTL/CMOS	No
BII2160 Series	100 Hz to 1 MHz.	-20 to 80 dB.	None, AGC	Yes
BII2170 Series	10 Hz to 10 MHz.	20, 40, 60, 80dB.	Digital, TTL/CMOS	Yes

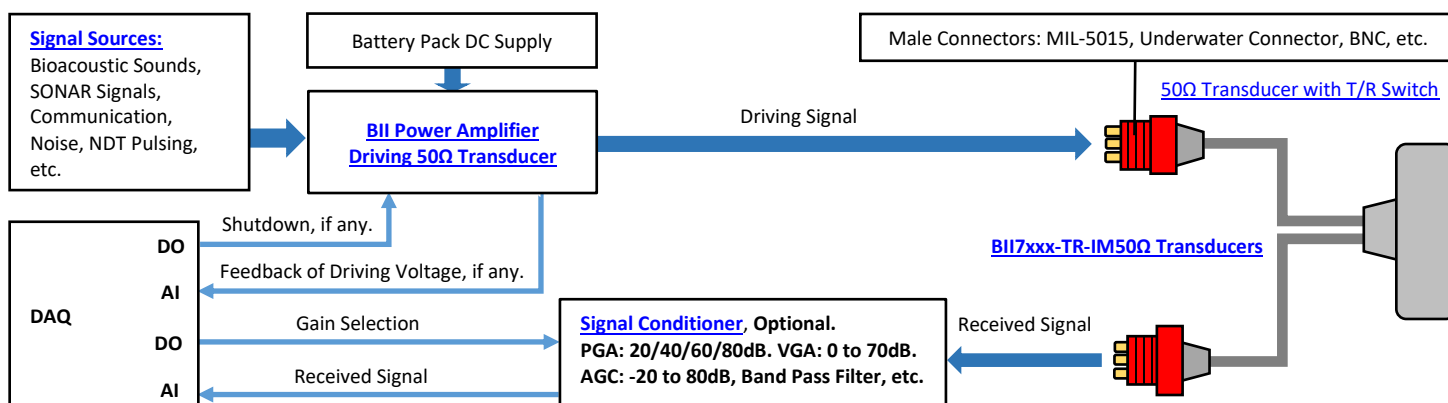
Acoustic Receivers on Underwater Sounds, NDT and AE.

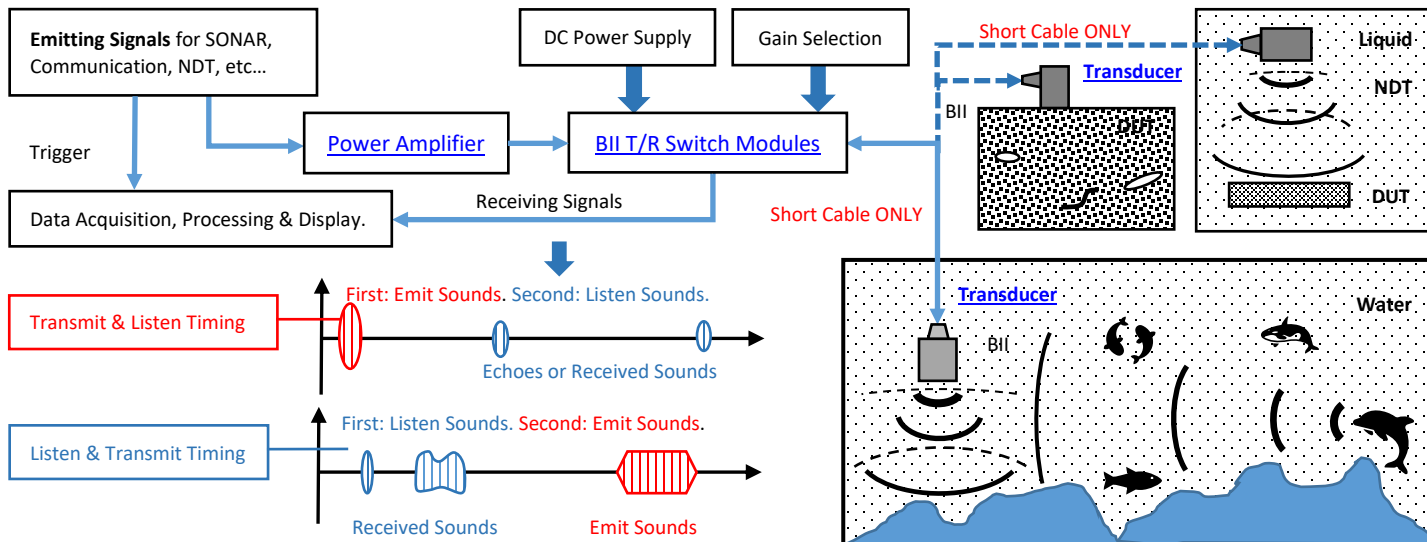


ACOUSTIC SYSTEM CONFIGURATION of GENERATION SOUNDS and WAVES.



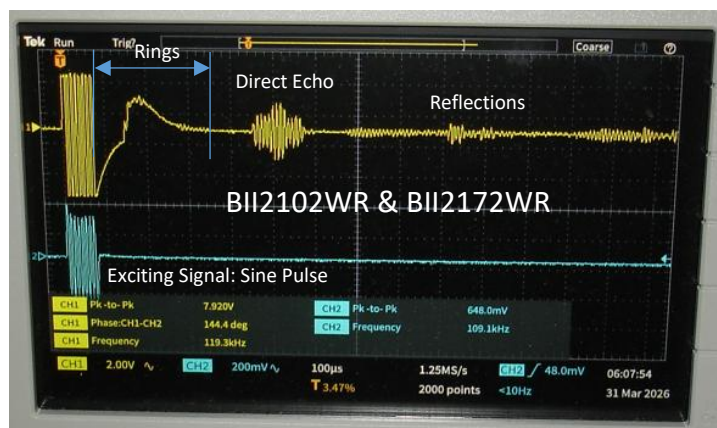
ACOUSTIC SYSTEM CONFIGURATION of TRANSMITTING and RECEIVING SOUNDS AND WAVES.



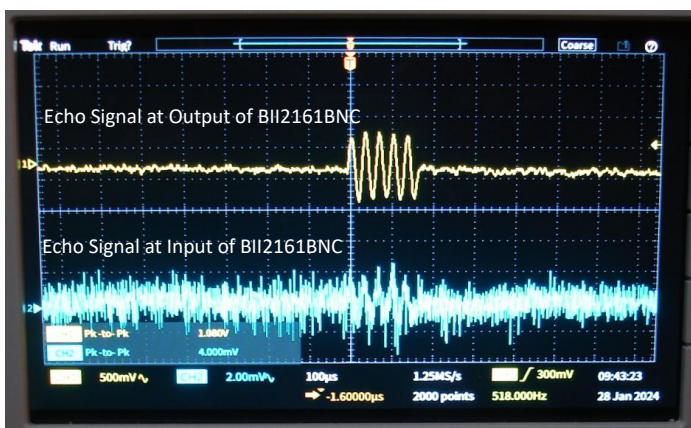


Acoustic Pulse-Echo Systems in a Small-body Water Tank

(1) 110 kHz Sine Pulse, System is Based on BII2102WR or BII2171WR.



(2) 30 kHz Sine Pulse, System is Based on BII2161BNC.



How to Choose BII TR Switch and Transducers (tuned or untuned) according to my acoustic project?

Acoustic Project I: Sub-Bottom Profiling.

Requisites: Electrical Average Power to Transducer $P_{avg} = 1000W$ during 1ms pulse duration; Operating Frequency: 8 kHz.

Transducer Options: (1) A untuned transducer $G_{max} = 0.73mS$, $B = 0.8mS$ at $f_s = 8kHz$.

(2) A series-tuned transducer $R_{TX} = 400\Omega$ at $f_s = 8kHz$.

Step 1. **Signal Types.** Referring to **Typical Signals for Active SONAR...** in BII's **Acoustic Transducers and Measurement Handbook**.

Step 2. Refer to **BII210x Specs:** Max. Transmit Current, Pulse Power Capacity during PD and Maximum Peak Pulse Voltage as function of Pulse Duration and Duty Cycle. Both BII2102WR and BII2102MIL meet limitations above.

Signal Types	Exciting Voltage to Transducer	TR Switch	PD and D
Sine Pulse (SP) to untuned transducer achieving 1000W P_{avg} :	$V_{tx} = \sqrt{8 * P_{avg} / G_{max}} = 3310 V_{pp}$.	BII2102WR	PD = 1mS and $D \leq 1\%$
Burst Square Signal (BSS) to untuned transducer achieving 1000W P_{avg} :	$V_{tx} = \sqrt{4 * P_{avg} / G_{max}} = 2341 V_{pp}$.	BII2102WR	PD = 1mS and $D \leq 1\%$
Sine Pulse (SP) to series-tuned transducer achieving 1000W P_{avg} :	$V_{tx} = \sqrt{8 * P_{avg} * R_{TX}} = 1789 V_{pp}$.	BII2102WR	PD = 1mS, $D \leq 10\%$
Burst Square Signal (BSS) to series-tuned transducer achieving 1000W P_{avg} :	$V_{tx} = \sqrt{4 * P_{avg} * R_{TX}} = 1265 V_{pp}$.	BII2102WR	PD = 1mS, $D \leq 10\%$
		BII2102MIL	PD = 1mS, $D \leq 10\%$

Step 3. **Determine Receiving Bandwidth.** Operating f is 8kHz. Refer to **BII210x Specs:** Input Referred Noise en.

To reduce electronic noise, at same time, to avoid affecting 8kHz echo signal, -3dB bandpass filter of receiving can be set to 0.8 to 80 kHz.

Take BII2102WR and FFVS = -180.0 dBV/ μPa as an example, equivalent pressure noise level at input from electronics, approximately:

$P_n = -110.4 dBV - FFVS$ of the transducer = 69.6 dB μPa .

Acoustic Project II: Underwater Communication.

Requisites: (1) **Electrical Average Power to Spherical Transducer** $P_{avg} = 300 \text{ W}$ during 10 ms pulse duration, $D=10\%$;
(2) Transmit and Receive **Communication Carrier Frequency:** 35 kHz.

Transducer Options: (1) **Untuned spherical transducer** BII7523 $G_{max} = 10.8 \text{ mS}$, $B = 8 \text{ mS}$ at $f_s = 35 \text{ kHz}$.
(2) **Series-tuned spherical transducer** BII7523ST $R_{TX} = 60 \Omega$ at $f_s = 35 \text{ kHz}$.

Step 1. **Signal Types.** Referring to [Typical Signals for Active SONAR...](#) in BII's [Acoustic Transducers and Measurement Handbook](#).

Step 2. Refer to [BII210x Specs](#): Max. Transmit Current, Pulse Power Capacity during PD and Maximum Peak Pulse Voltage as function of Pulse Duration and Duty Cycle.
BII2102WR, BII2102MIL, BII2102BNC, BII2103BNC, meet limitations above.

Signal Types	Exciting Voltage to Transducer	TR Switch	PD and D
Sine Pulse (SP) to untuned transducer achieving 300W P_{avg} :	$V_{tx} = \sqrt{8 * P_{avg} / G_{max}} = 471.4 \text{ V}_{pp}$.	BII2102WR, BII2102MIL, BII2102BNC, BII2103BNC.	PD = 10mS, D = 10%.
Burst Square Signal (BSS) to untuned transducer achieving 300W P_{avg} :	$V_{tx} = \sqrt{4 * P_{avg} / G_{max}} = 333.4 \text{ V}_{pp}$.		
Sine Pulse (SP) to series-tuned transducer achieving 300W P_{avg} :	$V_{tx} = \sqrt{8 * P_{avg} * R_{TX}} = 379.5 \text{ V}_{pp}$.		
Burst Square Signal (BSS) to series-tuned transducer achieving 300W P_{avg} :	$V_{tx} = \sqrt{4 * P_{avg} * R_{TX}} = 268.4 \text{ V}_{pp}$.		

Step 3. **Determine Receiving Bandwidth.** Carrier frequency $f = 35 \text{ kHz}$. Refer to [BII210x Specs](#): Input Referred Noise en.

To reduce electronic noise, at same time, to avoid affecting 35kHz communication signal, -3dB bandpass filter of receiving can be set to 3.5 to 350 kHz.

Take BII2102BNC and BII7523 FFVS = -197.0 dBV/ μPa as an example, equivalent pressure noise level at input from electronics, approximately:

$P_n = -104.0 \text{ dBV} - \text{FFVS of the transducer} = 93.0 \text{ dB } \mu\text{Pa}$.

Acoustic Project III: 1MHz NDT.

Requisites: (1) **Electrical Peak Power to NDT Transducer** $P_{peak} = 2000 \text{ W}_{peak}$, Positive or Negative Voltage Spike: 50 nS to 1 μs , $D \leq 1\%$;
(2) Transmit and Receive **Frequency:** 1 MHz.

Transducer Options: 50 Ω Transducer, $Z = 50 \Omega$, 1 MHz.

Step 1. **Signal Types.** Referring to [Typical Signals for Active SONAR...](#) in BII's [Acoustic Transducers and Measurement Handbook](#).

Step 2. Refer to [BII217x Specs](#): Max. Transmit Current, Pulse Power Capacity during PD and Maximum Peak Pulse Voltage as function of Pulse Duration and Duty Cycle.
BII2174BNC, meet limitations above.

Signal Types	Approximately Exciting Voltage to Transducer	TR Switch	PD and D
Voltage Spike (VS) to 50 Ω transducer achieving 2000W P_{peak} :	$V_{tx} = \sqrt{P_{peak} * R_{TX}} = 316.2 \text{ V}_{peak}$.	BII2174BNC.	PD = 50 nS to 1 μs ; $D \leq 1\%$.

Step 3. **Determine Receiving Noise Level at input.** Refer to [BII217x Specs](#): Input Referred Noise en.

-3dB bandpass filter of receiving is fixed to 10 kHz to 17 MHz.

Take BII2104BNC and FFVS of NDT Transducer = -206.0 dBV/ μPa as an example, equivalent pressure noise level at input from electronics, approximately:

$P_n = -95.6 \text{ dBV} - \text{FFVS of the transducer} = 110.3 \text{ dB } \mu\text{Pa}$.

Acoustic Project IV: 300 kHz Short Distance Underwater Echo Sounding.

Requisites: (1) **Electrical Average Power to NDT Transducer** $P_{avg} = 500 \text{ W}$, Sine Pulse Duration: 30 μs , $D \leq 0.1\%$;
(2) Transmit and Receive **Frequency:** 300 kHz.

Transducer Options: 50 Ω Transducer, $Z = 50 \Omega$, 300 kHz.

Step 1. **Signal Types.** Referring to [Typical Signals for Active SONAR...](#) in BII's [Acoustic Transducers and Measurement Handbook](#).

Step 2. Refer to [BII2161 Specs](#): Max. Transmit Current, Pulse Power Capacity during PD and Maximum Peak Pulse Voltage as function of Pulse Duration and Duty Cycle.
BII2161MIL and BII2161BNC meet limitations above. But BII2161BNC has lower noise level.

Signal Types	Approximately Exciting Voltage to Transducer	TR Switch	PD and D
Sine Pulse (SP) to 50 Ω transducer achieving 500W P_{avg} :	$V_{tx} = \sqrt{8 * P_{avg} * R_{TX}} = 447.2 \text{ V}_{pp}$.	BII2161MIL. BII2161BNC.	PD = 30 μs ; $D \leq 0.1\%$.

Step 3. **Determine Receiving Noise Level at input.** Refer to [BII2161 Specs](#): Input Referred Noise en.

-3dB bandpass filter of receiving set to 50 kHz to 1 MHz.

Take BII2161BNC and FFVS of 50 Ω Transducer = -198.0 dBV/ μPa as an example, equivalent pressure noise level at input from electronics, approximately:

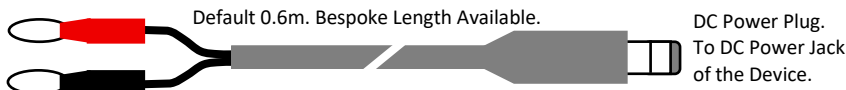
$P_n = -99.0 \text{ dBV} - \text{FFVS of the transducer} = 99.0 \text{ dB } \mu\text{Pa}$.

Accessories of Power Amplifiers, Transducers, or Acoustical Transmitting and Receiving Instruments/Modules:

DC Supply Cable Pair: Part Number **DCBP20**.

To Terminals of DC Supply:

- One Red 4mm Banana Plug.
- One Black 4mm Banana Plug.



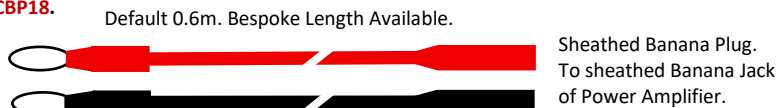
Red Banana Plug: +VDC. Black Banana Plug: Common.

One 0.6m DC supply cable. One end of the cable is with DC Power Plug, another end is Red and Black Banana Plugs.

DC Supply Cable Pair: Part Number **DCBP18**.

To Terminals of DC Supply:

- One Red 4mm Banana Plug.
- One Black 4mm Banana Plug.

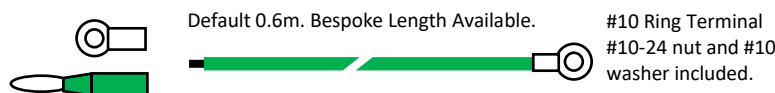


Two 0.6m DC supply cables. Red and Black. One end of the cable is Red and Black Banana Plugs, another end is Sheathed Banana Plug.

Grounding Cable and Terminals

Terminal to buyer's Grounding Terminal:

- Default: Wire Lead
- One #10 Ring Terminal
- One 4mm Banana Plug

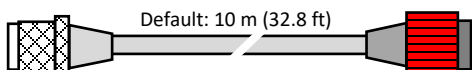


Grounding Cable, Part Number: **GWL18**, Support Single-Point Grounding with Multiple Devices.

One 0.6m AWG 18 Green Wire (or Green with a Yellow Stripe) with #10 Ring Terminal and Wire Lead. One #10 Ring Terminal and one 4mm Banana Plug (Green) are included. Depending on buyer's grounding terminal type, buyer assembles #10 Ring Terminal, 4mm Banana Plug, or other type of connectors to grounding cable at buyer's cost.

Accessory A9: **MIL-UMCF3S**, MIL-5015 (3 Pins) to UMC3S (Underwater Connector, 3 Sockets, Locking Sleeve: DLSA-F, Size: $\Phi 35.5 \times 33.5$ mm)

MIL-5015
3 Pin

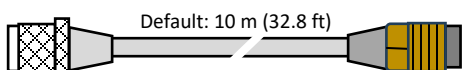


Underwater Connector, 3 Sockets.
Contact 2: Signal.
Contact 1: Common.
Contact 3: Shielding and Grounding.

Transducer with 3-Pin Underwater Connector and DLSA-M Locking Sleeve.

Accessory A10: **MIL-SUMC3S**, MIL-5015 (3 Pins) to Small UMC3S (Underwater Connector, 3 Sockets, Thread Locking, Size: $\Phi 22 \times 28$ mm)

MIL-5015
3 Pin

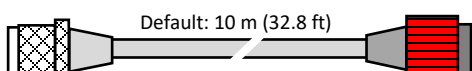


Underwater Connector, 3 Sockets.
Contact 2: Signal.
Contact 1: Common.
Contact 3: Shielding and Grounding.

Transducer with 3-Pin Underwater Connector and MCDLS-F Locking Sleeve.

Accessory A11: **MIL-UMCF2S**, MIL-5015 (3 Pins) to UMC2S (Underwater Connector, 2 Sockets, Locking Sleeve: DLSA-F, Size: $\Phi 35.5 \times 33.5$ mm)

MIL-5015
3 Pin

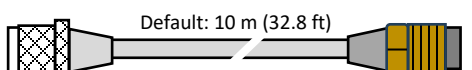


Underwater Connector, 2 Sockets.
Contact 2: Signal.
Contact 1: Common.

Transducer with 2-Pin Underwater Connector and DLSA-M Locking Sleeve.

Accessory A12: **MIL-SUMC2S**, MIL-5015 (3 Pins) to Small UMC2S (Underwater Connector, 2 Sockets, Thread Locking, Size: $\Phi 22 \times 28$ mm)

MIL-5015
3 Pin



Underwater Connector, 2 Sockets.
Contact 2: Signal.
Contact 1: Common.

Transducer with 2-Pin Underwater Connector and MCDLS-F Locking Sleeve.

Accessory A13: **BNC-MIL3S**, BNC Male (Pin) to MIL3S (MIL-5015 type 3-Socket Connector Thread Fastening)

BNC Male

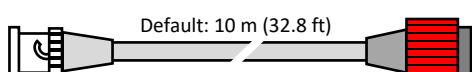


MIL-5015 Connector, 3 Sockets.
Contact C or G: Signal.
Contact B: Common.
Contact A: Shielding and Grounding.

Transducer with 3-Pin MIL-5015 Connector.

Accessory A14: **BNC-UMCF3S**, BNC Male to UMC3S (Underwater Connector, 3 Sockets, Locking Sleeve: DLSA-F, Size: $\Phi 35.5 \times 33.5$ mm)

BNC Male

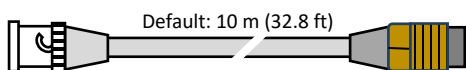


Underwater Connector, 3 Sockets.
Contact 2: Signal.
Contact 1: Common.
Contact 3: Shielding and Grounding.

Transducer with 3-Pin Underwater Connector and DLSA-M Locking Sleeve.

Accessory A15: BNC-SUMC3S, BNC Male to Small UMC3S (Underwater Connector, 3 Sockets, Thread Locking, Size: $\Phi 22 \times 28 \text{ mm}$)

BNC Male



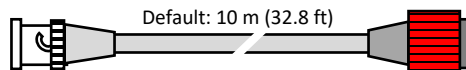
Underwater Connector, 3 Sockets.
Contact 2: Signal.
Contact 1: Common.
Contact 3: Shielding and Grounding.



Transducer with 3Pin Underwater Connector and MCDLS-F Locking Sleeve.

Accessory A16: BNC-UMCF2S, BNC Male to UMC2S (Underwater Connector, 2 Sockets, Locking Sleeve: DLSA-F, Size: $\Phi 35.5 \times 33.5 \text{ mm}$)

BNC Male



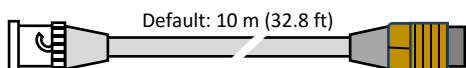
Underwater Connector, 2 Sockets.
Contact 2: Signal.
Contact 1: Common.



Transducer with 2-Pin Underwater Connector and DLSA-M Locking Sleeve.

Accessory A17: BNC-SUMC2S, BNC Male to Small UMC2S (Underwater Connector, 2 Sockets, Thread Locking, Size: $\Phi 22 \times 28 \text{ mm}$)

BNC Male



Underwater Connector, 2 Sockets.
Contact 2: Signal.
Contact 1: Common.



Transducer with 2-Pin Underwater Connector and MCDLS-F Locking Sleeve.

Are 50 Ω Power Amplifiers suitable to drive non-50 Ω transducers?

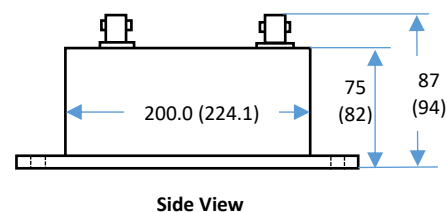
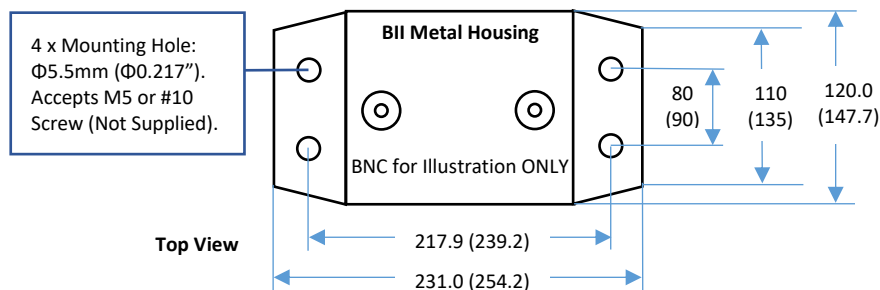
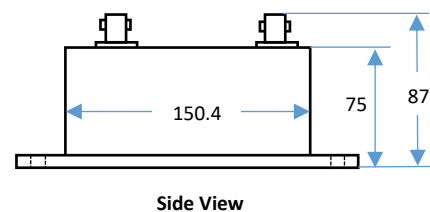
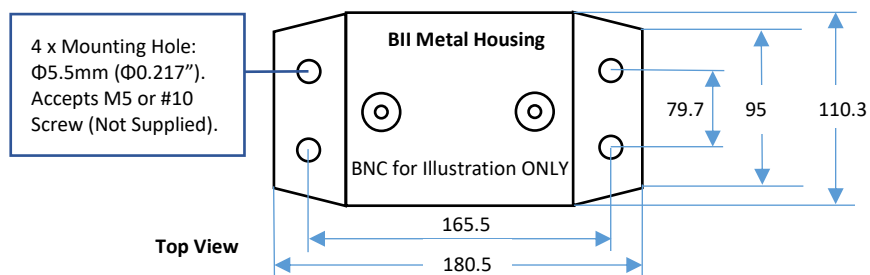
- (1) if the impedance of a transducer is greater than 50 Ω at operating frequency, the 50 Ω Power Amplifiers can drive this non-50 Ω transducer, but the power delivered to non-50 Ω transducer is reduced.
- (2) It is NOT recommended for 50 Ω Power Amplifiers to drive the transducer whose impedance is less than 50 Ω . The power amplifier is most likely to be damaged by overvoltage, exceeding rated power or overheat.

How do I connect the wire of BII Power Amplifier to my devices?

WARNING: HIGH VOLTAGES MAY BE PRESENT AT THE OUTPUT OF THIS UNIT. DO NOT TOUCH THE DEVICE, ITS WIRES, CABLES, AND CONNECTORS BEFORE THE POWER SUPPLIES AND SIGNAL SOURCES ARE SHUT DOWN.

1. **Wire Splicing Methods: Soldering or Crimp.** 2. **Proper Insulation for Safety:** All exposed bare wires, metal wires, wire leads, solders, and joints are insulated with insulation material such as heat shrink tubing, fully insulated wire splice connector, etc. The insulation voltage must be greater than twice the maximum voltage of the device.
3. **Grounding the device** (including metal chassis and/or metal housing, cable shield, etc.) firmly for operation safety.

Typical Metal Housings, Outline Dimensions (mm), Illustration only, the scale is not 1:1.



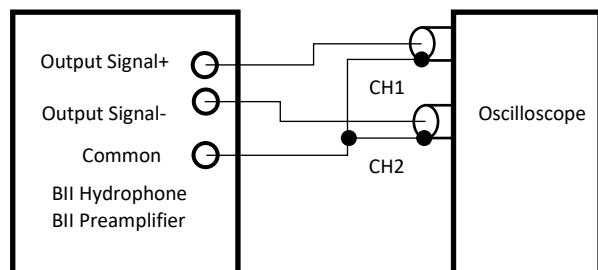
Wiring Information

Wirings of BII Hydrophones (Sound Receiver, AE Sensors, NDT Receivers, etc.)

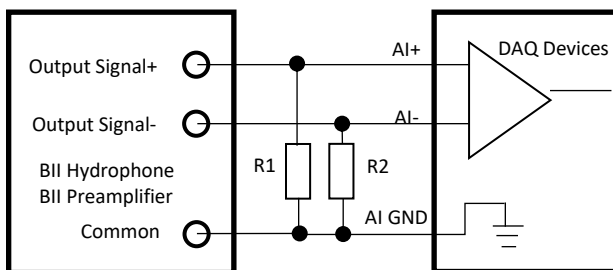
1. **Preamplifier Wirings to DAQ (Data Acquisition):** DAQ: Data Acquisition Hardware; AI: Analog Input; CH: Channel; GND: Ground.

R1 and R2 resistors are NOT necessary for most applications. If DAQ saturation occurs, use $R1 = R2 = 10k\Omega$ to $1M\Omega$ resistors.

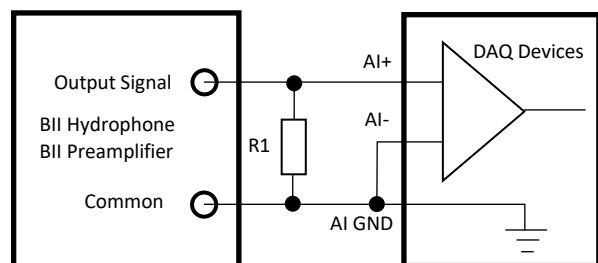
BII's Differential Output to BNC Input of an Oscilloscope



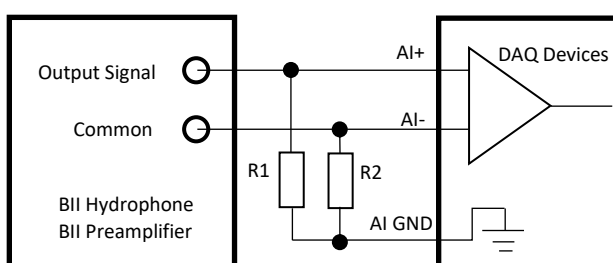
BII's Differential Output to Differential Input of a DAQ



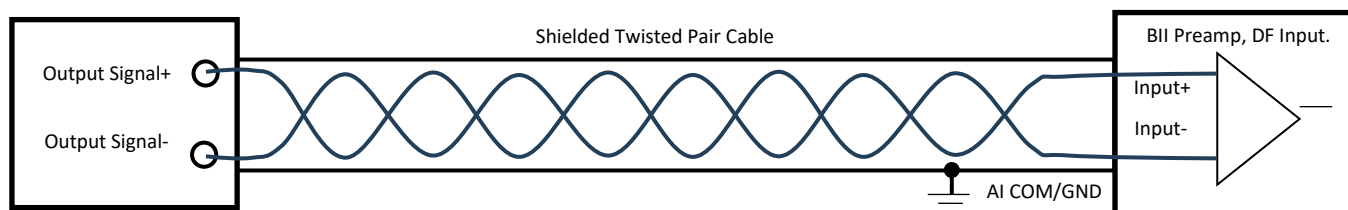
BII's Single-Ended Output to Single-Ended Input of a DAQ



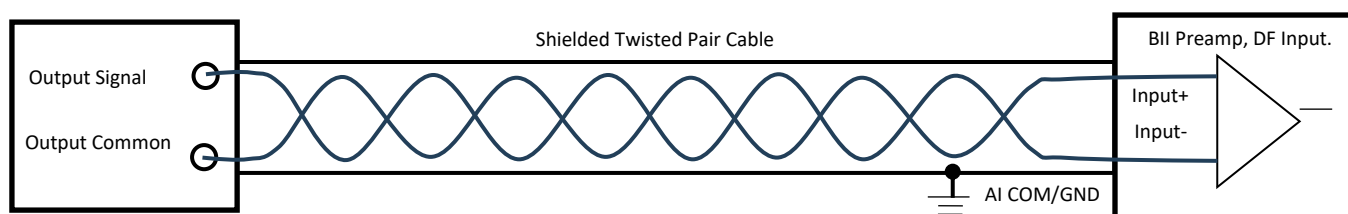
BII's Single-Ended Output to Differential Input of a DAQ



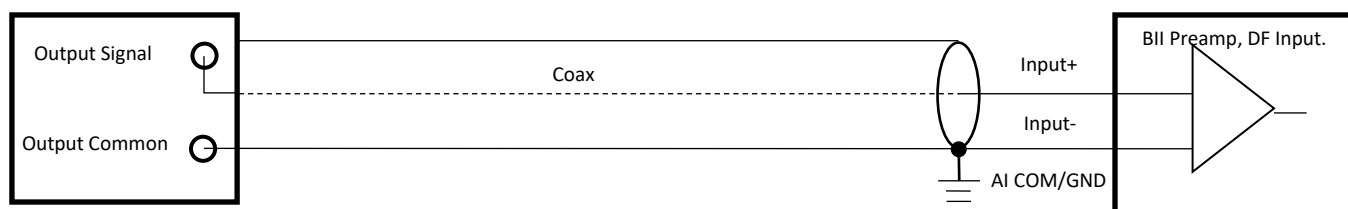
BII's Differential Sound Receiver to Differential Input of a BII Preamp (Signal Conditioner)



BII's Single-ended Receiver to Differential Input of a BII Preamp



BII's Single-ended Receiver to Differential Input of a BII Preamp



Driving 100Ω Balanced Twisted Pair Cable/Wires and 50 or 75 Ω Coax.

(1) Impedance of most Balanced Twisted Pair Cable/Wire is from 100Ω to 150Ω.

BII preamp has 100Ω output impedance or bespoke impedance to match the impedance of Balanced Twisted Pair Cable/Wires.

(2) Impedance of most Coax is 50Ω or 75Ω.

BII preamp has 50Ω output impedance or bespoke 75Ω impedance to match the impedance of coaxes.

2. Hydrophone (Sound Receiver, AE Sensor, NDT Receiver, etc.) Wiring without Built-in Preamplifier. Refer to [Abbreviation List for Ordering](#).

Differential Output:	Wire Leads	UMC3P	DIN3	TRS	XLR3
Signal +	White or Red	Pin 2	Pin 3	Tip, Positive/Hot	Pin 2, Positive/Hot.
Signal -	Black	Pin 1	Pin 1	Ring, Negative/Cold	Pin 3, Negative/Cold.
Common & Shielding	Shield	Pin 3	Pin 2	Sleeve, Ground/Common	Pin 1, Shield/Ground.
Single Ended Output:	Wire Leads	UMC3P	DIN3	BNC/SMA/SMC	Coax with Wire Leads
Signal	White or Red	Pin 2	Pin 3	Center Contact	Coax Center Contact
Signal Common	Black	Pin 1	Pin 1	Shield	Coax Shield
Shielding	Shield	Pin 3	Pin 2	Shield	Coax Shield
Wiring of Unshielded Cable:	Wire Leads WL	UMC2P (0.6m USC Cable originally coming from manufacturer of the connector, Fixed.). Locking Sleeve: DLSA-M.			
Signal	White	Contact 2			
Signal Common	Black	Contact 1			

3. Hydrophone (Sound Receiver, AE Sensor, NDT Receiver, etc.) Wiring with Built-in Preamplifier. Refer to [Abbreviation List for Ordering](#).

(1) Wiring Information of Hydrophones with Fixed-gain Preamps:

Single Ended Output:	Wire Leads	BNC + 9V BS	UMC4P/XLR4	DIN4	XLR3 + 9V BS	TRS + 9V BS
+VDC	Red	Female Snap	Pin 3	Pin 4	Female Snap	Female Snap
Common	Black	Male Snap	Pin 1	Pin 1	Male Snap	Male Snap
Signal	White	Center Pin	Pin 2	Pin 3	DIN Pin 3	TRRS Tip
Signal Common	Blue, Green, or Yellow.	BNC Shield	Pin 4	Pin 2	DIN Pin 1 & 2	TRRS Ring and Sleeve
Shielding	Shield	N/A	Metal Shell	Metal Shell	DIN and XLR Metal Shell	N/A
Differential Output:	Wire Leads	UMC4P/XLR4P	DIN4P	DIN3/XLR3 + 9V BS	BNC + 9V BS	TRS + 9V BS
+VDC	Red	Pin 3	Pin 4	Female Snap	Female Snap	Female Snap
Common	Black	Pin 1	Pin 1	Male Snap	Male Snap	Male Snap
Signal+	White	Pin 2	Pin 3	DIN3 Pin 3	XLR3 Pin 2	#1 BNC Center
Signal-	Blue, Green, or Yellow	Pin 4	Pin 2	DIN3 Pin 1	XLR3 Pin 3	#2 BNC Center
Signal Common	N/A	Pin 1	Pin 1	DIN3 Pin 2	XLR3 Pin 1	BNC Shell
Shielding	Shield	Metal Shell	Metal Shell	DIN3 and XLR3 Metal Shell	N/A	N/A
4mm Banana Plug Pair: Red Plug for +VDC, Black Plug for Common of the DC power supply.						

(2) Wiring Information of Hydrophones with One-Bit-Word Programmable Gain Preamps:

Single Ended Output:	Wire Leads	UMC6P/XLR6	DIN6	BNC + 9V BS	DIN3/XLR3 + 9V BS	TRS + 9V BS
+VDC	Red	Pin 3	Pin 4	Battery Female Snap	Battery Female Snap	Battery Female Snap
Common	Black	Pin 1	Pin 1	Battery Male Snap	Battery Male Snap, DIN Pin 2 or XLR Pin 1.	Battery Male Snap, TRS Sleeve.
Output Signal	White	Pin 2	Pin 3	BNC Center	DIN Pin 3	TRRS Tip
Signal Common	Green	Pin 4	Pin 2	BNC Shield	DIN Pin 1	TRRS Ring
Digital A0 (FFVS Selection)	Blue	Pin 6	Pin 5	Blue	Blue	Blue
Digital Common	Yellow or Brown	Pin 5	Pin 6	Yellow or Brown	Yellow or Brown	Yellow or Brown
Shielding	Shield	Metal Shell	Metal Shell	Shield	XLR Metal Shell	N/A
Differential Output:	Wire Leads	UMC6P/XLR6	DIN6	BNC + 9V BS	DIN3/XLR3 + 9V BS	TRS + 9V BS
+VDC	Red	Pin 3	Pin 4	Battery Female Snap	Battery Female Snap	Battery Female Snap
Common	Black	Pin 1	Pin 1	Battery Male Snap, BNC Shield.	Battery Male Snap, DIN Pin 2 or XLR Pin 1.	Battery Male Snap, TRS Sleeve.
Output Signal+	White	Pin 2	Pin 3	"1" BNC Center Pin	DIN Pin 3	TRRS Tip
Output Signal -	Green	Pin 4	Pin 2	"2" BNC Center Pin	DIN Pin 1	TRRS Ring
Digital A0	Blue	Pin 6	Pin 5	Blue	Blue	Blue
Digital Common	Yellow or Brown	Pin 5	Pin 6	Yellow or Brown	Yellow or Brown	Yellow or Brown
Shielding	Shield	Metal Shell	Metal Shell	BNC Shield	Metal Shell	N/A
4mm Banana Plug Pair: Red Plug for +VDC, Black Plug for Common of the DC power supply.						

Selecting Sensitivity of One-Bit-Word Digitally Programmable

FFVS Selection Wire A0	BII7XXPGDF Sensitivity FFVS at 1kHz.				BII7XXPGSE Sensitivity FFVS at 1kHz.	
0 (Logic Low)	xxx + 6 dB V/ μ Pa	xxx + 10 dB V/ μ Pa	xxx + 20 dB V/ μ Pa	xxx + 30 dB V/ μ Pa	xxx + 20 dB V/ μ Pa	xxx + 6 dB V/ μ Pa
1 (Logic High)	xxx + 46 dB V/ μ Pa	xxx + 40 dB V/ μ Pa	xxx + 50 dB V/ μ Pa	xxx + 60 dB V/ μ Pa	xxx + 60 dB V/ μ Pa	xxx + 46 dB V/ μ Pa

(3) Wiring Information of Hydrophones with Two-bit Programmable Gain Preamps:

Single Ended Output:	Wire Leads	BNC + 9V BS	UMC6P/XLR6	DIN6	DIN3/XLR3 + 9V BS	TRS + 9V BS
+VDC	Red	Female Snap	Pin 3	Pin 4	Female Snap	Female Snap
Common	Black	Male Snap	Pin 1	Pin 1	Male Snap	Male Snap
Digital Common		Black			Black	Black
Output Signal	White	BNC Center	Pin 2	Pin 3	DIN Pin 3	TRRS Tip
Signal Common	Green	BNC Shield	Pin 4	Pin 2	DIN Pin 1 & 2	TRRS Ring and Sleeve

Digital A1	Yellow or Brown	Yellow or Brown	Pin 5	Pin 6	Yellow or Brown	Yellow or Brown
Digital A0	Blue	Blue	Pin 6	Pin 5	Blue	Blue
Shielding	Shield	Shield	Metal Shell	Metal Shell	Metal Shell	N/A
Differential Output:	Wire Leads	UMC6P/XLR6	DIN6	BNC + 9V BS	DIN3/XLR3 + 9V BS	TRS + 9V BS
+VDC	Red	Pin 3	Pin 4	Female Snap	Female Snap	Female Snap
Common	Black	Pin 1	Pin 1	Male Snap	Male Snap, DIN Pin 2 or XLR Pin 1.	Male Snap, TRS Sleeve.
Digital Common				BNC Shield	Black	Black
Output Signal +	White	Pin 2	Pin 3	"1" Center Pin	DIN Pin 3	TRS Tip
Output Signal -	Green	Pin 4	Pin 2	"2" Center Pin	DIN Pin 1	TRS Ring
Digital A1	Yellow or Brown	Pin 5	Pin 6	Yellow or Brown	Yellow or Brown	Yellow or Brown
Digital A0	Blue	Pin 6	Pin 5	Blue	Blue	Blue
Shielding	Shield	Metal Shell	Metal Shell	BNC Shield	Metal Shell	N/A

4mm Banana Plug Pair: Red Plug for +VDC, Black Plug for Common of the DC power supply.

Selecting Sensitivity FFVS of Two-bit Digitally Programmable

FFVS Selection Wire A1	FFVS Selection Wire A0	Hydrophone Sensitivity at 1kHz
0 (Logic Low)	0 (Logic Low)	FFVS of Sensing Element + 0 V/μPa
0 (Logic Low)	1 (Logic High)	FFVS of Sensing Element + 20 V/μPa
1 (Logic High)	0 (Logic Low)	FFVS of Sensing Element + 40 V/μPa
1 (Logic High)	1 (Logic High)	FFVS of Sensing Element + 60 V/μPa

Wirings of BII Transducers (Projectors)

WARNING: DANGER — HIGH VOLTAGE on wires. Wires shall be insulated for safety. DO NOT TOUCH THE WIRES BEFORE THE DRIVING SIGNAL IS SHUT DOWN. Cable shield must be grounded firmly for safety.

for 50Ω BNC connector, it is buyer's sole responsibility to make sure that the BNC shield of the signal source is firmly grounded for operating safety before hooking up transducer/hydrophone to the signal source. Coax with BNC is not intended for hand-held use at voltages above 30Vac/60Vdc.

1. Wiring Information of a Transducer.

Cables will be labelled with #1, #2, #3, #4, #5 ...for multiple arrays inside a transducer.

Transducer Wiring:	Shielded Cable	Coax, BNC.	UMC3P, Locking Sleeve: DLSA-M.	MIL3P	DIN3P	XLR3P
Signal:	White or Red	Center Contact	Contact 2	Contact C or G	Pin 3	Pin 2
Signal Common:	Black	Shield	Contact 1	Contact B	Pin 1	Pin 3
Shielding and Grounding	Shield	Shield	Contact 3	Contact A	Pin 2	Pin 1
Please contact us for bespoke wirings of differential transducers such as dipole, quadrupole, multimode rings, and flextensional sources.						
Wiring of Unshielded Cable:	Wire Leads WL	UMC2P (0.6m USC Cable originally coming from manufacturer of the connector, Fixed.). Locking Sleeve: DLSA-M.				
Signal	White	Contact 2				
Signal Common	Black	Contact 1				

Wiring Information of Temperature Signal, if any.

Temperature Sensor Wiring:	Shielded Cable	Coax, BNC	Underwater Connector UMC2P. Locking Sleeve: DLSA-M.	DIN3S	TRS Plug
Signal:	White or Red	Center Contact	Contact 2	Socket 3	Tip
Signal Common:	Black	Shield	Contact 1	Socket 1	Ring
Shielding and Grounding	Shield	Shield	N/A	Socket 2	Sleeve

2. Wiring Information of Transmitting Sounds of a Transducer with T/R Switch.

Transducer Wiring:	Shielded Cable	Coax, BNC.	UMC3P, Locking Sleeve: DLSA-M.	MIL3P	DIN3P	XLR3P
Signal:	White or Red	Center Contact	Contact 2	Contact C or G	Pin 3	Pin 2
Signal Common:	Black	Shield	Contact 1	Contact B	Pin 1	Pin 3
Shielding and Grounding	Shield	Shield	Contact 3	Contact A	Pin 2	Pin 1
Please contact us for bespoke wirings of differential transducers such as dipole, quadrupole, multimode rings, and flextensional sources.						
Wiring of Unshielded Cable:	Wire Leads WL	UMC2P (0.6m USC Cable originally coming from manufacturer of the connector, Fixed.). Locking Sleeve: DLSA-M.				
Signal	White	Contact 2				
Signal Common	Black	Contact 1				

Wiring Information of Receiving Sounds of a Transducer with T/R Switch.

Differential Output:	Wire Leads	UMC4P/XLR4P	DIN4P	DIN3P/XLR3P + 9V BS	TRS + 9V BS
+VDC	Red	Pin 3	Pin 4	Battery Female Snap	Battery Female Snap
Common	Black	Pin 1	Pin 1	Battery Male Snap	Battery Male Snap
Signal+	White	Pin 2	Pin 3	DIN Pin3	TRS Tip
Signal-	Blue, Green, or Yellow	Pin 4	Pin 2	DIN Pin1	TRS Ring
Signal Common	N/A	Pin 1	Pin 1	DIN Pin2	TRS Sleeve
Shielding	Shield	Metal Shell	Metal Shell	Metal Shell	N/A
Optional DC Supply Connector: 4mm Banana Plug Pair, Red Plug for +VDC, Black Plug for Common of the DC power supply.					

Do-It-Yourself or BII Repair and Maintenance

Following information are with the assumption that the products are repairable. Some kinds of damages are unrepairable such as the piezoelectric material is overheated and destroyed.

General Information for Do-It-Yourself Repair and Maintenance of Transducers.

Before you do it yourself to repair and maintenance, please refer to BII's [IMPORTANT NOTICE AND DISCLAIMER](#). Repairman (Buyer or End User) is responsible and liable to the repair and maintenance.

If a transducer has minor damages such as dents, scratches, or cracks on housing, and broken cable, end user may fix these issues by Do-It-Yourself.

	Do-It-Yourself Repair and Maintenance	BII Repair and Maintenance
Advantage	Fast, convenient, and save money.	1. Re-calibration after repair. 2. Free repair and warranty extension if warranty is NOT expired or not voided. 3. BII is responsible to the repair specified in BII sales terms and conditions in the quote.
Disadvantage	1. Know well about the materials, processes, tools, and equipment. 2. Warranty is voided even if the original warranty is NOT expired. 3. Purchase repair compounds, tools, and equipment from third-party company. 4. Repairman (Buyer or End User) are liable and responsible to the repair and maintenance.	1. Shipping cost to BII, and 2 to 4 weeks Repair Period. 2. Repair fees if warranty is expired or voided.

BII Transducers	Repair Compounds
Metal Housing	Excellent adhesion and water-proof sealing with marine epoxy, polyurethane, and rubber potting compounds.
Plastic Housing	Excellent adhesion and water-proof sealing with marine epoxy, polyurethane, and rubber potting compounds.
Rubber Housing	Good adhesion and water-proof sealing with polyurethane and other rubber potting compounds.
Cable	Good adhesion and water-proof sealing with flexible marine epoxy, polyurethane, and rubber potting compounds. Cable jackets are PVC, EPDM, PUR, or other rubbers.
Connectors	Excellent adhesion and water-proof sealing with marine epoxy, polyurethane, and rubber potting compounds.
Repair Processes	
Before repair and maintenance, it is strongly recommended to test the adhesion between cured compounds and BII products by applying a small portion of compounds to a specific small repair area, and then following the curing instruction of the compounds.	
Surface Preparation:	1. Roughen surface with files or coarse sandpapers to provide the best adhesion. 2. Clean surface area of dirt, grease, oil, or loose debris. Use detergent or degreaser such as Isopropyl Alcohol to clean the surface to achieve best adhesion. 3. Some potting compounds might need primer to achieve better adhesion on rubber housing and cable.
Weighting and Mixing:	Following instructions of potting compounds.
Curing:	Following curing instructions of potting compounds. 1. It is strongly recommended to cure mixed compounds with temperature $\leq 60^{\circ}\text{C}$ or 140°F. 2. Ensure enough curing time. 3. After turning off oven, keep devices in oven and ensure devices to cool down to room temperature by themselves.

Repair and Maintenance of Electronic Products or Instruments.

BII does not suggest Do-It-Yourself for electronic products or instruments.

1. Contact BII for RMA number. BII will provide tech support and determine if or not the goods are repairable.

If the goods are repairable, BII will issue RMA number, and send quote to buyer if the warranty is expired.

2. After receiving RMA number, send electronic products or instruments back to BII for repair.

BII Repair and Maintenance of Transducers.

1. Contact BII for RMA number. BII will provide tech support and determine if or not the transducer is repairable.

If the transducer is repairable, BII will issue RMA number, and send quote to buyer if the warranty is expired.

2. After receiving RMA number, send the transducer back to BII for repair.

Acoustic Transducer Installation/Mounting

1. **Maximum Depths** listed in mounting parts description are the ratings of mounting/installation parts which are NOT the maximum depth ratings of transducers and hydrophones. Please refer to datasheet of respective transducers/hydrophones for their maximum depth underwater.
2. **Threadlockers** are recommended to prevent threaded fasteners from loosening due to shock and vibration. Threadlockers is NOT provided by BII.
3. **Maximum Diameter** listed in this table is the size of the mounting part, is NOT the size of the transducer (hydrophone and projector).
4. If the connector at the cable end is larger than the mounting hole, BII does NOT assemble the connector to cable end, and BII ships it to buyer. It is buyer's responsibility to assemble the connector to cable end at buyer's cost.

Cable Options

Cable Types: Wires, Coax Cables, Shielded Multi Conductor Cables, Shielded Cable with Twisted Pair, Unshielded Cable.

Cable Jacket: PVC, Polyurethane, EPDM rubber, FEP, PTFE, etc...

Cable Diameter Range: $\Phi D = \Phi 1.4$ to $\Phi 15.3$ mm

Voltage and Current Rating: Up to 600 Vrms, 10A.

Service Temperature Range: Up to -40° to 840° F or -40° to 449° C.

Connector

WL: Wire Leads

BNC: BNC Male (or Plug) 50 Ω

BNCF: BNC Female (or Jack) 50 Ω

SMA: Plug, Male Pin, 50 Ω

SMC: Plug, Female Socket, 50 Ω

LEMO: PLUG MALE 3 Pins.

TRS: TRS Plug

TRS35: 3.5mm or 1/8" TRS Plug

XLR: XLR Plug

5015: MIL-C-5015 style, Pins.

SP: Solder Pins

BS: +9V Battery Snap

UMC: Underwater Mateable Connector

Custom: Customized Connector

Note: Underwater Connectors are for uses underwater, other connectors and wire leads listed above are for dry uses ONLY in air.

Maximum Diameter Size of Connectors: useful to choose mounting/installation parts of the transducers/hydrophones, Unit: mm.

Connector:	Male BNC	SMA (Plug, Male Pin)	SMC (Plug, Female Socket)	LEMO (PLUG MALE 3 Pins)	1/8" (3.5mm) TRS	XLR Plug	MIL-C-5015	9V Battery Snap	Underwater Connector
Max. Size:	$\Phi 14.3$	$\Phi 9.24$	$\Phi 6.4$	$\Phi 9.5$	$\Phi 10.5$	$\Phi 20.2$	$\Phi 30$	$\Phi 13$	$\Phi 21.5$ or $\Phi 35$

Free-hanging (FH) with Cable Gland

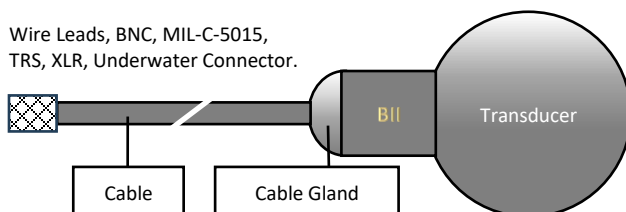
Maximum Depth: Plastics 300 m; Stainless Steel 2000 m.

Cable-Out: By default, the cable goes out of the device from the end face. To save space and have the device shorter, the cable can go out from the side wall transducers. Specify this customization when ordering.

Maximum diameters of typical free-hanging mounting parts (mm): 1. **Plastics:** $\Phi 16$, $\Phi 21$, $\Phi 25$, $\Phi 37$..., 2. **Stainless Steel:** $\Phi 22$.



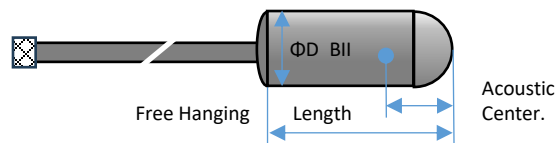
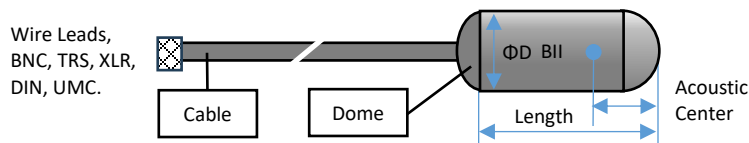
Wire Leads, BNC, MIL-C-5015, TRS, XLR, Underwater Connector.



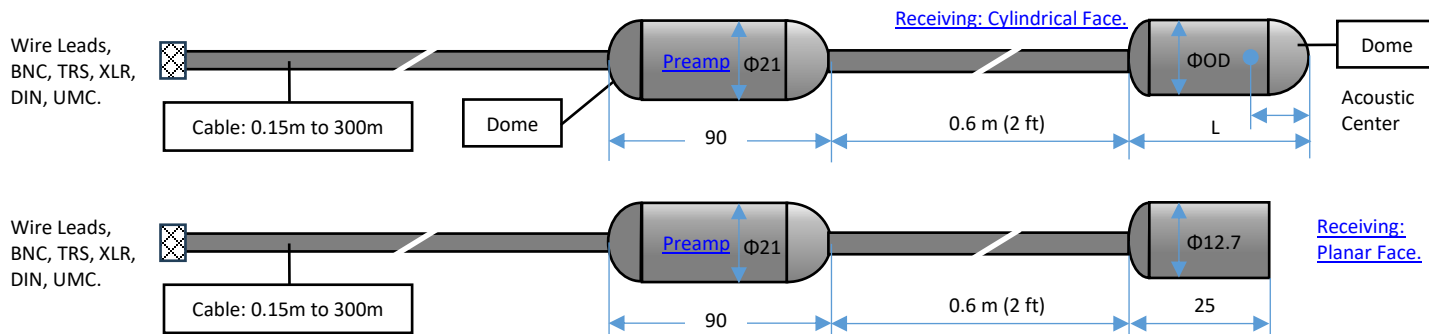
The streamlined hemispherical domes minimize drag forces and hydrodynamic noises caused by the hydrophone in motion or the flow past the hydrophone.

[BII7011/BII7012 Hydrophones:](#) Default Free Hanging with Smooth Domes.

[BII7011/BII7012 Hydrophones:](#) Free Hanging with Short Length.



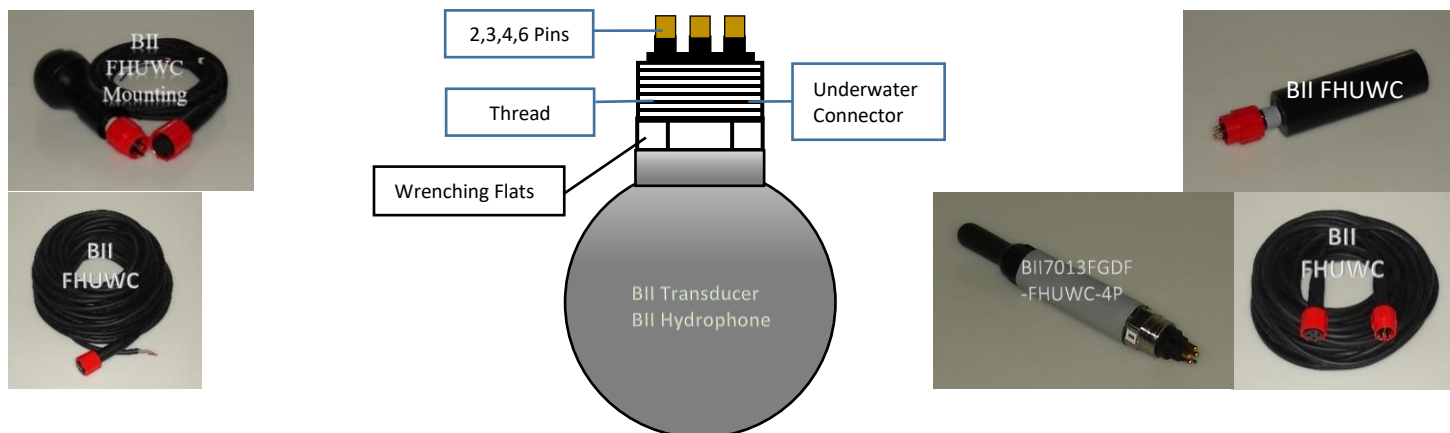
[Hydrophone as Array Element:](#) Free Hanging with Smooth Domes for Linear/Planar/Conformal Discrete Arrays. Illustration Only, Size Scale is NOT 1:1.



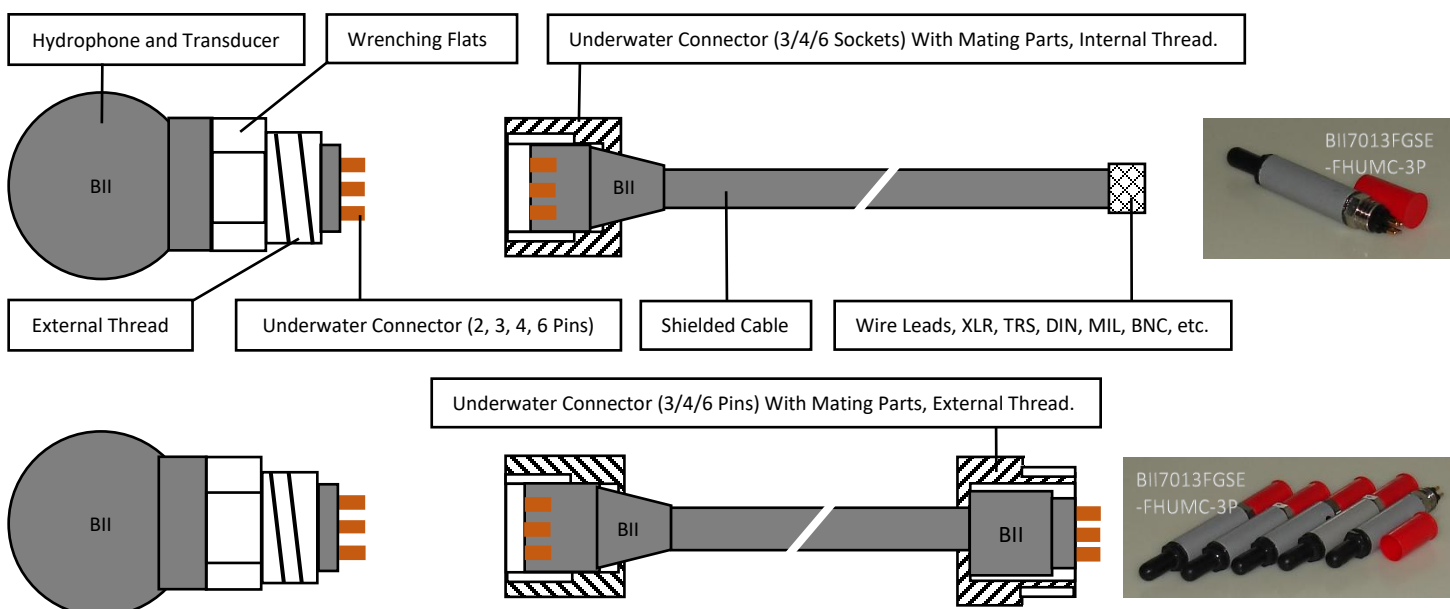
[Go to Catalogue](#)

Free-hanging with Underwater Connector (FHUWC) (Pin)

Maximum Depth: 2000m. Transducers/Hydrophone with Underwater Connectors: 3 pins, 4 pins and 6 pins. External thread for cable splice.



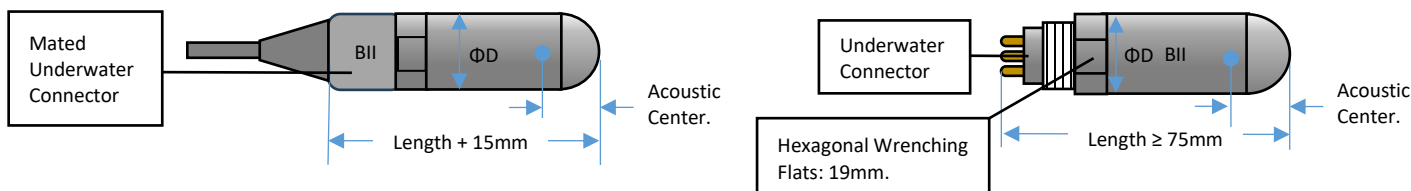
Free Hanging with Underwater Connector (FHUWC) and Accessories



Underwater Connector and Accessories: Hookup to Data Acquisition System, Cable Extension, etc.	
Part Number:	Description (P: Pin, S: Socket.)
FHUWC-2P	An Underwater Connector (2 Pins, External Thread) are encapsulated on Transducers/Hydrophones.
FHUWC-3P	An Underwater Connector (3 Pins, External Thread) are encapsulated on Transducers/Hydrophones.
FHUWC-4P	An Underwater Connector (4 Pins, External Thread) are encapsulated on Transducers/Hydrophones.
FHUWC-6P	An Underwater Connector (6 Pins, External Thread) are encapsulated on Transducers/Hydrophones.
Mating Connector and Cable	UWC-Cable Length-Connector: Underwater Connector with Socket insert and Internal-Thread Mating Parts, customized-length shielded cable, a Connector (WL, XLR, TRS, DIN, MIL, UMC, etc.) to DAQ devices or digital recorders.
	How to order cable with mating underwater connector? for example: UMC4S-20m-WL: 20 m cable with Underwater Mateable Connector 4 Sockets (UMC4S) on one end and wire leads (WL) on other end. UMC4S-20m-XLR3/BS: 20 m cable with and Underwater Mateable Connector 4 Sockets (UMC4S) on one end and XLR Receptacle with 3 Male Pins (XLR3) and Two +9V Battery Snaps on other end.

Hydrophone: Free-hanging with Underwater Connector (FHUWC), 3 Pins (No Preamplifier), 4 Pins (Fixed Sensitivity), and 6 Pins (Programmable Sensitivity).

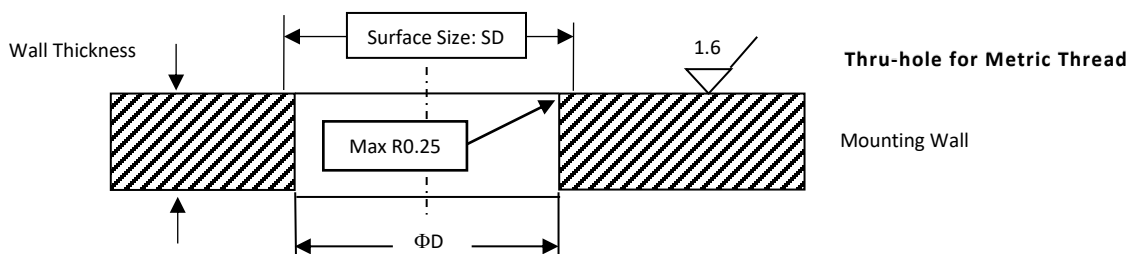
Free-hanging with Underwater Connector FHUWC-4P, 4 Pins (Fixed Sensitivity); FHUWC-6P, 6 Pins (Programmable Sensitivity).



Thru-hole Mounting (Metric Thread) with Single O-ring Sealing (Dimension Unit: mm)

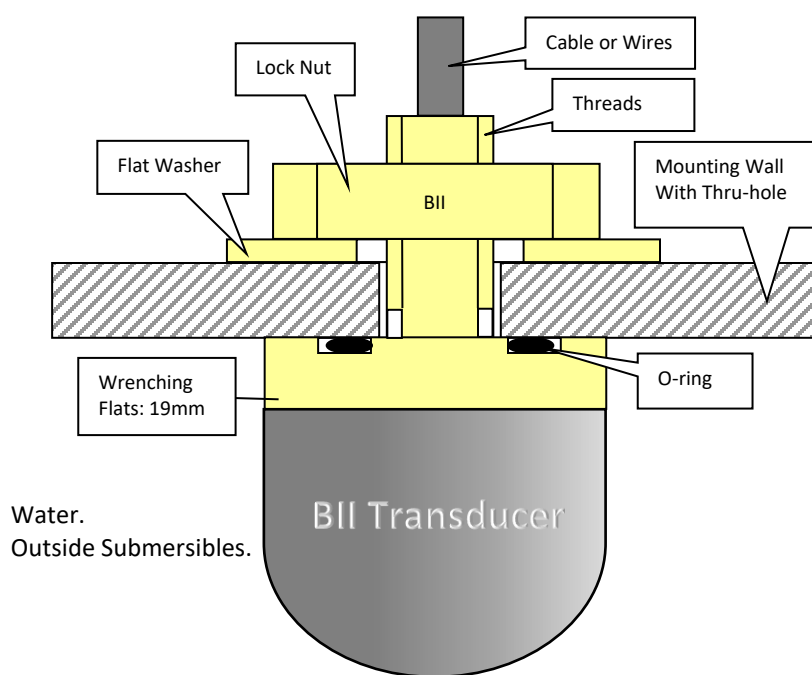
Part Number:	N/A	THM-M10x22	THM-M10x15
Thread Size:	Thread: M5x0.8x15 (Right Hand)	Thread: M10x1.5x22 (Right Hand)	Thread: M10x1.5x15 (Right Hand)
Hydrophone or Projector:	Single ended Signal ONLY	Single-ended or Differential	Single-ended or Differential
Mounting Hole Size ØD:	Ø5.21 mm ±0.1	Ø10.26 mm ±0.1	Ø10.26 mm ±0.1
Thickness of Mounting Wall:	≤9 mm with standard nut.	≤12 mm with standard nut.	≤5 mm with standard nut.
Maximum Diameters:	Ø11.5	Ø21	Ø21
Surface Size ØSD:	Ø12.5	Ø23	Ø23
O-ring Included:	ID x OD = Ø6.1 x Ø9.3 mm	ID x OD = Ø12 x Ø16 mm	ID x OD = Ø12 x Ø16 mm
Four Wrenching Flats:	None	19 mm	19 mm
Fastening Torque:	≤2 Nm	≤10 Nm	≤10 Nm
Materials of the Body:	Stainless Steel	Stainless Steel	Stainless Steel
Accessories Included:	Nut, and Flat Washer.	Nut, and Flat Washer.	Nut, and Flat Washer.
Service Temperature:	1. Default: -40 to 140 °F or -40 to 60 °C. 2. Bespoke: -40 to 250 °F or -40 to 121 °C. 3. Bespoke: -40 to 400 °F or -40 to 204.4 °C.		
Maximum Depth: 2500 m			
A transducer/hydrophone with thru-hole mount can also be installed into threaded hole of a wall. It is buyer's responsibility and liability to take measures at buyer's cost to prevent the installation from being loose or detached.			
Threadlockers are recommended to prevent threaded fasteners from loosening due to shock and vibration. NOT provided by BII.			
If the connector at the cable end is larger than the mounting hole, BII does NOT assemble the connector to cable end, and BII ships it to buyer. It is buyer's responsibility to assemble the connector to cable end at buyer's cost.			

Recommended Mounting Hole for Metric Thread Right Hand (Single O-ring)



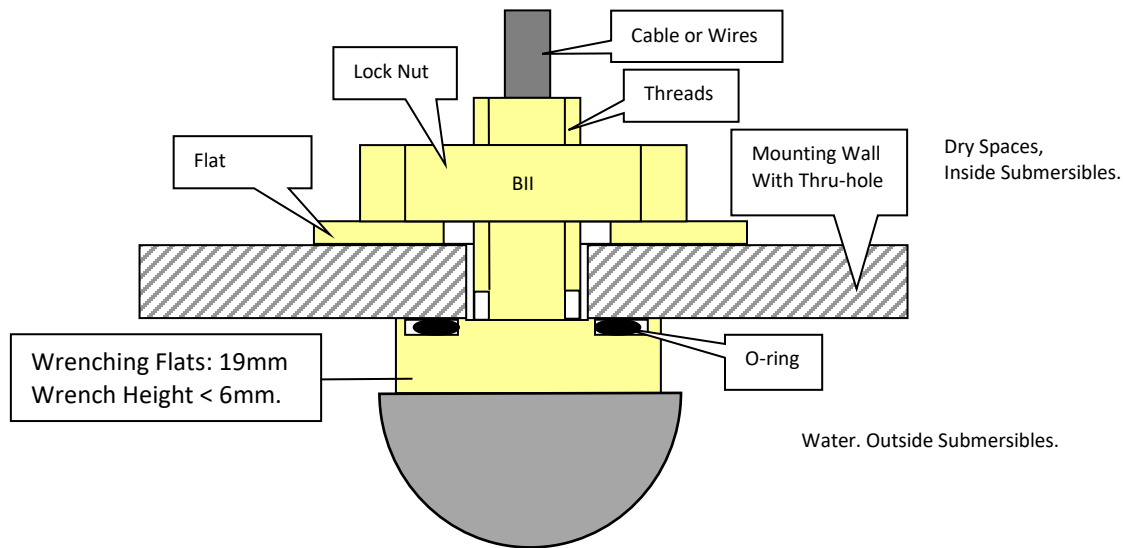
Metric Thru-hole Installation with Nut and Washer

It is buyer's responsibility and liability to take measures (such as thread lockers, etc...) at buyer's cost to prevent the installation from being loose or detached.



Dry Spaces,
Inside Submersibles.

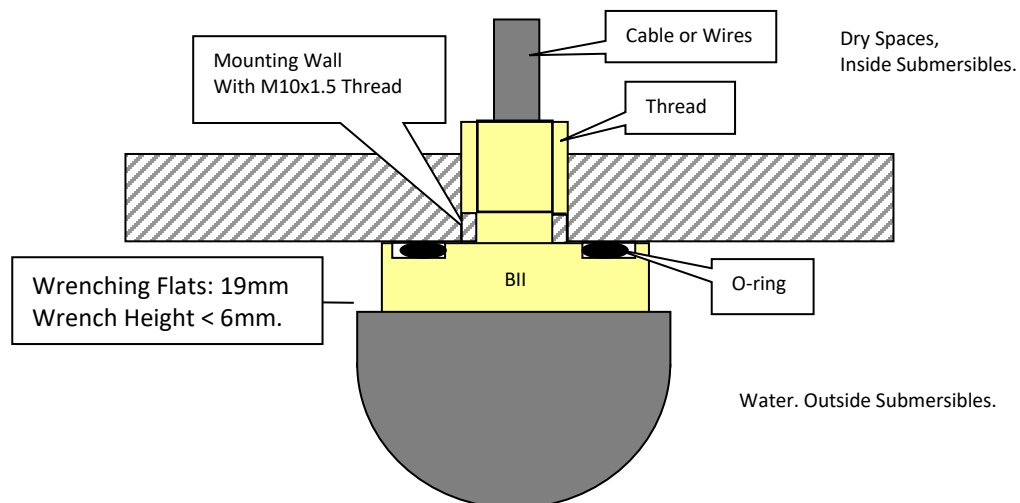
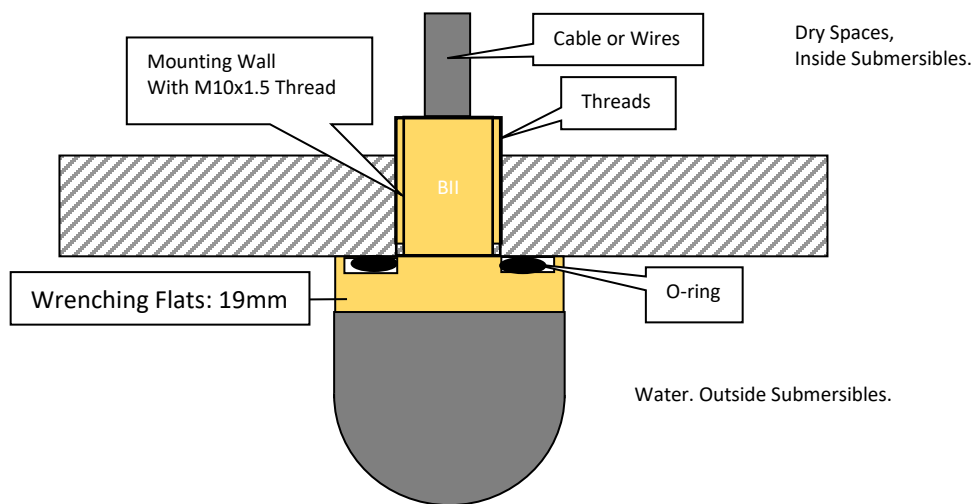




Metric Thread Installation without Nut and Washer

Refer to [Thread Sealing Tips](#).

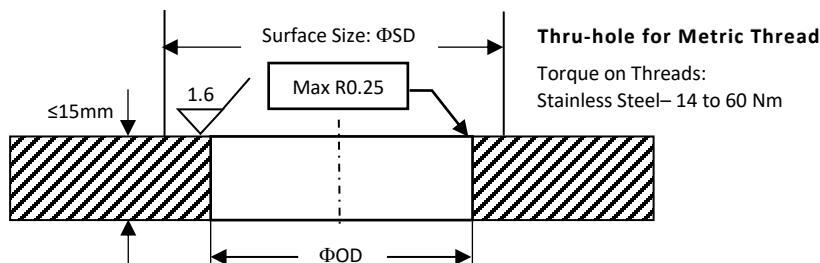
It is buyer's responsibility and liability to take measures (such as threadlockers etc...) at buyer's cost to prevent the installation from being loose or detached.



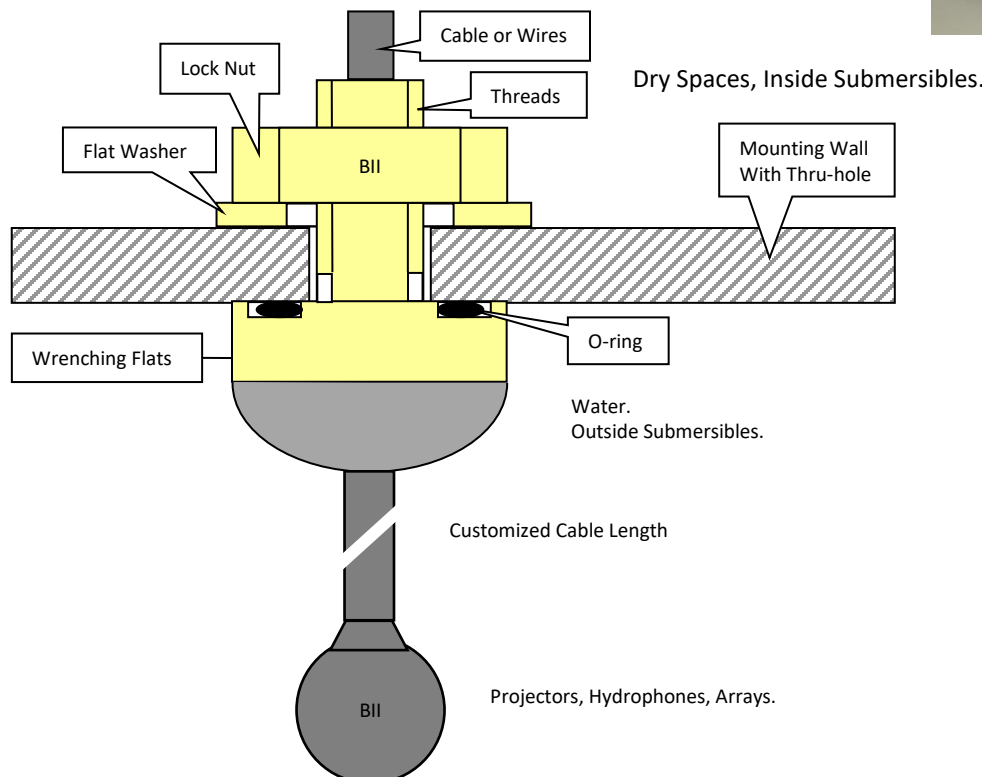
Thru-hole Mounting + Free Hanging for Moorings, Buoys, Drifters and Ocean Observatories, (Dimension Unit: mm).

Part Number:	THFH-M10	THFH-7/16"	THFH-1/2"	THFH-5/8"	THFH-1"
Thread (Right Hand):	M10x1.5x22	7/16"-20x22 UNF-2A	1/2"-13x22 UNC-2A	5/8"-18x27 UNF-2A	1"-14x25.4 UNS-2A
Application:	Projector or Hydrophone			Arrays	
Material:	Stainless Steel	Stainless Steel	Anodized Aluminum	Stainless Steel	Stainless Steel
O-ring (ΦID x C/S in mm):	Φ12 x 2	014 (Φ12.42 x 1.78)	113 (Φ13.94 x 2.62)	017 (Φ17.17 x 1.78)	122 (Φ28.24 x 2.62)
Flat Washer and Nut:	Included				
Maximum Diameter:	Φ21 mm	Φ22 mm	Φ24 mm	Φ27 mm	Φ38 mm
Wrenching Flats:	Four, 19 mm.	None	None	None	None
Fastening Torque:	≤10 Nm	≤30 Nm	≤30 Nm	≤50 Nm	≤60 Nm
Thickness of Mounting Wall:	≤12mm with nut.	≤10mm with nut.	≤9mm with nut.	≤12mm with nut.	≤10mm with nut.
Height of Thin Nut:	5 mm	¼" or 6.35 mm	N/A	3/8" or 9.525 mm	35/64" or 13.89 mm
Height of Standard Nut:	8 mm	3/8" or 9.525 mm	7/16" or 10.94 mm	35/64" or 13.89 mm	Not Used.
Surface Size ΦSD:	Φ23.0 mm	Φ23.0 mm	Φ23.0 mm	Φ28.0 mm	Φ39.0 mm
Mounting Hole Size ΦD:	Φ10.26mm ± 0.1	Φ11.3 mm ± 0.1	Φ12.8 mm ± 0.1	Φ16.0 mm ± 0.1	Φ26 mm ± 0.1
Service Temperature:	1. Default: -40 to 140 °F or -40 to 60 °C. 2. Bespoke: -40 to 250 °F or -40 to 121 °C. 3. Bespoke: -40 to 400 °F or -40 to 204.4 °C.				
Maximum Depth:	300 m to 500 m, depending on diametral clearance.				
If the connector at the cable end is larger than the mounting hole, BII does NOT assemble the connector to cable end, and BII ships it to buyer. It is buyer's responsibility to assemble the connector to cable end at buyer's cost.					
Threadlockers are recommended to prevent threaded fasteners from loosening due to shock and vibration. NOT provided by BII.					

Mounting Hole:



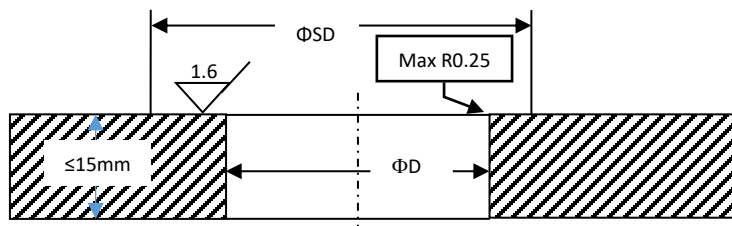
Installation:



Thru-hole Mounting (Inch Thread) with Single O-ring Sealing (Dimension Unit: mm)

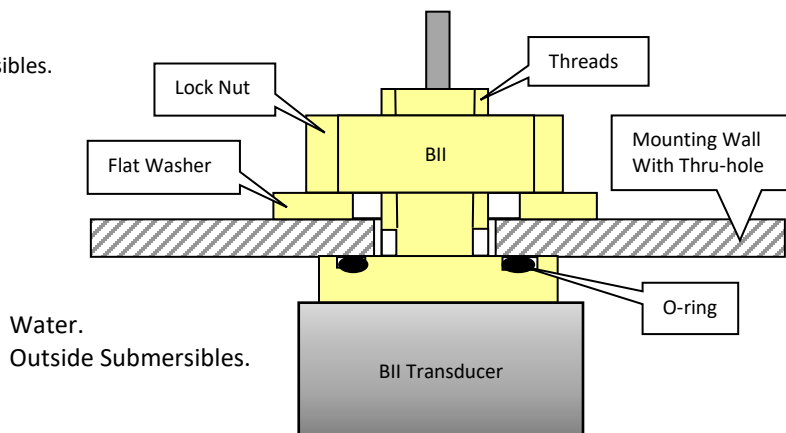
Part Number:	THM-7/16"	THM-1/2"	THM-5/8"	THM-1"
Thread (Right Hand):	7/16"-20x22 UNF-2A	1/2"-13x22 UNC-2A	5/8"-18x27 UNF-2A	1" 14x25.4 UNS-2A
Application:	Projector or Hydrophone			
Material:	Stainless Steel	Anodized Aluminum	Stainless Steel	Stainless Steel
O-ring (ΦID x C/S in mm):	014 (Φ12.42 x 1.78)	113 (Φ13.94 x 2.62)	017 (Φ17.17 x 1.78)	122 (Φ28.24 x 2.62)
Flat Washer and Nut:	Included			
Maximum Diameter:	Φ22 mm	Φ24 mm	Φ27 mm	Φ38 mm
Fastening Torque:	≤30 Nm	≤30 Nm	≤50 Nm	≤60 Nm
Thickness of Mounting Wall:	≤10 mm with nut.	≤9mm with nut.	≤12mm with nut.	≤10mm with nut.
Height of Thin Nut:	¼" or 6.35 mm	N/A	3/8" or 9.525 mm	35/64" or 13.89 mm
Height of Standard Nut:	3/8" or 9.525 mm	7/16" or 10.94 mm	35/64" or 13.89 mm	Not Used.
Surface Size ΦSD:	Φ23.0 mm	Φ23.0 mm	Φ28.0 mm	Φ39.0 mm
Mounting Hole Size ΦD:	Φ11.3 mm ± 0.1	Φ12.8 mm ± 0.1	Φ16.0 mm ± 0.1	Φ26 mm ± 0.1
Service Temperature:	1. Default: -40 to 140 °F or -40 to 60 °C. 2. Bespoke: -40 to 250 °F or -40 to 121 °C. 3. Bespoke: -40 to 400 °F or -40 to 204.4 °C.			
Maximum Depth:	2500 m, depending on diametral clearance.			
If the connector at the cable end is larger than the mounting hole, BII does NOT assemble the connector to cable end, and BII ships it to buyer. It is buyer's responsibility to assemble the connector to cable end at buyer's cost.				
Threadlockers are recommended to prevent threaded fasteners from loosening due to shock and vibration. NOT provided by BII.				

Recommended Mounting Hole Size

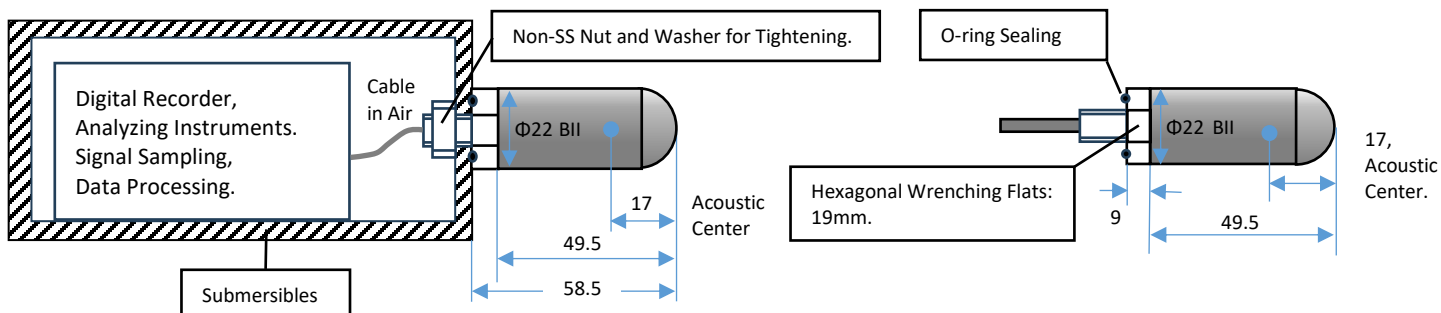


Transducer/Hydrophone Installation:

Dry Spaces,
Inside Submersibles.



Thru-hole Mounting (Inch Thread) with Single O-ring Sealing THM-7/16" (7/16"-20x22 UNF-2A).

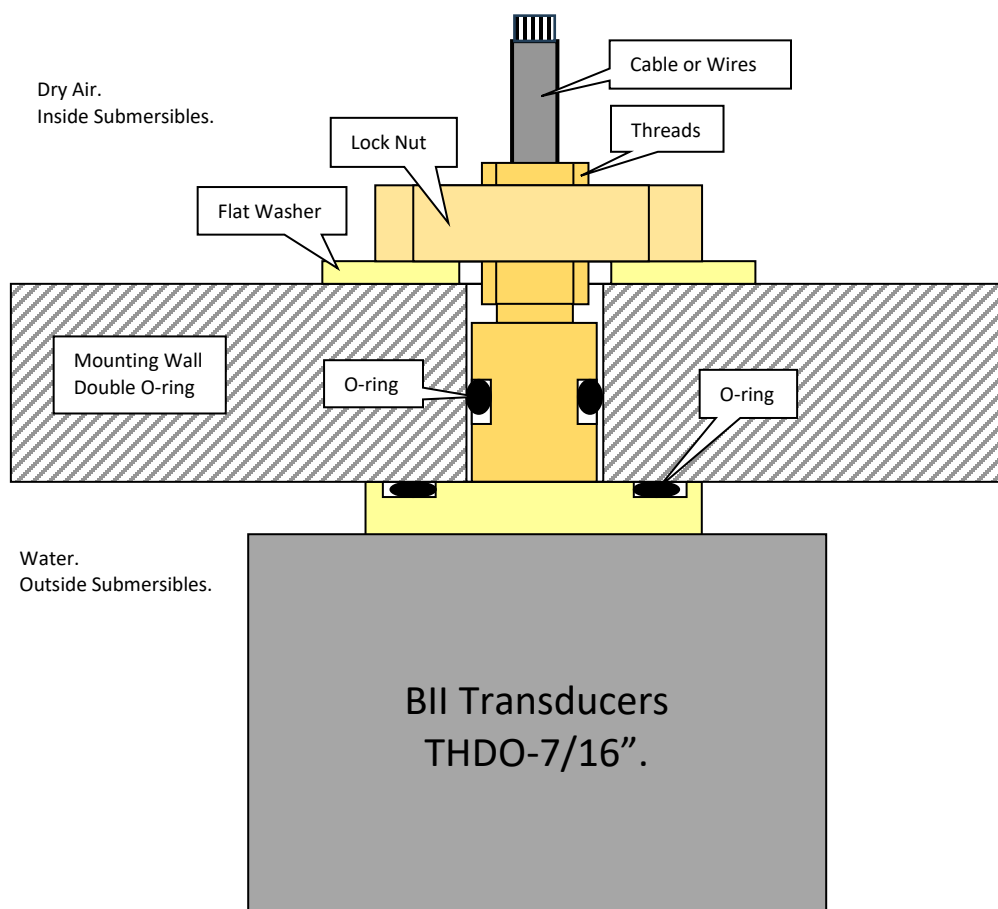
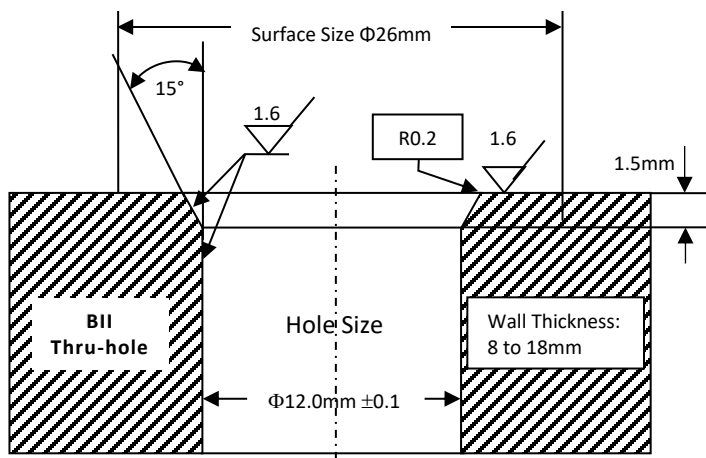


Thru-hole Mounting (Inch Thread) with Double O-ring Sealing (Dimension Unit: mm)

Part Number: THDO-7/16".

Maximum Depth: 2500 m.	Thread: 7/16"-20x25 UNF-2A (Right Hand).	Maximum Diameter: Ø25 mm	Fastening Torque: ≤30 Nm.
O-rings (O12 and O15), flat washer and nut are included.		Thickness of Mounting Wall: 8 to 18 mm.	
Service Temperature:	1. Default: -40 to 140 °F or -40 to 60 °C. 2. Bespoke: -40 to 250 °F or -40 to 121 °C. 3. Bespoke: -40 to 400 °F or -40 to 204.4 °C.		
If the connector at the cable end is larger than the mounting hole, BII does NOT assemble the connector to cable end, and BII ships it to buyer. It is buyer's responsibility to assemble the connector to cable end at buyer's cost.			
Threadlockers are recommended to prevent threaded fasteners from loosening due to shock and vibration. NOT provided by BII.			

Recommended Mounting Hole (Double O-ring)



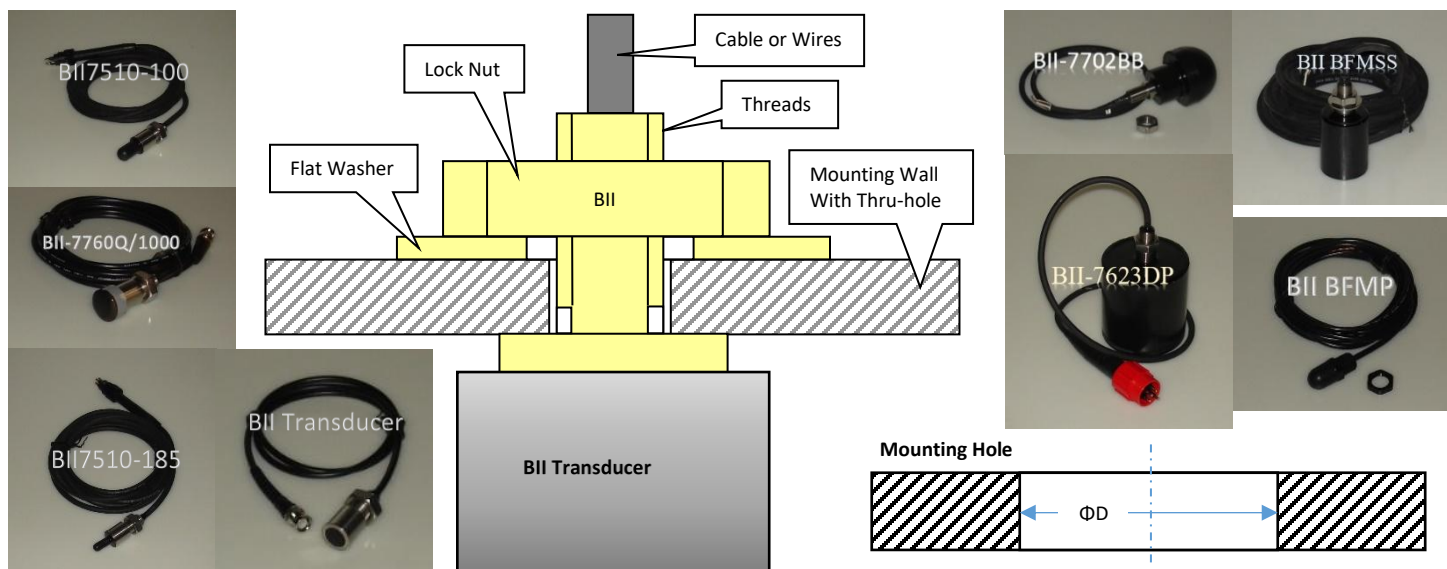
Bolt-Fastening Mount (Dimension Unit: mm)

Part Number	BFM-M6	BFM-7/16"	BFM-5/8"	BFM-FH	BFMP-M12	BFMP-NPT3/8"
Application: (Portable Apparatus)	Small Device, Array Element.	Single Device, Array Element.	Single Device, Array Element.	Portable Devices	Small Light Device, Array Element.	Small Light Device, Array Element.
Material:	Stainless Steel	Stainless Steel	Stainless Steel	Stainless Steel	Plastics	Plastics
Thread (Right Hand):	M6x1x16	7/16"-20x22 UNF	5/8"-18x22 UNF	3/8"-16 x 1.0" 1/4"-20 x 0.8"	M12x1.5x8mm	NPT3/8"x15mm
Maximum Diameter:	Φ9.25	Φ22	Φ27	Φ18 Φ14	Φ20.5	Φ24
Maximum Depth:	2000 m	2000 m	2000 m	500 m	300 m	300 m
Accessories:	Flat Washer and Nut Flat washer: included ONLY for heavy transducer (weight > 5 kg).			Hex Nut	Lock nut Height: 5mm	Lock nut Height: 5mm
Fastening Torque:	≤ 3 Nm	≤ 30 Nm	≤ 50 Nm	≤ 10 Nm	≤ 1.5 Nm	≤ 2.5 Nm
Thickness of Mounting Wall:	≤ 9 mm	≤ 10 mm	≤ 10 mm	≤ 10 mm	≤ 3 mm	≤ 8 mm
Mounting Hole Size ΦD:	Φ6.25 mm ± 0.1	Φ11.3 mm ± 0.1	Φ16.0 mm ± 0.1	Φ9.7 mm ± 0.15 Φ6.8 mm ± 0.15	Φ12.1 to Φ13.0 mm	Φ17.0 mm ± 0.1
Service Temperature:	1. Default: -40 to 140 °F or -40 to 60 °C. 2. Bespoke: -40 to 250 °F or -40 to 121 °C. 3. Bespoke: -40 to 400 °F or -40 to 204.4 °C.				-4 to 176 °F or -20 to 80 °C.	-4 to 176 °F or -20 to 80 °C.

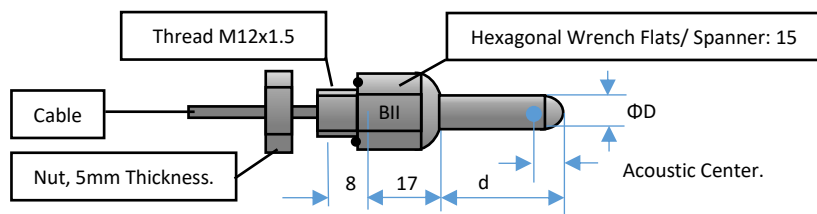
If the connector at the cable end is larger than the mounting hole, a slot cut on mounting apparatus is a must for cable passing though.

Threadlockers are recommended to prevent threaded fasteners from loosening due to shock and vibration. NOT provided by BII.

Bolt Fastening Mounting

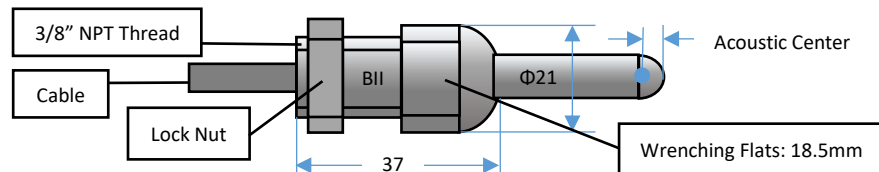


Bolt-fastening Mounting (Plastics) BFMP-M12 (300m Depth or 3MPa Ratings) or Thread Mounting into a submersible enclosure (IP68, tighten with o-ring).



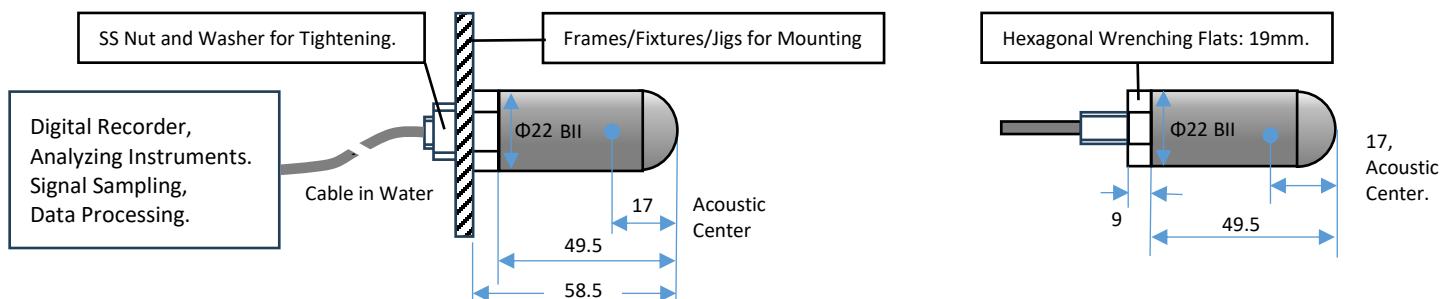
Bolt-Fastening Mount (Plastics) (BFMP-M12x1.5)
Thread: M12x1.5, Lock nut: included.
Maximum Depth: 300 m.
Mounting Hole: Φ12.1 to Φ13.0 mm.
Fastening Torque: ≤ 1.5 Nm.
Mounting Wall Thickness: ≤ 3 mm.

Physical Size (Dimensional Unit: mm) with BFMP-NPT3/8" Mounting Part:



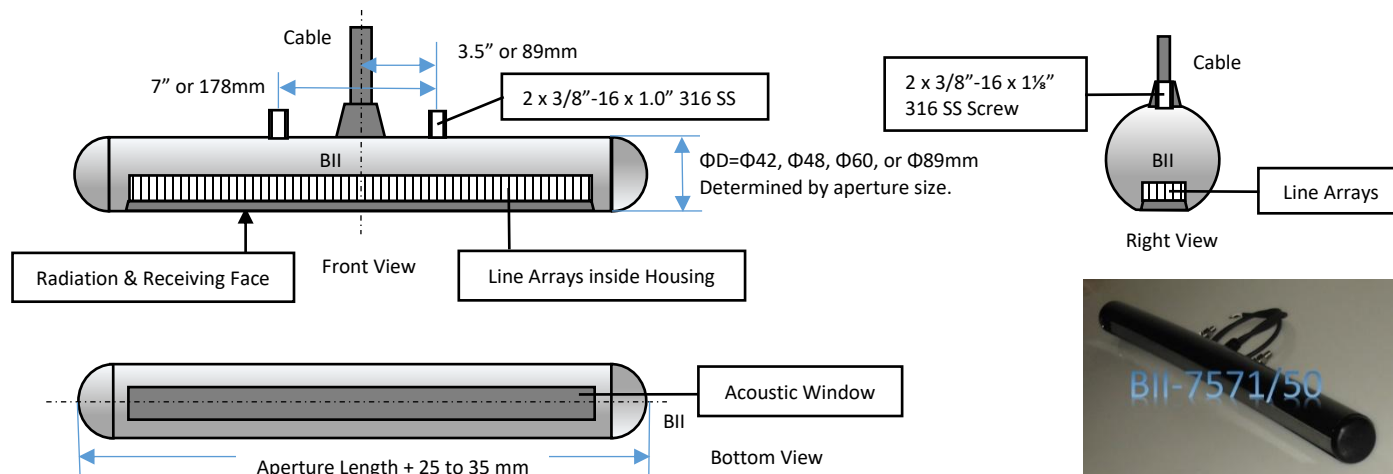
Bolt-Fastening Mount (Plastics) (BFMP-NPT3/8")
Thread: NPT-3/8", Lock nut: included.
Maximum Depth: 300 m.
Mounting Hole: Φ17.0±0.1 mm.
Fastening Torque: ≤ 2.5 Nm.
Mounting Wall Thickness: ≤ 10 mm.

Bolt-Fastening Mounting BFM-7/16" (7/16"-20x22 UNF-2A).

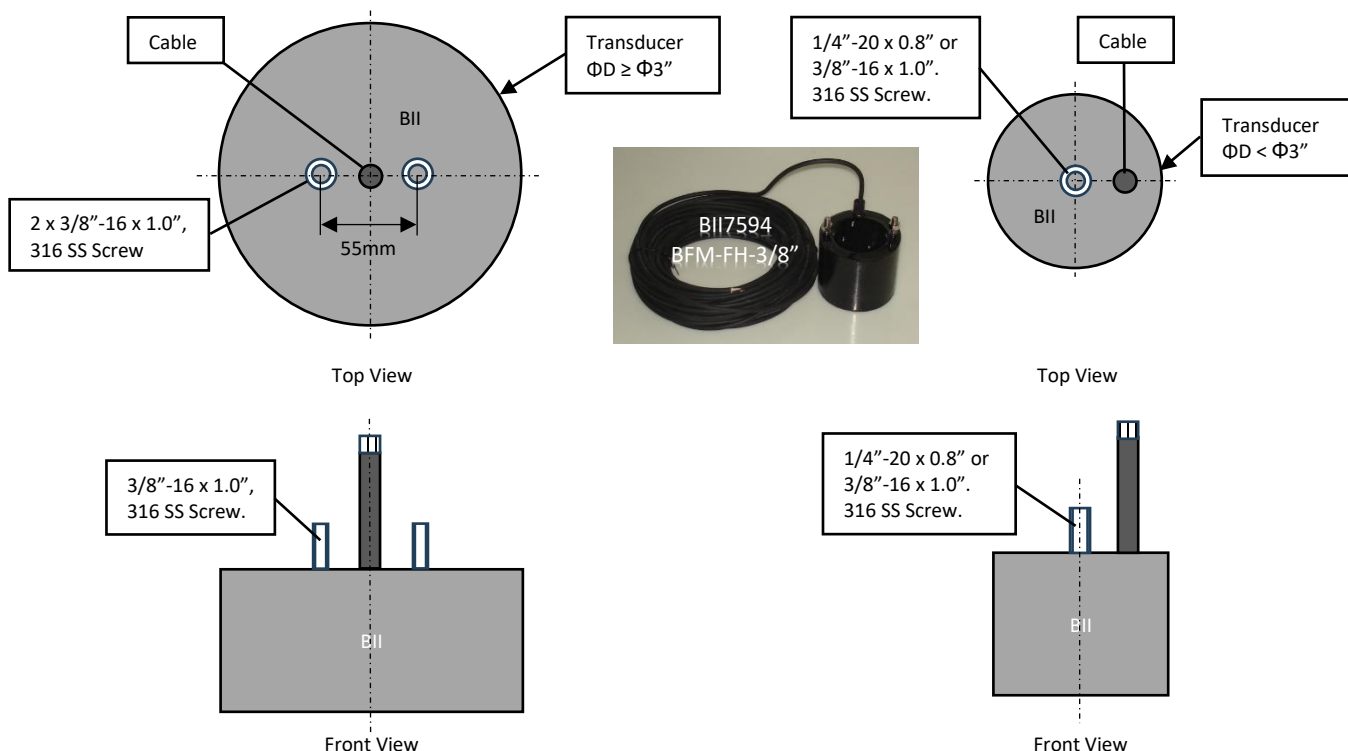


Bolt-Fastening Mounting with Free Hanging BFM-FH

(1) BFM-FH-3/8" for Line (Rectangular) Array. Overall Size (Dimensional Unit: mm). Following transducer structures are for illustration ONLY.



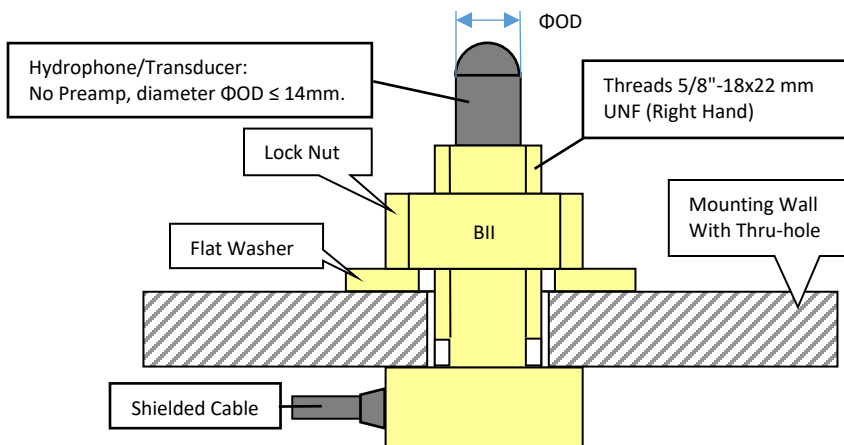
(2) BFM-FH-1/4", BFM-FH-3/8" for Spherical, Hemispherical, Cylindrical, and Planar Transducers. Available Thread: 1/4"-20 x 0.8" and 3/8"-16 x 1.0").



Bolt-Fastening Front Mount (Dimension Unit: mm)

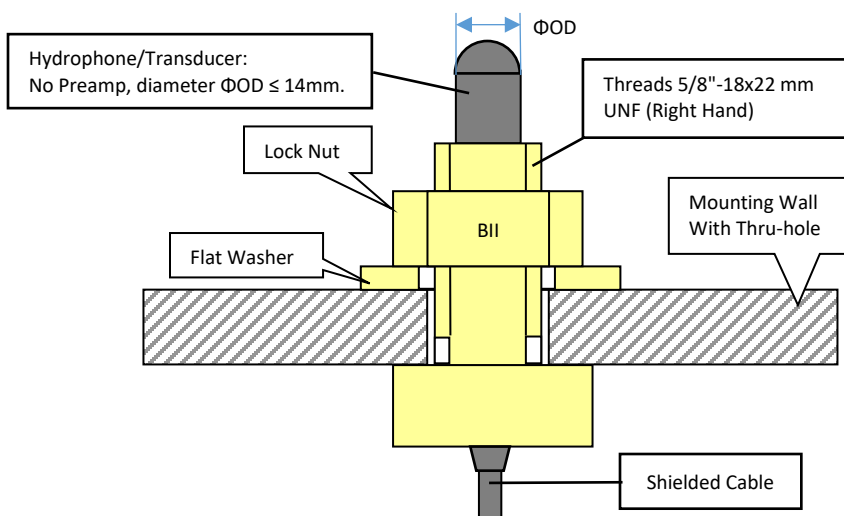
Part Number:	BFFMRA	BFFMSA
Description:	Bolt-Fastening Front Mount with 90° Right Angle.	Bolt-Fastening Front Mount with 180° Straight Angle.
Material:	Stainless Steel	
Thread (Right Hand):	5/8"-18x22 UNF	
Application:	Portable Apparatus, Single Device Array Element, Hydrophone (Receiver) Mooring Deployments. Diameter of sensing (active) elements of hydrophones or transducers should be less than 14mm.	
Maximum Diameter:	Φ27	
Maximum Depth:	500 m	
Accessories:	Flat Washer and Nut	
Fastening Torque:	≤ 20 Nm	
Thickness of Mounting Wall:	≤ 8 mm	
Mounting Hole Size ΦD:	Φ16.0 mm ± 0.1	
Service Temperature:	1. Default: -40 to 140 °F or -40 to 60 °C. 2. Bespoke: -40 to 250 °F or -40 to 121 °C. 3. Bespoke: -40 to 400 °F or -40 to 204.4 °C.	
If the connector at the cable end is larger than the mounting hole, a slot cut on mounting apparatus is a must for cable passing though.		
Threadlockers are recommended to prevent threaded fasteners from loosening due to shock and vibration. NOT provided by BII.		

Bolt Fastening Front Mounting with 90° Right Angle (BFFMRA)



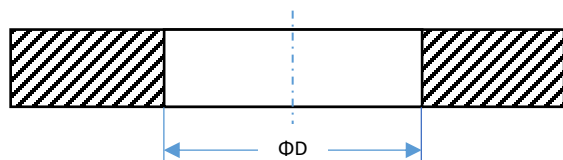
Sensing Element ≤ Φ14mm.
Mounting Hole: Φ16.0±0.1mm.
Fastening Torque: ≤30Nm.
Mounting Thickness: ≤12.7mm.

Bolt Fastening Front Mounting with 180° Straight Angle (BFFMSA)



Sensing Element ≤ Φ14mm.
Mounting Hole: Φ16.0±0.1mm.
Fastening Torque: ≤30Nm.
Mounting Thickness: ≤12.7mm.

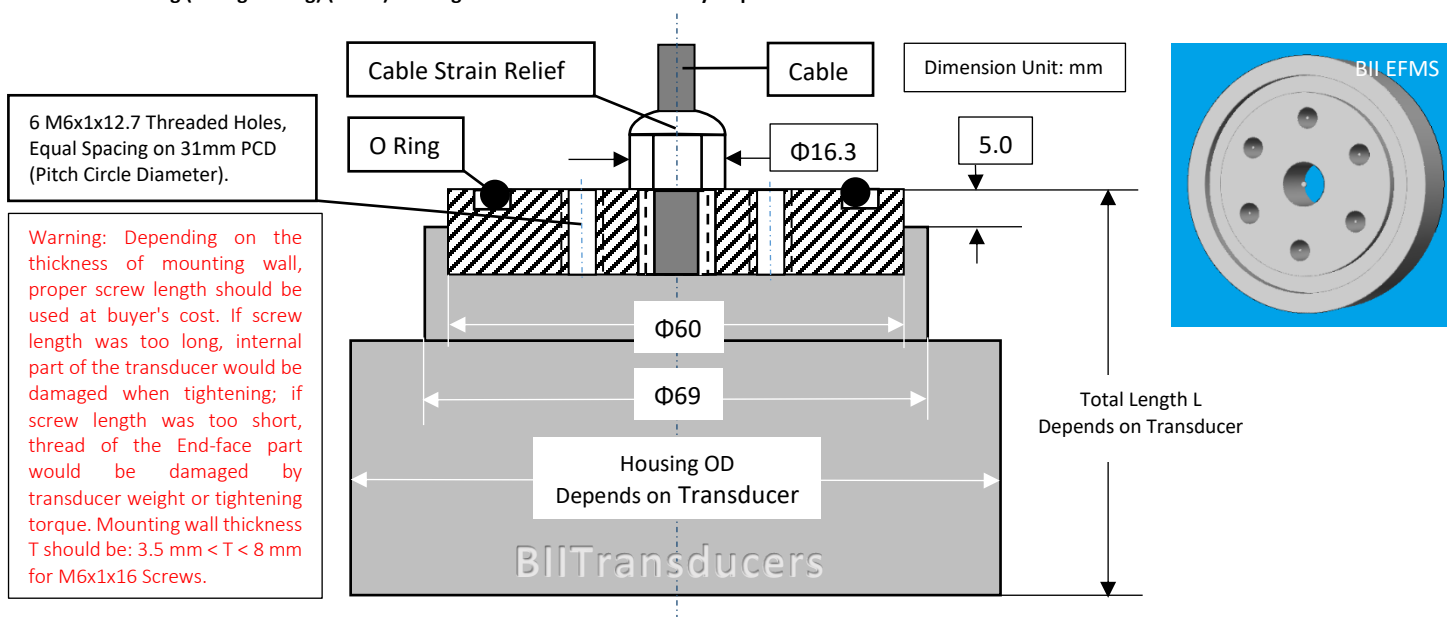
Mounting Hole



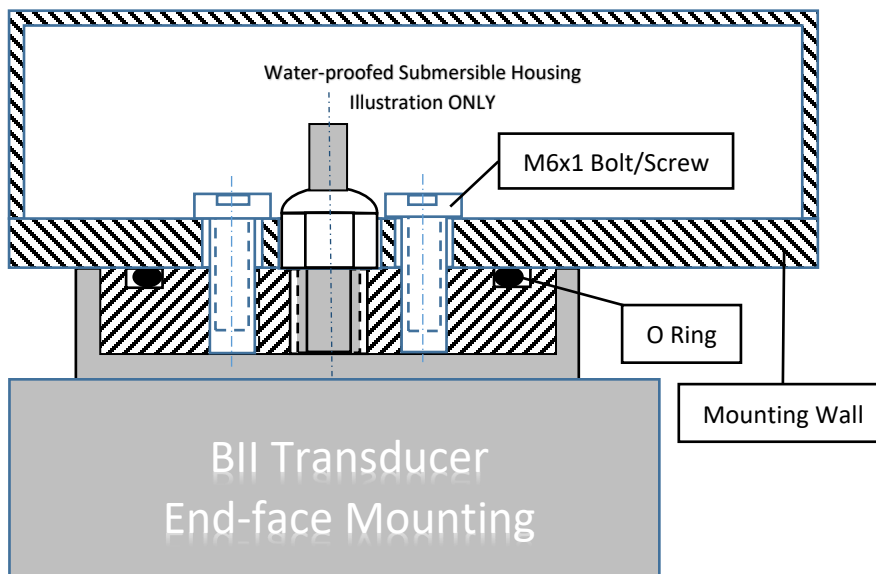
End Face Mount with O-ring Sealing (Dimension Unit: mm)

Part Number:	EFMS	EFMM
Description	End Face Mount, O-ring Sealing.	End Face Mount, O-ring Sealing.
Material:	Anodized Aluminum	Anodized Aluminum
Application:	Single Transducer or Hydrophone	Array, for Multi Cables (Cable Bundle)
Outside Diameter:	OD = $\Phi 60$ mm, Maximum.	
Center Hole Diameter:	ID = $\Phi 10.5$ mm, for Cable Through.	ID = $\Phi 26.0$ mm, for Cable Bundle Through.
Threaded Mounting Holes:	6 M6x1x12.7 Threaded Holes, Equal Spacing on 31mm PCD.	6 M6x1x12.7 Threaded Holes, Equal Spacing on 37mm PCD.
	1. PCD: Pitch Circle Diameter. 2. Range of Engagement Length of Screw Threads: 8 to 12.5 mm.	
	Warning: Depending on the thickness of mounting wall, proper screw length should be used at buyer's cost. If screw length was too long (Engagement Length of Screw Threads > 12.5mm), internal part of the transducer would be damaged during tightening; if screw length was too short (Engagement Length of Screw Threads < 8mm), thread of the End-face part would be damaged by transducer weight or tightening torque.	
Accessories:	6 M6x1x16 Screws and O-ring ID x CS = $\Phi 43 \times 3$.	6 M6x1x16 Screws, and O-ring ID x CS = $\Phi 48 \times 3$.
Mounting wall thickness T:	Mounting wall thickness T should be: $3.5 \text{ mm} < T < 8 \text{ mm}$ for M6x1x16 Screws. Refer to Warning .	
Fastening Torque:	Choose proper screw length with known wall thickness T: $8 \text{ mm} + T < \text{Screw Length} < 12.5 \text{ mm} + T$.	
Service Temperature:	$\leq 1 \text{ Nm}$	
Maximum Depth:	1. Default: -40 to 140°F or -40 to 60°C . 2. Bespoke: -40 to 250°F or -40 to 121°C . 3. Bespoke: -40 to 400°F or -40 to 204.4°C .	
Threadlockers are recommended to prevent threaded fasteners from loosening due to shock and vibration. NOT provided by BII.		

End-Face Mounting (O-ring Sealing) (EFMS) for Single-channel Transducer or Hydrophone

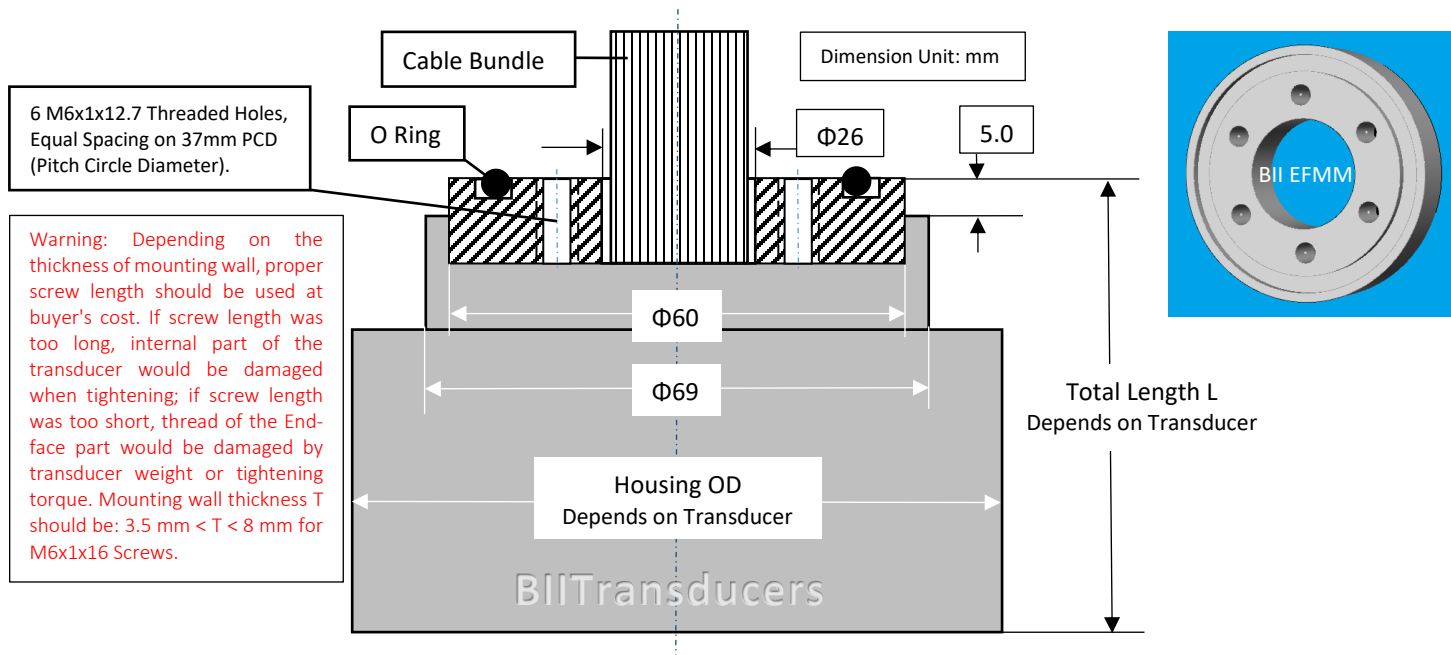


Installation on Submersible

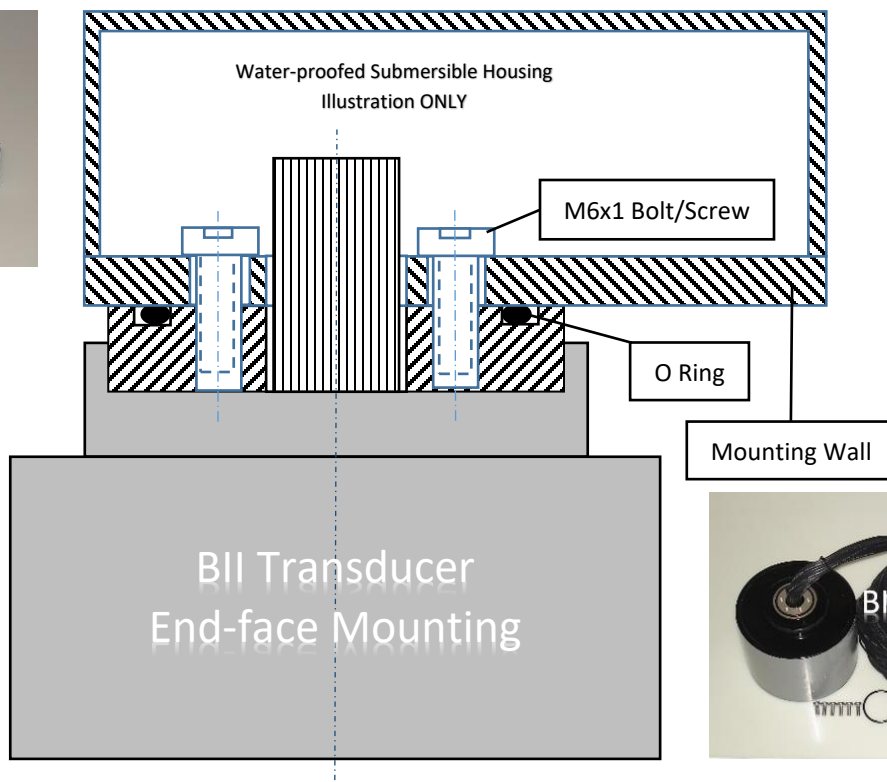


End-Face Mounting with O-ring Sealing for Cable Bundle of Multi-channel Array Transducer (EFMM) (Dimension Unit: mm)

Material: Anodized Aluminum.



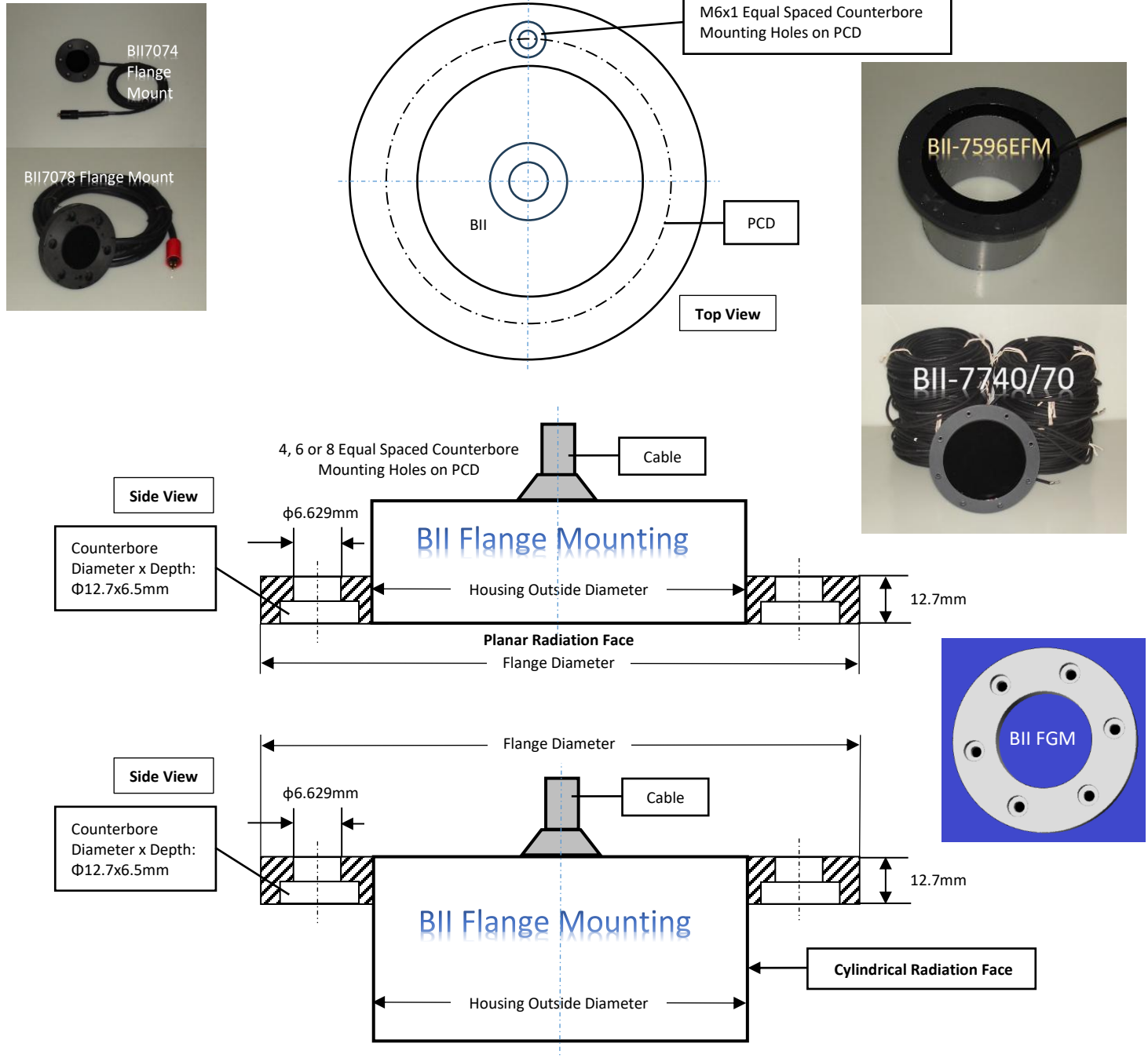
Installation on Submersible



Flange Mount (Marine Sealant) (FGM) (Dimension Unit: mm)

Maximum Depth: 100 m	Material: Plastics.	M6x1 Screws and Nut: Not Included.	Marine Sealant: Not Included.
Service Temperature:	-40 to 140 °F or -40 to 60 °C.		
Counterbored Mounting Holes for M6X1, Equal Spacing on PCD. PCD: Pitch Circle Diameter.			
Threadlockers are recommended to prevent threaded fasteners from loosening due to shock and vibration. NOT provided by BII.			

Flange Mounting



Part Number	Flange Diameter (mm)	Pitch Circle Diameter PCD (mm)	Housing Outside Diameter (mm)	M6x1 Mounting Hole Number on PCD	Flange Thickness (mm)
FGM-Φ220	Φ220	Φ195	Φ168	8	12.7
FGM-Φ190	Φ190	Φ165	Φ141	8	12.7
FGM-Φ165	Φ165	Φ140	Φ114	6	12.7
FGM-Φ140	Φ140	Φ115	Φ89	6	12.7
FGM-Φ110	Φ110	Φ85	Φ60	6	12.7
FGM-Φ80	Φ80	Φ58	Φ33	4	12.7

Flush Mount (Marine Sealant or Gasket) (FSM) (Dimension Unit: mm)

Part Number	Acoustic Aperture	Material	Thread	Housing Length L	Flange Diameter ΦD	Mounting Wall Thickness	Fastening Torque
FSM-M10	$\leq \Phi 5$ mm	Stainless Steel	M10x1.5	24.75	$\Phi 18$	$\leq (L - 14)$	≤ 20 Nm
FSM-M14	$\leq \Phi 10$ mm	Anodized Aluminum	M14x1.5	26.75	$\Phi 22$	$\leq (L - 16)$	≤ 10 Nm
FSM-M35	$\leq \Phi 27$ mm	Anodized Aluminum	M35x1.5	29.75, 50, 80.	$\Phi 59$	$\leq (L - 13)$	≤ 20 Nm
FSM-M36	$\leq \Phi 27$ mm	Anodized Aluminum	M36x4	29.75, 50, 80.	$\Phi 59$	$\leq (L - 13)$	≤ 20 Nm
FSM-M56	$\leq \Phi 47$ mm	Anodized Aluminum	M55x5.5	29.75, 50.	$\Phi 70$	$\leq (L - 13)$	≤ 30 Nm
FSM-M72	$\leq \Phi 62$ mm	Anodized Aluminum	M72x4	29.75, 50.	$\Phi 85$	$\leq (L - 13)$	≤ 40 Nm

Hex Nut: Included, for dry use ONLY. Material: Steel. **Moisture-Resistant Grease is recommended to resist moisture to prevent corrosion if necessary.**

BII does NOT provide sealing materials such as marine sealants and gaskets. Buyer may buy these materials from buyer's local stores of adhesives, boats, automobiles, and industry suppliers.

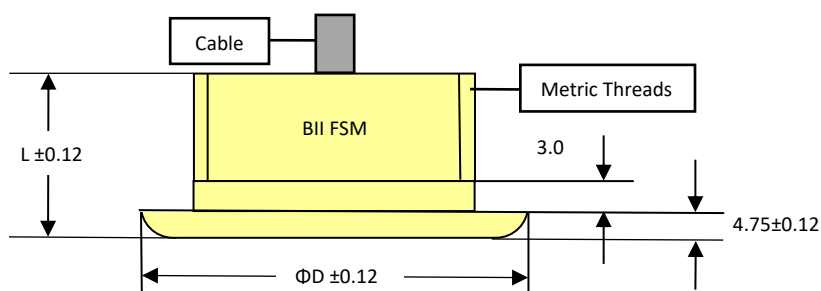
Surface Finish of the flange against the mounting wall: 50.8 microns Ra, Linear tolerance ± 0.12 mm. Counterbored Mounting Hole is the best.

Service Temperature: 1. Default: -40 to 140 °F or -40 to 60 °C. 2. Bespoke: -40 to 250 °F or -40 to 121 °C. 3. Bespoke: -40 to 400 °F or -40 to 204.4 °C.

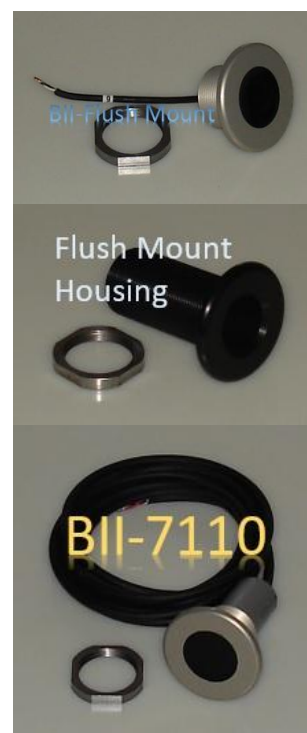
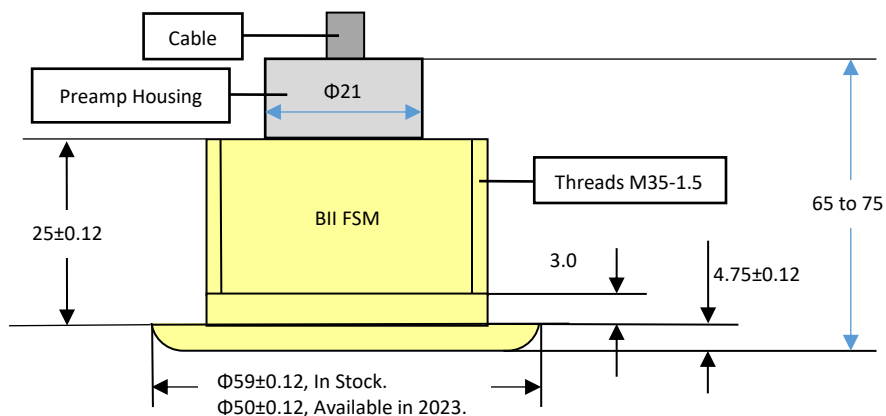
Maximum Operating Depth: 100 m to 300 m, limited by the performance of the sealing materials. For deeper underwater deployment (maximum 300m), one option is that O-ring grooves are cut on the mounting wall and O-rings are used besides marine sealants or casting sealants.

Threadlockers are recommended to prevent threaded fasteners from loosening due to shock and vibration. NOT provided by BII.

Flush Mounting Parts for Transducers/Projectors and Hydrophones (No Built-in Preamp), Physical Size (Dimensional Unit: mm):

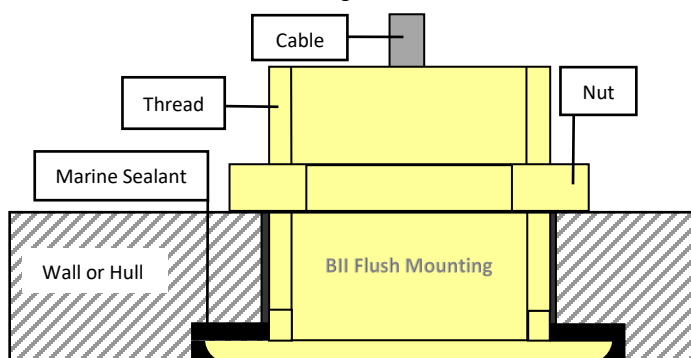


Flush Mounting Parts for Hydrophones with Built-in Preamp, Physical Size (Dimensional Unit: mm):

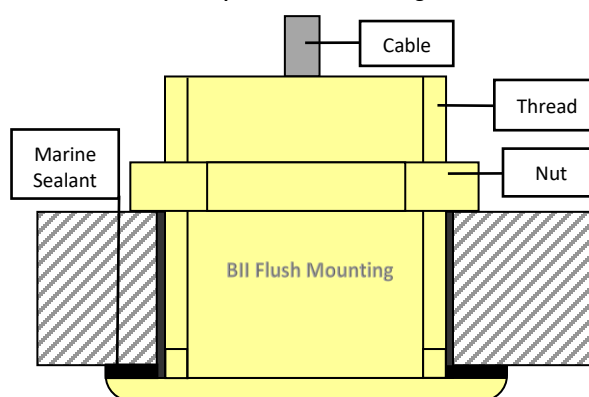


Installation/Mounting

Flush Mounting with Counterbore Hole.



Low-profile Flush Mounting

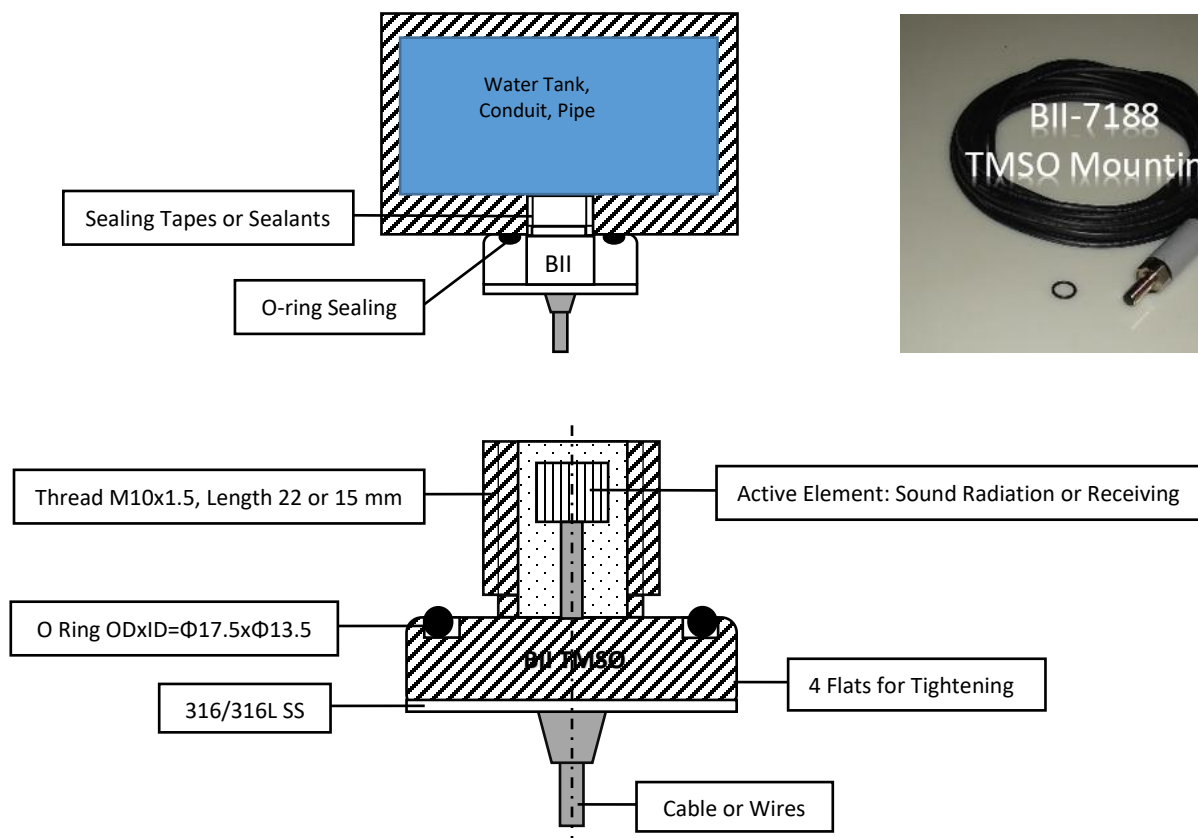


Thread Mounting with Single O-Ring (TMSO) (Dimension Unit: mm)

Part Number:	TMSO-M10x22	TMSO-M10x15
TMSO Options:	Thread: M10x1.5, Length=22 (Right Hand)	Thread: M10x1.5, Length=15 (Right Hand)
	Metric Screw Thread: M Profile, or United Screw Threads (UTS).	
Hydrophone or Projector:	Single-ended or Differential	Single-ended or Differential
	Suitable frequency for hydrophones: 0.1 Hz to 10 MHz. Suitable frequency for projectors: 50 kHz to 10 MHz.	
Protection Seal:	1. With O-ring provided by BII: 70 PSI, 5 Bar or 0.5 MPa. 2. With proper sealing tape or sealant: Maximum 1450 PSI or 10 MPa with full engagement length of the thread.	
Thickness of Mounting Wall:	Warning: thickness of mounting wall affects Engagement Length of Threads. If Engagement Length of Screw Threads were too short, threads would be damaged by pressure.	
Maximum Body Diameters:	Φ21	
Accessories:	1. O-ring Included: ID x OD = Φ13.5 x Φ17.5 mm. Durometer 70 A. 2. Thread Sealant Tape is available upon request when ordering. Specify the sealant tape types according to your applications.	
Four Wrenching Flats:	FlatxH = 19x10 mm	FlatxH = 19x5 mm
Fastening Torque:	≤10 Nm	
Materials of the Body:	Stainless Steel 316/316L	
Service Temperature:	1. Default: -40 to 140 °F or -40 to 60 °C. 2. Bespoke: -40 to 250 °F or -40 to 121 °C. 3. Bespoke: -15 to 400 °F or -26.1 to 204.4 °C.	
Threadlockers are recommended to prevent threaded fasteners from loosening due to shock and vibration. NOT provided by BII. It is buyer's responsibility and liability to take measures at buyer's cost to prevent the installation from being loose or detached.		

Thread Sealing Tips. (Followings sealants and sealing tapes are NOT provided by BII by default. Some of them are available upon request when ordering.)

- High-Density Thread Sealant Tape:** This tape has a high density for a better seal between threads. Made of PTFE, it has a slippery surface that keeps threads from binding to make parts easy to assemble.
- Thread Sealant Tape for Stainless Steel Threads:** This tape contains nickel to prevent seizing and galling associated with stainless steel threaded connections. It's also good for use with other metals. This tape has a high density for a better seal between threads. Made of PTFE, it has a slippery surface that keeps threads from binding to make parts easy to assemble.
- Submersible Sealants:** Marine grade, stand up to water, corrosion from salt spray, mildew, and UV light.
- Structural Submersible Sealants:** The strongest submersible sealants, these combine the mechanical strength of an adhesive with excellent sealing properties and high flexibility. They're marine grade, so they stand up to water, corrosion from salt spray, mildew, and UV light.
- Cured in Place Seals and Formed in Place Seals and Gaskets.** If these sealants are necessary to end-user's specific applications, BII recommends end-user to purchase these sealants from local industrial suppliers in end-user's country. BII does not provide curing sealants.



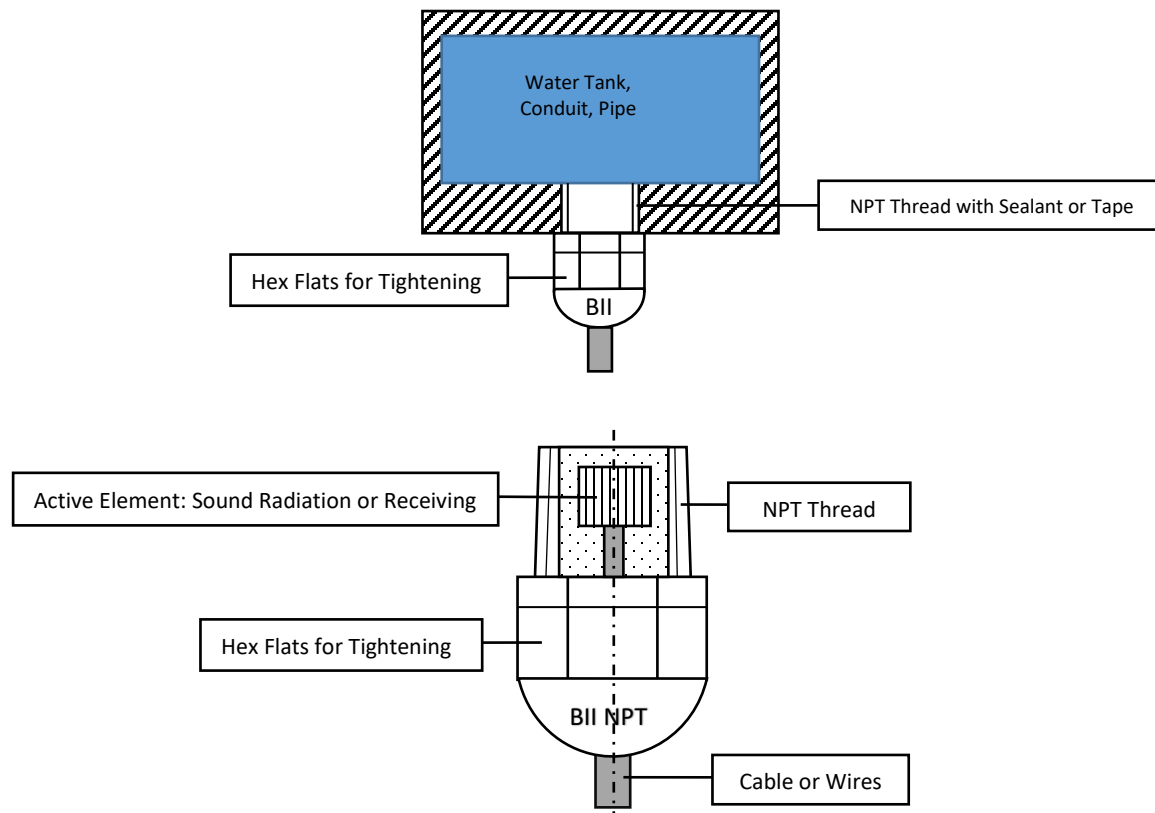
NPT Thread Mounting (NPT) (Dimension Unit: mm)

Part Number:	NPT-3/8"P; NPT-1/2"P; NPT-3/4"P; ...	NPT-3/8"SS; NPT-1/2"SS; NPT-3/4"SS, NPT-1"SS
NPT Thread Options:	Thread: 3/8", 1/2", 3/4", 1", 1-1/4", 1-1/2", 2". American National Taper Pipe Threads (NPT).	Thread: 3/8", 1/2", 3/4", 1".
Hydrophone or Projector:	Single-ended or Differential Suitable frequency for hydrophones: 0.1 Hz to 10 MHz. Suitable frequency for projectors: 40 kHz to 10 MHz.	Single-ended or Differential
Materials of the Body:	Plastics	Stainless Steel
Protection Seal:	70 PSI, 5 Bar or 0.5 MPa.	1450 PSI or 10 MPa
Thickness of Mounting Wall:	Warning: thickness of mounting wall affects Engagement Length of Threads. If Engagement Length of Screw Threads were too short, threads would be damaged by pressure. To achieve maximum pressure rating, please use full engagement length of the thread.	
Maximum Body Diameters:	Depends on thread options.	
Wrenching Flats:	Hex	
Fastening Torque:	≤1 Nm	≤10 Nm
Sealant:	Sealant (tape or paste) should be used for sealing. 1. Default: BII does not provide sealant. 2. Thread Sealant Tape is available upon request when ordering. Specify the sealant tape types according to your applications.	
Service Temperature:	-40 to 176 °F or -40 to 80 °C.	-40 to 250 °F or -40 to 121 °C.
Threadlockers are recommended to prevent threaded fasteners from loosening due to shock and vibration. NOT provided by BII. It is buyer's responsibility and liability to take measures at buyer's cost to prevent the installation from being loose or detached.		

Thread Sealing Tips. (Followings sealants and sealing tapes are NOT provided by BII by default. Some of them are available upon request when ordering.)

- High-Density Thread Sealant Tape:** This tape has a high density for a better seal between threads. Made of PTFE, it has a slippery surface that keeps threads from binding to make parts easy to assemble.
- Thread Sealant Tape for Stainless Steel Threads:** This tape contains nickel to prevent seizing and galling associated with stainless steel threaded connections. It's also good for use with other metals. This tape has a high density for a better seal between threads. Made of PTFE, it has a slippery surface that keeps threads from binding to make parts easy to assemble.
- Submersible Sealants:** Marine grade, stand up to water, corrosion from salt spray, mildew, and UV light.
- Structural Submersible Sealants:** The strongest submersible sealants, these combine the mechanical strength of an adhesive with excellent sealing properties and high flexibility. They're marine grade, so they stand up to water, corrosion from salt spray, mildew, and UV light.
- Cured in Place Seals and Formed in Place Seals and Gaskets.** If these sealants are necessary to end-user's specific applications, BII recommends end-user to purchase these sealants from local industrial suppliers in end-user's country. BII does not provide curing sealants.

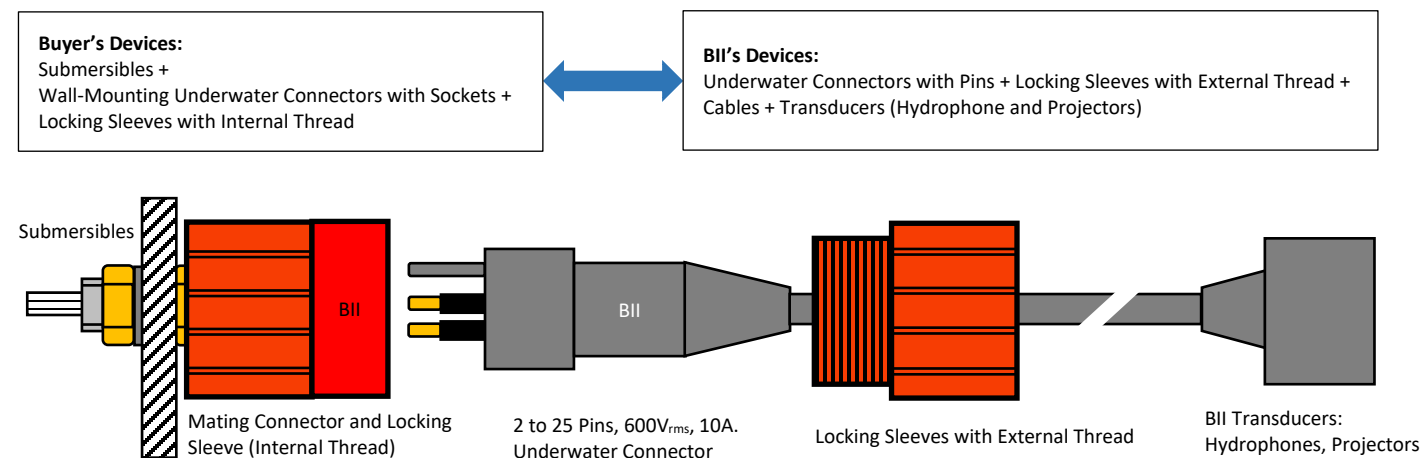
Thread (NPT) Mount



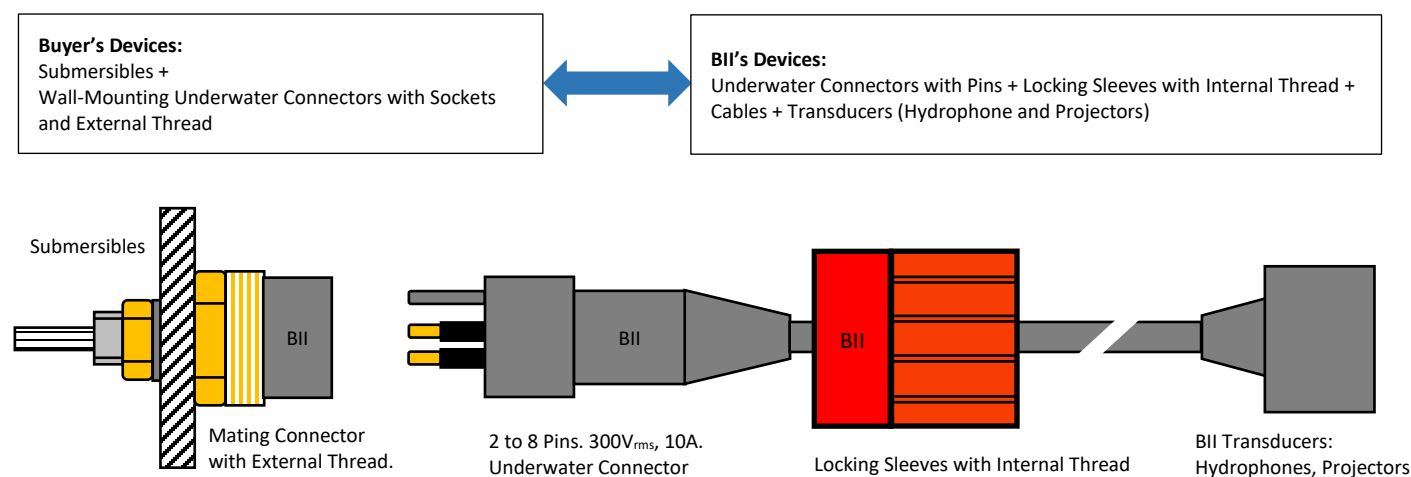
Underwater Connector Assembly

BII purchases underwater connectors from international underwater connector manufacturers who may have sales branches in buyers' countries. A buyer may purchase mating connectors from manufacturers directly or from BII as an integral part of BII transducers (hydrophones and transducers). Please contact BII for more information of the connectors such as manufacturers' name, part numbers, etc...

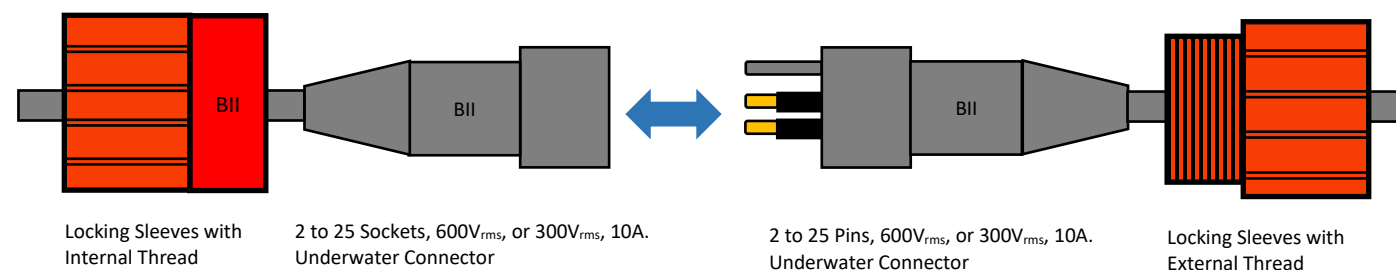
1. Standard Circular Series Underwater Connectors



2. Micro Circular Series Underwater Connectors



3. In-line Mating of Underwater Connectors



Export, Import, HS Code, Export Compliance, End-Use Statement, Shipping & Delivery

Harmonized System (HS) Code Reference of BII (Benthowave Instrument Inc.) Products			
Description of BII Products	HS Code	Description of BII Products	HS Code
Hydrophone as Parts and Accessories for Underwater (Oceanography) Acoustic Instruments	90159000	Power Amplifier as Parts and Accessories for Acoustic Instruments	85184000
Underwater Transducer as Parts and Accessories for Underwater (Oceanography) Acoustic Instruments	90159000	Electronic Preamplifier as Parts and Accessories for Acoustic Instruments	85185000
High Intensity Focused Ultrasound Transducer as Parts and Accessories for Acoustic Instruments	90279000	Electronic Amplifier (Transmit/Receive Switch, Time-Variable Gain, Automatic Gain Control, Band Pass Filter, Envelope Detection, Array Beamforming) as Parts and Accessories for Acoustic Instruments	85185000
Non-Destructive Testing Transducer as Parts and Accessories for Acoustic Measuring Instruments	90279000	Pulsed Signal Generator as Parts and Accessories for Acoustic Measuring Instruments	85432000
Acoustic Emission Transducer as Parts and Accessories for Acoustic Measuring Instruments	90279000	Signal Pulser & Receiver as Parts and Accessories for Acoustic Measuring Instruments	85432000
Ultrasonic Air Transducer as Parts and Accessories for Acoustic Instruments	90279000	Underwater Sound Recorder for Measuring and Analyzing Underwater Sounds	90278900
Impedance Matching Transformer as Parts and Accessories for Acoustic Instruments	85043100	Scientific Echo Sounder as Rangefinder in Oceanography	90151000

Sales and Technical Support (BII: Benthowave Instrument Inc.)

Selecting products, request quotes and technical support (Communication Language: English, in plain text or PDF): info@benthowave.com

Please check out your all email folders and/or check out your email-server's setting if you do NOT receive BII's response over 1 week, and ensure that your Email Server can receive emails from our company.

Basic Receiving Email Server Settings: 1. Log into email server. 2. Check out settings: Mail Filter, Blocked Senders, Safe Senders, Security, Block Images, etc. Pay attention to default handling of suspected Spam and Virus. Some business email may be treated as spams or virus and be deleted automatically by the email server.

Commercial Information: DO NOT email or send any confidential information such as trade secrets and Intellectual Property to BII. BII does not manufacture customized (bepsoko) products which utilize Consignee's intellectual properties.

"Ship to" Address Format: Contact Person's Name, Company/Organization Name (and branch or department if any), Street Number and Street Name, City, Province, Postal Code, Country, Phone number and E-mail address, **VAT** (Value Added Tax Identification) Number, **EORI** Number (EU, Northern Ireland, Norway and Switzerland). **EIN** (Employer Identification Number of USA Companies). Please use ", " or ";" to clearly separate address information and keep the address as compact and short as possible.

Consignee and End User: In most cases, consignee and end user are same person, company, research center, or entity. If an end user has a representative or consignee to purchase goods from BII, the representative or consignee should be in end user's country.

Sales: Benthowave sells and ships products directly to Consignees, Buyers, and End Users all over the world.

Payments: Buyer should notify BII once the buyer makes the payment.

Shipping and Delivery

Shipping Date: from January 5 to December 15. No shipping from December 16 to January 4 of the next year.

Shipping Updates: BII requests carrier's email-notifications service for shipping updates. BII emails the tracking number to buyer/consignee once BII creates waybill label. New online updates from courier website will be available about 5 to 24 hrs later.

Shipping Package: Cartons, maximum weight: 30 kg of each carton package.

Shipping Terms: BII prepares commercial invoice, fills out the FTA if any (Free Trade Agreement between buyer's country and Canada), and submit export-declarations-forms to Canada Border Services Agency (CBSA) in Canada.

1. **DAP**, ship with BII's courier account. BII strongly recommends that Buyers/Consignees choose this option for fast and trouble-free shipping.
2. **EXW-Ex Works**, Buyer's Freight Forwarder picks up goods (package) at BII. Buyer/Consignee arranges the pickup at BII address.

Allocations of Costs to Buyer (Consignee) and BII (Benthowave Instrument Inc.) according to Incoterms.

DAP: Insurable risk is for the Benthowave's shipping account such as DHL, UPS or FedEx.				
Both EXW-Ex and FOB: Insurable risk is for the Consignee's shipping account.				
Incoterm		1. Insurance (Shipment Protection). 2. Carriage to Port of Export. 3. Unloading of Truck in Port of Export. 4. Loading on Airplane in Port of Export. 5. Carriage (Air) to Port of Import. 6. Unloading in Port of Import. 7. Loading on Truck in Port of Import. 8. Carriage to Place of Destination.	1. Import Customs Clearance 2. Import Duties and Taxes	Unloading at Destination
DAP	BII (Benthowave)	BII (Benthowave)	Consignee or Buyer	Consignee or Buyer
EXW-Ex	BII (Benthowave)	Consignee or Buyer	Consignee or Buyer	Consignee or Buyer
FOB	BII (Benthowave)	Consignee or Buyer	Consignee or Buyer	Consignee or Buyer
DDP	BII (Benthowave)	BII (Benthowave)	BII (Benthowave)	Consignee or Buyer

Question: Why does not BII use buyer/Consignee's courier account to ship orders?

BII does NOT use buyer/Consignee courier account to ship because this may cause issues resulting from various reasons.

If BII shipped goods with buyer/Consignee's courier account, depending on carrier's terms, the carrier may still have rights to charge BII (Shipper) if carrier thought that it did not receive payment of the shipment from buyer/consignee. The worst case BII experienced was that BII received a surprise bill from a carrier four months later and BII spent about one year with numerous calls and emails to solve the issue among BII, Buyer/Consignee and Carrier.

Sales Tax and Custom Brokerage Services

No Canadian sales tax to international customers. GST or HST is applicable to Canadian customers ONLY.

For international customers: Consignee is responsible to import customs clearance, and pay import duties, tariffs, brokerage fees, taxes imposed by consignee's country and other costs related to shipment.

1. **If consignee uses Carrier's brokerage services for customs clearance**, the carrier pays import taxes on behalf of consignee. The consignee should repay or reimburse the carrier and pay the brokerage charges (if any) at time of delivery. The consignee may contact carrier's local customer service on how to repay or reimburse the carrier.
2. **If consignee has his own custom broker for customs clearance**, consignee should notify the broker to do customs clearance of the shipment once BII emails consignee the shipment information.

To Consignees/Buyers in EU, Northern Ireland, Norway and Switzerland: EORI number (Economic Operator Registration and Identification number of VAT-registered businesses). In Norway it's known as an 'Organization number', while in Switzerland it's referred to as the 'Unique Identification (UID) number'. EU countries are Austria, Belgium, Bulgaria, Croatia, Cyprus, Czechia, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, and Sweden.

To Consignees/Buyers in USA: for shipment value $\geq$ USD\$2500, consignee's Tax Identification Number (TIN), which can be either a Social Security Number (SSN) or an Employer Identification Number (EIN), is required by USA customs.

Sales Terms and Conditions: The Sales Terms and Conditions is enclosed in BII official quote. Consignee may refer to www.exportcontrols.gc.ca for more information on Area Control List, Export Prohibitions and Sanctions.

Hours of Operation

9:00a.m. to 5:00p.m., Monday to Friday.

Closed on Saturday, Sunday, and [Public Holidays](#) in Canada.

Please note that BII might shut down during severe weathers such as storms in summer and snow squalls in winter. Generally, there are many severe weathers in our area in December, January and February.

Benthowave Instrument Inc.

89 Kells Crescent,

Collingwood, Ontario, L9Y 0B4

Canada

Phone: +1 705 444 0187

Ordering, Manufacturing and Shipping Process

BII directly sells products to buyers/consignees all over the world and BII ships products directly to buyers/consignees all over the world. Except Canada, BII does NOT have any branch and/or sales representative in any other country.

1. First, buyer/consignee specifies the products in detail by referring to **<How to Order>** in respective datasheet. Email BII consignee's complete address with Postal Code and phone number, and payment methods (**USA & Canada: Cheque, Bank Draft or Wire Transfer; Other Countries: Wire Transfer through Bank**). **BII does not accept credit card payment.**
2. BII issues official quote to buyer which enclose BII's sales terms and conditions. Buyer/consignee reads the quote in depth. Contact BII for any question and update.
3. Buyer places order with email confirmation or purchase order.
4. BII issues Proforma Invoice to buyer, which encloses BII's bank information. Buyer starts to make payment to BII, and BII prepares to manufacture buyer's order.
5. BII manufactures, tests, and calibrates the products.
6. BII prepares export documents and complete export declaration to Canadian Customs.
7. BII ships goods to buyer with DHL, UPS, FedEx, or Buyer's Freight forwarder, and notifies buyer with waybill/tracking number and/or commercial invoice.
8. Buyer tracks the shipment online at carrier's website, complete the import customs clearances at buyer/consignee's country, and pay the taxes and or duties, if any, imposed by buyer/consignee's country.
 - a. Generally (for most of buyers), the courier will conduct customs clearance on behalf of buyer/consignee, the buyer/consignee just need to provide necessary information to courier and pay taxes and/or duties, if any, to courier before delivery.
 - b. If buyer/consignee has his/her own customs brokers, the buyer/consignee should notify the customs broker to conduct customs clearance once BII emails the shipping information to buyer/consignee.
9. Buyer/Consignee receives the shipment.

Estimated Processing Time of Quotations and Approximate manufacturing time for Bespoke Products.

Processing Time: The time, after buyer releases complete and accurate bespoke info to BII, necessary for BII to complete corresponding design, simulation, manufacturing processes, and estimated or simulated specs.

Products	Bespoke Parameters and/or Parts	Processing Time of Quote	Approximate manufacturing time ⁽¹⁾
Hydrophone AE Sensor NDT Sensor	Shorter Cable, Connector , others are same to standardized hydrophones.	1-5 days	BII checks out inventory. In stock: 2 to 3 days. Out of Stock: 3 to 6 weeks.
	Longer Cable, Connector , others are same to standardized hydrophones.	1-5 days	3 to 6 weeks.
	Output Type, Mounting, Cable Types, Cable Length, and/or Connectors.	1-5 days	4 to 7 weeks.
	-3dB Cut-off frequencies of High Pass, Low Pass, or Band Pass Filters.	3 days to 2 weeks.	4 to 7 weeks.
	Low Power Consumption of Receiver with Preamps	3 days to 2 weeks.	4 to 10 weeks.
	Free-field Voltage Sensitivity (FFVS) and/or Pressure Noise Density.	2 to 4 weeks	6 to 10 weeks.
	-3dB Beam Angle, Directivity Response , and/or Sidelobe Level.	2 to 4 weeks	6 to 10 weeks.
	Linear/Line Array: FFVS, -3dB Beam Angle and/or Sidelobe Level.	2 to 4 weeks	7 to 10 weeks.
	Depth Rating in Water of Ocean, Lake, or River.	2 to 4 weeks	Shallow Water: 3 to 6 weeks. Deep Water: 8 to 12 weeks.
	Service Temperature in Liquid, in Air or on Solid Surface.	2 to 4 weeks	4 to 10 weeks.
Sound Projector	Shorter Cable, Connector , others are same to standardized projectors.	1-5 days	BII checks out inventory. In stock: 2 to 3 days. Out of Stock: 3 to 6 weeks.
	Longer Cable, Connector , others are same to standardized projectors.	1-5 days	3 to 6 weeks.

	Mounting, Cable Types, Cable Length, and/or Connectors.	1-5 days	4 to 7 weeks.
	Impedance matching to non-50Ω	1-5 days	6 to 9 weeks.
	Transmitting Voltage Response (TVR), Power Rating.	N/A	N/A
	-3dB Beam Angle, Directivity Response, and/or Sidelobe Level.	2 to 4 weeks	6 to 10 weeks.
	Linear/Line Array: FFVS, -3dB Beam Angle and/or Sidelobe Level.	2 to 4 weeks	7 to 10 weeks.
	Depth Rating in Water of Ocean, Lake, or River.	2 to 4 weeks	Shallow Water: 3 to 6 weeks. Deep Water: 8 to 12 weeks.
	Service Temperature in Liquid, in Air or on Solid Surface.	2 to 4 weeks	4 to 12 weeks.
Transducer (Transmit & Receive)	Shorter Cable, Connector, others are same to standardized projectors.	1-5 days	BII checks out inventory. In stock: 2 to 3 days. Out of Stock: 6 to 9 weeks.
	Longer Cable, Connector, others are same to standardized projectors.	1-5 days	6 to 9 weeks.
	Mounting, Cable Types, Cable Length, and/or Connectors.	1-5 days	6 to 9 weeks.
	Impedance matching to non-50Ω	1-5 days	6 to 9 weeks.
	Transmitting Voltage Response (TVR), Power Rating.	N/A	N/A
	Free-filled Voltage Sensitivity (FFVS) and/or Pressure Noise Density.	2 to 4 weeks	6 to 12 weeks.
	-3dB Beam Angle, Directivity Response, and/or Sidelobe Level.	2 to 4 weeks	6 to 10 weeks.
	Linear/Line Array: FFVS, -3dB Beam Angle and/or Sidelobe Level.	2 to 4 weeks	6 to 10 weeks.
	Depth Rating in Water of Ocean, Lake, or River.	2 to 4 weeks	Shallow Water: 6 to 9 weeks. Deep Water: 8 to 12 weeks.
	Service Temperature in Liquid, in Air or on Solid Surface.	2 to 4 weeks	6 to 12 weeks.

Note ⁽¹⁾: Approximate manufacturing time depends on the complexity of various transducers. If bespoke transducer is a new one and/or some parts are out of stock from BII and/or our suppliers, the manufacturing time is much longer.

Disclaimer: BII does NOT guarantee the bespoke manufacturing time, and buyer may cancel the delayed order during manufacturing when the delay is not acceptable to buyer any more.

1. BII is NOT responsible to buyer's any loss caused by the delay of delivery of BII goods.
2. BII is NOT responsible to buyer's any loss caused by the failure of BII manufacturing the goods.
3. After buyer cancel the delayed order, the ONLY liability of BII is to refund the buyer with the amount Buyer paid BII for the purchase.

EXPORT AND SANCTIONS LAWS AND COMPLIANCE. BII Products are subject to control under the authority of the Export and Import Act administered by the Export Control Division of Foreign Affairs, Trade and Development Canada (www.exportcontrols.gc.ca). Without export permit issued by the minister of Foreign Affairs under the Export and Import Act of Canada, Products may not be exported or re-exported to any country, any person or entity in Area Control List (ACL), or in Export Prohibitions and Sanctions List imposed by Canada government, and Products on the Export Control List (ECL) may not be exported to customer. Customer agrees it will comply with the export laws, trade prohibitions, and sanctions of all applicable countries and will not export, re-export or transfer Products purchased from BII without the required authorization, including an export or re-export license issued by the Canada authorities, or to any prohibited destination or for a prohibited end-use. Products may also require export license(s) issued by the applicable authorities before being returned to BII, and the repaired products may also need export permit for returning it back to the customer.

The issuance of a Quote, a sales order acknowledgment (proforma invoice), or a RMA (Return Material Authorization) by BII does not constitute export authorization. BII reserves the right to refuse and/or cancel any order if, at any time, BII believes that any export controls may be violated. If a customer purchases Products on Export Control List, BII shall request an End User Statement from customer to apply for export license or permit. Orders requiring export licensing cannot be confirmed prior to receipt of approved export license from appropriate governing authority. Permits normally require 2-3 weeks to obtain.

End-Use Statement for Export Controlled Goods. A non-Canadian buyer shall email End-Use Statements to BII if the products are on the Canadian Export Control List (**Most BII products are NOT on Canadian export control list.**). The Export Control information is enclosed in official quote. An End-Use Statement must be dated (preferably within 6 months from the time the export permit application is submitted) and written on company letterhead in either English or French and should:

1. Identify the items, including quantities, which are the subject of the export permit application.
2. Identify the end-user and the location where the items will be delivered.
3. Identify any consignees, agents, or other entities involved in the export from Canada.
4. State the purpose and end-use of the products, including a statement of whether the intended end-use of the items is civilian (commercial) or military.
5. State whether or not the goods or technology are intended for re-sale, re-transfer or re-export to another party or parties and, if so, describe the circumstances; or state that the goods or technology will not be resold, re-transferred or re-exported.
6. Declare whether or not the goods and technology will be used for any purpose associated with the development or production of chemical, biological or nuclear weapons, or their delivery systems (such as missiles).
7. State that the goods or technology will not be transhipped to other destinations or otherwise diverted from what has been previously described; and
8. Identify the name and title of the person signing the End-Use Statement, his/her address, telephone number, fax number, e-mail address and corporate website.

Refer to [End-use statement template](#).

IMPORTANT NOTICE AND DISCLAIMER

BII PROVIDES TECHNICAL AND RELIABILITY DATA (INCLUDING DATASHEETS), DESIGN RESOURCES (INCLUDING REFERENCE DESIGNS), APPLICATION OR OTHER DESIGN ADVICE, SAFETY INFORMATION, AND OTHER RESOURCES "AS IS" AND WITH ALL FAULTS, AND DISCLAIMS ALL WARRANTIES, EXPRESS AND IMPLIED, INCLUDING WITHOUT LIMITATION ANY IMPLIED WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE OR NON-INFRINGEMENT OF THIRD PARTY INTELLECTUAL PROPERTY RIGHTS.

These resources are intended for skilled developers designing with BII products. You are solely responsible for (1) selecting the appropriate BII products for your application, (2) designing, validating and testing your application, and (3) ensuring your application meets applicable standards, and any other safety, security, or other requirements. These resources are subject to change without notice. BII grants you permission to use these resources only for development of an application that uses the BII products described in the resource. Other reproduction and display of these resources is prohibited. No license is granted to any other BII intellectual property right or to any third party intellectual property right. BII disclaims responsibility for, and you will fully indemnify BII and its representatives against, any claims, damages, costs, losses, and liabilities arising out of your use of these resources.

BII's products and services are provided subject to BII's **Terms and Conditions of Sale** enclosed in official quote.

Careers at BII

Acoustics is an indispensable element of civilization which help understanding the world and improve livings. Benthowave staffs manufacture acoustical components and instruments to fulfill due contributions as acoustics benefits human beings.

While working at BII, we are challenged daily to meet and exceed various customers' requirements and solve the challenges during design, manufacturing, calibrations, and quality assurance with experiences and textbooks on mathematics, physics, chemistry, acoustics, mechanics, and electronics, and are rewarded for outstanding performances.

Proud tradition at BII is to emphasize individuality, respect, growth, and creativity. BII is a workplace where the staff can learn, contribute, innovate, and be rewarded as a career and not just a job, and values the staff as most important factors of success.

We believe in fair compensation that reflects the hard work and expertise our team members bring, we aim to provide opportunities for growth, and we believe the optimum match among wage, manufacturing capability, quality, and price is a successfully cooperating bridge between global customers and our business.

Following knowledge and experiences are our companions in daily working tasks:

Electroacoustics, Ultrasonics, Underwater Acoustics.	Electrical & Electronics.	Manufacturing/Mechanical Engineering.
Materials, Solid State Chemistry & Physics.	Ceramics, Polymer, Composite.	Marine/Ocean Engineering.

If you're available and capable of following positions, Benthowave'd be grateful for your assistance.

NOC 2021	Job Title and Responsibilities	Job Types	Job Vacancy	Contact
94201	1. Electronics Assemblers, Fabricators, Inspectors and Testers. 2. Necessary Training Provided by Benthowave.	Full Time	Please refer to: Careers at BII.	hr@benthowave.com
22310	Electronics Engineering Technologists and Technicians	Full Time		
22301	Mechanical Engineering Technologists and Technicians	Full Time		
12200	Accounting Technicians and Bookkeepers	Full Time		
21301	Acoustics (Electroacoustics) Engineers (Mechatronic Engineering)	Full Time		
21322	Materials Engineers	Full Time		
90010	Manufacturing Managers (Mechatronic Engineering)	Full Time		

Note: Please send resume (PDF Format is preferred) to apply the job available.

1. Work time with 30-minute meal break: 9:00 AM to 5:30 PM, no shifts, Monday to Friday.

2. Hourly pay with a bonus every half-year (semi-annually).

Career Training at BII

BII provides necessary training to staff to fit for tasks of acoustic components and systems.

News in 2025: New expansion of 200m² facility increases BII's capabilities of production and indoor calibration.

END-USER STATEMENT - REQUEST AND REQUIREMENTS:

Please note that this document must be completed by the final user of the goods, in English or French, and be completed on company letterhead. This document cannot be completed by the exporter, an intermediate consignee, a broker/freight forwarder or their agents.

Date: Day-Month-Year

Canada Border Services Agency
Export Control Unit

END-USER STATEMENT (Sample)

We hereby declare that:

- Name of End-User (Individual/Company)

Are purchasing/acquiring:

- Complete list of the goods contained in the shipment.
- Location where the goods will be used.
- Identify any consignees (both intermediary and final), agents, or other entities involved in the export from Canada.
- State the purpose and end-use of the products, including a statement of whether the intended end-use of the items is civilian (commercial) or military.
- State whether or not the goods or technology are intended for re-sale, re-transfer or re-export to another party or parties and, if so, describe the circumstances; or state that the goods or technology will not be resold, re-transferred or re-exported.
- Declare whether or not the goods and technology will be used for any purpose associated with the development or production of chemical, biological or nuclear weapons, or their delivery systems (such as missiles).
- State that the goods or technology will not be transhipped to other destinations or otherwise diverted from what has been previously described/declared.

Signature/Authority:

- Name and title of the person signing the End-Use Statement.
- Complete address, telephone number and fax number.
- e-mail address and corporate website.

DÉCLARATION D'UTILISATEUR FINAL - DEMANDE ET CONDITIONS :

Veuillez noter que ce document doit être complété par l'utilisateur final des marchandises, en français ou en anglais, et doit être complété sur papier à en-tête de l'entreprise. Ce document ne peut être complété par l'exportateur, un destinataire intermédiaire, un courtier/transitaire ou leurs agents.

Date : jour-mois-année

Agence des services frontaliers du Canada
Section des contrôles à l'exportation

DÉCLARATION D'UTILISATEUR FINAL (Échantillon)

Nous déclarons par la présente :

- Nom de l'utilisateur final (Individu/Entreprise)

Nous achetons/acquérons :

- Liste complète des marchandises contenues dans la cargaison.
- Endroit où les marchandises seront utilisées.
- Désigner les destinataires (intermédiaires et finaux), les agents ou toute autre entité intervenant au cours du processus d'exportation au départ du Canada.
- Décrire la fin et l'utilisation finale des produits et déclarer si l'utilisation finale prévue des articles est civile (commerciale) ou militaire.
- Indiquer si les marchandises ou la technologie sont destinées ou non à être revendues, retransférées ou réexportées à une ou plusieurs autres parties et, le cas échéant, décrire les circonstances de ces opérations; ou déclarer que les marchandises ou la technologie ne seront pas revendues, retransférées ou réexportées.
- Déclarer si les marchandises et les technologies seront utilisées ou non à des fins associées au développement ou à la production d'armes chimiques, biologiques ou nucléaires ou de leurs vecteurs (comme des missiles).
- Déclarer que les marchandises ou la technologie ne seront pas réexpédiées vers d'autres destinations ou autrement détournées de ce qui a été précédemment décrit.

Signature/autorisation :

- Nom et titre de la personne qui signe la déclaration d'utilisation finale.
- Adresse, numéro de téléphone et numéro de télécopieur.
- Adresse courriel et adresse du site Web de l'entreprise.

MANUFACTURER'S AUTHORIZATION

<End-User Address Here>

BENTHOWAVE INSTRUMENT INC. (BII) is an official manufacturer of acoustic products (Underwater, Air, AE, NDT, and HIFU) at < **BII Official Address**>. (www.benthowave.com. Contact: info@benthowave.com.)

<Buyer's company name and address here> will directly purchase <Product Part Number Here> from BII.

BII (BENTHOWAVE INSTRUMENT INC.) will provide full warranty and technical support to the end user of BII products during the warranty period based on BII sales terms and conditions. After Warranty expired, BII will continue to provide technical support to end users of BII products. Information of **original buyer's name, BII's quote number (or invoice number), and BII product number** are necessary during technical support. Without this information, postponed and delayed support is expected, and BII might refuse to provide technical support.

This Authorization consist of one Pages, including all attachments.

This Authorization is valid till <input a date here>. Validity of one year ONLY starting from the issue date of this authorization.		
Authorization Signature:	Company: Benthowave Instrument Inc.	
Name:	Title: Sales Manager	
Date:	Telephone: BII Phone Number	Email: info@benthowave.com